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ON THE THEORY OF LINEAR NOISY SYSTEMS

ON THE THEORY OF LINEAR NOISY SYSTEMS

PROEFSCHRIFT

TER VERKRIJGING VAN DE GRAAD VAN DOCTOR IN DE TECHNISCHE WETENSCHAPPEN AAN DE TECHNISCHE HOGESCHOOL TE EINDHOVEN OP GEZAG VAN DE RECTOR MAGNIFICUS, DR. K. POSTHUMUS, HOOGLERAAR IN DE AFDELING DER SCHEIKUNDIGE TECHNOLOGIE, VOOR EEN COMMISSIE UIT DE SENAAT TE VERDEDIGEN OP DINSDAG 31 JANUARI 1967, DES NAMIDDAGS TE 4 UUR

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GEBOREN TE GROOTEGAST

DIT PROEFSCHRIFT IS GOEDGEKEURD DOOR DE PROMOTOR
PROF. DR. IR. A. A. Th. M. VAN TRIER

Aan Ir. G. de Vries

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1. INTRODUCTION

The notion that arbitrary fluctuations about the equilibrium state of a physical system exist is quite old. Although these fluctuations had been observed even earlier, as early as 1828 Brown published his classical account of what is now called the Brownian motion of a relatively large particle suspended in a fluid [1,2]). However, nearly a century was to elapse before the phenomenon was analyzed by Einstein [3,4]), Von Smoluchovski [5]), and others [6,7]), and confirmed experimentally by Perrin [8]). A good account of these early investigations has been given by De Haas-Lorentz [9]).

Despite the effort subsequently put into the subject by many physicists, from an engineering point of view it was considered merely curious and not very interesting. A major breakthrough occurred in the second and third decades of this century, when communication engineers reached the threshold beyond which small signals could not be detected easily, if at all. This led them to the investigation of what in general is now called noise. This field of scientific investigation therefore changed from being largely the province of physicists to becoming that of electrical engineers. In consequence a large part of modern noise theory is phrased in terms familiar to the electrical or electronics engineer, in which he states his problems and presents the solutions [10]). Voltage and current, impedance and admittance are the concepts, frequency analysis and Fourier transforms are the methods used to attack noise problems. Also new concepts to characterize noise properties of systems, such as noise factor or noise figure, available power, and equivalent resistance, have been introduced by electronics engineers to meet their specific problems. Although Nyquist [11]) used a physical argument to arrive at the quantitative explanation of the experiments performed by Johnson on thermal noise of a resistor [12]), his final results were worded in terms of an effective voltage.

Since the concept of noise far beyond its acoustic origin now includes all spurious and continuous random fluctuations of all kinds of physical systems, it will be clear that the electrical-noise terminology which paid off so much in electrical-noise problems, is not always adapted to provide answers as easily to similar questions in other fields. This is even true within the field of electricity and electronics itself. In many branches of these sciences voltages and currents are not the most appropriate variables to describe the phenomena. For example, the variables in which electronic-beam theories [13,14]) are described are most often density- and velocity-modulation waves. Radio propagation too is stated in terms of waves. Moreover, in wave-guiding systems one mostly resorts to incident and reflected waves, while the properties of components of such systems are expressed in reflection and transmission coefficients rather than in impedances or admittances.

More reasons exist why the general utility of electrically defined noise quantities may be questionable. During the last decade with the advent of new types of low-noise amplifiers, such as masers, parametric and tunnel-diode devices, a growing uneasiness about the general adequacy of conventional methods of noise description can be detected in the literature [15-17]. The difficulties are of two kinds.

A first problem of minor importance is concerned with the quantitative value of noise quantities [18]. The definitions of many conventional noise quantities, such as noise factor and effective input noise temperature, imply standard input terminations. In defining the noise factor the standard termination is thought to be passive and to have a uniform temperature of 290 °K. Very often such definitions are not well suited to practical situations.

A second and much more important difficulty is due to the fact that in a system noise and signal properties are intimately interwoven. The noise performance of a given component in a system depends strongly on the other components of the system [17]. To investigate such difficulties Haus and Adler [15] made a general analysis of the performance of a noisy linear n-port, imbedding it in a lossless transforming network. Varying the latter network, they investigated the power-density spectra of the noise quantities of the former. They defined a characteristic-noise matrix [19] and found that certain quantities, the eigenvalues of that matrix, are invariant under variations of the imbedding network. Furthermore, they showed that these eigenvalues are identical with the extreme values of the noise measure, a figure introduced by them [19] to characterize the noise of a 2-port. The smallest of the extreme values determines the optimum noise performance achievable with the n-port.

The object of the present investigation is to make a further contribution to a general, concise, and consistent theory of linear noisy systems. It should be applicable not only in electronics but in any field of investigation. We believe that one of the goals of such a theory must be that the noise properties and the signal properties of any multiport are represented separately. This formal separation of noise properties from signal properties ought to be independent of the specific variables which have been chosen to describe noise and signal phenomena in a system.

In the present treatise a mathematical formalism is developed for a general coordinate description of the exchange of noise and signal power between different parts, or components, of a system. Some care is taken to prove the consistency of the formalism with the laws of thermodynamics.

Prior to the formulation of the formalism in a general coordinate system a discussion with respect to a particular coordinate system is given, viz. with respect to the scattering-matrix representation of multiports. This is done for two reasons.

First, in theories concerning general coordinate systems it is convenient to

refer the general reference frame to a particular one and to give the formulas which govern the transformation from the particular to the general coordinate system. Then, starting from the particular reference frame, the representations and theorems can easily be given in the general one. For analogous reasons theories about spaces mostly start with a discussion of the Cartesian reference frame. Rather arbitrarily we have adopted the scattering-matrix representation as the primary coordinate system.

Secondly, as has been observed above, many results of noise theories and experiments are formulated exclusively in terms of impedance- or admittance-matrix representations. However, many systems cannot be described easily and lucidly in terms of those coordinate systems. Furthermore, many phenomena can be interpreted physically equally well or even much better in terms of waves than of voltages and currents.

We have developed a formalism which is general to the extent that a formal separation of intrinsic noise properties and signal properties of a multiport has been introduced. It will be shown that the power-density matrix of the primary noise coordinates of a multiport can be written as a product of two matrices. One of these matrices, the noise-temperature matrix, describes the intensity of internally excited noise. The eigenvalues of that temperature matrix are invariant to coordinate transformations. They can be given a clear and definite interpretation as the characteristic noise temperatures of the multiport. The other matrix, the noise-distribution matrix, depends on the signal properties of the multiport only. It governs the internal distribution of the internally generated noise power over the ports of the multiport. The division of the noise properties into two matrix factors is itself invariant to coordinate transformations. The noise-temperature matrix is transformed by a similarity transformation and the noise-distribution matrix by a Hermitian congruence transformation. Both these transformations involve the same transformation matrix: the noise-transformation matrix.

It turns out that many of our considerations run parallel with the work of Haus and Adler [15]) and deepen it at several places. To be more specific, the characteristic-noise matrix of Haus and Adler is equivalent with the noise-temperature matrix. The work done by the above authors on simultaneous diagonalization of noise and signal performance of a multiport is therefore reviewed. A different and more constructive proof of the existence of the simultaneous diagonalization is given. In particular it is shown that noise and signal performance cannot simultaneously be diagonalized when noise sources of both a dissipative part and an active part of a multiport are completely correlated. This situation occurs for example with triodes and transistors in which the noise is purely shot noise. In this exceptional case the externally manifest noise can be reduced to zero by means of a lossless feedback circuit without the signal performance being changed, at least over a small bandwidth.

If a multiport is described in a coordinate system where both signal and noise performance appear in diagonal form, the noise-temperature matrix can be interpreted clearly and definitely.

For considering noise phenomena two different fundamental methods exist. A first one is used when one is interested mainly in the behaviour of fluctuating quantities as functions of time. Calculations and observations are then concerned with the auto- and cross-correlation functions of the fluctuating quantities.

A second method is employed when one wishes to know the frequency response of a system and the amounts of noise power which, being contained in a small frequency band, on the average flow in that system. These average noise powers are expressed in the self- and cross-power-density spectra of the fluctuating variables. The relationship between the two methods is governed by the Wiener-Khintchine theorem. We shall not pay further attention to this subject because it has extensively been discussed elsewhere [20]). The present treatise is confined to the second method.

2. NOISE AND SIGNAL PERFORMANCE

2.1. Signal and noise waves

2.1.1. *Signals and noise*

Although it is not intended to give a lengthy discussion of the elementary meanings of signal and noise, at the very outset it should be made clear what we understand by these concepts. All time-dependent deviations from the static equilibrium state of a system will be called signals.

Very often signals are man-made. They then mostly contain intentionally induced information. In so far as the induced information is known in advance, signals can be thought to be predictable and to cohere in time. Because of the content of information such signals are of primordial interest. In the present treatise, however, we are not interested in the way information is contained. Neither will we investigate how information can be induced, handled, and extracted.

Noisy fluctuations about the equilibrium state occur at random. They cohere in time very poorly, i.e. they can be predicted over short times only. Noise does not contain useful deliberately induced information. It is true, noise contains some information about the amount of disorder in, and the physical nature of, a noisy system but we are not now interested in physical causes of noise. We are only interested how in linear systems energy carried by noisy disturbances is transported and distributed. More specifically, we are interested in the time-average values of these flows of noise power.

When a signal is periodic in time, Fourier analysis is a useful tool to analyze it and to determine its behaviour in a system. When a disturbance is not periodic in time, as is the case with most signals, in particular with noise, Fourier analysis can still be applied formally by means of a limiting process for periods tending to infinity. As a result any signal, whether it is noise or not, can be decomposed into a set of frequency components. Each of the components can be described by a frequency, an amplitude, and a phase.

An intentionally induced signal is in principle known at all times, at least at some place in a system. Hence, frequencies, amplitudes, and phases of its components can be determined. This is not the case with noise. Over long intervals of time amplitudes and phases of the components of a noisy disturbance vary at random. It is this fundamental randomness which — in the frequency domain — distinguishes noise from signals proper.

In a noisy system noise components are present in any given band of frequencies. The spectral distribution of these components is in general infinitely dense. As a group the components transmit an amount of power. This power can be averaged over time. Since we will be interested in stationary

systems only, we assume that this average power does not vary in time. It is the transmission in a restricted bandwidth of time-average power which will be the main subject of this treatise. This will be so because the ratio of the average power of noise to the power of an intentionally induced signal constitutes the main measure of the hindrance noise puts up to the detection of wanted signals.

2.1.2. *Signals and signal representations*

To know the signal performance of a system and, more specifically, of the components of a system it suffices to determine the behaviour of a harmonic signal component in that system. At any particular frequency the stationary signal state of a linear network can be described by a set of variables. The network may be electrical, acoustical, etc. The variables may or may not be interpretable physical quantities. Mostly, they are present in pairs at the entrances or ports of the network, but it is also possible that they are linear combinations of quantities at different ports. A network with n ports shall be called an n-port or, more generally, a multiport.

Depending on the system under consideration, several choices of physical quantities can be made for the variables, e.g. voltages and currents, pressures and velocities, incident and reflected electromagnetic-wave amplitudes, etc. Each choice determines a particular reference frame for the signal state.

At each frequency the signal performance of a multiport can be described by a set of simultaneous linear algebraic equations. The equations relate the above variables expressing half of them in terms of the other half. Thus, the variables are divided into dependent and independent ones. For each choice of variables this partition can still be made in various different ways.

It will be clear that the elements of the coefficient matrix of the above set of equations are determined by the multiport, and vice versa. Depending on the choice of variables and on the partition in dependent and independent ones, one is led to the concept of impedance matrix, transfer matrix, scattering matrix, and so on. These matrices describe the signal performance of the multiport. We therefore call them signal-performance matrices.

A particular choice and division of variables leads to what we call a signal representation or coordinate system of multiports. Different representations are related by linear transformations. For 2-terminal pair networks such transformations have been described by Belevitch [21]). Squabbling about a particular choice of representation may be a rather futile occupation. However, in a given situation one choice may be more advantageous than another [22]) (cf. ch. 1). In the present treatise we start rather arbitrarily from the scattered-wave formalism.

In a system the components, i.e. the composing multiports, are thought to be interconnected by homogeneous lossless transmission paths. The lengths of

these paths can be small or even zero. In the following, transmission paths will be called lines for short *).

When in a linear system signals propagate along lines, one can speak about waves in terms of voltage, velocity, electric-field intensity, and so on. As has been remarked above, when investigating the signal performance of multiports, we can confine the observations to waves which are harmonic in time. In fact, the considerations will be restricted to travelling harmonic waves. For any travelling wave we can define a "complex amplitude" which determines both amplitude and phase. Hereafter, that complex quantity will be called the wave amplitude or, simply, the amplitude for short.

The modulus of the wave amplitude is related to the power carried by the wave along the line. In general, lines are characterized by wave impedances. The need to carry these quantities along through all computations can be circumvented by having the wave amplitudes normalized with respect to them [23,24]). Then, if P be the transmitted power and a the wave amplitude, the equation

$$P = a\,a^* \tag{2.1}$$

can be defined, where the asterisk denotes the complex conjugate. It should be noted that, when eq. (2.1) is defined in this way, the magnitude of a is equal to what conventionally is called the normalized effective value of the amplitude of the wave. Since wave impedances may be different for different lines incorporated in a system, normalization must be carried out for each line separately [23,24]).

The argument of the wave amplitude determines the phase of the wave. The phase depends on the choice of time origin and on the point of observation in the line. The dependence on the origin of time implies merely an additive constant. Knowledge of phases is of importance only for the evaluation of interference effects which occur when two or more waves of the same frequency are present in the same line travelling in the same direction. Then one is in essence interested in phase differences. As the additive constants are the same for all waves of the same frequency, they are not at all important and we may forget about the choice of time origin.

The dependence of the phase on the position in a line is of more concern. In order to relate the phases a wave has at different positions, knowledge of the propagation constant, or the wave length, of the line at the frequency of the wave is indispensable. Let $a(l_1)$ and $a(l_2)$ be the wave amplitudes at the positions l_1 and l_2, respectively, and let $\beta = 2\pi/\lambda$ be the propagation constant.

*) Introduction of the word line for the general concept of transmission path does not imply that the results which will be obtained are valid for electrical systems only, for which systems the word line or transmission line is commonly used. In fact, the study made here applies to all kinds of linear systems whatever their physical nature may be. If inhomogeneous, lossy, or anomalously dispersive, lines are incorporated in a system, they can better be considered as 2-ports than as lines.

The relation between $a(l_1)$ and $a(l_2)$ is then given by *)

$$a(l_2) = \exp\{i\beta(l_2 - l_1)\}\, a(l_1). \tag{2.2}$$

In a line waves can propagate in two directions. Therefore, to cover the complete signal state of a system at a particular frequency two waves must be given in every line; one for each direction.

Sometimes it is possible to give a linearized matrix description of systems in which interaction between signals of different frequencies plays a dominant role, such as mixers and parametric devices [25,26]). In such systems various waves of different frequencies may propagate through the same transmission path. Furthermore, in the same physical transmission structure different waves of the same frequency may propagate in the same direction but in different modes. In both cases one can assign to each frequency band and to each mode its own pair of oppositely directed waves. One may consider the one physical transmission path as being split up into a number of lines [23,24]).

2.1.3. *Spot-noise theory*

In sec. 2.1.1 it has been stated that we are interested in the transmission of time-average noise power contained in a restricted bandwidth. There it has been assumed that through any line of a linear noisy system and contained in any small but finite frequency band a finite amount of noise power is transmitted in each direction. In actual systems these partial flows of noise power can be measured and averaged over time if sufficiently long time intervals are available. If the bandwidth is taken smaller the interval of time over which the averaging is performed must be longer.

Let us consider an arbitrarily small bandwidth δf at a frequency f. Let δP be the time-average power contained in δf and transmitted through a line in a given direction. For the flow of noise power we define a power density $G(f)$ at f by

$$G(f) = \delta P/\delta f. \tag{2.3}$$

In general, $G(f)$ is a function of f. If $G(f)$ does not depend on f, the noise is called white.

All relevant quantities in a system are functions of frequency. To circumvent dependence on the frequency of signal performances of multiports and to avoid dispersion effects in lines the bandwidth δf can be taken so small that all important quantities can be considered constant over it. Under this restriction one speaks of "spot noise". In fact, the major part of the present treatise will be

*) In the wave amplitude the omitted time factor is thought to be $\exp(-i2\pi ft)$. For waves which travel in the direction of increasing values of l the wave number β is positive and for opposite waves negative. Below, when defining the concept of noise wave, the validity of the phase relation, eq. (2.2), will be crucial.

confined to spot-noise theory. A discussion of the conditions under which spot-noise theory is valid will be postponed until a later section (sec. 2.2.2).

2.1.4. *The concept of noise waves*

In association with flows of time-average noise power we will now introduce quantities which are defined sufficiently well to make a consistent spot-noise theory possible but not more precisely than is needed for that purpose. The quantities must offer the possibility of evaluating the influence of the multiports of a system on the flows of average noise power in that system. Hence, interference effects and, therefore, correlation of two or more flows of noise power must be coped with.

In sec. 2.1.2 it has been shown that a harmonic signal wave can be represented by a complex number. In the same way we will now represent a noisy disturbance at a frequency f by an element A of a Hilbert space.

Since in what follows we only need some main axioms and theorems pertaining to Hilbert spaces, we will be satisfied by mentioning only those properties which apply to our considerations *).

For our purposes Hilbert space can be described as an infinite-dimensional complex linear space in which a positive definite inner product of any pair of two elements is defined. The elements can be considered to be infinite-dimensional vectors.

With any two elements A and B the sum, denoted by $(A + B)$, is defined and is again an element. The summation is commutative and associative.

The multiplication of an element A by a complex number λ is defined and gives again an element. This product is denoted by λA. It is commutative, associative with respect to complex numbers, i.e.

$$(\lambda\mu)\, A = \lambda\, (\mu A), \tag{2.4}$$

and distributive with respect to both complex numbers and elements of Hilbert space, i.e.

$$(\lambda + \mu)\, A = \lambda A + \mu A, \tag{2.5}$$

$$\lambda\, (A + B) = \lambda A + \lambda B. \tag{2.6}$$

It is this product operation which makes a Hilbert space a complex linear space, in which a zero element O is defined.

The inner-product operation is obtained by subjoining to any pair of elements A and B a complex number $\{A, B\}$. The subjunction is defined as a positive definite Hermitian bilinear form. Hence we have

$$\{B, A\} = \{A, B\}^*, \tag{2.7}$$

*) For complete analyses of Hilbert spaces we may refer to appropriate textbooks. The properties we need can be found in literature on linear algebra such as refs 27 and 28.

$$\{\lambda A, \mu B\} = \lambda^* \mu \{A, B\} \tag{2.8}$$

and

$$\{A, B + C\} = \{A, B\} + \{A, C\}. \tag{2.9}$$

From eq. (2.7) it is clear that in general the inner product is not commutative.

In connection with a Hilbert space of elements A a dual Hilbert space of conjugate elements A^+ can be defined [26,29]). The superscript plus indicates the conjugation. We need not define the conjugation in detail but it suffices to observe that it is a one-to-one correspondence between the conjugate elements A and A^+ of the dual spaces, which is reciprocal and antilinear. We have

$$(A^+)^+ = A, \tag{2.10}$$

$$(A + B)^+ = A^+ + B^+, \tag{2.11}$$

$$(\lambda A)^+ = \lambda^* A^+. \tag{2.12}$$

We further define a commutative product operation of an element A and a dual element B^+ by *) [29])

$$AB^+ = B^+ A = \{B, A\}. \tag{2.13}$$

Due to eq. (2.7), we then have **)

$$A^+ B = (AB^+)^*. \tag{2.14}$$

From eq. (2.9) it is readily seen that the product operation is distributive. Since the product is concerned essentially with two elements, one cannot speak about it as associative ***). It is easy to verify that eqs (2.4)-(2.14) are consistent.

When the product AB^+ is equal to zero, the elements A and B are called orthogonal. In the dual space the conjugates A^+ and B^+ are then orthogonal, too.

From eq. (2.14) we obtain

$$(A^+ A)^* = A^+ A. \tag{2.15}$$

The product of an element and its own conjugate is a real number. Since the bilinear form has been defined above to be positive definite, that number is non-negative. For all elements A the relation

$$A^+ A \geqslant 0 \tag{2.16}$$

---

*) It should be noted that a product AB of two elements A and B is in general not defined.

**) In the dual Hilbert space an inner product of two elements A^+ and B^+ is given by $\{A^+, B^+\} = AB^+ = B^+ A$. From eq. (2.14) it is seen that $\{A^+, B^+\} = \{A, B\}^* = \{B, A\}$. This equation makes the one-to-one correspondence between the dual spaces an isomorphism [28]). This means that in abstracto the dual spaces are identical. It is not necessary that the identity is also valid in concreto. When that, too, is the case, one speaks of an automorphism and, then, the conjugation maps the space onto itself.

***) It should be noted that $(A^+ B)\, C^+$ is in general not equal to $A^+(BC^+)$. A necessary and sufficient condition for these expressions to be identical is that A and C are proportional, i.e. $C = \lambda A$. This is easy to prove.

is valid where equality applies only when A is the zero element O. The positive square root of $\mathsf{A}^+\mathsf{A}$ is called the norm of A. This quantity is reminiscent of the absolute value of a complex number and of the magnitude of a real vector. It is the concept of a norm, i.e. the content of eq. (2.16), which provides the key to the representation of average noise-power flows by elements of Hilbert space introduced above.

At the frequency f an element A is now subjoined to the power density $G(f)$, as it has been defined by eq. (2.3), such that

$$\mathsf{A}^+\mathsf{A} = \mathsf{A}\,\mathsf{A}^+ = G(f). \tag{2.17}$$

Combining this with eq. (2.3), we obtain the basic relation

$$\delta P = \mathsf{A}\,\mathsf{A}^+\delta f. \tag{2.18}$$

Since power is inherently non-negative, the subjunction of A to $G(f)$ and δP in this way justifies the restriction to positive definite inner products as has been made above.

We will call the element A amplitude function or noise-wave amplitude. The term amplitude is used because the norm of A is reminiscent of the (real) amplitude of a harmonic signal wave. There exists a close resemblance to quantities which in conventional noise theories are used to indicate "amplitudes" of noisy disturbances. This can best be made clear perhaps by means of an example. The amplitude of a harmonic signal wave can be expressed in terms of a voltage v which is normalized with respect to the characteristic impedance of the line the wave is travelling along (cf. sec. 2.1.2). For a flow of noise power through that line one often defines a mean-square value of an equivalent voltage by [10,11])

$$\overline{vv^*} = \delta P = G(f)\,\delta f. \tag{2.19}$$

The bar denotes the time-average value. Due to eqs (2.18) and (2.19), we have

$$\overline{vv^*} = \mathsf{A}\,\mathsf{A}^+\delta f, \tag{2.20}$$

from which the correspondence can be seen. While $\overline{vv^*}$ depends on the magnitude of δf, $\mathsf{A}\,\mathsf{A}^+$ does not. The conjugate A^+ on the right-hand side finds its counterpart in the complex conjugate v^* on the left-hand side. The complex quantity is often interpreted as the (complex) amplitude of a harmonic signal wave which is equivalent to the flow of noise power contained in δf. This equivalence means only that on the average the signal wave and the noisy disturbance carry the same amount of power. It does not mean that the noisy disturbance can be associated with a phase equal to the argument of v and that two noisy disturbances v_1 and v_2 interact in the same way as two harmonic waves v_1 and v_2.

The term function is used to emphasize that at different frequencies a noisy disturbance is represented by different A's.

In what follows A will be considered to be the amplitude of a wave. At the frequency f this abstract wave, the noise wave, represents the flow of noise power. For that reason the term noise-wave amplitude has been introduced above. Henceforth a noise wave will be represented by the corresponding amplitude function, its amplitude *). The non-negative real number $A\,A^+$ is called the power of the noise wave **).

The above formalism being adopted, the phase equation, eq. (2.2), should also apply to noise waves. Therefore, we assume that, when $A(l_1)$ and $A(l_2)$ are the amplitudes of the same noise wave at two different positions l_1 and l_2 in a line, the equation

$$A(l_2) = \exp\{i\beta\,(l_2 - l_1)\}\,A(l_1) \tag{2.21}$$

is valid. This assumption does not violate any one of the above definitions and equations. In particular, by virtue of eqs (2.8) and (2.21), subsequent substitution of $A(l_2)$ and $A(l_1)$ into eq. (2.18) yields

$$\delta P(l_2) = A(l_2)\,A^+(l_2)\,\delta f = A(l_1)\,A^+(l_1)\,\delta f = \delta P(l_1). \tag{2.22}$$

At both places l_1 and l_2 the average powers contained in δf are the same as they should be in one line in a linear system.

Furthermore, we assume that a noise wave incident upon a linear multiport is reflected, transmitted, and absorbed by it in the same way as a harmonic signal wave is scattered. In analogy to the concept of signal state we can speak of the noise state of an n-port. It is described by a set of $2n$ noise waves, two at each port; an incident and a reflected one. This assumption implies that the $2n$ noise waves are related by the same set of simultaneous linear equations with complex coefficients as holds for the harmonic signal waves which describe the signal state. That this assumption is justified will become clear in subsequent sections where it is shown not to lead to contradictions with respect to the laws of thermodynamics.

When we compare the complex amplitude a of a harmonic signal wave and the amplitude function A of a noise wave, we see that both include the concept of a real effective amplitude, i.e. the magnitude and the norm, respectively. On the other hand, whereas a uniquely determines a phase, i.e. the argument, an equivalent quantity is not contained in A. Noise waves cannot be associated with definite phases.

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*) It should be kept in mind that, since noise-wave amplitudes or, briefly, noise waves depend on frequency, relations between them can exist only if they all refer to the same frequency.

**) It has not the dimension of power but of power per unit bandwidth. It is a power density.

2.1.5. *Correlation of noise waves*

An important aspect of noise has not yet been considered. Two or more flows of noise power can be correlated. They may depend on each other.

In a strictly formal manner one of two correlated noisy disturbances in δf can be thought to consist of two parts. At all times the amplitude distribution of one part equals that of the other noisy disturbance except for a complex factor which in general depends on frequency when a finite bandwidth is considered. The amplitude distribution of the other part has no relations whatsoever. The dependent part of a noise-power flow and the noisy disturbance it depends on are said to cohere in time, to be completely correlated. The other part does neither depend on the former part nor on the second disturbance. It is called independent or uncorrelated.

The formal division of a correlated flow of noise power into a completely correlated and an uncorrelated part is arbitrary. That particular model suggests that part of a correlated noisy disturbance originates from the same physical noise source the other noisy disturbance is emanating from, while the other part, the uncorrelated portion, is emitted by a completely different source. This model is arbitrary of course. The division can be made in a great many different ways. If nothing is known about physical sources, as often is the case, no criterion exists to make such a division in agreement with whatever physical reality. For our purposes it would not be very interesting either.

The argument of the complex factor which relates completely correlated noisy disturbances refers to the phase difference, or relative phase, of those disturbances. It can be thought to be caused by the difference in phase lengths of the covered transmission paths.

When two or more noisy disturbances in δf are simultaneously present in the same line and travelling in the same direction, the power of the resultant disturbance need not simply be the sum of the powers of the constituent disturbances. In fact, that would be so when the constituent disturbances were mutually uncorrelated, and under very specific conditions only when they are correlated. When two correlated noisy disturbances propagate simultaneously along the same line, the resultant power is affected by interference of the coherent parts. The interference can be destructive, resulting in a power which is less than the sum, or constructive, causing a larger resultant power. Whether the interference is destructive or constructive depends on the relative phases of the coherent parts *).

*) This discussion reflects the fact that two synchronous sources which radiate into the same region of space are coupled by means of their radiation fields. Depending on the relative phase, that coupling forces the sources to radiate less or more power than would be the case when they were solitary. We need not consider these radiation-field problems in detail. As will become clear, the formalism of waves, introduced above, and of noise-wave sources, to be introduced in the next section, is consistent with these phenomena and, in fact, takes care of them by means of interference calculations. At least this is true as long as we are interested in flows of power only.

From the above discussion it will be clear that, to cover interference effects, further restrictions must be put on the bandwidth δf over which the formalism can be applied. The bandwidth δf must be so small that the relative phases of completely correlated disturbances or parts of disturbances are the same or nearly the same for all frequencies in δf. The interference effects are then the same also for all frequencies. A further discussion of the conditions for δf will be given in secs 2.2.2 and 2.4.6.

Let A and B be two simultaneous noise waves at a frequency f. We define the resultant noise wave W as the sum of A and B, i.e.

$$W = A + B. \tag{2.23}$$

When A and B are uncorrelated, the resultant power δP_W contained in δf is equal to the sum of the powers δP_A and δP_B of the two independent noisy disturbances contained in δf. Hence, by virtue of eq. (2.18), we have

$$W W^+ = A A^+ + B B^+. \tag{2.24}$$

On the other hand evaluation of $W W^+$ from eq. (2.23) yields

$$W W^+ = A A^+ + B B^+ + A B^+ + B A^+. \tag{2.25}$$

Comparing these two results for $W W^+$, we find that for uncorrelated noise waves A and B the equation

$$A B^+ + B A^+ = 0 \tag{2.26}$$

must apply. However, as will be seen below, this equation is not yet sufficient to define non-correlation of A and B, since under special conditions it also applies to correlated noise waves A and B. We rather define two noise waves A and B to be uncorrelated if the inner product $A B^+$ and, hence, the complex conjugate product $B A^+$ are equal to zero, i.e.

$$A B^+ = B A^+ = 0. \tag{2.27}$$

Two uncorrelated noise waves are orthogonal, and vice versa.

When A and B are correlated, eq. (2.26) does not apply in general. Let A consist of two parts A' and A'', i.e.

$$A = A' + A'', \tag{2.28}$$

such that A' is completely correlated with B, and A'' not at all. We assume that A' can be written in the form

$$A' = D B, \tag{2.29}$$

where D is a complex number. Since A'' is uncorrelated with and, hence, orthogonal to B, it is so with respect to A'. From these observations we obtain

$$A\,B^{+} = (A' + A'')\,B^{+} = D\,B\,B^{+}. \tag{2.30}$$

In order to specify $A\,B^{+}$ in more detail we now proceed as follows. The power of A is equal to the sum of the powers of A' and A'', i.e.

$$A\,A^{+} = (A' + A'')\,(A'^{+} + A''^{+}) = A'A'^{+} + A''A''^{+}. \tag{2.31}$$

The terms on the right-hand side can be written as fractions of $A\,A^{+}$, i.e.

$$A'A'^{+} = CC^{*}A\,A^{+}, \tag{2.32}$$

$$A''A''^{+} = (1 - CC^{*})\,A\,A^{+} \tag{2.33}$$

with, since both these powers are non-negative,

$$0 \leqslant CC^{*} \leqslant 1. \tag{2.34}$$

From eqs (2.29) and (2.32) we obtain *)

$$A'A'^{+} = CC^{*}A\,A^{+} = DD^{*}B\,B^{+}. \tag{2.35}$$

The argument of C, which has not yet been given a fixed meaning, can still be chosen freely. We define it to be equal to that of D, i.e.

$$\arg(C) = \arg(D). \tag{2.36}$$

From eq. (2.35) we then obtain

$$C = D\,(B\,B^{+}/A\,A^{+})^{1/2}. \tag{2.37}$$

Elimination of D from eqs (2.30) and (2.37) yields

$$C = A\,B^{+}/(A\,A^{+}\,.\,B\,B^{+})^{1/2}. \tag{2.38}$$

The arguments of C, D, and $A\,B^{+}$ are thus identical to each other.

Conversely, if B is split up into two parts B' and B'', where B' is completely correlated with A, and B'' is not, it is easy to show that the equations

$$B'B'^{+} = CC^{*}B\,B^{+} \tag{2.39}$$

and

$$B''B''^{+} = (1 - CC^{*})\,B\,B^{+} \tag{2.40}$$

hold.

The complex number C will be called correlation factor [10]). Together with $A\,B^{+}$ it can be interpreted as follows. The arguments of these quantities can be described as the relative phase of the completely correlated parts. The modulus of $A\,B^{+}$ relative to the square root of the product of $A\,A^{+}$ and $B\,B^{+}$ is equal to that of C. This relative magnitude of $A\,B^{+}$ refers to that part of one of

*) It should be noted that, although it seems plausible in analogy with eq. (2.29), it is not permitted to write $A' = C\,A$. This equation would imply that A' is completely correlated with A which is not the case.

two noise waves A and B, which is completely correlated with the other noise wave.

From eqs (2.33) and (2.40) it is clear that A and B are completely correlated when $C\,C^*$ is equal to unity, that is when

$$A\,B^+ \,.\, A^+B = A\,A^+ \,.\, B\,B^+ \tag{2.41}$$

is satisfied. This can be so only if A and B are collinear, i.e.

$$A = D\,B. \tag{2.42}$$

Substitution of eq. (2.38) in eq. (2.25) yields

$$W\,W^+ = A\,A^+ + B\,B^+ + (C + C^*)\,(A\,A^+ \,.\, B\,B^+)^{1/2}. \tag{2.43}$$

The resultant power depends not only on the magnitude of C, but also strongly on its argument. The relative phase of the completely correlated parts of A and B plays its important role through interference. If that relative phase is equal to $\pm\,\pi/2$, the last term on the right-hand side of eq. (2.43) is equal to zero. That equation then reduces to eq. (2.24), which in this special case holds also for correlated noise waves. It will therefore now be clear that, above, eq. (2.26) could not have been used for defining the inner product of correlated noise waves, but that recourse had to be taken to eq. (2.27).

2.1.6. *Noise-wave sources*

Any noisy 1-port radiates noise power into the line it is connected to. Further along that line part of the power, or even the total power, can be reflected. When the 1-port is not matched to the line, the reflected power is thereupon again reflected, partly or completely, by the 1-port, and so on. Multiple reflections may occur.

The net primary noise power emitted by the 1-port is called the primary noise power of the 1-port. It can be defined more precisely as the net noise power injected by the 1-port into an infinitely long, or reflectionlessly terminated, line which is at absolute zero temperature.

The condition of absolute zero temperature is imposed to exclude noise power radiated by the line and its termination. Despite the conditions on temperature and matching of line and termination, the adjective "net" is included in the definition to avoid such questions as whether the zero-point energy of the combination of 1-port and line must be considered as flowing back and forth between 1-port and line or not. In fact, in the present treatise zero-point energy will not be considered.

The signal performance of a 1-port is determined at a reference plane in the line connected to the 1-port. The position of the reference plane is rather arbitrary. If the scattering-matrix signal representation is used to describe multiports, the signal performance of a 1-port is uniquely determined by a

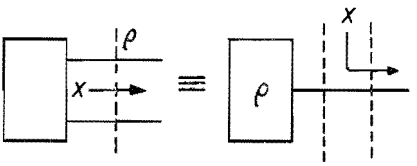


Fig. 1. Noisy 1-port. The emitted noise can be represented by a noise-wave source. The 1-port and the noise-wave source can be considered separately.

reflection coefficient ϱ at the reference plane. This quantity depends not only on the properties of the 1-port itself but also on the wave impedance of the line (cf. sec. 2.1.2) and on the position of the reference plane.

At any frequency the primary flow of noise power can be represented by a noise wave: the primary noise wave. We now define a noise-wave source *) as the source which emits the primary noise wave into the line. This source is located at the reference plane. It will be represented by the amplitude of the primary noise wave.

A formal separation of signal and noise properties of noisy 1-ports can now be achieved. A noisy 1-port can be considered as consisting of an equivalent noiseless 1-port with the same reflection coefficient and of a noise-wave source at the reference plane. This is illustrated in fig. 1 where a symbol for the noise-wave source is used as introduced by Bauer and Rothe [31]). Such a formal separation of signal performance and noise properties is well known in noise literature [32–34]). It affords the possibility to evaluate separately the signal state of, and the noise-power exchange in, a linear system.

2.2. Noise-wave equations

2.2.1. *Two 1-ports connected by a lossless line*

Within a small band δf exchange of noise power between two 1-ports connected to the two ends of a lossless line of finite length can be calculated by means of algebraic equations. In fig. 2 a sketch of the configuration is given.

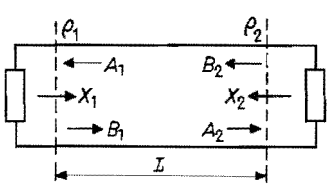


Fig. 2. Two passive 1-ports connected by a line of finite length. When the bandwidth is small, noise-wave equations can be formulated.

*) The concept of noise-wave source is related to the concept of equivalent wave source (Ersatzwellenquelle), which has been introduced by Butterweck [30]) when formulating Thévenin's theorem in terms of the scattered-wave formalism.

At their respective reference planes the 1-ports are described by reflection coefficients ϱ_1 and ϱ_2 and noise-wave sources X_1 and X_2. We assume the noise of the 1-ports to be mutually independent. Hence, due to eq. (2.27) we have

$$X_1 X_2^+ = 0. \tag{2.44}$$

Let A_1 and A_2 be the noise waves which are incident from the line upon the reference planes and B_1 and B_2 the noise waves which at the reference planes propagate into the line. If L is the length of the line, by virtue of eq. (2.21), the phase relations

$$A_2 = \exp(i\beta L)\, B_1 \tag{2.45}$$

and

$$B_2 = \exp(-i\beta L)\, A_1 \tag{2.46}$$

are valid.

At the reference planes, we can formulate the relations

$$B_1 = X_1 + \varrho_1 A_1 \tag{2.47}$$

and

$$B_2 = X_2 + \varrho_2 A_2. \tag{2.48}$$

These equations will be called the noise-wave equations of the configuration. The noise waves A_1 and B_1 — or A_2 and B_2 — can be considered as the unknowns. Substitution of eqs (2.45) and (2.46) yields

$$-\varrho_1 A_1 + B_1 = X_1, \tag{2.49}$$

$$A_1 - \varrho_2 \exp(i\,2\beta L)\, B_1 = \exp(i\beta L)\, X_2. \tag{2.50}$$

If the condition

$$\varrho_1 \varrho_2 \exp(i\,2\beta L) \neq 1 \tag{2.51}$$

is satisfied, A_1 and B_1 can be solved in terms of X_1 and X_2. When this condition is not fulfilled, the system is resonant at the frequency f. Apart from the flow of noise power a harmonic-signal state with an arbitrary amplitude can then exist. When only power is measured over δf, the harmonic signal cannot be distinguished from the noise. Hence, in this "resonant" case the noise-wave solution is not unique. In practice, however, the addition of a small amount of attenuation — in the line for instance — removes the singularity so that this case may be ignored.

The solutions for A_1 and B_1 are given by

$$A_1 = \frac{\exp(i\beta L)\, X_2 + \varrho_2 \exp(i\,2\beta L)\, X_1}{1 - \varrho_1 \varrho_2 \exp(i\,2\beta L)} \tag{2.52}$$

and

$$B_1 = \frac{X_1 + \varrho_1 \exp(i\beta L)\, X_2}{1 - \varrho_1 \varrho_2 \exp(i\,2\beta L)}. \tag{2.53}$$

By virtue of eqs (2.18) and (2.44), the noise powers transmitted through the line respectively to the left and to the right and contained in δf are found to be

$$A_1A_1^+\delta f = \frac{(X_2X_2^+ + \varrho_2\varrho_2^*\, X_1X_1^+)\,\delta f}{\{1 - \varrho_1\varrho_2 \exp(i\,2\beta L)\}\{1 - \varrho_1^*\varrho_2^* \exp(-i\,2\beta L)\}} \tag{2.54}$$

and

$$B_1B_1^+\delta f = \frac{(X_1X_1^+ + \varrho_1\varrho_1^*\, X_2X_2^+)\,\delta f}{\{1 - \varrho_1\varrho_2 \exp(i\,2\beta L)\}\{1 - \varrho_1^*\varrho_2^* \exp(-i\,2\beta L)\}}. \tag{2.55}$$

Since the primary amounts of noise power which are emitted into the line by the 1-ports are respectively $X_1X_1^+\delta f$ and $X_2X_2^+\delta f$, the portion of the primary power of the first 1-port, which is dissipated in the second, is given by

$$\delta P_{21} = \frac{(1 - \varrho_2\varrho_2^*)\, X_1X_1^+\,\delta f}{\{1 - \varrho_1\varrho_2 \exp(i\,2\beta L)\}\{1 - \varrho_1^*\varrho_2^* \exp(-i\,2\beta L)\}}. \tag{2.56}$$

The denominator of the quotient expresses the magnification of the primary power by multiple reflections. The numerator represents that part of the magnified power which is absorbed in the second 1-port. The remainder of the primary power emitted by the first 1-port is absorbed again in that 1-port. This portion is given by

$$\delta P_{11} = \left\{1 - \frac{1 - \varrho_2\varrho_2^*}{\{1 - \varrho_1\varrho_2 \exp(i\,2\beta L)\}\{1 - \varrho_1^*\varrho_2^* \exp(-i\,2\beta L)\}}\right\} X_1X_1\delta f. \tag{2.57}$$

The distribution of the primary power of the second 1-port over the two 1-ports can be evaluated analogously. The same result can be obtained by exchange of the subscripts 1 and 2 in the preceding equations.

It should be noted that the power-magnification factor

$$[\{1 - \varrho_1\varrho_2 \exp(i2\beta L)\}\{1 - \varrho_1^*\varrho_2^* \exp(-i2\beta L)\}]^{-1}$$

is symmetrical in the subscripts 1 and 2. Hence, the magnification is the same for the two primary powers.

The results of the present section are also valid for reflection coefficients with $|\varrho| > 1$. It signifies that the formalism can also be applied to 1-port amplifiers, such as negative-resistance devices.

2.2.2. *Bandwidth condition*

In order that eqs (2.45) and (2.46) hold over δf, it is necessary that the condition

$$L\,\delta\beta \ll 2\pi \tag{2.58}$$

is satisfied, where $\delta\beta$ is the increase of β over δf. Since the effects of multiple reflections have been considered implicitly above, a still better condition

would be

$$\nu L\,\delta\beta \ll 2\pi, \tag{2.59}$$

where ν is the number of significant subsequent reflections. In sec. 2.4.6 the meaning of ν is illustrated with an example. It is also shown there how ν can be estimated. It now suffices to observe that ν strongly depends on the magnitude of ϱ_1 and ϱ_2. This can be seen from the results of the previous section. Variations of β over δf make themselves manifest only through the magnification factor. The effect of variations of β is the stronger the larger the magnitude of the product $\varrho_1\varrho_2$. On the other hand, multiple reflections persist longer, that is ν is larger, when the magnitude of $\varrho_1\varrho_2$ is larger.

If δf is small, $\delta\beta$ is given by

$$\delta\beta = (2\pi/v_g)\,\delta f, \tag{2.60}$$

where v_g is the group velocity in the line at the frequency f. Hence, eq. (2.59) can be written in the form

$$\delta f \ll 1/\nu\tau, \tag{2.61}$$

where the transit time τ is given by

$$\tau = L/v_g. \tag{2.62}$$

Beyond the reference plane a 1-port may contain one or more long sections of line. If the above restriction for δf includes internal transit times, too, this does not alter the formalism since internal delays are then accounted for. To state it more precisely, the bandwidth δf must be chosen very small with respect to the inverse of the longest total transit time of the complete system, the number of significant subsequent multiple reflections being included. Throughout the present treatise δf will be assumed to have been chosen in accordance with this condition.

2.2.3. *Two 1-ports connected directly*

In practice the elements of a system are often connected to each other without the intervention of lines. In the limit of a vanishingly short line the two reference planes are identical and so are the noise waves A_1 and B_2 and the noise waves B_1 and A_2. In fig. 3 this has been illustrated. The identical waves are denoted by

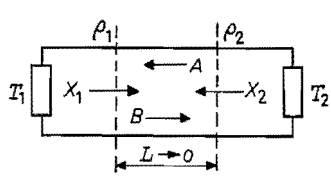


Fig. 3. Two passive 1-ports, at temperatures T_1 and T_2, connected directly.

respectively A and B. In the configuration of sec. 2.2.1 the line can also be thought to form an integral part of one of the two 1-ports. If, for example, the line belongs to the second 1-port, the reflection coefficient and the noise-wave source of that 1-port are $\varrho_2 \exp(i2\beta L)$ and $\exp(i\beta L)\, X_2$, respectively.

For the vanishingly short line the noise-wave equations can be derived from eqs (2.49) and (2.50) for L tending to zero. They can also be formulated at once with the help of fig. 3. In either way, we have

$$-\varrho_1 A + B = X_1 \tag{2.63}$$

and

$$A - \varrho_2 B = X_2. \tag{2.64}$$

All relevant quantities can be evaluated from the corresponding results of sec. 2.2.1 for L tending to zero or, alternatively, directly from the solution of the above equations. For uncorrelated noise-wave sources X_1 and X_2 the powers of the noise waves A and B are found to be respectively given by

$$A A^+ = \frac{X_2 X_2^+ + \varrho_2 \varrho_2^* X_1 X_1^+}{(1 - \varrho_1 \varrho_2)(1 - \varrho_1^* \varrho_2^*)} \tag{2.65}$$

and

$$B B^+ = \frac{X_1 X_1^+ + \varrho_1 \varrho_1^* X_2 X_2^+}{(1 - \varrho_1 \varrho_2)(1 - \varrho_1^* \varrho_2^*)}. \tag{2.66}$$

The product $A B^+$, given by

$$A B^+ = \frac{\varrho_2 X_1 X_1^+ + \varrho_1^* X_2 X_2^+}{(1 - \varrho_1 \varrho_2)(1 - \varrho_1^* \varrho_2^*)}, \tag{2.67}$$

expresses the correlation of A and B. Only in the case of both ϱ_1 and ϱ_2 being equal to zero, is the cross-product equal to zero and are the noise waves uncorrelated. Indeed, then, A and B have no source in common.

Henceforth, we will use the direct method to formulate noise-wave equations which are valid at the common reference planes at the interconnections of system components.

2.3. Thermal noise

2.3.1. *Consequences of the laws of thermodynamics*

A 1-port is called passive if the power reflected — or re-radiated — by it is not larger than the power incident upon it and if it does not radiate harmonic-signal power spontaneously. The reflection coefficient ϱ of a passive 1-port must satisfy the condition

$$\varrho \varrho^* \leqslant 1. \tag{2.68}$$

If the equality symbol applies, the 1-port reflects all incident power. Then it is

lossless and called reactive *). In all other cases the incident power is absorbed partly or completely. The 1-port is then lossy and called dissipative. Those 1-ports which do not satisfy eq. (2.68) reflect more power than is incident upon them. They are called active. In order that the first law of thermodynamics be not violated, active 1-ports must contain power sources. Two types of active 1-ports can be distinguished. The power sources are either spontaneous (generators) or they react to stimulation only (amplifiers). The latter type, which we shall mostly be concerned with, is characterized by a reflection coefficient ϱ with a magnitude larger than unity, i.e. $|\varrho| > 1$. The reflection coefficient of a spontaneously active 1-port has an absolute value which may or may not satisfy eq. (2.68). It is characterized by the fact that it can be represented by a 1-port with a signal-wave source at its terminals. Apart from its use as a signal source we will not be interested in that type of active 1-ports.

Let us consider a passive system which is in temperature equilibrium. By a passive system we understand a system that consists of passive components only. In such a system any dissipative component thermally radiates noise power towards the other components. On the other hand, it absorbs noise power which emanates from those other components. The second law of thermodynamics requires that, if the system is in temperature equilibrium, for each component the net absorbed power be equal to the net radiated power. Since we are interested only in linear systems and do not consider frequency-converting components, the principle of detailed balancing can be applied over any band δf. This principle states that the exchange of noise power in a system which is in equilibrium must be balanced at every possible mechanism of power transmission **). Hence the above balance of radiated and absorbed powers applies over any band δf. Thus, if the two 1-ports of the previous chapter are dissipative and at the same temperature, the equation

$$\delta P_{12} = \delta P_{21} \tag{2.69}$$

must be valid. Since for any dissipative 1-port the quantity $(1 - \varrho\varrho^*)$ is positive, by virtue of eq. (2.56) and its cyclic version, this results in

$$\frac{X_1 X_1^+}{1 - \varrho_1 \varrho_1^*} = \frac{X_2 X_2^+}{1 - \varrho_2 \varrho_2^*}. \tag{2.70}$$

This equation applies to any combination of two arbitrary dissipative 1-ports at the same temperature. Hence, for dissipative 1-ports at uniform temperatures the ratio $X X^+/(1 - \varrho\varrho^*)$ does not depend on the signal-performance properties. The ratio can depend only on temperature and frequency. Therefore,

*) In a reactive element energy can be stored. In a stationary system, however, averaged stored energy does not vary.

**) As will be seen below (see sec. 2.6.7) this does not necessarily imply that there is a balance of radiation in each line of a multiport system.

we introduce a function $F(T,f)$ in such a way that the noise-wave-source power of any dissipative 1-port at the uniform absolute temperature T can be written in the form

$$\mathsf{X}\mathsf{X}^{+} = F(T,f)\,(1 - \varrho\varrho^{*}). \tag{2.71}$$

The function $F(T,f)$ will be called the noise-intensity function. It is a function of temperature and frequency only.

A passive reactive 1-port reflects all power incident upon it. It does not absorb power. Hence, by virtue of eq. (2.69), it cannot radiate power either and its noise-wave source is identically zero. Thus, both the quantities $\mathsf{X}\mathsf{X}^{+}$ and $(1 - \varrho\varrho^{*})$ are equal to zero. Both sides of eq. (2.71) are zero so that, formally, that equation can be thought to apply also to passive reactive 1-ports.

A further consequence of the second law of thermodynamics is the fact that the net flow of thermal noise power from a warmer 1-port towards a colder one is larger than the opposite net flow. Let T_1 and T_2 be the respective absolute temperatures of the two 1-ports. If the inequality

$$T_1 < T_2 \tag{2.72}$$

applies, the inequality

$$\delta P_{21} < \delta P_{12} \tag{2.73}$$

and, which is the same, the inequality

$$\frac{\mathsf{X}_1\mathsf{X}_1^{+}}{1 - \varrho_1\varrho_1^{*}} < \frac{\mathsf{X}_2\mathsf{X}_2^{+}}{1 - \varrho_2\varrho_2^{*}} \tag{2.74}$$

are valid. The latter implies that $F(T,f)$ is a monotonically increasing function of temperature.

The way primary flows of noise power and noise-wave sources have been defined (sec. 2.1.6) implies that, if a passive 1-port is at zero absolute temperature, the power of its noise-wave source is equal to zero. By virtue of eq. (2.71), this leads to

$$F(0,f) = 0. \tag{2.75}$$

In a strictly formal manner eq. (2.71) can also be generalized for 1-port amplifiers. These active 1-ports are characterized by

$$(1 - \varrho\varrho^{*}) < 0. \tag{2.76}$$

Since the power of noise-wave sources is inherently non-negative, in order that eq. (2.71) be formally valid, for 1-port amplifiers the noise-intensity function must be non-positive. We now assume $F(T,f)$ to exist also for negative values

of T *). To preserve the monotonic character and to maintain the validity of eq. (2.75), $F(T,f)$ must attain negative values in that region of T. Now, a negative temperature can be attributed to a noisy 1-port amplifier, viz. that negative value of T **) for which $F(T,f)$ has the appropriate negative value. This negative temperature will be called the characteristic noise temperature ***) of the amplifier.

The concept of characteristic noise temperature can be generalized. A dissipative passive 1-port may not have a uniform temperature. Moreover, an arbitrary 1-port may emit noise power which is not uniquely due to thermal causes. Anyhow, whatever the frequency f, the noise of a 1-port can be characterized by a noise-wave source X. Since negative power does not exist, $X X^+$ is always non-negative. The ratio of $X X^+$ to $(1 - \varrho\varrho^*)$, having the sign of the latter quantity, determines the value the noise-intensity function has for the particular 1-port. The value of T at which this function attains that value is called the characteristic noise temperature of the 1-port at the frequency f.

2.3.2. *Thermal noise in a long lossless line*

To derive the noise-intensity function we consider the noise waves in a long section of lossless line, which is incorporated in a closed passive system in temperature equilibrium. Through such a line noise energy is transported in both directions. The two noise waves, that is the radiation field, in the line must be in equilibrium with the radiation field in the remainder of the system. Hence the powers of the noise waves are governed by the temperature of the system. For evaluation of these powers Nyquist's argument [11]) is used in a form which is generalized to include dispersive lines.

When at a certain moment of time the line is disconnected at both ends from the system, e.g. by short-circuiting, an amount of energy is trapped in the line. In order that this energy be given by the two noise waves in accordance with eq. (2.18) many resonance frequencies of the short-circuited line must be contained in δf. Therefore, the travel time of the line must be long as compared with the inverse of the bandwidth and with the inverse of the frequency, i.e.

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*) Although we have not yet derived $F(T,f)$, which will be done in sec. 2.3.3, we now suppose that $F(T,f)$ is an analytic function for positive values of T which can be continued for negative values.

**) If, for negative values of T, the analytic continuation of $F(T,f)$ is supposed to apply, it can be shown [35,36]) that the negative temperature derived from it for an ideal maser-type amplifier is equal to the negative temperature which is a measure of the population inversion of the significant energy levels. We consider a maser-type amplifier to be ideal if its noise is caused only by the spontaneous transitions from the higher towards the lower energy level, and not by thermal radiation of dissipative parts of the circuitry.

***) It is not called equivalent or effective noise temperature, because in literature on noise theories [37]) these names are generally used for other noise-characterizing quantities with temperature dimension. Besides, as will become clear below, further good reasons exist to make use of the adjective characteristic.

the conditions

$$\tau \gg 1/\delta f \tag{2.77}$$

and

$$\tau \gg 1/f \tag{2.78}$$

must be fulfilled. It should be noted that the former condition is quite contrary to the requirement of eq. (2.61) which applies to considerations in which relative phases are significant. When the above conditions are satisfied, the trapped radiation field is in temperature equilibrium itself. Then the powers of the two opposite noise waves are equally large, for, if they were not, on the average more power would flow in one direction than in the other and that would contradict the self-equilibrium of the trapped radiation. Thus, if A and B are the two noise waves, the equation

$$\mathsf{A}\,\mathsf{A}^{+} = \mathsf{B}\,\mathsf{B}^{+} \tag{2.79}$$

is valid. The trapped energy δE contained in δf is given by

$$\delta E = 2\tau\,\mathsf{A}\,\mathsf{A}^{+}\delta f. \tag{2.80}$$

The line short-circuited at both ends has several modes of oscillation and corresponding natural frequencies. It is well known that each frequency component of such a closed radiation field has two degrees of freedom. Consequently, if the number of natural frequencies is large, in accordance with Planck's formula every frequency component has an average energy content as given by *)

$$\overline{E_i} = hf\{\exp(hf/kT) - 1\}^{-1}, \tag{2.81}$$

where h and k are, Planck's and Boltzmann's constants, respectively. Let f_1 be the first natural frequency in δf and f_2 the first one beyond it. Let further β_1 and β_2 be the corresponding propagation constants of the line. The number n_1 of nodes at f_1, including the nodes at the short-circuited ends, is given by

$$n_1 = 1 + L\beta_1/\pi. \tag{2.82}$$

At f_2 the number of nodes n_2 is given by

$$n_2 = 1 + L\beta_2/\pi. \tag{2.83}$$

The number δn of natural frequencies contained in δf is found to be

$$\delta n = n_2 - n_1 = L(\beta_2 - \beta_1)/\pi. \tag{2.84}$$

Provided that δf is small, the group velocity of the line is expressed by

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*) The average energy $\overline{E_i}$ should be augmented with an additional term $\frac{1}{2}hf$ for the zero-point energy but, as stated in sec. 2.1.6, we will ignore this in the present treatise.

$$v_g = \frac{\mathrm{d}\omega}{\mathrm{d}\beta} = 2\pi \frac{\delta f}{\beta_2 - \beta_1}. \tag{2.85}$$

By virtue of eq. (2.62), substitution of v_g in eq. (2.84) yields

$$\delta n = 2\tau \delta f, \tag{2.86}$$

so that in conjunction with eq. (2.77) there are a large number of modes of oscillation. The trapped energy is found to be given by

$$\delta E = 2\tau h f \, \delta f \{\exp(hf/kT) - 1\}^{-1}. \tag{2.87}$$

Equating the results of eqs (2.80) and (2.87) yields for the powers of the two noise waves

$$A\,A^+ = B\,B^+ = hf\{\exp(hf/kT) - 1\}^{-1}. \tag{2.88}$$

Concluding, we can formulate the following statement. In lossless lines, which are long with respect to the wavelengths considered and which are incorporated in passive systems in temperature equilibrium, two opposite noise waves exist, the powers of which are equal and given by Planck's formula.

2.3.3. *The noise-intensity function*

Let one end of the long line, considered in the previous section, be terminated by a passive 1-port. Let further the 1-port be at the uniform temperature T and have a reflection coefficient ϱ (fig. 4). The noise waves A and B, travelling respectively towards and from the 1-port, satisfy eq. (2.88). We assume the noise-wave source X of the 1-port to be uncorrelated with A. At the reference plane the noise-wave equation

$$B = X + \varrho A \tag{2.89}$$

can be formulated. Evaluation of the power of B and substitution of eq. (2.88) yield

$$X\,X^+ = (1 - \varrho\varrho^*)\, hf\,\{\exp(hf/kT) - 1\}^{-1}. \tag{2.90}$$

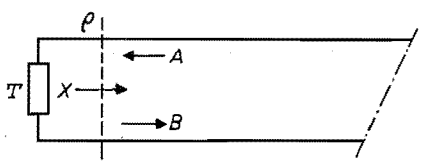


Fig. 4. Passive 1-port at a temperature T connected to an infinitely long line.

Comparison with eq. (2.71) shows that the noise-intensity function is given by *)

$$F(T,f) = hf\,\{\exp(hf/kT) - 1\}^{-1}. \tag{2.91}$$

When the frequency is relatively low and/or the temperature relatively high, that is when the condition

$$hf \ll kT \tag{2.92}$$

applies, $F(T,f)$ is given approximately by

$$F(T,f) = kT. \tag{2.93}$$

In this case, which occurs very often in practice, the powers of noise-wave sources do not depend on frequency and one speaks of white noise. In the following we shall mostly consider this case only. Instead of eq. (2.90) we can then write

$$XX^{+} = kT(1 - \varrho\varrho^{*}). \tag{2.94}$$

In the configuration of sec. 2.2.3, where two 1-ports are interconnected directly (cf. fig. 3), let the temperatures of the 1-ports be T_1 and T_2, respectively. By virtue of eqs (2.65) and (2.66), the noise power emanating from the first 1-port and dissipated in the second is readily evaluated. We find

$$\delta P_{21} = \frac{(1 - \varrho_1\varrho_1^{*})(1 - \varrho_2\varrho_2^{*})}{(1 - \varrho_1\varrho_2)(1 - \varrho_1^{*}\varrho_2^{*})}\, kT_1\, \delta f. \tag{2.95}$$

The opposite flow of noise power δP_{12} is found by exchange of the subscripts 1 and 2. It is readily seen that, if the temperatures are equal, $T_1 = T_2$, these exchanged powers are equal, as they should be. The net flow of noise power from the left towards the right (see fig. 3) is given by

$$\delta P_{21} - \delta P_{12} = (BB^{+} - AA^{+})\,\delta f = \frac{(1 - \varrho_1\varrho_1^{*})(1 - \varrho_2\varrho_2^{*})}{(1 - \varrho_1\varrho_2)(1 - \varrho_1^{*}\varrho_2^{*})}\, k\,(T_1 - T_2). \tag{2.96}$$

It is proportional to the temperature difference.

In the more general case, where eq. (2.91) applies, the net power exchange is given by

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*) The noise-intensity function as given by eq. (2.91) is discontinuous at $T = 0$ making a jump of hf. Nevertheless we suppose eq. (2.91) to be valid also for negative values of T. The discontinuity reflects the fact that extremely small, both positive and negative, values of T are unrealistic. An extremely small negative value of T means an explosion of activity. In the approximation of eq. (2.93) the discontinuity does not occur, but it should be realized that eq. (2.93) is not valid for small T.

$$(BB^+ - AA^+)\,\delta f = \frac{(1-\varrho_1\varrho_1^*)\,(1-\varrho_2\varrho_2^*)\,hf\sinh\left(\dfrac{hf(T_1-T_2)}{2\,k\,T_1T_2}\right)}{(1-\varrho_1\varrho_2)\,(1-\varrho_1^*\varrho_2^*)\,2\sinh\left(\dfrac{hf}{2\,kT_1}\right)\sinh\left(\dfrac{hf}{2\,kT_2}\right)}. \tag{2.97}$$

Since the hyperbolic sine is an odd function, the net flow of noise power is from the hotter towards the colder 1-port. This is in agreement with the second law of thermodynamics (cf. sec. 2.3.1).

All observations made thus far still apply formally when one of the temperatures (say T_2) is negative and the modulus of the corresponding reflection coefficient (say ϱ_2) is greater than unity. Both the eqs (2.96) and (2.97) then yield negative results. Hence, the net flow of power is from the second towards the first 1-port. This can be understood from the fact that the power emitted by the first 1-port is amplified and reflected by the second. The result expresses the point of view that a system with a formally negative temperature can be considered to be hotter than hot [38]).

2.4. Finite bandwidth

2.4.1. *Power over finite bandwidths*

Thus far we have investigated noise-power exchange in linear systems as it is contained in small bandwidths δf. In sec. 2.2.2 a condition for δf has been formulated such that the interference effects of correlated noisy disturbances are the same for all frequencies in δf. As a result these effects can be accounted for collectively by means of algebraic noise-wave equations. However, in practice one is often interested in noise-power exchange which occurs over a larger bandwidth. Then, in general, the condition of sec. 2.2.2 is not satisfied. Therefore, as an interlude, in the present section we will investigate how the flow of noise power over a finite bandwidth can be calculated. This investigation will lead to the formulation of conditions by which some interesting cases can be distinguished.

Let Δf be the finite bandwidth of interest. Many practical systems contain such large delays that the phase lags at different frequencies in Δf cannot be considered equally large any more. In such systems interference effects at different frequencies cannot be evaluated collectively. Hence the formalism developed in the preceding sections cannot be applied. The complete bandwidth Δf can, however, be subdivided into a large number of small bandwidths δf over which the formalism can be used. The complete noise power contained in Δf is equal to the sum of the powers contained in the δf's.

In sec. 2.1.4 a noise wave A has been defined so that the power of that part

of a noisy disturbance which is contained in δf is given by the expression $\mathsf{A}\mathsf{A}^+\delta f$. Let the power contained in Δf be denoted by $P_A(\Delta f)$. If f_1 and f_2 are the end frequencies of Δf, we can write down the equation

$$P_A(\Delta f) = \int_{f_1}^{f_2} \mathsf{A}\mathsf{A}^+ \mathrm{d}f. \tag{2.98}$$

Let a noisy disturbance contained in Δf consist of two parts. Let the corresponding noise wave in the small sub-bandwidth δf be W and the noise waves which correspond to the two parts be A and B. From eqs (2.25) and (2.43) we then have, by virtue of eq. (2.37),

$$\begin{aligned} \mathsf{W}\mathsf{W}^+ &= \mathsf{A}\mathsf{A}^+ + \mathsf{B}\mathsf{B}^+ + \mathsf{A}\mathsf{B}^+ + \mathsf{B}\mathsf{A}^+ = \\ &= \mathsf{A}\mathsf{A}^+ + \mathsf{B}\mathsf{B}^+ + (C + C^*)(\mathsf{A}\mathsf{A}^+ \cdot \mathsf{B}\mathsf{B}^+)^{1/2} = \\ &= \mathsf{A}\mathsf{A}^+ + \mathsf{B}\mathsf{B}^+ + (D + D^*)\,\mathsf{B}\mathsf{B}^+. \end{aligned} \tag{2.99}$$

Integrating eq. (2.99) over Δf, we obtain for the power $P_W(\Delta f)$ contained in Δf the equation

$$P_W(\Delta f) = P_A(\Delta f) + P_B(\Delta f) + P_{A,B}(\Delta f), \tag{2.100}$$

where

$$\begin{aligned} P_{A,B}(\Delta f) &= \int_{f_1}^{f_2} \mathsf{A}\mathsf{B}^+ \mathrm{d}f + \int_{f_1}^{f_2} \mathsf{B}\mathsf{A}^+ \mathrm{d}f = \\ &= \int_{f_1}^{f_2} (C + C^*)(\mathsf{A}\mathsf{A}^+ \cdot \mathsf{B}\mathsf{B}^+)^{1/2} \mathrm{d}f = \\ &= \int_{f_1}^{f_2} (D + D^*)\,\mathsf{B}\mathsf{B}^+ \mathrm{d}f. \end{aligned} \tag{2.101}$$

Due to eq. (2.14) the integrands of the two integrals in the second member are complex conjugate so that $P_{A,B}(\Delta f)$ is real, as it should be. If we introduce coefficients c and d by

$$c = \int_{f_1}^{f_2} C\,(\mathsf{A}\mathsf{A}^+ \cdot \mathsf{B}\mathsf{B}^+)^{1/2} \mathrm{d}f \Big/ \left[\int_{f_1}^{f_2} \mathsf{A}\mathsf{A}^+ \mathrm{d}f \cdot \int_{f_1}^{f_2} \mathsf{B}\mathsf{B}^+ \mathrm{d}f \right]^{1/2} \tag{2.102}$$

and

$$d = \int_{f_1}^{f_2} D\,\mathsf{B}\mathsf{B}^+ \mathrm{d}f \Big/ \int_{f_1}^{f_2} \mathsf{B}\mathsf{B}^+ \mathrm{d}f, \tag{2.103}$$

respectively, we obtain from eq. (2.101) the equations

$$P_{A,B}(\Delta f) = (c + c^*)\,[P_A(\Delta f)\,P_B(\Delta f)]^{1/2} = \\ = (d + d^*)\,P_B(\Delta f). \tag{2.104}$$

From eqs (2.102), (2.103), and (2.37) we readily see that c and d are related by

$$d\,[P_B(\Delta f)]^{1/2} = c\,[P_A(\Delta f)]^{1/2}. \tag{2.105}$$

When $P_{A,B}(\Delta f)$ is zero, the power of the complete noisy disturbance is equal to the sum of the powers of the parts:

$$P_W(\Delta f) = P_A(\Delta f) + P_B(\Delta f). \tag{2.106}$$

In special situations this is so, when c and d are imaginary. The noise waves A and B can then be considered to be on the average over Δf in "phase quadrature". The parts of the noisy disturbance cannot then be considered uncorrelated, because one can imagine that by shifting the average phase of one of them by $\pi/2$, eq. (2.106) does not apply any more.

The cases, where

$$c = d = 0 \tag{2.107}$$

applies, are more general and more important. They occur among others, when $C = 0$ or, what is the same, when $D = 0$. Moreover, when $(C + C^*)$ and, hence, $(D + D^*)$ are strongly oscillating functions of frequency or, in other words, when the arguments of C and D vary rapidly over Δf, c and d tend to zero. In all cases where eq. (2.107) applies, we will consider the parts of the noisy disturbance uncorrelated over Δf.

Two noisy disturbances can thus be called uncorrelated over Δf notwithstanding the fact that the corresponding noise waves may be correlated. Physically, this means that over Δf the interference of the noise waves oscillates between constructive and destructive characteristics in such a manner that on the average the resultant power is simply the sum of the constituent powers (cf. footnote next section). This is in particular the case when the nature of the interference varies rapidly over Δf and independently of the magnitude of Δf. Although two noisy disturbances can have much internal relation — they may be correlated over small bands of frequency δf —, it need not be apparent over larger bands Δf over which the correlation then cancels to zero. Indeed, to know correlation in detail a coarse power measurement over a wide band Δf is not sufficient, but more refined measurements over narrow bands δf are required.

When one is not satisfied by a mere spot-noise theory as described in preceding sections, all significant results of that theory which are stated in terms of power must be integrated with respect to frequency. In general the integrands are functions of frequency. For two 1-ports connected to each other this is

easy to see from eqs (2.65) and (2.66). In most such cases the reflection coefficients ϱ_1 and ϱ_2 depend on frequency. Moreover, if the 1-ports are not at a uniform temperature or if they are not passive, even the powers $X_1X_1^+$ and $X_2X_2^+$ of the noise-wave sources are likely to be functions of frequency.

The integration over Δf may be a quite cumbersome procedure. In certain important cases, however, the integration will turn out to be simple. Before we proceed to distinguish those simple cases, we will first investigate whether transmission of noise power over finite bands of frequency can be described by a kind of noise waves.

2.4.2. *Equivalent noise waves*

In sec. 2.1.4 a noisy disturbance contained in δf has been represented by an element A of a Hilbert space. We will now ask the question whether noisy disturbances contained in Δf can analogously be represented by elements of a further Hilbert space.

Let be given a Hilbert space of elements a and the dual space of elements a^+. Let an element a be subjoined to a noisy disturbance in Δf, with noise wave A, in such a manner that aa^+ is equal to the power $P_A(\Delta f)$. From eq. (2.98) we then have

$$aa^+ = P_A(\Delta f) = \int_{f_1}^{f_2} AA^+ \mathrm{d}f. \tag{2.108}$$

In analogy to two uncorrelated noise waves A and B which in sec. 2.1.5 have been defined orthogonal (cf. eq. (2.27)), two uncorrelated noisy disturbances a and b should be orthogonal, i.e.

$$ab^+ = a^+b = 0. \tag{2.109}$$

Let a noisy disturbance w consist of two parts a and b, i.e.

$$w = a + b. \tag{2.110}$$

The power of w is then given by

$$ww^+ = (a + b)(a + b)^+ = aa^+ + bb^+ + ab^+ + ba^+. \tag{2.111}$$

When the two parts a and b are uncorrelated, application of eq. (2.109) gives the required result that the complete power is the sum of the constituent powers (cf. eq. (2.106)). When the two parts a and b are correlated, in analogy to eq. (2.28) we suppose that one of them, say a, consists of two parts a' and a'', i.e.

$$a = a' + a''. \tag{2.112}$$

It is further assumed that one of these parts, say a', is completely correlated with b and can be written in the form

$$a' = d'b, \tag{2.113}$$

where d' is an, as yet undetermined, complex coefficient. The other part a'' is considered to be uncorrelated with both a' and b. Substitution of eqs (2.112) and (2.113) in eq. (2.111) gives

$$\begin{aligned} ww^+ &= (a''a''^+ + d'd'^* bb^+) + bb^+ + (d' + d'^*) bb^+ = \\ &= aa^+ + bb^+ + (d' + d'^*) bb^+. \end{aligned} \tag{2.114}$$

Comparing this result with eqs (2.100) and (2.104), we see that d' can be identified with the coefficient d as it has been defined by eq. (2.103). From eqs (2.112), (2.113), (2.103) and (2.30) we get, by virtue of this identity, for the product ab^+ the equations

$$ab^+ = d\,bb^+ = \int_{f_1}^{f_2} D\,BB^+ \mathrm{d}f = \int_{f_1}^{f_2} AB^+ \mathrm{d}f. \tag{2.115}$$

Equation (2.105) now reads

$$d\,(bb^+)^{1/2} = c\,(aa^+)^{1/2}. \tag{2.116}$$

It is furthermore readily verified that the powers $a'a'^+$ and $a''a''^+$ of the correlated und uncorrelated parts a' and a'' are respectively given by *)

$$a'a'^+ = dd^*\,bb^+ = cc^*\,aa^+, \tag{2.117}$$

$$a''a''^+ = (1 - cc^*)\,bb^+. \tag{2.118}$$

These expressions correspond to those given by eqs (2.32) and (2.33) for the powers of the parts A' and A'' of the noise wave A. In the literature on phenom-

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*) Although it seems logical, $a'a'^+$ and $a''a''^+$ cannot be derived from an integration of $A'A'^+$ and $A''A''^+$. Equations such as

$$a'a'^+ = \int_{f_1}^{f_2} A'A'^+ \mathrm{d}f = \int_{f_1}^{f_2} CC^*\,AA^+\,\mathrm{d}f$$

or

$$cc^* = \int_{f_1}^{f_2} CC^*\,AA^+\,\mathrm{d}f \Big/ \int_{f_1}^{f_2} AA^+\,\mathrm{d}f$$

are not valid. The latter would determine the square magnitude of c as the average of the square magnitude of C and that is not consistent with the definition of c as contained in eq. (2.102). In fact, equations of the above type determine an average correlation content. One might it call intrinsic correlation. However, we are not interested in an average content of correlation, but in the average effects of correlation on power transmission as caused by interference phenomena. These are more accurately described by equations as they are discussed in the text. To distinguish the latter type of overall correlation from intrinsic correlation, it could be called extrinsic correlation.

enological noise theory [10]) the complex number c is called the correlation coefficient of the noisy disturbances a and b.

From the above discussion we may conclude that representing a noisy disturbance over Δf by an element a of a Hilbert space in such a manner that aa^+ determines the power of the disturbance leads to a consistent formalism. The quantity a can be considered as the amplitude of a wave which transports the same power as the noisy disturbance does on the average. We will call this wave the equivalent noise wave *). The quantity a, however, does not denote a noise wave in the sense as it has been introduced in sec. 2.1.4. Because different parts of an equivalent noise wave that are contained in different small bandwidths δf have different phase lags upon travelling the same length of line, the phase relation of eq. (2.21) does not now apply. In sec. 2.1.4 it has moreover been stated that the set of simultaneous linear algebraic equations, which describes the signal properties of a multiport, also relates the amplitudes of the noise waves at the ports of that multiport. Such equations are valid at a distinct frequency. An equivalent noise wave, however, can be thought to be smeared out over a band of frequencies. Hence signal-performance equations cannot be applied to equivalent noise waves. As a matter of fact, this is due to the same cause which forbids application of the phase relation. The only item of an equivalent noise wave a, which has physical interest, is the power expression aa^+ as defined by the integral of eq. (2.108).

2.4.3. *Line length and bandwidth*

In preceding sections it has been stated that the noise powers contained in fairly large bands of frequencies Δf can be calculated from solutions of noise-wave equations. The results of spot-noise theory must then be integrated with respect to the frequency. In two extreme, but practically important, cases, of which one is in essence the spot-noise situation, the integration turns out to be simple. It can then be omitted. In the present section the extreme cases will be investigated for a particular configuration. The conditions pertinent to them will be formulated.

Let us consider once more the system of two 1-ports which are connected to a section of line of finite length (see fig. 2). In sec. 2.2.1 for the two opposite noise waves A_1 and B_1, eqs (2.54) and (2.55) have been derived for the respective noise-wave powers $A_1A_1^+$ and $B_1B_1^+$. If the reflection coefficients ϱ_1 and ϱ_2 are written in the form

$$\varrho_j = R_j \exp(i\,\varphi_j), \qquad j = 1, 2, \tag{2.119}$$

*) The nomenclature has been adopted from convention. In conventional noise theories, e.g. $\overline{ee^*}$ represents the squared effective e.m.f. of an equivalent noise voltage source. That source would cause the same distribution of power as on the average and in the bandwidth Δf the actual noisy disturbance does. In our notation such a squared effective e.m.f. is denoted by ee^+.

for the noise-wave powers we can write down

$$A_1A_1^+ = (X_2X_2^+ + R_2^2X_1X_1^+)/D(R,\varphi) \tag{2.120}$$

and

$$B_1B_1^+ = (X_1X_1^+ + R_1^2X_2X_2^+)/D(R,\varphi), \tag{2.121}$$

with

$$D(R,\varphi) = 1 - 2\,R_1R_2\cos(2\,\beta L + \varphi_1 + \varphi_2) + R_1^2R_2^2. \tag{2.122}$$

To obtain the powers $a_1a_1^+$ and $b_1b_1^+$ of the corresponding equivalent noise waves the noise-wave powers $A_1A_1^+$ and $B_1B_1^+$ must be integrated over Δf.

Since we are interested only in underlying principles, and not in specific solutions, we will now make some simplifying assumptions. First, we assume that the internal transit times of the 1-ports are very short. Inherent lengths of actual 1-ports can be thought to be incorporated in the line. Secondly, we assume that neither the reflection coefficients ϱ_1 and ϱ_2 nor the noise-wave-source powers $X_1X_1^+$ and $X_2X_2^+$ vary with the frequency over Δf. When the 1-port characteristics depend on the frequency, the total bandwidth can often be divided into sub-bandwidths over which the latter assumption is approximately valid.

These assumptions having been made, the only frequency-dependent element in the above equations is the propagation constant β. Hence, on integration of eqs (2.120) and (2.121) only the integral

$$\int_{f_1}^{f_2}\{D(R,\varphi)\}^{-1}\mathrm{d}f = \int_{f_1}^{f_2}\{1 - 2\,R_1R_2\cos(2\,\beta L + \varphi_1 + \varphi_2) + R_1^2R_2^2\}^{-1}\mathrm{d}f \tag{2.123}$$

is retained as such. In general this integral must be evaluated numerically. However, two cases of practical interest in which the evaluation is fairly simple can be distinguished. Let the propagation constants be given at the end frequencies f_1 and f_2, respectively, by β_1 and β_2. Then, the variation of β over Δf is given by

$$\Delta\beta = \beta_2 - \beta_1. \tag{2.124}$$

Hence, the variation of the argument of the cosine in the integrand is given by

$$\Delta\varphi = 2\,L\,\Delta\beta. \tag{2.125}$$

Now, the integration becomes simple when this last variation is either very small or very large. Hence the two extreme cases of interest are determined by the conditions

(a) $$L\,\Delta\beta \ll 2\,\pi, \tag{2.126}$$

(b) $$L\,\Delta\beta \gg 2\,\pi. \tag{2.127}$$

The distinction can be made in another way. The number n of wavelengths

which span the line at a frequency f is given by

$$n = \beta L/2\pi. \tag{2.128}$$

The variation of n over Δf is given by

$$\Delta n = n_2 - n_1 = \Delta\beta L/2\pi, \tag{2.129}$$

so that eqs (2.126) and (2.127) are respectively equivalent to the conditions

(a) $$\Delta n \ll 1, \tag{2.130}$$

(b) $$\Delta n \gg 1. \tag{2.131}$$

The two extreme cases thus apply if the length of the line, expressed in number of wavelengths, varies either slightly or very much over the band.

In the former case, the length of the line in terms of wavelengths may be large, although this is not very likely since it would require a very small Δf. In the latter case, the line length in terms of wavelengths cannot be small for all frequencies contained in Δf since, due to Δn being large, at least n_2 must be a large number.

Case (a)

The condition for the former case can be rewritten in the form

$$L\,\Delta f \ll v_g. \tag{2.132}$$

This is equivalent to the condition of eq. (2.61) for spot-noise theory. If either L or Δf or both are small, the spot-noise theory applies over the whole bandwidth Δf. This signifies that the effective phase lengths of the system are nearly the same at all frequencies in Δf. Hence, the constructive or destructive nature of the interference is the same at all frequencies so that the solutions of noise-wave equations can be used throughout.

In this case the denominator $D(R,\varphi)$ does not depend on frequency. The integral of eq. (2.123) simplifies to the form

$$\int_{f_1}^{f_2} \{D(R,\varphi)\}^{-1}\,\mathrm{d}f = \Delta f/D(R,\varphi). \tag{2.133}$$

Furthermore, the relations

$$aa^+ = \mathsf{A}\mathsf{A}^+\,\Delta f, \tag{2.134}$$

$$bb^+ = \mathsf{B}\mathsf{B}^+\,\Delta f \tag{2.135}$$

and

$$ab^+ = \mathsf{A}\mathsf{B}^+\,\Delta f \tag{2.136}$$

are valid.

Case (b)

The condition for the other extreme case can be rewritten in the form

$$L\,\Delta f \gg \overline{v_g}, \tag{2.137}$$

where $\overline{v_g}$ is the mean group velocity in Δf. This condition is equivalent to the one of eq. (2.77) for the Nyquist derivation to be valid. In appendix A it is shown that in this case the integral of eq. (2.123) assumes the simple form

$$\int_{f_1}^{f_2} \{D(R,\varphi)\}^{-1}\,\mathrm{d}f = \frac{\Delta f}{1 - R_1{}^2 R_2{}^2}\,. \tag{2.138}$$

The powers of the equivalent noise waves a and b are easy to evaluate, resulting in

$$aa^+ = (x_2x_2{}^+ + R_2{}^2x_1x_1{}^+)/(1 - R_1{}^2R_2{}^2) \tag{2.139}$$

and

$$bb^+ = (x_1x_1{}^+ + R_1{}^2x_2x_2{}^+)/(1 - R_1{}^2R_2{}^2), \tag{2.140}$$

with

$$x_1x_1{}^+ = X_1X_1{}^+\Delta f \tag{2.141}$$

and

$$x_2x_2{}^+ = X_2X_2{}^+\Delta f. \tag{2.142}$$

To obtain the mean cross-product ab^+ the expression AB^+ must be integrated over Δf. Analogous to the evaluation of eq. (2.138) it can be shown that, due to the large variations in the arguments of the terms in the denominator of the integrand, the result of that integration tends to zero (cf. sec. 2.4.2):

$$ab^+ = 0. \tag{2.143}$$

This means that in this case the correlation of equivalent noise waves is zero.

If the two 1-ports are passive and have the same uniform temperature T, by virtue of eq. (2.94), the general result of sec. 2.3.2 is readily verified, yielding

$$aa^+ = bb^+ = kT\,\Delta f. \tag{2.144}$$

In intermediate cases, if neither eq. (2.126) nor eq. (2.127) applies, the results of the noise-wave-equation procedure, that is of the spot-noise theory, must actually be integrated. This can be quite cumbersome. Therefore, it is common practice to omit integrating and to be satisfied by giving only "spot-noise" characteristics.

2.4.4. *Noise-power equations*

In broadband systems which contain large delays (case (b) of the previous section) the correlation of two simultaneously present noisy disturbances is externally cancelled over large bandwidths. The resultant power is equal to the

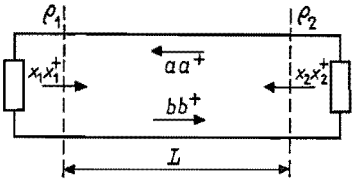


Fig. 5. Two 1-ports connected by a long line. When the bandwidth is large, noise-power equations can be formulated.

sum of the constituent powers. In that case the noise-wave-equation procedure can be circumvented. One can resort at once to what we will call noise-power equations.

For two 1-ports connected to the two ends of a long line (see fig. 5) the power equations

$$-\varrho_1\varrho_1^* aa^+ + bb^+ = x_1x_1^+ \tag{2.145}$$

and

$$aa^+ - \varrho_2\varrho_2^* bb^+ = x_2x_2^+ \tag{2.146}$$

can be formulated directly. The solution of these equations is equivalent to eqs (2.139) and (2.140).

2.4.5. *Equivalence of matching and delay effects*

Let one of the two 1-ports be reflectionless in the configuration of sec. 2.2.1 (see fig. 2), say

$$\varrho_2 = 0. \tag{2.147}$$

Then, the solution of the noise-wave equations is considerably simplified. Equations (2.54) and (2.55) reduce to the simple forms

$$A_1A_1^+\delta f = X_2X_2^+\delta f \tag{2.148}$$

and

$$B_1B_1^+\delta f = X_1X_1^+\delta f + \varrho_1\varrho_1^* X_2X_2^+\delta f. \tag{2.149}$$

Since frequency-dependent arguments do not occur any more, these equations can be simply integrated over finite bandwidths. The result is given by

$$aa^+ = x_2x_2^+ \tag{2.150}$$

and

$$bb^+ = x_1x_1^+ + \varrho_1\varrho_1^* x_2x_2^+. \tag{2.151}$$

This result is the same as would have been obtained by solving the noise-power equations under the condition of eq. (2.147). We conclude that, if at least one of the two 1-ports is reflectionless, the noise-wave equations and the noise-power equations lead to the same result. This is even true when the conditions for the latter type of equations are not fulfilled.

The configuration discussed above is a particular case of a more general type of systems. The above conclusion can therefore be generalized. Let us define a quasi-matched system to be a system in which no (noise) waves, or parts thereof, pass a given point more than once in the same direction and in which no multi-path transmission occurs. Evaluating the transmission of power through a line in such a system, interference need not be taken into account. This is precisely the condition which makes noise-power equations applicable. Hence we observe that in any quasi-matched system the powers of equivalent noise waves can be calculated directly from noise-power equations.

In many observations about noise performance of microwave-transmission systems only matched, or nearly matched, configurations are considered. The total power of equivalent noise waves is mostly calculated by simply adding the powers of the constituent equivalent noise waves. This is quite permissible because of the absence of interference effects as mentioned above but it is often done without justification [55]).

2.4.6. *A simple example*

Above it has been shown that the total transmitted power of simultaneous noisy disturbances can be calculated by different methods. In this section some of these methods will be illustrated by a simple example. As in fig. 6, two resistors R_1 and R_2 are connected by means of a line of characteristic impedance R and length L. The line is assumed to be lossless. Let the first resistor have a uniform absolute temperature T, while the other resistor is at absolute zero temperature. Then, only R_1 emits noise power. Part of this power is dissipated in R_2, the remainder being reabsorbed by R_1. In what follows the former part will be calculated in three different ways.

2.4.6.1. Noise-voltage source

As shown in fig. 7, a noise-voltage source E can be associated with the resistor R_1. The power $\mathsf{E}\mathsf{E}^+$ of that source is given by *)

$$\mathsf{E}\mathsf{E}^+ = 4\,kTR_1. \tag{2.152}$$

The resistor R_1 is loaded by an impedance Z given by

$$Z = R\,(R_2 - i\,R\tan\beta L)\,(R - i\,R_2\tan\beta L)^{-1} =$$

$$= R\frac{RR_2\,(1 + \tan^2\beta L)}{R^2 + {R_2}^2\tan^2\beta L} + i\,R\frac{({R_2}^2 - R^2)\tan\beta L}{R^2 + {R_2}^2\tan^2\beta L}. \tag{2.153}$$

*) It should be noted that $\mathsf{E}\mathsf{E}^+$ is a mean-square effective voltage per unit bandwidth. Conventionally, the mean-square effective voltage a noise-voltage source has over a small bandwidth δf is given by [10]) $\overline{ee^*} = 4\,kTR_1\,\delta f$, while, in our notation, we have $\overline{ee^*} = \mathsf{e}\mathsf{e}^+ = \mathsf{E}\mathsf{E}^+\,\delta f$ (cf. secs 2.4.2 and 2.4.3).

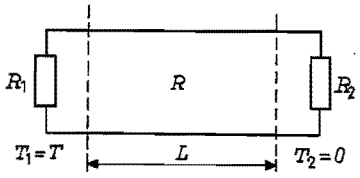


Fig. 6. Two resistors connected by a line of finite length.

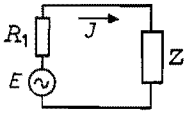


Fig. 7. A noise-voltage source drives a noise current.

The noise current J flowing through R_1 and Z is given by

$$J = \frac{1}{R_1 + Z} E. \tag{2.154}$$

For the power of this noise current we obtain

$$JJ^+ = EE^+ \frac{R^2 + R_2{}^2 \tan^2 \beta L}{R^2 (R_1 + R_2)^2 + (R_1 R_2 + R^2)^2 \tan^2 \beta L}. \tag{2.155}$$

By virtue of eq. (2.152), we get for the density G_{21} of the power P_{21} absorbed in R_2 the equation

$$G_{21} = JJ^+ \operatorname{Re}[Z] =$$

$$= 4\,kT \frac{R^2 R_1 R_2}{R^2 (R_1 + R_2)^2 + (R^2 - R_1{}^2)(R^2 - R_2{}^2) \sin^2 \beta L}. \tag{2.156}$$

2.4.6.2. Noise-wave source

As seen from the line, the resistors R_1 and R_2 are 1-ports with reflection coefficients ϱ_1 and ϱ_2, respectively given by

$$\varrho_1 = (R_1 - R)/(R_1 + R) \tag{2.157}$$

and

$$\varrho_2 = (R_2 - R)/(R_2 + R). \tag{2.158}$$

As shown in fig. 8, a noise-wave source X can be assigned to R_1. Due to eq. (2.94) the power XX^+ of that source is given by

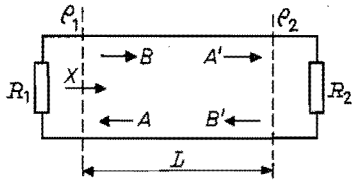


Fig. 8. A noise-wave source excites noise waves.

$$XX^+ = kT(1 - \varrho_1{}^2). \tag{2.159}$$

At the two ends of the line the noise-wave equations

$$B = X + \varrho_1 A \tag{2.160}$$

and

$$B' = \varrho_2 A' \tag{2.161}$$

can be formulated. Moreover, the equations

$$A' = B \exp(i\beta L) \tag{2.162}$$

and

$$B' = A \exp(-i\beta L) \tag{2.163}$$

apply. Solving these equations for A and B, we get

$$B = \frac{X}{1 - \varrho_1\varrho_2 \exp(i\,2\beta L)} \tag{2.164}$$

and

$$A = \frac{\varrho_2 \exp(i\,2\beta L)\,X}{1 - \varrho_1\varrho_2 \exp(i\,2\beta L)}. \tag{2.165}$$

The density G_{21} of the power which is dissipated in R_2 is given by

$$G_{21} = BB^+ - AA^+ = \\ = \frac{(1 - \varrho_2{}^2)\,XX^+}{\{1 - \varrho_1\varrho_2 \exp(i\,2\beta L)\}\,\{1 - \varrho_1\varrho_2 \exp(-i\,2\beta L)\}}. \tag{2.166}$$

With eqs (2.157), (2.158), and (2.159) it is easy to verify that this is identical to eq. (2.156).

2.4.6.3. Noise-power source

When the line is long or when the waves are reflected back and forth between R_1 and R_2 many times before they are attenuated appreciably, the influence of interference on the total power is cancelled over finite bandwidths Δf. The method of noise-power equations, discussed in sec. 2.4.4, can then be employed.

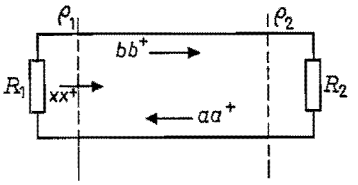


Fig. 9. A noise-power source causes flows of noise power.

It is illustrated in fig. 9. From R_1 an equivalent noise wave x with a power

$$xx^+ = kT(1 - \varrho_1^2)\,\Delta f \tag{2.167}$$

is injected into the line. In the line equivalent noise waves a and b are present. The noise-power equations now read

$$bb^+ = xx^+ + \varrho_1^2 aa^+ \tag{2.168}$$

and

$$aa^+ = \varrho_2^2 bb^+. \tag{2.169}$$

The solutions for aa^+ and bb^+ are given by

$$aa^+ = \frac{\varrho_2^2\, xx^+}{1 - \varrho_1^2\varrho_2^2} \tag{2.170}$$

and

$$bb^+ = \frac{xx^+}{1 - \varrho_1^2\varrho_2^2}. \tag{2.171}$$

The power dissipated in R_2 is given by

$$P_{21} = bb^+ - aa^+ = \frac{(1 - \varrho_2^2)\, xx^+}{1 - \varrho_1^2\varrho_2^2} \tag{2.172}$$

or, due to eqs (2.157), (2.158), and (2.167), by

$$P_{21} = \frac{4\,kT\,\Delta f\,RR_1R_2}{(R_1 + R_2)(R_1R_2 + R^2)}. \tag{2.173}$$

2.4.6.4. Discussion

The results of the first two methods are equivalent but, in most cases, the third one gives a different result. Only when at least one of the resistors, say R_1, is matched to the line, i.e. $R_1 = R$ or $\varrho_1 = 0$, it is easy to show that all three methods give the "classical" result

$$P_{21} = 4\,kT\,\Delta f \frac{RR_2}{(R + R_2)^2}. \tag{2.174}$$

Any noise wave then passes a given cross-section of the line only once in the same direction. This quasi-matched case has been discussed in sec. 2.4.5.

In general, it is not clear without investigation which method gives a reliable result over a given bandwidth. The bandwidth over which the first two methods can be relied upon may sometimes reduce to quite small values. In fact, they are valid inherently at one frequency only. To obtain the power dissipated in R_2 their results must therefore be integrated over the actual bandwidth. The question arises whether that integration is simple or not. This problem hinges on whether the power density G_{21}, as given by eqs (2.156) and (2.166), is constant or varies rapidly over Δf.

To estimate the bandwidth Δf over which G_{21} can be considered constant, we first observe that in the example discussed above the propagation constant β is the only frequency-dependent element in G_{21}. We next rewrite eq. (2.166) in the form

$$G_{21} = \frac{(1 - \varrho_2^2)\, XX^+}{1 - 2\,\varrho_1\varrho_2 \cos 2\beta L + \varrho_1^2\varrho_2^2}. \tag{2.175}$$

From the logarithmic derivative of G_{21} with respect to β we find for the magnitude of the relative variation of G_{21} the equation

$$\left|\frac{\Delta G_{21}}{G_{21}}\right| = \left|\frac{2\,\varrho_1\varrho_2 \sin 2\beta L}{1 - 2\,\varrho_1\varrho_2 \cos 2\beta L + \varrho_1^2\varrho_2^2}\right| 2\,L\,\Delta\beta, \tag{2.176}$$

where $\Delta\beta$ is the total variation of β over Δf. The first factor in the right-hand side attains a maximum value when β satisfies

$$\cos 2\,\beta L = 2\,\varrho_1\varrho_2(1 + \varrho_1^2\varrho_2^2)^{-1}; \tag{2.177}$$

the cases $\varrho_1\varrho_2 = 0$ (see above) and $L = 0$ are trivial. Hence, for all values of β we have

$$\left|\frac{\Delta G_{21}}{G_{21}}\right| \leqslant 4\,L\,\frac{\varrho_1\varrho_2}{1 - \varrho_1^2\varrho_2^2}\,\Delta\beta. \tag{2.178}$$

If for the determination of G_{21}, or of P_{21}, a relative accuracy better than ε, $\varepsilon \ll 1$, is required, the variation $\Delta\beta$ is restricted by

$$\Delta\beta < \frac{1 - \varrho_1^2\varrho_2^2}{4\,L\,\varrho_1\varrho_2}\,\varepsilon. \tag{2.179}$$

The dependence of $\Delta\beta$ on Δf is a characteristic of the line. We have

$$\Delta\beta = (2\,\pi/v_g)\,\Delta f, \tag{2.180}$$

where v_g is the group velocity. This gives

$$\Delta f < \frac{\varepsilon}{2\,\pi}\;\frac{1 - \varrho_1^2\varrho_2^2}{4\,\varrho_1\varrho_2}\;\frac{v_g}{L}. \tag{2.181}$$

We conclude that spot-noise theory gives a good approximation of the power dissipated in R_2 over a bandwidth which is within the bounds set by the above inequality.

At the frequencies considered, (L/v_g) is the transit time τ of the line (eq. (2.62)). Hence, from eq. (2.181) we have

$$\Delta f \ll \frac{1 - \varrho_1^2 \varrho_2^2}{4\,\varrho_1 \varrho_2} (1/\tau) . \tag{2.182}$$

If we compare this with eq. (2.61), we see that the inverse of the first factor of the right-hand side gives an estimate for the number ν of significant subsequent reflection, i.e.

$$\nu = \nu(\varrho_1 \varrho_2) = \frac{4\,\varrho_1 \varrho_2}{1 - \varrho_1^2 \varrho_2^2} . \tag{2.183}$$

The number ν tends to infinity when $\varrho_1 \varrho_2$ tends to unity, i.e. when both resistors are nearly completely reflecting. In that case Δf must be extremely small. In sec. 2.2.1 the unstable resonant case $\varrho_1 \varrho_2 = 1$ has been excluded from the considerations (cf. appendix B).

When L tends to zero, Δf is almost unrestricted. This is the situation which is very often supposed to exist, and, indeed, in usual electronics it occurs frequently.

When $\varrho_1 \varrho_2$ is very small, i.e.

$$\varrho_1 \varrho_2 \ll 1, \tag{2.184}$$

the number ν becomes very small. Then, one cannot properly speak any more of significant subsequent reflections. In that case spot-noise theory gives adequate results over rather broad bandwidths. As has been observed above, the results are then the same as those of the noise-power equations. In fact, eq. (2.184) presents the condition for the quasi-matched situation.

A further criterium for the noise-power-equations method is obtained from the observation that, for that method to be applicable, it is necessary that over Δf the power density G_{21}, given by eq. (2.175), oscillates many times between its maximum and minimum values. This is the case if the variation of $2\beta L$ over Δf is very large. With eq. (2.180) this leads to (cf. eq. (2.137))

$$\Delta f \gg \overline{v_g}/L = 1/\overline{\tau}, \tag{2.185}$$

where $\overline{v_g}$ and $\overline{\tau}$ are the mean values of v_g and τ, averaged over Δf.

2.5. Multiports and noise

2.5.1. *Multiports and noise-wave sources*

The procedure of ascribing noise sources to noisy 1-ports introduced in

sec. 2.1.6 can be extended to noisy multiports. To each port of a noisy multiport an associate noise-wave source can be assigned. These sources radiate their noise waves into the connected lines. As above, the powers of these noise waves are equal to the average densities of the primary noise powers radiated into the connected, infinitely long or non-reflectingly terminated, absolutely cold lines. Noise-wave sources can be defined only in connection with definite lines. This is analogous to the requirement that the connected lines must also be specified, when signal properties of multiports are described by means of scattering matrices.

There is a complication with multiports. All noise-wave sources together represent the internal physical sources of noise of the multiport. Moreover, different noise-wave sources depend mostly on the same internal causes. In general, the noise-wave sources of an n-port are mutually correlated. For a complete description of the noise properties of a multiport these correlations must be included.

From what has been said above it is easy to comprehend that a number of n noise-wave sources is necessary to describe the noise of an n-port. It is also sufficient, as can be seen as follows. When the termination of an n-port is not reflectionless, it reflects the primary noise waves partly or completely. These reflected noise waves are again incident upon the n-port. They are subsequently transmitted to the other ports where, then, they are simultaneously present with the primary waves with which they may interfere. Since, as has been supposed above, all mutual correlations are known, the interference effects can, however, be fully described. Moreover, in this and the next sections it will be seen that, by describing the noise of an n-port by n mutually correlated noise-wave sources, a formalism is obtained which is consistent with the laws of thermodynamics.

It is not necessary that all the sources emit their noise waves into the lines. It is quite possible that a portion of the n noise-wave sources are thought to radiate towards the n-port. Alternatively, one can associate with some ports two sources; one source for each direction. A corresponding number of ports can then be considered sourceless so that the total number of sources remains n. In this way Bauer and Rothe [31,39]) described the noise properties of a 2-port by means of two wave sources located at the input, the output being sourceless.

The authors mentioned above accounted for the correlation by inserting two "correlation" admittances in parallel with the input, one before and one behind the pair of wave sources which were considered to be uncorrelated themselves. We shall not adopt such a procedure. In order to achieve agreement with sec. 2.1.5, we will represent the correlation of two noise-wave sources by the inner product of their amplitudes.

2.5.2. *Amplitude vector and power matrix*

To meet the requirements presented in the previous section and to obtain a

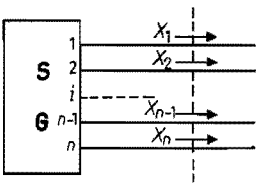


Fig. 10. Noisy multiport. The emitted noise can be described by n noise-wave sources.

convenient notation, we now introduce the concepts of noise-wave "amplitude vector" and "power matrix". As illustrated in fig. 10, to the ith port of an n-port a noise-wave source X_i is ascribed. The density G_{ii} of the noise power radiated primarily from the ith port into the connected line, that is the power of the ith noise-wave source, is given by

$$G_{ii} = X_i X_i^+. \tag{2.186}$$

The correlation between the ith and the jth noise-wave source, X_i and X_j, respectively, is determined by the inner product

$$G_{ij} = X_i X_j^+ \tag{2.187}$$

or its complex conjugate G_{ji}.

Now, we define the noise-wave amplitude vector $\mathbf{X}$ to be the column vector

$$\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix}. \tag{2.188}$$

It should be noted that the components of this vector are not complex numbers but elements of the Hilbert space of noise-wave amplitudes. Furthermore, we define the noise-wave power matrix $\mathbf{G}$ to be the $n \times n$-matrix

$$\mathbf{G} = \begin{pmatrix} G_{11} & G_{12} & \dots & G_{1n} \\ G_{21} & G_{22} & \dots & G_{2n} \\ \vdots & \vdots & & \vdots \\ G_{n1} & G_{n2} & \dots & G_{nn} \end{pmatrix}. \tag{2.189}$$

The elements of this matrix are complex numbers. By virtue of eq. (2.14), it is clear from the defintion of G_{ij}, eqs (2.186) and (2.187), that $\mathbf{G}$ is an Hermitian matrix. The equation

$$\mathbf{G}^\times = \mathbf{G} \tag{2.190}$$

applies where the superscript cross denotes the Hermitian conjugate or adjugate, that is the complex conjugate of the transposed matrix. In the case of vectors with elements of a Hilbert space as components — such as $\mathbf{X}$ of eq.

(2.188) — instead of the complex conjugate one should take the conjugate element of the dual space. The matrix **G** is related to the vector **X** via the dyadic product *)

$$\mathbf{G} = \mathbf{X}\mathbf{X}^{\times}. \tag{2.191}$$

The noise performance of any linear multiport, passive or active, reciprocal or non-reciprocal, can now be completely described by the scattering matrix **S** and the noise-wave-source power matrix **G**. The noise-wave-source amplitude vector **X** is used in calculating the behaviour of the multiport in relation with other transmission systems. The elements of **S** and **G** must be determined by measurement or calculation, but that will not be the subject of the present treatise.

2.5.3. *Noise-wave equations*

In order to formulate noise-wave equations, in sec. 2.2.3 a line of vanishingly small length was thought to be inserted between two interconnected 1-ports (cf. fig. 3). This procedure can also be applied to more complex transmission systems. The most general termination of an n-port is another n-port. Between any port of an n-port and the corresponding port of the termination a short line can be thought to be inserted. For each of these lines two equations relating the noise waves in those lines can be formulated. From these $2n$ equations the amplitudes of the $2n$ noise waves travelling through the n lines can be obtained; one wave in each direction. These amplitudes are expressed in terms of the amplitudes of the noise-wave sources of both the n-port and its termination. The powers and the correlation products of the waves can be calculated. They are found in terms of the scattering matrices and the power matrices of both the n-port and its termination.

In the configuration of fig. 11 let **S** and **Q** be the nth-order scattering matrices of the two n-ports mentioned above. Further, let **X** and **Y** be the amplitude

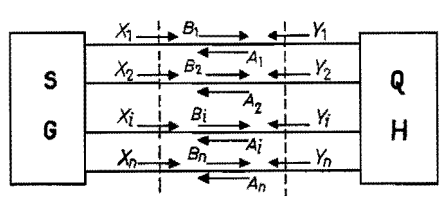


Fig. 11. A noisy n-port with a noisy load, which itself is a noisy n-port.

*) This dyadic product is formal only. Normally the rank of a dyadic is equal to unity, but here, due to the way inner products $X_i X_j^+$ have been defined, it need not be so. In fact, mostly it is not.

vectors and **G** and **H** be the power matrices of the noise-wave sources of respectively the n-ports **S** and **Q**. The equations

$$\mathbf{G} = \mathbf{X}\mathbf{X}^{\times} \tag{2.192}$$

and

$$\mathbf{H} = \mathbf{Y}\mathbf{Y}^{\times} \tag{2.193}$$

hold. Furthermore, we assume that the noise-wave sources of **S** are uncorrelated with those of **Q**. This is expressed by

$$\mathbf{X}\mathbf{Y}^{\times} = \mathbf{O}, \tag{2.194}$$

where **O** is the zero matrix. Further, let **A** and **B** be the amplitude vectors of the sets of noise waves A_i and B_i, respectively incident upon the n-ports **S** and **Q** as is illustrated in fig. 12.

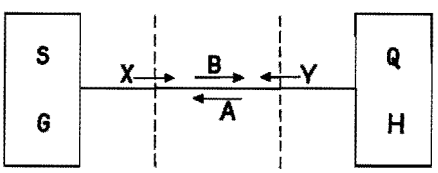


Fig. 12. Two connected noisy multiports. The noise-wave equations can be formulated directly in matrix form.

The noise-wave equations can be written down at once in matrix notation. We have

$$-\mathbf{S}\mathbf{A} + \mathbf{B} = \mathbf{X} \tag{2.195}$$

and

$$\mathbf{A} - \mathbf{Q}\mathbf{B} = \mathbf{Y}. \tag{2.196}$$

Solving these equations *), we obtain for **A** and **B** the expressions

$$\mathbf{A} = (\mathbf{I} - \mathbf{Q}\mathbf{S})^{-1}(\mathbf{Y} + \mathbf{Q}\mathbf{X}) \tag{2.197}$$

and

$$\mathbf{B} = (\mathbf{I} - \mathbf{S}\mathbf{Q})^{-1}(\mathbf{X} + \mathbf{S}\mathbf{Y}), \tag{2.198}$$

where **I** is the identity matrix. By virtue of eqs (2.192), (2.193), and (2.194), the power matrices of the sets of noise waves are given by

*) These equations can only be solved if the matrices $(\mathbf{I} - \mathbf{QS})$ and $(\mathbf{I} - \mathbf{SQ})$ are non-singular. If they are singular, they cannot be inverted. In appendix B it is shown that singularity of these matrices implies physically the occurrence of undamped resonances (cf. sec. 2.2.1). As the resonances can be circumvented easily, as is also shown in appendix B, we will not pay any further attention to these particular cases.

$$\mathbf{AA}^{\times} = (\mathbf{I} - \mathbf{QS})^{-1} (\mathbf{H} + \mathbf{QGQ}^{\times}) (\mathbf{I} - \mathbf{S}^{\times}\mathbf{Q}^{\times})^{-1} \tag{2.199}$$

and

$$\mathbf{BB}^{\times} = (\mathbf{I} - \mathbf{SQ})^{-1} (\mathbf{G} + \mathbf{SHS}^{\times}) (\mathbf{I} - \mathbf{Q}^{\times}\mathbf{S}^{\times})^{-1}. \tag{2.200}$$

When the scattering matrices and noise-wave power matrices of the composing multiports are known, the noise-power transmission in any linear transmission system can be calculated completely. This can be done in detail for every (short) connecting line and for every component. Even the correlation between the noise waves can be evaluated. The mutual correlation of the **A**-waves is given by the off-diagonal elements of $\mathbf{AA}^{\times}$ and of the **B**-waves by the same elements of $\mathbf{BB}^{\times}$. The correlation between **A**- and **B**-waves is described by the elements of the matrix

$$\mathbf{AB}^{\times} = (\mathbf{I} - \mathbf{QS})^{-1} (\mathbf{QG} + \mathbf{HS}^{\times}) (\mathbf{I} - \mathbf{Q}^{\times}\mathbf{S}^{\times})^{-1} \tag{2.201}$$

and its Hermitian conjugate. The density of the total power radiated from **Q** towards **S** equals the trace of $\mathbf{AA}^{\times}$, and the density of the total power flowing in the opposite direction is given by the trace of $\mathbf{BB}^{\times}$.

2.5.4. *Definite character of the power matrix*

The powers $A_i A_i^{+}$ and $B_i B_i^{+}$ of the individual noise waves A_i and B_i are inherently non-negative. Hence the diagonal elements of the matrices $\mathbf{AA}^{\times}$ and $\mathbf{BB}^{\times}$ must be non-negative under all circumstances. Let us suppose that the n-port **Q** is noiseless. Then the matrix **H** is identically the zero matrix **O**. The diagonal elements of the matrix

$$\mathbf{BB}^{\times} = (\mathbf{I} - \mathbf{SQ})^{-1} \mathbf{G} (\mathbf{I} - \mathbf{Q}^{\times}\mathbf{S}^{\times})^{-1} \tag{2.202}$$

must be non-negative for all terminations of **S**, i.e. for all possible matrices **Q**. Further, let $\mathbf{U}_i$ denote the ith row vector of the matrix $(\mathbf{I} - \mathbf{SQ})^{-1}$. Then the scalar inequality

$$B_i B_i^{+} = \mathbf{U}_i \mathbf{G} \mathbf{U}_i^{\times} \geqslant 0 \tag{2.203}$$

must be valid for all possible row vectors $\mathbf{U}_i$. This is the condition for the matrix **G** to be positive semi-definite.

From eqs (2.199) and (2.200) it is easy to see that both the matrices $\mathbf{AA}^{\times}$ and $\mathbf{BB}^{\times}$ are sums of Hermitian congruence, or equivalence, transformations of the matrices **H** and **G**. In matrix algebra Sylvester's law of inertia implies that a Hermitian congruence transformation preserves the definite character of Hermitian matrices [40]. Hence, the matrices $\mathbf{AA}^{\times}$ and $\mathbf{BB}^{\times}$ are positive semi-definite. Since, furthermore, the diagonal elements of positive semi-definite Hermitian matrices are non-negative, the above-stated requirement is fulfilled.

2.6. Passive multiports and thermal noise

2.6.1. *Passivity and the first law of thermodynamics*

A multiport is defined to be passive if it is impossible to extract power from it other than thermally radiated noise power. The power conveyed to a passive multiport by incident waves can partly be absorbed by the multiport. The other part of that power is scattered, that means transmitted and reflected. If the scattering matrix of a passive noiseless multiport is **S** and if the incident and scattered waves are represented by the column vectors **A** and **B**, respectively, the equation

$$\mathbf{B} = \mathbf{S}\,\mathbf{A} \tag{2.204}$$

applies. The input power P_i and output power P_o are respectively given by

$$P_i = \mathbf{A}^{\times}\mathbf{A} \tag{2.205}$$

and

$$P_o = \mathbf{B}^{\times}\mathbf{B} = \mathbf{A}^{\times}\mathbf{S}^{\times}\mathbf{S}\mathbf{A}. \tag{2.206}$$

The power P_{abs}, absorbed in the multiport, is given by

$$P_{abs} = P_i - P_o = \mathbf{A}^{\times}\mathbf{P}_S\,\mathbf{A}, \tag{2.207}$$

where

$$\mathbf{P}_S = \mathbf{I} - \mathbf{S}^{\times}\mathbf{S}. \tag{2.208}$$

The Hermitian matrix $\mathbf{P}_S$ will be called the power-absorption matrix of the multiport **S** (cf. sec. 3.1.7). Together with the definition of passivity the first law of thermodynamics now requires that the power absorbed in a passive multiport be non-negative. Thus, if a multiport **S** is passive, the condition

$$\mathbf{A}^{\times}\,\mathbf{P}_S\,\mathbf{A} \geqslant 0 \tag{2.209}$$

holds for all non-zero vectors **A**, the zero vector being trivial. This is also the condition for the Hermitian matrix $\mathbf{P}_S$ to be positive semi-definite. Hence we may conclude that a passive multiport has a positive semi-definite power-absorption matrix *).

If a passive multiport is lossless, it cannot absorb power at all. The output power is equal to the input power. Then, in eq. (2.209) the equality symbol applies for all possible vectors **A** and that is the case if $\mathbf{P}_S$ is identically the zero matrix. Application to eq. (2.208) shows that, then, the scattering matrix **S** must satisfy the unitary relation

$$\mathbf{S}^{\times}\mathbf{S} = \mathbf{I}. \tag{2.210}$$

*) The passivity condition is given here in a form which is slightly different from the usual form [23,41]). In conventional theories on multiports the considerations are mostly confined only to reciprocal systems and, hence, to symmetrical scattering matrices. Then the passivity can be stated by the positive semi-definite character of the matrix $(\mathbf{I} - \mathbf{S}^{*}\mathbf{S})$ instead of by that of the matrix $(\mathbf{I} - \mathbf{S}^{\times}\mathbf{S})$, as has been done here.

Thus a lossless passive multiport has a unitary scattering matrix.

In appendix C *), theorem (C.II), it is derived that for any pair of quadratic matrices **M** and **K** the matrices $(\mathbf{I} - \mathbf{MK})$ and $(\mathbf{I} - \mathbf{KM})$ have the same eigenvalues with the same multiplicity. This is even true when both the matrices **M** and **K** are singular. As a corrollary, theorem (C.VII), it can be stated that the positive (semi-)definiteness of the matrix $(\mathbf{I} - \mathbf{MM}^{\times})$ ensues upon that of the matrix $(\mathbf{I} - \mathbf{M}^{\times}\mathbf{M})$, and vice versa. Hence passivity of a multiport **S** can be guaranteed by the positive semi-definite character of the matrix $(\mathbf{I} - \mathbf{SS}^{\times})$ as well as by that of the matrix $(\mathbf{I} - \mathbf{S}^{\times}\mathbf{S})$. For these reasons and analogous to the definition of eq. (2.208) we now introduce the matrix $\mathbf{N}_S$ by

$$\mathbf{N}_S = \mathbf{I} - \mathbf{SS}^{\times}. \tag{2.211}$$

The matrix $\mathbf{N}_S$ is equivalent to the matrix $\mathbf{P}_S$ in that it contains the same information. For reasons which will become clear below, $\mathbf{N}_S$ will be called noise-distribution matrix (cf. sec. 2.6.5).

One may ask whether there is much sense in stating these different forms of the passivity condition separately. The answer is yes, and for two reasons. First, in the case of non-reciprocal devices calculation of the second form from the first one sometimes requires a lot of cumbersome algebra. Secondly, in practice positive semi-definiteness of $\mathbf{N}_S$ can often be employed with much more convenience than positive semi-definiteness of $\mathbf{P}_S$, for instance, to demonstrate the possibility or the impossibility of existence of a proposed passive device [42,43]) (cf. next section).

2.6.2. *Some implications of passivity*

To shed some light on the difference in the natures of the two equivalent forms of the passivity condition, in the present section some of the restrictions they impose directly on the signal performance of a multiport will be considered. The positive semi-definite character of a Hermitian matrix implies that its real diagonal elements are non-negative. Hence, by virtue of eq. (2.208), if $\mathbf{P}_S$ is positive semi-definite, among others the conditions

$$\sum_{j=1}^{j=n} S_{ji}^{*} S_{ji} \leqslant 1, \qquad i = 1, 2, \ldots, n, \tag{2.212}$$

hold where S_{ji} is the element in the jth row and in the ith column of the matrix **S**.

When one of the ports, say the ith, of an n-port is completely reflecting, the equation

$$S_{ii}^{*} S_{ii} = 1 \tag{2.213}$$

*) During the course of the present treatise some less known theorems of matrix algebra are used. Those of them which do not appear in the ordinary literature on matrix algebra have been collected in appendix C together with their proofs. In what follows these will be denoted by the capital C followed by an appropriate roman number.

is valid. Substitution in eq. (2.212) yields

$$\sum_{\substack{j=1 \\ (j \neq i)}}^{n} S_{ji}{}^{*} S_{ji} \leqslant 0. \tag{2.214}$$

Since all terms of this sum are real and non-negative this relation can be satisfied only if the equation

$$S_{ji} = 0, \qquad j \neq i \tag{2.215}$$

applies. This means:

> "*When a signal is incident upon a completely reflecting port of a linear and passive multiport, no transmission towards any of the other ports occurs*".

Of course, this is a rather obvious principle.

When a signal incident upon one of the ports, say the ith, of an n-port is completely transmitted towards another port, say the kth, the equation

$$S_{ki}{}^{*} S_{ki} = 1 \tag{2.216}$$

is valid. Analogously to the above derivation, this results in

$$S_{ji} = 0, \qquad j \neq k, \tag{2.217}$$

which means that all elements, but the kth, of the ith column are equal to zero. This leads to the quite trivial principle:

> "*When a signal, incident upon a given port of a linear and passive multiport, is transmitted completely towards another port, no transmission from the former towards a third port can take place*".

Summarizing, we may conclude that the passivity, as defined by the positive semi-definiteness of $\mathbf{P}_S$, directly imposes restrictions upon the way power, incident upon a given port, is distributed over all ports. In general this type of restriction is well recognized and thoroughly understood.

By virtue of eq. (2.211), positive semi-definiteness of $\mathbf{N}_S$ yields the conditions

$$\sum_{j=1}^{n} S_{ij} S_{ij}{}^{*} \leqslant 1, \qquad i = 1, 2, \ldots, n. \tag{2.218}$$

Substitution of eq. (2.213) gives ultimately the equation

$$S_{ij} = 0, \qquad j \neq i. \tag{2.219}$$

It can be formulated as follows:

> "*When a signal is incident upon a partly reflecting or a non-reflecting port of a linear and passive multiport, no transmission towards a completely reflecting port can take place*".

In the same way, substitution of eq. (2.216) leads to the equation

$$S_{ij} = 0, \qquad j \neq k. \tag{2.220}$$

It means:

> "*When a signal, incident upon a given port of a linear and passive multiport, is transmitted completely towards another port, no transmission from a third port towards the latter can take place*".

We may conclude that, if the passivity is defined by the positive semi-definite character of $\mathbf{N_S}$, it directly expresses conditions which restrict the way the output power of a given port is being built up from the input powers of all ports. This type of restriction is not as well known and is sometimes not recognized as may be clear from the following examples.

The first principle of the latter type implies that a reflection isolator cannot exist. A reflection isolator is a lossless passive 2-port which reflects completely a signal incident upon one port, while a signal incident upon the other port is transmitted completely. That the non-existence of a reflection isolator is not quite obvious will be clear from the discussions about the "paradox" of Lax and Button [42]) and from the fact that Ishii [43]) thought he had constructed such a device. The second principle of the latter type prohibits, for instance, the existence of a lossless passive device which would simultaneously feed the total powers of two not-phase-locked generators into the same line. Nevertheless, occasionally, the need of such a 3-port is felt and it is searched for.

2.6.3. *Passivity and net flow of noise power*

If in the configuration of fig. 12 the termination **Q** of the *n*-port **S** is passive and noiseless, the net total flow of noise power from **S** towards **Q** must be non-negative. The freedom of noise of the *n*-port **Q** implies that its noise-wave power matrix **H** is identically zero. Then, by virtue of theorem (C.III), eqs (2.199) and (2.200) can be written in the form

$$\mathbf{A}\mathbf{A}^{\times} = \mathbf{Q}\,(\mathbf{I} - \mathbf{S}\mathbf{Q})^{-1}\,\mathbf{G}\,(\mathbf{I} - \mathbf{Q}^{\times}\mathbf{S}^{\times})^{-1}\,\mathbf{Q}^{\times} \tag{2.221}$$

and

$$\mathbf{B}\mathbf{B}^{\times} = (\mathbf{I} - \mathbf{S}\mathbf{Q})^{-1}\,\mathbf{G}\,(\mathbf{I} - \mathbf{Q}^{\times}\mathbf{S}^{\times})^{-1}. \tag{2.222}$$

The density of the total noise power flowing from **S** towards **Q** and given by the trace of $\mathbf{B}\mathbf{B}^{\times}$ must not be smaller than the density of the total noise power flowing in the opposite direction and given by the trace of $\mathbf{A}\mathbf{A}^{\times}$, i.e.

$$\text{trace}\,(\mathbf{B}\mathbf{B}^{\times} - \mathbf{A}\mathbf{A}^{\times}) \geqslant 0. \tag{2.223}$$

From eqs (2.221) and (2.222) it can be seen that this can be written in the form

$$\text{trace}\,(\mathbf{B}\mathbf{B}^{\times} - \mathbf{Q}\mathbf{B}\mathbf{B}^{\times}\mathbf{Q}^{\times}) \geqslant 0. \tag{2.224}$$

In sec. 2.5.4 it has been shown that $\mathbf{BB}^{\times}$ is a positive semi-definite matrix. Moreover, the passivity of the n-port $\mathbf{Q}$ implies that the matrix $(\mathbf{I}-\mathbf{QQ}^{\times})$ is also positive semi-definite. Then, by virtue of theorem (C.VIII), eq. (2.224) is confirmed.

One might wonder whether the above principle must be valid in all individual connecting lines, that is, whether the relation

$$B_i B_i^{+} \geqslant A_i A_i^{+} \tag{2.225}$$

must hold for all values of i. However, at least when non-reciprocal components are involved, this need not be true (cf. sec. 2.6.7).

2.6.4. *Diagonal multiport*

Since it is convenient for future derivations and considerations, we will now introduce the concept of a diagonal n-port. We define a diagonal n-port to be a combination of n separate, uncoupled 1-ports with mutually uncorrelated noise-wave sources. It is illustrated in fig. 13.

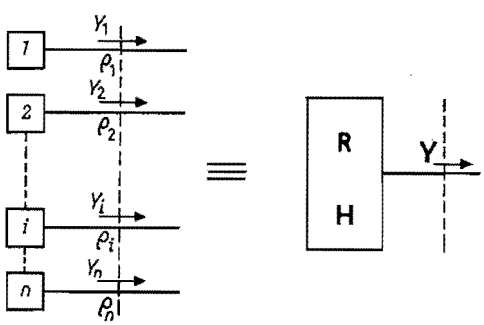


Fig. 13. Diagonal multiport. A noisy diagonal n-port consists of n noisy 1-ports.

Let ϱ_i be the reflection coefficient of the ith 1-port. Then the scattering matrix $\mathbf{R}$ of the diagonal n-port is the diagonal matrix diag (ϱ_i). Its elements are given by

$$R_{ii} = \varrho_i, \tag{2.226}$$

$$R_{ij} = 0, \qquad i \neq j. \tag{2.227}$$

It is readily seen that the equations

$$\mathbf{I}-\mathbf{RR}^{\times} = \mathbf{I}-\mathbf{R}^{\times}\mathbf{R} = \text{diag}\,(1-\varrho_i\varrho_i^{*}) \tag{2.228}$$

are valid. Hence, both the power-absorption matrix $\mathbf{P}_R$ and the noise-distribution matrix $\mathbf{N}_R$ are diagonal. If a diagonal n-port is passive, the latter matrices are positive semi-definite. This implies that the scalar quantity $(1-\varrho_i\varrho_i^{*})$ is non-negative for all values of i so that, by virtue of eq. (2.68), all constituent 1-ports must be passive, which is quite obvious.

Let Y_i be the amplitude of the noise-wave source ascribed to the ith 1-port. Then, by virtue of eq. (2.94), the power of that source can be written in the form

$$Y_i Y_i^+ = kT_i (1 - \varrho_i \varrho_i^*), \tag{2.229}$$

where T_i is the characteristic noise temperature of the ith 1-port. Since the noise-wave sources are uncorrelated, the products $Y_i Y_j^+$ $(i \neq j)$ are equal to zero. Hence the noise-wave power matrix of a diagonal n-port can be written in the form

$$\mathbf{H} = k\mathbf{T}_R (\mathbf{I} - \mathbf{R}\mathbf{R}^\times) = k\mathbf{T}_R\mathbf{N}_R, \tag{2.230}$$

where $\mathbf{T}_R$ is a diagonal matrix given by

$$\mathbf{T}_R = \text{diag}\,(T_i). \tag{2.231}$$

The matrix $\mathbf{T}_R$ is called the temperature matrix of $\mathbf{R}$. It is easy to see that the matrix $\mathbf{H}$, too, is diagonal.

If a passive diagonal n-port is at a uniform temperature T, the matrix $\mathbf{T}_R$ reduces to the scalar matrix $T\mathbf{I}$, i.e.

$$\mathbf{T}_R = \text{diag}\,(T) = T\mathbf{I}. \tag{2.232}$$

In this case the noise-power matrix is given by

$$\mathbf{H} = kT(\mathbf{I} - \mathbf{R}\mathbf{R}^\times) = kT\mathbf{N}_R. \tag{2.233}$$

2.6.5. *Passive multiport at uniform temperature*

We next consider an arbitrary passive n-port $\mathbf{S}$ which is at a uniform absolute temperature T. Let this n-port be terminated by the passive diagonal n-port $\mathbf{R}$ introduced in the previous section, as is illustrated in fig. 14. This is a particular case of the general configuration discussed in sec. 2.5.3.

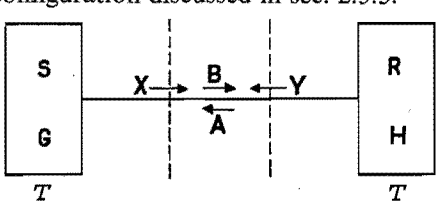


Fig. 14. Noisy n-port loaded by n noisy 1-ports. The complete load is a noisy diagonal n-port.

If $\mathbf{R}$, too, is uniformly at the temperature T, thermal equilibrium must exist in any bandwidth δf. The ports of $\mathbf{R}$ being mutually uncoupled, the second law of thermodynamics requires that in all connection lines between $\mathbf{S}$ and $\mathbf{R}$ the two opposite flows of power are in balance since, otherwise, some 1-ports would warm up while others would be getting colder. The equation

$$A_i A_i^+ = B_i B_i^+ \tag{2.234}$$

must be valid for all values of i.

Let us now write the noise-wave power matrix $\mathbf{G}$ of $\mathbf{S}$ in the form

$$\mathbf{G} = kT\mathbf{M}, \tag{2.235}$$

where $\mathbf{M}$ is a matrix which is still to be determined. Replacement of $\mathbf{Q}$ by $\mathbf{R}$ and substitution of eq. (2.233) into eqs (2.199) and (2.200) yield

$$\mathbf{A}\mathbf{A}^{\times} = kT\,(\mathbf{I} - \mathbf{R}\mathbf{S})^{-1}\,(\mathbf{I} - \mathbf{R}\mathbf{R}^{\times} + \mathbf{R}\mathbf{M}\mathbf{R}^{\times})\,(\mathbf{I} - \mathbf{S}^{\times}\mathbf{R}^{\times})^{-1} \tag{2.236}$$

and

$$\mathbf{B}\mathbf{B}^{\times} = kT\,(\mathbf{I} - \mathbf{S}\mathbf{R})^{-1}\,(\mathbf{M} + \mathbf{S}\mathbf{S}^{\times} - \mathbf{S}\mathbf{R}\mathbf{R}^{\times}\mathbf{S}^{\times})\,(\mathbf{I} - \mathbf{R}^{\times}\mathbf{S}^{\times})^{-1}. \tag{2.237}$$

After some algebra we obtain

$$\mathbf{A}\mathbf{A}^{\times} - \mathbf{B}\mathbf{B}^{\times} = kT\,(\alpha - \beta), \tag{2.238}$$

where the matrices α and β are given by

$$\alpha = (\mathbf{I} - \mathbf{R}\mathbf{S})^{-1} + (\mathbf{I} - \mathbf{S}^{\times}\mathbf{R}^{\times})^{-1} - (\mathbf{I} - \mathbf{S}\mathbf{R})^{-1} - (\mathbf{I} - \mathbf{R}^{\times}\mathbf{S}^{\times})^{-1} \tag{2.239}$$

and

$$\beta = \gamma - \mathbf{R}\gamma\mathbf{R}^{\times}, \tag{2.240}$$

with

$$\gamma = (\mathbf{I} - \mathbf{S}\mathbf{R})^{-1}\,(\mathbf{M} - \mathbf{I} + \mathbf{S}\mathbf{S}^{\times})\,(\mathbf{I} - \mathbf{R}^{\times}\mathbf{S}^{\times})^{-1}. \tag{2.241}$$

The condition of eq. (2.234) requires that corresponding diagonal elements of $\mathbf{A}\mathbf{A}^{\times}$ and $\mathbf{B}\mathbf{B}^{\times}$ are the same. Hence the equation

$$\alpha_{ii} = \beta_{ii} \tag{2.242}$$

must apply. Since $\mathbf{R}$ is a diagonal matrix, by virtue of theorem (C.VI), the diagonal elements α_{ii} are all equal to zero for all values of i. The diagonal elements β_{ii} are given by

$$\beta_{ii} = (1 - \varrho_i\varrho_i^{*})\,\gamma_{ii}. \tag{2.243}$$

As in the general case $(1 - \varrho_i\varrho_i^{*})$ is not equal to zero, the condition of eq. (2.242) requires that γ_{ii} be zero. If $\mathbf{U}_i$ is the ith row vector of the matrix $(\mathbf{I} - \mathbf{S}\mathbf{R})^{-1}$, we hence have

$$\gamma_{ii} = \mathbf{U}_i\,(\mathbf{M} - \mathbf{I} + \mathbf{S}\mathbf{S}^{\times})\,\mathbf{U}_i^{\times} = 0 \tag{2.244}$$

for all possible row vectors $\mathbf{U}_i$. This can be so only if the relation

$$\mathbf{M} = \mathbf{I} - \mathbf{S}\mathbf{S}^{\times} \tag{2.245}$$

is valid. This is the defining relation eq. (2.211) of the matrix $\mathbf{N}_S$. The noise-wave-power matrix $\mathbf{G}$ of a passive multiport $\mathbf{S}$ at the uniform temperature T can now be written in the form

$$\mathbf{G} = kT\,(\mathbf{I} - \mathbf{S}\mathbf{S}^{\times}) = kT\,\mathbf{N}_S. \tag{2.246}$$

The matrix $\mathbf{N}_S$ describes how the noise power generated in **S** is distributed over the ports of **S**. In sec. 2.6.1 $\mathbf{N}_S$ has therefore been called the noise-distribution matrix.

2.6.6. *Two passive multiports at uniform temperatures*

In the previous section having only considered a particular termination of a passive n-port, we have still to verify the general validity of eq. (2.246). Therefore, in the next three sections, an investigation into the consistency of that result with the second law of thermodynamics will be carried out.

In the general configuration of fig. 12, described in sec. 2.5.3, let the n-ports **S** and **Q** both be passive and have the uniform temperatures T_S and T_Q, respectively. Let us assume that the respective noise-wave power matrices can be written in the forms

$$\mathbf{G} = kT_S(\mathbf{I} - \mathbf{S}\mathbf{S}^\times) \tag{2.247}$$

and

$$\mathbf{H} = kT_Q(\mathbf{I} - \mathbf{Q}\mathbf{Q}^\times). \tag{2.248}$$

Substitution into eq. (2.199) yields for the power matrix of the noise waves which travel from **Q** towards **S**,

$$\begin{aligned}\mathbf{A}\mathbf{A}^\times = {} & kT_Q(\mathbf{I} - \mathbf{Q}\mathbf{S})^{-1}(\mathbf{I} - \mathbf{Q}\mathbf{Q}^\times)(\mathbf{I} - \mathbf{S}^\times\mathbf{Q}^\times)^{-1} + \\ & + kT_S(\mathbf{I} - \mathbf{Q}\mathbf{S})^{-1}\mathbf{Q}(\mathbf{I} - \mathbf{S}\mathbf{S}^\times)\mathbf{Q}^\times(\mathbf{I} - \mathbf{S}^\times\mathbf{Q}^\times)^{-1}.\end{aligned} \tag{2.249}$$

This can be written in the more convenient form

$$\mathbf{A}\mathbf{A}^\times = \tfrac{1}{2}k(T_Q + T_S)\mathbf{K}_{S,Q} + \tfrac{1}{2}k(T_Q - T_S)\mathbf{L}_{S,Q}, \tag{2.250}$$

with

$$\begin{aligned}\mathbf{K}_{S,Q} &= (\mathbf{I} - \mathbf{Q}\mathbf{S})^{-1}(\mathbf{I} - \mathbf{Q}\mathbf{S}\mathbf{S}^\times\mathbf{Q}^\times)(\mathbf{I} - \mathbf{S}^\times\mathbf{Q}^\times)^{-1} = \\ &= (\mathbf{I} - \mathbf{Q}\mathbf{S})^{-1} + (\mathbf{I} - \mathbf{S}^\times\mathbf{Q}^\times)^{-1} - \mathbf{I}\end{aligned} \tag{2.251}$$

and

$$\mathbf{L}_{S,Q} = (\mathbf{I} - \mathbf{Q}\mathbf{S})^{-1}(\mathbf{I} - 2\,\mathbf{Q}\mathbf{Q}^\times + \mathbf{Q}\mathbf{S}\mathbf{S}^\times\mathbf{Q}^\times)(\mathbf{I} - \mathbf{S}^\times\mathbf{Q}^\times)^{-1}. \tag{2.252}$$

Of course, as can easily be seen, the matrices $\mathbf{K}_{S,Q}$ and $\mathbf{L}_{S,Q}$ are Hermitian. Equation (2.250) divides the power of the noise waves travelling from **Q** towards **S** into two parts. The first term of the right-hand side determines the mutual exchange of noise power between **S** and **Q**. This part is proportional to the mean temperature $\frac{1}{2}(T_Q + T_S)$. The second term refers to the overflow of noise power. This overflow is proportional to the temperature difference $(T_Q - T_S)$.

In the same way for the opposite noise waves the power matrix turns out to be

$$\mathbf{B}\mathbf{B}^\times = \tfrac{1}{2}k(T_S + T_Q)\mathbf{K}_{Q,S} + \tfrac{1}{2}k(T_S - T_Q)\mathbf{L}_{Q,S}, \tag{2.253}$$

where the matrices $\mathbf{K}_{Q,S}$ and $\mathbf{L}_{Q,S}$ can be derived from eqs (2.251) and (2.252), respectively, by exchange of **S** and **Q**.

2.6.7. *Passivity and the second law of thermodynamics at temperature equilibrium*

The configuration investigated above can be considered to be a general collection of interconnected arbitrary passive multiports. One of the members of this collection is represented by **S** and all the others by **Q**.

When the temperatures of all the multiports are the same, say T, we have the equation

$$T_S = T_Q = T. \tag{2.254}$$

In this case those terms of eqs (2.250) and (2.253), which refer to the overflow of noise power, are equal to zero. On account of these equations the power matrices of the noise waves travelling from the one multiport **S** to all the others **Q** and of those in the opposite direction are respectively given by

$$\mathbf{B}\mathbf{B}^{\times} = kT\,\mathbf{K}_{Q,S} \tag{2.255}$$

and

$$\mathbf{A}\mathbf{A}^{\times} = kT\,\mathbf{K}_{S,Q}. \tag{2.256}$$

For the exchange of power the diagonal elements only are meaningful. The densities of the net exchange of power per line and from **S** towards **Q** are given by the diagonal elements of the matrix

$$\mathbf{B}\mathbf{B}^{\times} - \mathbf{A}\mathbf{A}^{\times} = kT\,(\mathbf{K}_{Q,S} - \mathbf{K}_{S,Q}). \tag{2.257}$$

The density of the total net power exchange from **S** towards **Q** is equal to the trace of this matrix.

Now, the second law of thermodynamics requires that, in the case of temperature equilibrium, the total net exchange of noise power between one multiport of a collection and all the others is equal to zero. This means that the trace of the matrix of eq. (2.257) must be equal to zero. In appendix C theorem (C.V) states that, if they exist, the traces of the matrices $(\mathbf{I} - \mathbf{QS})^{-1}$ and $(\mathbf{I} - \mathbf{SQ})^{-1}$ are equal. If this theorem together with its Hermitian congruent version is applied to eq. (2.251) and its cyclic version, it is found that the equation

$$\text{trace}\,(\mathbf{K}_{Q,S}) = \text{trace}\,(\mathbf{K}_{Q,S}) \tag{2.258}$$

is valid so that, indeed, the second law is satisfied.

The above observations can be made in even more detail. Let us suppose that the multiport **Q** consists of a number of separate uncoupled multiports. Then its scattering matrix contains an equally large number of non-zero principal submatrices $\mathbf{Q}_{jj}$ while all other elements are equal to zero, i.e.

$$\mathbf{Q} = \begin{pmatrix} \mathbf{Q}_{11} & \mathbf{O} & \dots & \mathbf{O} \\ \mathbf{O} & \mathbf{Q}_{22} & \dots & \mathbf{O} \\ \vdots & \vdots & & \vdots \\ \mathbf{O} & \mathbf{O} & \dots & \mathbf{Q}_{ll} \end{pmatrix}. \tag{2.259}$$

By an argument analogous to the one used in the proof of theorem (C.VI), the equation

$$[(\mathbf{I}-\mathbf{QS})^{-1}]_{jj}\,\mathbf{Q}_{jj} = \mathbf{Q}_{jj}\,[(\mathbf{I}-\mathbf{SQ})^{-1}]_{jj} \tag{2.260}$$

can be derived. The double subscript jj refers to the jth principal submatrix. From this equation it can be seen that the matrix $[(\mathbf{I}-\mathbf{QS})^{-1}]_{jj}$ can be obtained from the matrix $[(\mathbf{I}-\mathbf{SQ})^{-1}]_{jj}$ by a similarity transformation. Thus, being similar, these matrices have the same eigenvalues and, hence, the same trace. Now, in the same way as eq. (2.258) has been deduced, it can be shown that, if temperature equilibrium exists, the flows of noise power to and from any separate uncoupled multiport, which forms part of a general termination, are in balance. This detailed balance, too, conforms to the requirements of the second law. The configuration of n separate uncoupled 1-ports discussed in sec. 2.6.5. is a particular case.

One may now ask a question analogous to the one posed at the end of sec. 2.6.3. One may wonder whether in every line the opposite flows of noise power balance. This would be even more than is dictated by the second law of thermodynamics. That law demands only the balance of opposite total noise-power flows to and from a multiport. In fact, if one or both of the multiports are non-reciprocal, that is, if one or both of the matrices **S** and **Q** are not symmetric, the power balance per line need not exist. This can be verified readily, for example, by a configuration of two 2-ports one of which is an ideal isolator, the other being a section of line. However, if all multiports are reciprocal, the balance of power per line exists always. Then, both the matrices **S** and **Q** are symmetric, i.e.

$$\mathbf{S}^{\mathrm{T}} = \mathbf{S} \tag{2.261}$$

and

$$\mathbf{Q}^{\mathrm{T}} = \mathbf{Q}, \tag{2.262}$$

the superscript T denoting the transpose. By virtue of these equations, we have

$$\{(\mathbf{I}-\mathbf{SQ})^{-1}\}^{\mathrm{T}} = (\mathbf{I}-\mathbf{Q}^{\mathrm{T}}\mathbf{S}^{\mathrm{T}})^{-1} = (\mathbf{I}-\mathbf{QS})^{-1}. \tag{2.263}$$

Since transposition of a matrix does not alter the diagonal elements, it is readily seen that the matrices $(\mathbf{I}-\mathbf{SQ})^{-1}$ and $(\mathbf{I}-\mathbf{QS})^{-1}$ have the same diagonal elements and, hence, so have the matrices $\mathbf{K}_{\mathrm{Q,S}}$ and $\mathbf{K}_{\mathrm{S,Q}}$. This proves the above statement of balance of power per line in reciprocal systems. It should be noted that this result is a consequence of reciprocity alone and not of the laws of thermodynamics.

2.6.8. *Passivity and the second law of thermodynamics at different multiport temperatures*

When the uniform temperatures T_{S} and T_{Q} are different, the density of the

net total flow of noise power from S towards Q is no longer given by the trace of the matrix of eq. (2.257), but is equal to the trace of the matrix

$$\mathbf{BB}^{\times} - \mathbf{AA}^{\times} = \tfrac{1}{2} k (T_S + T_Q)(\mathbf{K}_{Q,S} - \mathbf{K}_{S,Q}) + \\ + \tfrac{1}{2} k (T_S - T_Q)(\mathbf{L}_{Q,S} + \mathbf{L}_{S,Q}). \tag{2.264}$$

The first term of the right-hand side has been investigated in the previous section. Its trace is always equal to zero, even when the multiport temperatures are different. Therefore the first term represents the balanced part of the noise-power exchange. The second term describes the net overflow of noise power.

After some algebraic manipulations, we obtain from eq. (2.252) and its cyclic version

$$\mathbf{L}_{Q,S} + \mathbf{L}_{S,Q} = (\mu - \mathbf{S}\mu\mathbf{S}^{\times}) + (\nu - \mathbf{Q}\nu\mathbf{Q}^{\times}), \tag{2.265}$$

where

$$\mu = (\mathbf{I} - \mathbf{QS})^{-1} \mathbf{N}_Q (\mathbf{I} - \mathbf{S}^{\times}\mathbf{Q}^{\times})^{-1} \tag{2.266}$$

and

$$\nu = (\mathbf{I} - \mathbf{SQ})^{-1} \mathbf{N}_S (\mathbf{I} - \mathbf{Q}^{\times}\mathbf{S}^{\times})^{-1}. \tag{2.267}$$

The matrices μ and ν are Hermitian congruence transformations of the matrices $\mathbf{N}_Q$ and $\mathbf{N}_S$, respectively. Due to passivity, the latter matrices are positive semi-definite. Since an Hermitian congruence transformation preserves this property, the matrices μ and ν are positive semi-definite. Hence, by virtue of theorem (C.VIII), the traces of both terms of the right-hand side of eq. (2.265) are non-negative. Thus the relation

$$\text{trace}\,(\mathbf{L}_{Q,S} + \mathbf{L}_{S,Q}) \geqslant 0 \tag{2.268}$$

holds. Application of this result to eq. (2.264) shows readily that more power is radiated from a hotter passive multiport towards a colder one than in the opposite direction. Herewith a further requirement of the second law of thermodynamics is fulfilled.

Under the approximate condition $hf \ll kT$ of eq. (2.92) the total overflow of noise power is proportional to the temperature difference. Otherwise, it is proportional to the difference of the values the noise function $F(T,f)$ has for T_S and T_Q, respectively (cf. sec. 2.3.3). Due to the monotonically increasing character of that function, the above statement about the direction of power exchange is quite general.

It is worth while to point out that in the preceding sections it has been shown that the formalism developed thus far is consistent in that it satisfies all conditions which are imposed by the laws of thermodynamics.

At this stage we may conclude that a consistent and adequate formalism to describe the spot-noise performance of linear systems has been obtained.

2.6.9. *Alternative description of noise-source correlation*

It has been found that the noise of a multiport can be described by assigning a noise-wave source to each port. The powers of the sources are given by the diagonal elements of the Hermitian matrix **G** defined by eq. (2.189). The correlation of the sources is described by the off-diagonal elements of that matrix. However, one can think of alternative methods of describing correlation of a set of noise-wave sources. In this section an example of such an alternative method will be given.

Let X_i be the amplitude of the noise-wave source assigned to the ith port. Starting with the mth port, the following set of n equations can be written down:

$$\left.\begin{aligned} X_m &= A_m, \\ X_{m+1} &= A_{m+1} + D_{m+1,m} A_m, \\ &\vdots \\ X_{m+p} &= A_{m+p} + \sum_{q=m}^{m+p-1} D_{m+p,q} A_q, \\ &\vdots \end{aligned}\right\} \tag{2.269}$$

These equations indicate that the $(m+p)$th noise-wave source X_{m+p} can be thought to consist of a part A_{m+p}, which is not correlated with the preceding A_q's, and of p parts $D_{m+p,q} A_q$, which are completely correlated successively with the p preceding A_q's, and so on for every source X_i. For every value of m a set of n such equations can be formulated. Hence, this method can be employed, whatever port is started with.

In order that this procedure corresponds to a physically permissible model, it is necessary that the powers of the part-sources A_i are all non-negative. Hence, the condition

$$A_i A_i^+ \geqslant 0 \tag{2.270}$$

must be satisfied for all values of i. Successive substitutions in eqs (2.269) lead to the condition

$$X_i X_i^+ \geqslant \sum_{q=m}^{i-1} D_{i,q} D_{i,q}^* A_q A_q^+, \tag{2.271}$$

which must be valid for all values of m and all values of i which cyclically follow after m. Substitution of eqs (2.269) in eqs (2.186) and (2.187) yields $A_i A_i^+$ and $D_{i,q}$ in terms of the elements of the matrix **G**. Inserting that result into eq. (2.271) then shows that those conditions are equivalent to the requirement that the determinants of all principal submatrices of **G** be non-negative and that is true because **G** is positive semi-definite as has been shown in sec. 2.5.4.

Thus a description of the noise of an n-port is also possible in terms of n powers AA^+ of uncorrelated noise-wave sources together with a set of

$\frac{1}{2}n(n-1)$ complex correlation factors $D_{i,q}$ $(i > q$; cf. sec. 2.1.5). For this description as well as for the one of the matrix $\mathbf{G}$ to be complete n^2 real numbers must be given. Whatever procedure is used, n^2 is the smallest amount of real numbers necessary to describe the noise of an n-port.

Since, in most cases, the procedure developed in the preceding sections is more efficient and more concise than others, it will be used throughout the present treatise.

2.7. Noise-temperature matrix

2.7.1. *Definition*

In investigations of noise performance of non-lumped transmission systems it is common practice to characterize the noise contribution of components by means of effective temperatures. Such "noise" temperatures are used in calculations as if they were equivalent with flows of energy transmitted by the components of the system. However, this is permissible only for matched and quasi-matched configurations. Only there powers of noise waves can be considered to be proportional and equivalent to the temperatures or the effective temperatures of the emitting components. In other systems, where reflections occur, matters are not so simple. Then interference effects, that is correlation, must be accounted for.

In sec. 2.6.4, where the concept of a diagonal multiport was introduced, a temperature matrix has been defined. This matrix, given by eq. (2.231), describes the temperatures or the characteristic temperatures of all constituent 1-ports. Such an introduction of a noise-temperature matrix also in the general case of arbitrary noisy multiports could probably maintain the concept of noise temperature and, perhaps, even deepen its meaning. Therefore we now proceed by defining a noise-temperature matrix analogously to the introduction of such a matrix by eq. (2.230).

When $\mathbf{G_S}$ is the noise-wave power matrix of an arbitrary noisy multiport and $\mathbf{S}$ and $\mathbf{N_S}$ are its scattering matrix and noise-distribution matrix, respectively, we define the noise-temperature matrix $\mathbf{T_S}$ by

$$\mathbf{G_S} = k\,\mathbf{T_S}\,(\mathbf{I} - \mathbf{S}\mathbf{S}^{\times}) = k\,\mathbf{T_S}\mathbf{N_S} \tag{2.272}$$

or by

$$\mathbf{T_S} = k^{-1}\,\mathbf{G_S}\mathbf{N_S}^{-1}. \tag{2.273}$$

In some special cases of singular matrices $\mathbf{N_S}$ the second defining equation is meaningless. As will be seen in sec. 4.2.2, then one can still rely on the first equation. Moreover, by a method analogous to that described in appendix B these special situations can also be met by introduction of small perturbing losses and evaluation of $\mathbf{T_S}$ by a limiting process for a vanishingly small per-

turbation *). For the time being we will not deal with these extreme cases.

2.7.2. *Motivation*

The definition of the noise-temperature matrix may seem to be a very impractical abstraction. For most multiports its elements do not represent real temperatures. Moreover, generally the elements are complex quantities. Only in some particular cases $\mathbf{T}_S$ has a distinct physical meaning. When a multiport is passive and at the uniform temperature T, by virtue of eqs (2.246) and (2.273), $\mathbf{T}_S$ reduces to the scalar matrix $T\mathbf{I}$, which can be interpreted quite readily. For a diagonal multiport the physical meaning of $\mathbf{T}_S$ is clear from the discussion in sec. 2.6.4. Furthermore, for a 1-port the noise-temperature matrix as well as the scattering matrix, the noise-wave-source power matrix and the noise-distribution matrix reduce to single numbers, that is to 1×1-matrices. Then, the noise-temperature matrix can be identified with the characteristic noise temperature introduced in sec. 2.3.1. In nearly all other cases $\mathbf{T}_S$ cannot be interpreted as easily.

Other attempts to describe the noise of multiports in terms of quantities with temperature dimension have been made. Vlaardingerbroek, Knol and Hart [44]) described the noise properties of a 2-port in terms of four "temperatures".

The mere fact that in some simple situations the noise-temperature matrix can be interpreted is not a sufficient justification for its introduction. Better motives however exist. In chapter 3 of the present treatise it will be shown that a change of coordinate systems transforms the noise-temperature matrix by a similarity transformation. Hence, the eigenvalues are invariants. In chapter 4 it will further be shown that these characteristic values are real numbers. They can be called characteristic noise temperatures of the multiport. In fact, they constitute the respective characteristic noise temperatures of the 1-ports which result when a multiport is transformed into a diagonal form. As a set, the characteristic noise temperatures together represent the intensity of the noise of a multiport. Furthermore, in chapter 5 a discussion of the work of Haus and Adler [15]) will be given. It will be shown there that the noise-temperature matrix is equivalent to the characteristic-noise matrix introduced by these authors and that the characteristic noise temperatures can be interpreted as a generalization of the extreme values of the noise measure, and vice versa. With all these observations to be made, the introduction of the concept of noise-temperature matrix will, in our opinion, be justified sufficiently.

2.8. Broadband systems with delay

2.8.1. *Noise-power equations*

In sec. 2.4.3 it has been found that under certain conditions the evaluation

*) See also the remark on passive reactive 1-ports, made in sec. 2.3.1.

of noise-power exchange over finite bandwidths Δf in broadband one-line systems which contain large delays is greatly simplified. Then, as shown in sec. 2.4.4, noise-power equations can directly be used. The necessary condition, given by eq. (2.137), is that the transit time be very large with respect to the inverse of Δf. If this condition is satisfied for all possible transmission paths in a multiport system, such a system can also be investigated directly by means of noise-power equations.

Furthermore, in sec. 2.4.5 it has been argued that the same simple procedure applies to quasi-matched one-line systems. The same argument is valid for quasi-matched multiport systems. Hence, if no wave or part of it travels more than once past the same point in the same direction and if no multipath transmission occurs, noise-power equations can also advantageously be used for multiport systems.

In actual systems various components may have internal delay times which are not short in comparison with the reciprocal of the bandwidth; e.g. travelling-wave tubes, differential-phase shifters, Faraday-rotation circulators, etc. The elements of the scattering matrices of such components have amplitudes that are fairly independent of frequency, but the arguments of those elements are sensitive to frequency variations. In systems which consist of such components, it is possible to think of equivalent "short" multiports with additional entrance lines. Instead of those systems one can investigate equivalent systems which consist of "short" multiports interconnected by "lines". The elements of the "short" scattering matrices have frequency-independent arguments. In the "lines" the effects of interference, constructive at some frequencies and destructive at others, can cancel when averaged over the whole band. This is even more probable in such systems as radar receivers, etc., where in many cases the components are separated by lines of substantial lengths. As shown in secs 2.4.1 and 2.4.2 equivalent noise waves in such systems are then uncorrelated since, due to the large variations of the arguments of the correlation factors C, the correlation coefficients c become zero.

Being interested in power transmission only, we do not need the scattering matrix as such in such "long-line" systems but the matrix which relates the powers of the scattered waves to the powers of the incident waves. This matrix, which we will denote by the appropriate lower case letter, i.e. by $\mathbf{s}$, will be called the power-scattering matrix. Its elements are equal to the squared magnitudes of the corresponding elements of the scattering matrix, i.e.

$$s_{ij} = |S_{ij}|^2 = S_{ij}S_{ij}^*. \tag{2.274}$$

Let $a_i a_i^+$ be the power of the wave or of all the waves contained in the band Δf and incident upon the ith port of an n-port. The complete set of incident powers can be denoted by the column vector $\mathbf{a}$ given by

$$\mathbf{a} = \begin{pmatrix} a_1 a_1^+ \\ a_2 a_2^+ \\ \vdots \\ a_n a_n^+ \end{pmatrix}. \tag{2.275}$$

The same formalism can be adopted for the scattered waves. This leads to the vector **b**. Then,

$$\mathbf{b} = \mathbf{sa}. \tag{2.276}$$

When exchange of noise power is considered, instead of the signal-wave power, the power $a_i a_i^+$ of the ith incident equivalent noise wave should be taken. Subsequently this distinction between equivalent signal waves and noise waves will not explicitly be made any more. The set of equivalent-noise-wave-power sources of an n-port can be represented by the vector **x** given by

$$\mathbf{x} = \begin{pmatrix} x_1 x_1^+ \\ x_2 x_2^+ \\ \vdots \\ x_n x_n^+ \end{pmatrix}. \tag{2.277}$$

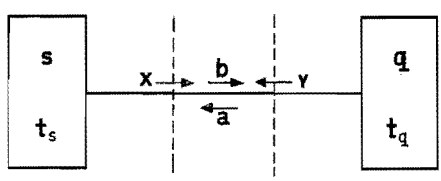


Fig. 15. Two connected noisy multiports. When the bandwidth is large, noise-power equations can be formulated in matrix form.

When two multiports are interconnected by means of long lines as illustrated in fig. 15, the following pair of noise-power equations can be written down at once in matrix notation:

$$-\mathbf{sa} + \mathbf{b} = \mathbf{x} \tag{2.278}$$

and

$$\mathbf{a} - \mathbf{qb} = \mathbf{y}, \tag{2.279}$$

where **s** and **q** are the power-scattering matrices and **x** and **y** the noise-power vectors of the two multiports, respectively. These equations can be solved for **a** and **b** in terms of **x** and **y**, yielding *)

$$\mathbf{a} = (\mathbf{I} - \mathbf{qs})^{-1} (\mathbf{y} + \mathbf{qx}) \tag{2.280}$$

and

$$\mathbf{b} = (\mathbf{I} - \mathbf{sq})^{-1} (\mathbf{x} + \mathbf{sy}). \tag{2.281}$$

*) The solution is straightforward only if the matrix $(\mathbf{I} - \mathbf{qs})$ is non-singular. When it is singular, analogous considerations apply as are discussed in appendix B.

If a multiport is passive and at uniform temperature, by virtue of eq. (2.246), a relation between **x**, **s** and the multiport temperature T_S exists. Only the powers $x_i x_i^+$ of the equivalent-noise-wave sources have to be known, the correlation being zero. Integration in the sense of eq. (2.98) of the diagonal elements of the matrix **G** given by eq. (2.246) and over Δf yields the equation

$$x_i x_i^+ = kT_S \left(1 - \sum_{j=1}^{n} S_{ij} S_{ij}^* \right) \Delta f. \tag{2.282}$$

Bearing eq. (2.274) in mind, one can write this in the matrix form

$$\mathbf{x} = kT_S \Delta f (\mathbf{I} - \mathbf{s})\,(\mathbf{i}), \tag{2.283}$$

where (**i**) is the column vector all elements of which are equal to unity. Substitution of this result respectively for **x** and **y** in eqs (2.280) and (2.281) yields

$$\mathbf{a} = k\,\Delta f (\mathbf{I} - \mathbf{qs})^{-1} \{T_Q (\mathbf{I} - \mathbf{q}) + T_S \mathbf{q} (\mathbf{I} - \mathbf{s})\}\,(\mathbf{i}) \tag{2.284}$$

and

$$\mathbf{b} = k\,\Delta f (\mathbf{I} - \mathbf{sq})^{-1} \{T_S (\mathbf{I} - \mathbf{s}) + T_Q \mathbf{s} (\mathbf{I} - \mathbf{q})\}\,(\mathbf{i}). \tag{2.285}$$

Rearranging the terms algebraically, we can write these equations in the forms

$$\mathbf{a} = \tfrac{1}{2} k\,\Delta f (T_Q + T_S)\,(\mathbf{i}) + \tfrac{1}{2} k\,\Delta f (T_Q - T_S)\,\mathbf{m}_{S,Q}\,(\mathbf{i}) \tag{2.286}$$

and

$$\mathbf{b} = \tfrac{1}{2} k\,\Delta f (T_S + T_Q)\,(\mathbf{i}) + \tfrac{1}{2} k\,\Delta f (T_S - T_Q)\,\mathbf{m}_{Q,S}\,(\mathbf{i}), \tag{2.287}$$

where the square matrices $\mathbf{m}_{S,Q}$ and $\mathbf{m}_{Q,S}$ are given by respectively

$$\mathbf{m}_{S,Q} = (\mathbf{I} - \mathbf{qs})^{-1} (\mathbf{I} - 2\mathbf{q} + \mathbf{qs}) \tag{2.288}$$

and

$$\mathbf{m}_{Q,S} = (\mathbf{I} - \mathbf{sq})^{-1} (\mathbf{I} - 2\mathbf{s} + \mathbf{sq}). \tag{2.289}$$

The net power exchange, that is the overflow of power from **s** towards **q**, is represented by the vector

$$\mathbf{b} - \mathbf{a} = k\,\Delta f \tfrac{1}{2} (T_S - T_Q)\,(\mathbf{m}_{S,Q} + \mathbf{m}_{Q,S})\,(\mathbf{i}). \tag{2.290}$$

We now define a matrix **m** by

$$\mathbf{m} = (\mathbf{I} - \mathbf{q})\,(\mathbf{I} - \mathbf{sq})^{-1}\,(\mathbf{I} - \mathbf{s}) = (\mathbf{I} - \mathbf{s})\,(\mathbf{I} - \mathbf{qs})^{-1}\,(\mathbf{I} - \mathbf{q}), \tag{2.291}$$

the equality of the second and third members being covered by theorem (C.X). It is readily shown that eq. (2.290) can be written in the simplified form

$$\mathbf{b} - \mathbf{a} = k\,\Delta f (T_S - T_Q)\,\mathbf{m}\,(\mathbf{i}). \tag{2.292}$$

Let $\Sigma \mathbf{v}$ denote the sum of all elements of the vector **v**. Then, it is easy to show that, if both the multiports **S** and **Q** are passive, the relation

$$\Sigma \mathbf{m}\,(\mathbf{i}) \geqslant 0 \tag{2.293}$$

must apply. Hence the net power exchange is from the hotter towards the colder multiport and, as one might expect, proportional to the temperature difference of the two multiports.

It is furthermore easy to show that at temperature equilibrium,

$$T_S = T_Q = T, \tag{2.294}$$

the equations

$$\mathbf{a} = \mathbf{b} = kT\,\Delta f\,(\mathbf{i}) \tag{2.295}$$

apply. This is even the case when one or both of the multiports are non-reciprocal (cf. sec. 2.6.7). Hence, in all the (long) lines of a broadband system with delay and at temperature equilibrium, two equivalent noise waves exist, one in each direction and the powers of which are equal to $kT\,\Delta f$. Of course, this result is in full agreement with observations as made in sec. 2.3.

2.8.2. *Apparent port-temperature matrix*

Analogously, as was done in sec. 2.7.1, a kind of temperature matrix can be introduced. If $\mathbf{x}$ and $\mathbf{s}$ are respectively the equivalent-noise-power vector and the power-scattering matrix of an arbitrary noisy multiport, a temperature matrix $\mathbf{t}_s$ can be defined by

$$\mathbf{x} = k\,\mathbf{t}_s\,(\mathbf{I} - \mathbf{s})\,(\mathbf{i})\,\Delta f. \tag{2.296}$$

It can easily be shown that $\mathbf{t}_s$ need only be a diagonal matrix and that its ith element t_{ii} can be interpreted as the apparent temperature of the ith port. Therefore $\mathbf{t}_s$ is called the apparent port-temperature matrix.

3. COORDINATE TRANSFORMATIONS AND IMBEDDINGS

3.1. Signal representations

3.1.1. *The signal-state space*

At any given frequency the signal state of an arbitrary linear n-port, which may be passive or active, reciprocal or non-reciprocal, can be described uniquely by means of a number of $2n$ interrelated variables. These variables can be interpreted as the coordinates of a point in a $2n$-dimensional space: *the signal-state space* of n-ports [45]), or as the components of a $2n$-dimensional column vector: the *signal-state vector*. From a mathematical point of view the physical meaning of the variables is not significant, nor is it important whether they can be measured or not.

It should be noticed that the variables are either complex numbers, when harmonic signal waves are considered, or elements of Hilbert space, when pure noise is considered, or combinations of these quantities. In what follows harmonic waves and noise waves will often simultaneously be present in the same line and propagate in the same direction. It would, therefore, be convenient if both could be represented by the same quantities. A combination could then be denoted by one symbol only and be considered to be one wave. Whenever it is convenient and we are interested in powers only, we will adopt this convention. It will then always be clear from the context of what kind a wave is or which part of it is a harmonic wave and which part a noise wave.

To make the convention correct, the definitions which have led to the concept of noise waves must also apply to harmonic signals. Because harmonic waves are in principle more precisely defined than noise waves, it will be clear that we must necessarily forego part of the knowledge of harmonic waves. In fact, the absolute phases of harmonic waves will not be described by the formalism, the knowledge of relative phases being retained only. Since we are not interested in actual amplitudes and instantaneous values, but in power and power density, the knowledge of relative phases is, however, sufficient and that of relative phases not important. Hence, the less comprehensive formalism of describing harmonic waves, too, by elements A of the Hilbert space which has been introduced in sec. 2.1.4, instead of by complex numbers as in sec. 2.1.2, can adequately serve our purposes.

Let a harmonic wave a at a frequency f_a be represented by A. The conjugate element A^+ of the dual Hilbert space can be considered to be subjoined to a^*. In analogy to eq. (2.18), we define A in such a manner that for any given infinitesimal bandwidth δf which comprises the frequency f_a the equation

$$aa^* = AA^+ \, \delta f \tag{3.1}$$

applies *). The product ab^* of two non-zero complex numbers a and b^* is never equal to zero. Hence, the product AB^+ of two harmonic waves A and B which are at exactly the same frequency never vanishes. Two harmonic waves at the same frequency are considered to be completely correlated; the magnitude of their correlation factor C (cf. sec. 2.1.5) is unity and the argument of C defines the relative phase. If two harmonic waves are at different, even slightly different, frequencies, the product AB^+ is identically zero because, then, the time average of ab^* vanishes. The harmonic waves can then be called uncorrelated. All harmonic waves at a given frequency form a 1-dimensional subspace of Hilbert space and all harmonic waves at a different frequency constitute another 1-dimensional subspace **).

Any set of $2n$ independent linear combinations of the $2n$ variables can serve as another set of variables. Changing from one set to another corresponds to a coordinate transformation in the signal-state space. For example, the signal state of a 1-port can be described by an incident wave A and a reflected wave B. The pair of quantities A and B constitute one set of coordinates for the 2-dimensional signal-state space. An alternative set of independent linear combinations is given by ***)

$$V = A + B \tag{3.2}$$

and

$$J = A - B. \tag{3.3}$$

These quantities refer to the normalized versions of the voltage over and the current into the 1-port (cf. sec. 2.1.2).

An illustration of the signal-state space of 1-ports at d.c. is given in fig. 16. At d.c. the concept of absolute phase has become meaningless while the concept of relative phase has been reduced to that of polarity. Elements of Hilbert space can then be represented by real numbers, the Hilbert space itself having become identical with the 1-dimensional real space. The 2-dimensional signal-state space

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*) As a matter of fact, in order to obtain aa^* the squared norm AA^+ should be integrated over δf instead of being multiplied by δf as in eq. (3.1). The definition of eq. (3.1) yields, however, consistent results if, during the course of a given problem, it is implicitly understood that δf does not vary. On the other hand the results of integration over δf must be independent of δf as long as f_a is contained in δf. If f_a is not contained in δf, the integral is identically zero. It will be clear that AA^+ must have the nature of the Dirac delta function $aa^* \, \delta(f_a - f)$. In the same way the product AB^+ of two harmonic waves A and B at the frequencies f_a and f_b, respectively, behaves like $ab^* \, \delta_{f_a f_b} \, \delta(f_a - f)$, where $\delta_{f_a f_b}$, the Kronecker delta, is unity, when $f_a = f_b$, and is zero, when $f_a \neq f_b$. It is readily verified that all the properties of Hilbert space which have been mentioned in sec. 2.1.4 are satisfied.

**) Pursuing this line of thought, we can think of a noise wave as consisting of an infinite collection of harmonic waves of arbitrary infinitesimal amplitudes and at different frequencies.

***) Another set of combinations is given by Youla [63]), Penfield [64]), and Kurokawa [65]), who describe (power) waves in lines with complex wave impedances.

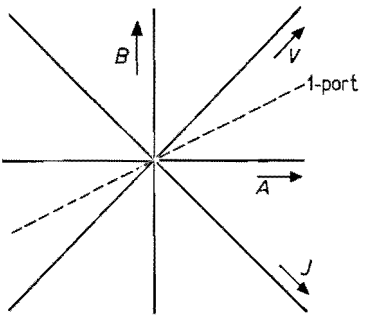


Fig. 16. Signal-state space of 1-ports at d.c. The signal state can be represented by various sets of coordinates, e.g. A, B and V, J. A 1-port defines a 1-dimensional subspace.

is, then, a 2-dimensional real space. In fig. 16 the, now real, A and B coordinate axes are respectively drawn horizontally and vertically. The V and J axes make angles of $\pi/4$ radians with the horizontal. With every point in the A,B-plane or, what is the same, in the V,J-plane, there corresponds a possible signal state of some 1-port at d.c.

The $2n$-dimensional signal-state space contains all possible signal states of all possible n-ports. However, for a particular n-port not all states are possible. Since one half of the coordinates depends linearly on the other half, any n-port defines an n-dimensional subspace of the signal-state space. This subspace contains all signal states possible for the given n-port. Hence, for any n-port there are only n linearly independent states. Of course the subspace defined by a given n-port is invariant under coordinate transformations. In fig. 16 the dashed line indicates such a subspace defined by a particular 1-port (a resistor in this case).

3.1.2. *Excitation and response vectors*

The n variables chosen out of the $2n$ variables to be independent can be considered as the components of an n-dimensional vector: the *excitation vector*. The remaining n variables, the dependent ones, form the components of another vector: the *response vector*. For the collection of all possible n-ports all excitation vectors subtend the n-dimensional *excitation space* and all response vectors the n-dimensional *response space*.

For a given n-port the choice of independent variables cannot always be completely arbitrary. Only those choices where all excitation vectors are actually possible for that n-port are permitted. For a particular n-port not all response vectors need be realizable. A linear n-port defines a linear dependence of the response vectors on the excitation vectors but this need not be a one-to-one correspondence. The dependence can be singular, i.e. two or more different excitations may cause the same response. We say, an n-port defines a mapping of the excitation space whether onto or into the response space, depending on

whether the relation between the excitation and response vectors is a one-to-one correspondence or not.

When **K** and **L** are the excitation and response vectors, a given n-port defines an nth-order matrix **M** such that

$$\mathbf{L} = \mathbf{M}\,\mathbf{K}. \tag{3.4}$$

This matrix will be called the signal-performance matrix with respect to the particular coordinate system. The matrix is non-singular or singular, depending on whether the n-port defines or does not define a one-to-one correspondence between **K** and **L**.

With **K** and **L** the signal-state vector can be written conveniently as the $2n$-dimensional vector *)

$$\mathbf{W}_{(\mathbf{M})} = \begin{pmatrix} \mathbf{L} \\ \hline \mathbf{K} \end{pmatrix}. \tag{3.5}$$

From eq. (3.4) it is then easy to show that the subspace of the signal-state space which pertains to the n-port **M** is determined by

$$(\mathbf{I} \mid -\mathbf{M})\,\mathbf{W}_{(\mathbf{M})} = \mathbf{O}, \tag{3.6}$$

where $(\mathbf{I} \mid -\mathbf{M})$ is an $n \times 2n$-matrix and **O** the n-dimensional zero vector. The signal-performance matrix determines the subspace of the signal-state space which pertains to the n-port.

When the above formalism is being adopted, one has to bear in mind that the notation of eq. (3.5) presupposes that a definite choice of dependent variables has been made. Henceforth, we will agree that the first half of the components of the signal-state vector form the dependent and the second half the independent variables.

3.1.3. *Primary coordinate system*

For further investigations it is advantageous that a particular one out of all possible coordinate systems of the signal-state space is considered to be the primary reference frame. All observations about coordinate transformations and the like can then be referred to that one system. Since in preceding sections much attention has been paid to the scattering-matrix signal representation, the corresponding choice of coordinates and the associate division into dependent and independent variables will be thought of as the primary one. We wish to emphasize that this choice is quite arbitrary. However, it is a convenient one due to the fact that the scattering-matrix representation has been explored thoroughly and that it allows for intuitive physical thinking.

For this primary system eqs. (3.4) and (3.6) read

$$\mathbf{B} = \mathbf{S}\mathbf{A} \tag{3.7}$$

*) A subscript formed by a signal-performance matrix between brackets does not refer to a particular multiport but to the coordinate system used, i.e. to the particular signal representation.

and

$$(\mathbf{I} \mid -\mathbf{S}) \left(\begin{array}{c} \mathbf{B} \\ \hline \mathbf{A} \end{array} \right) = \mathbf{O}. \tag{3.8}$$

The incident- and reflected-wave vectors **A** and **B** are respectively the excitation and response vectors while the scattering matrix **S** is the signal-performance matrix.

3.1.4. *Coordinate transformations*

The general signal representation, the **M**-representation, can be obtained from the primary signal representation, the **S**-representation, by means of a coordinate transformation performed in the signal-state space. Let $\mathbf{\Phi}$ be the $2n$th-order matrix of that transformation. Then the general and primary signal-state vectors are related by

$$\left(\begin{array}{c} \mathbf{L} \\ \hline \mathbf{K} \end{array} \right) = \mathbf{\Phi} \left(\begin{array}{c} \mathbf{B} \\ \hline \mathbf{A} \end{array} \right) = \left(\begin{array}{c|c} \mathbf{\Phi}_{11} & \mathbf{\Phi}_{12} \\ \hline \mathbf{\Phi}_{21} & \mathbf{\Phi}_{22} \end{array} \right) \left(\begin{array}{c} \mathbf{B} \\ \hline \mathbf{A} \end{array} \right), \tag{3.9}$$

where $\mathbf{\Phi}$ is given in a partitioned form in terms of nth-order submatrices $\mathbf{\Phi}_{ij}$. Since we are interested further on merely in non-singular coordinate transformations, we will consider non-singular $\mathbf{\Phi}$-matrices only. Let $\mathbf{\Psi}$ be the inverse of $\mathbf{\Phi}$. When $\mathbf{I}_{(2n)}$ is the $2n$th-order identity matrix, the equations

$$\mathbf{\Psi\Phi} = \mathbf{\Phi\Psi} = \mathbf{I}_{(2n)} \tag{3.10}$$

are valid. The inverse coordinate transformation is expressed by

$$\left(\begin{array}{c} \mathbf{B} \\ \hline \mathbf{A} \end{array} \right) = \mathbf{\Psi} \left(\begin{array}{c} \mathbf{L} \\ \hline \mathbf{K} \end{array} \right) = \left(\begin{array}{c|c} \mathbf{\Psi}_{11} & \mathbf{\Psi}_{12} \\ \hline \mathbf{\Psi}_{21} & \mathbf{\Psi}_{22} \end{array} \right) \left(\begin{array}{c} \mathbf{L} \\ \hline \mathbf{K} \end{array} \right). \tag{3.11}$$

From eq. (3.9) the new excitation vector **K** can be expressed in the old excitation and response vectors **A** and **B**. We have

$$\mathbf{K} = \mathbf{\Phi}_{21}\mathbf{B} + \mathbf{\Phi}_{22}\mathbf{A}, \tag{3.12}$$

or, with substitution of eq. (3.7),

$$\mathbf{K} = (\mathbf{\Phi}_{21}\mathbf{S} + \mathbf{\Phi}_{22})\mathbf{A} = \mathbf{Y}\mathbf{A}, \tag{3.13}$$

where

$$\mathbf{Y} = \mathbf{\Phi}_{21}\mathbf{S} + \mathbf{\Phi}_{22}. \tag{3.14}$$

The excitation vector is transformed linearly by the matrix $\mathbf{Y}$. For this reason $\mathbf{Y}$ will be called *excitation-transformation matrix*. It should be kept in mind that $\mathbf{Y}$ still depends on the particular n-port **S** which relates **B** to **A**. Evidently, if a coordinate transformation is to be meaningful for a given multiport, that is for a particular **S**, the criterion is that the matrix $\mathbf{Y}$ be non-singular (cf. appendix D).

Henceforth, the investigations will be confined to those transformations whic satisfy this condition with respect to the multiports under consideration.

Analogously, for the new response vector **L** we obtain

$$\mathbf{L} = (\mathbf{\Phi}_{11}\mathbf{S} + \mathbf{\Phi}_{12})\,\mathbf{A}. \tag{3.15}$$

If **S** is non-singular, **L** can be expressed directly in **B** and a response-transformation matrix could be defined. However, that would be relatively unimportant so that we will not pursue that subject.

Since Y is assumed non-singular, the inverse of eq. (3.13) is also valid. Substitution of this inverse equation in eq. (3.15) yields

$$\mathbf{L} = (\mathbf{\Phi}_{11}\mathbf{S} + \mathbf{\Phi}_{12})\,(\mathbf{\Phi}_{21}\mathbf{S} + \mathbf{\Phi}_{22})^{-1}\,\mathbf{K}. \tag{3.16}$$

Comparing with eq. (3.4), one finds that the new signal-performance matrix **M** is given by

$$\mathbf{M} = (\mathbf{\Phi}_{11}\mathbf{S} + \mathbf{\Phi}_{12})\,(\mathbf{\Phi}_{21}\mathbf{S} + \mathbf{\Phi}_{22})^{-1}. \tag{3.17}$$

The inverse coordinate transformation $\mathbf{\Psi}$, as defined by eq. (3.11), gives analogous converse results. We obtain

$$\mathbf{A} = (\mathbf{\Psi}_{21}\mathbf{M} + \mathbf{\Psi}_{22})\,\mathbf{K} = \mathbf{\Lambda}\mathbf{K}, \tag{3.18}$$

with

$$\mathbf{\Lambda} = \mathbf{\Psi}_{21}\mathbf{M} + \mathbf{\Psi}_{22} \tag{3.19}$$

and

$$\mathbf{S} = (\mathbf{\Psi}_{11}\mathbf{M} + \mathbf{\Psi}_{12})\,(\mathbf{\Psi}_{21}\mathbf{M} + \mathbf{\Psi}_{22})^{-1}. \tag{3.20}$$

The matrix $\mathbf{\Lambda}$, too, will be called excitation-transformation matrix. For a given multiport **M** these results are meaningful only, if the matrix $\mathbf{\Lambda}$ is non-singular. By virtue of eq. (3.10), substitution of eq. (3.17) for **M** in eq. (3.19) gives

$$\mathbf{\Lambda} = (\mathbf{\Phi}_{21}\mathbf{S} + \mathbf{\Phi}_{22})^{-1} = Y^{-1}. \tag{3.21}$$

Hence, when Y is non-singular, $\mathbf{\Lambda}$ is so, and vice versa. These matrices are each other's inverses, which is confirmed by eqs (3.13) and (3.18).

For future derivations it is convenient to rewrite the results of the present section in a terse notation. By virtue of eqs (3.13) and (3.18), from eqs (3.4) and (3.7) the general and primary signal-state vectors can be written respectively in the forms

$$\mathbf{W}_{(M)} = \begin{pmatrix} \mathbf{L} \\ \cdots \\ \mathbf{K} \end{pmatrix} = \begin{pmatrix} \mathbf{M} \\ \cdots \\ \mathbf{I} \end{pmatrix} \mathbf{K} = \begin{pmatrix} \mathbf{M} \\ \cdots \\ \mathbf{I} \end{pmatrix} Y\mathbf{A} \tag{3.22}$$

and

$$\mathbf{W}_{(S)} = \begin{pmatrix} \mathbf{B} \\ \cdots \\ \mathbf{A} \end{pmatrix} = \begin{pmatrix} \mathbf{S} \\ \cdots \\ \mathbf{I} \end{pmatrix} \mathbf{A} = \begin{pmatrix} \mathbf{S} \\ \cdots \\ \mathbf{I} \end{pmatrix} \mathbf{\Lambda}\mathbf{K}. \tag{3.23}$$

The first factors in the right-hand sides are $2n \times n$-matrices in partitioned

form. If we substitute these equations in eqs (3.9) and (3.11) and remember that the results must be valid for all vectors **K** or **A**, we obtain respectively

$$\left(\begin{array}{c}\mathbf{M}\\ \hline \mathbf{I}\end{array}\right) = \mathbf{\Phi}\left(\begin{array}{c}\mathbf{S}\\ \hline \mathbf{I}\end{array}\right)\mathbf{\Lambda} \tag{3.24}$$

and

$$\left(\begin{array}{c}\mathbf{S}\\ \hline \mathbf{I}\end{array}\right) = \mathbf{\Psi}\left(\begin{array}{c}\mathbf{M}\\ \hline \mathbf{I}\end{array}\right) Y. \tag{3.25}$$

By virtue of eq. (3.21), the former of these equations is identical with eqs (3.17) and (3.14) while the latter is identical with eqs (3.20) and (3.19).

3.1.5. *Alternative equations*

Alternative expressions for **M** in terms of **S** and for **S** in terms of **M** can be obtained by solving eqs (3.20) and (3.17) for **M** and **S**, respectively. These alternative equations can also be obtained by a direct method as follows. If we insert eq. (3.10) into eq. (3.8), by virtue of eq. (3.9), we get

$$(\mathbf{I}\,|\,-\mathbf{S})\,\mathbf{\Psi\Phi}\left(\begin{array}{c}\mathbf{B}\\ \hline \mathbf{A}\end{array}\right) = (\mathbf{\Psi}_{11}-\mathbf{S\Psi}_{21}\,|\,\mathbf{\Psi}_{12}-\mathbf{S\Psi}_{22})\left(\begin{array}{c}\mathbf{L}\\ \hline \mathbf{K}\end{array}\right) = \mathbf{O}. \tag{3.26}$$

Premultiplying by the inverse of the matrix $(\mathbf{\Psi}_{11}-\mathbf{S\Psi}_{21})$ and comparing the result with eq. (3.6), we obtain

$$\mathbf{M} = -(\mathbf{\Psi}_{11}-\mathbf{S\Psi}_{21})^{-1}(\mathbf{\Psi}_{12}-\mathbf{S\Psi}_{22}). \tag{3.27}$$

This would also have been obtained when eq. (3.20) had been solved for **M**. The result is valid only if the matrix $\mathbf{\Gamma}$, given by

$$\mathbf{\Gamma} = (\mathbf{\Psi}_{11}-\mathbf{S\Psi}_{21})^{-1}, \tag{3.28}$$

is non-singular.

Similarly, when inserting eq. (3.10) into eq. (3.6), premultiplying with the inverse of $(\mathbf{\Phi}_{11}-\mathbf{M\Phi}_{21})$, and comparing the result with eq. (3.8), we obtain

$$\mathbf{S} = -(\mathbf{\Phi}_{11}-\mathbf{M\Phi}_{21})^{-1}(\mathbf{\Phi}_{12}-\mathbf{M\Phi}_{22}). \tag{3.29}$$

This result is identical with the solution of eq. (3.17) for **S**. It is valid only if the matrix $\mathbf{\Delta}$, given by

$$\mathbf{\Delta} = (\mathbf{\Phi}_{11}-\mathbf{M\Phi}_{21})^{-1}, \tag{3.30}$$

is non-singular.

The alternative results can be written in the terse form

$$(\mathbf{I}\,|\,-\mathbf{M}) = \mathbf{\Gamma}\,(\mathbf{I}\,|\,-\mathbf{S})\,\mathbf{\Psi}, \tag{3.31}$$

$$(\mathbf{I}\,|\,-\mathbf{S}) = \mathbf{\Delta}\,(\mathbf{I}\,|\,-\mathbf{M})\,\mathbf{\Phi}. \tag{3.32}$$

When one of the matrices $\mathbf{\Gamma}$ and $\mathbf{\Delta}$ is non-singular, it is, analogously to the derivation of eq. (3.21), easy to show that the other one is its inverse, i.e.

$$\mathbf{\Gamma\Delta} = \mathbf{\Delta\Gamma} = \mathbf{I}. \tag{3.33}$$

The requirement that $\mathbf{\Gamma}$ and $\mathbf{\Delta}$ must be non-singular seems to be an additional condition for the coordinate transformation to be meaningful for a particular multiport. Multiplying eq. (3.19) from the right by $\mathbf{\Phi}_{21}$ and the inverse of eq. (3.30) from the left by $\mathbf{\Psi}_{21}$ and taking the sum of the results, we have, by virtue of eqs (3.10) and (3.33),

$$\mathbf{\Psi}_{21}\mathbf{\Gamma} + \mathbf{\Lambda\Phi}_{21} = \mathbf{O}. \tag{3.34}$$

In appendix D it is shown that, due to this relationship and the fact that the coordinate transformation is non-singular, the non-singularity of the pair $\mathbf{\Gamma}$ and $\mathbf{\Delta}$ entails that of the pair $\boldsymbol{Y}$ and $\mathbf{\Lambda}$, and vice versa. Hence, in addition to the requirement of non-singular $\boldsymbol{Y}$, there is no further condition of non-singular $\mathbf{\Gamma}$, but these two forms of the applicability criterion for coordinate transformations are identical. A further discussion of this condition will be given in sec. 3.6.7.

Below, in sec. 3.2.2, it shall be shown that $\mathbf{\Gamma}$ and $\mathbf{\Delta}$ transform the noise of an n-port in the same way as $\boldsymbol{Y}$ and $\mathbf{\Lambda}$ transform the excitation. The matrices $\mathbf{\Gamma}$ and $\mathbf{\Delta}$ will therefore be called *noise-transformation matrices* as, above, $\boldsymbol{Y}$ and $\mathbf{\Delta}$ have been called excitation-transformation matrices.

3.1.6. *Signal-state power-absorption matrix*

One can attribute to any point in the signal-state space a real number P_{abs} that represents the power absorbed in or, if the number is negative, emitted by an n-port at that signal state. This quantity depends neither on the coordinate system used nor on the properties of the particular n-port but is characteristic of the signal state proper. Although P_{abs} must be invariant to coordinate transformations it can be expressed in the coordinates of the signal state. In fact, for linear systems, P_{abs} is a real quadratic form of the signal-state vector. Haus and Adler [15]) define the $2n$th-order matrix of that quadratic form to be the power matrix $\mathbf{Q}_{(\mathbf{M})}$. For a signal state $\begin{pmatrix} \mathbf{L} \\ \hline \mathbf{K} \end{pmatrix}$ the absorbed power is given by

$$P_{\text{abs}} = \begin{pmatrix} \mathbf{L} \\ \hline \mathbf{K} \end{pmatrix}^{\times} \mathbf{Q}_{(\mathbf{M})} \begin{pmatrix} \mathbf{L} \\ \hline \mathbf{K} \end{pmatrix}. \tag{3.35}$$

Since P_{abs} is real, $\mathbf{Q}_{(\mathbf{M})}$ is an Hermitian matrix. We will call this matrix the signal-state power-absorption matrix, short **Q**-matrix, to distinguish it from the excitation power-absorption matrix, short **P**-matrix, which has been introduced in sec. 2.6.1 and which will be discussed further in the next section. Moreover, the **Q**-matrix will not simply be called power matrix because, in sec. 2.5.2, that

name has been given by us to the matrix **G** which describes the powers and the correlation products of the noise-wave sources of a multiport.

For the primary coordinate system, i.e. for the scattering-matrix signal representation, the power P_{abs} absorbed in an n-port is given by (cf. eq. (2.207))

$$P_{abs} = \mathbf{A}^{\times}\mathbf{A} - \mathbf{B}^{\times}\mathbf{B}. \tag{3.36}$$

This can be written in the form

$$P_{abs} = \begin{pmatrix} \mathbf{B} \\ \hline \mathbf{A} \end{pmatrix}^{\times} \mathbf{Q}_{(S)} \begin{pmatrix} \mathbf{B} \\ \hline \mathbf{A} \end{pmatrix}, \tag{3.37}$$

where *)

$$\mathbf{Q}_{(S)} = \left(\begin{array}{c|c} -\mathbf{I} & \mathbf{O} \\ \hline \mathbf{O} & \mathbf{I} \end{array}\right) = -\boldsymbol{\Omega}. \tag{3.38}$$

Equation (3.37) is the scattering-matrix version of eq. (3.35), while eq. (3.38) gives the actual definition of the signal-state power-absorption matrix for the scattering-matrix signal representation.

Inserting eq. (3.10) and its Hermitian conjugate version between the factors of the right-hand side of eq. (3.37), by virtue of eq. (3.9), we obtain

$$P_{abs} = \begin{pmatrix} \mathbf{L} \\ \hline \mathbf{K} \end{pmatrix}^{\times} (\boldsymbol{\Psi}^{\times}\mathbf{Q}_{(S)}\boldsymbol{\Psi}) \begin{pmatrix} \mathbf{L} \\ \hline \mathbf{K} \end{pmatrix}. \tag{3.39}$$

Comparing this result with eq. (3.35), we see that the transformation

$$\mathbf{Q}_{(M)} = \boldsymbol{\Psi}^{\times}\mathbf{Q}_{(S)}\boldsymbol{\Psi} \tag{3.40}$$

applies. The signal-state power-absorption matrix is transformed by the Hermitian congruence transformation defined by the inverse $\boldsymbol{\Psi}$ of the signal-state transformation matrix $\boldsymbol{\Phi}$.

3.1.7. *Excitation power-absorption matrix*

Due to eq. (3.6), any possible signal state of a given n-port is described completely by the excitation vector together with the signal-performance matrix. The power absorbed in or emitted by an n-port at a particular signal state must therefore be also a real quadratic form of that excitation vector. Substitution of eq. (3.22) in eq. (3.35) gives

$$P_{abs} = \mathbf{K}^{\times}\mathbf{P}_{M}\mathbf{K}, \tag{3.41}$$

where

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*) We introduce here the polarity matrix $\boldsymbol{\Omega}$, defined by eq. (3.38), because below it will be used in connection with the polarity of flows of power. In fact, in eqs (3.37) and (3.38) it serves the same purpose.

$$\mathbf{P}_M = \left(\frac{\mathbf{M}}{\mathbf{I}} \right)^{\times} \mathbf{Q}_{(M)} \left(\frac{\mathbf{M}}{\mathbf{I}} \right). \tag{3.42}$$

The nth-order Hermitian matrix $\mathbf{P}_M$ is a characteristic feature of the particular n-port $\mathbf{M}$. We will call it the excitation power-absorption matrix, or, as in sec. 2.6.1, simply the power-absorption matrix, of the n-port $\mathbf{M}$. Of course, this matrix is not invariant to coordinate transformations. For the scattering-matrix signal representation the excitation power-absorption matrix is readily evaluated by inserting eq. (3.38) into eq. (3.42). One obtains the expression, given in eq. (2.208),

$$\mathbf{P}_S = \mathbf{I} - \mathbf{S}^{\times}\mathbf{S}. \tag{3.43}$$

If we substitute eq. (3.18) in the expression for the absorbed power as it appears in the scattering-matrix signal representation, we have

$$P_{abs} = \mathbf{A}^{\times} (\mathbf{I} - \mathbf{S}^{\times}\mathbf{S}) \mathbf{A} = \mathbf{A}^{\times}\mathbf{P}_S\mathbf{A} = \mathbf{K}^{\times}\mathbf{\Lambda}^{\times}\mathbf{P}_S \mathbf{\Lambda} \mathbf{K}. \tag{3.44}$$

Comparing this with eq. (3.41) and remembering that both expressions must give identical results for all $\mathbf{K}$, we obtain

$$\mathbf{P}_M = \mathbf{\Lambda}^{\times}\mathbf{P}_S\mathbf{\Lambda}. \tag{3.45}$$

The excitation power-absorption matrix is transformed by the Hermitian congruence transformation defined by the signal-transformation matrix $\mathbf{\Lambda}$. This result could of course also have been obtained by insertion of eqs (3.24) and (3.40) into eq. (3.42).

3.2. Noise representations

3.2.1. *Dependent-coordinate noise representations*

In chapter 2 noise properties of multiports have been described in the scattering-matrix signal representation. The noise of a multiport has been represented by noise-wave sources which are ascribed to the ports; one source to each port. These sources emit noise waves out of the multiport and into the connected lines. Therefore the sources are assigned with appropriate directions, viz. the directions of the reflected waves at the corresponding ports. This procedure allows for a formal separation of noise and transmission properties of a multiport. In conventional noise theories such separations are made very often [33,34]).

To restate matters, the noise performance of a multiport has thus far been described by the mere addition of noise waves to the waves which are reflected by the corresponding "noiseless" multiport or, which is the same thing, by just adding noise quantities to the dependent components of the signal-state vector. In this way a noise-source vector has been introduced. It is identical with the

amplitude vector of the noise-wave sources as has been introduced in sec. 2.5.2. The noise-source vector must be added to the response vector.

The addition of noise waves to reflected waves, which themselves may be noise waves, harmonic signal waves, or a combination of the two, is one of the main reasons why in sec. 3.1.1 harmonic signal waves have been represented by elements of a subspace of Hilbert space.

Let **S** be the scattering matrix of a noisy multiport. The corresponding noiseless multiport is described by

$$\mathbf{B}' = \mathbf{SA}, \tag{3.46}$$

where **A** and **B**′ are the excitation and response vectors pertinent to that *noiseless* multiport. Let **X** be the noise-source vector (cf. sec. 2.5.2, eq. (2.188)) and **B** the response vector of the *noisy* multiport. We then have

$$\mathbf{B} = \mathbf{B}' + \mathbf{X} = \mathbf{SA} + \mathbf{X}. \tag{3.47}$$

This equation is a generalized form of eq. (3.7). Moreover, it is the equation which has been used in the discussions of chapter 2. In fig. 17 the ideas recalled in the present section are illustrated.

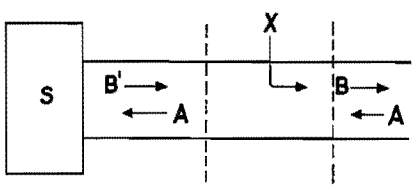


Fig. 17. Representation of a noisy multiport by a "noiseless" multiport and noise-wave sources.

The concepts which underly eq. (3.47) can also be applied to a general coordinate system of the signal-state space. The partition of the coordinates into dependent and independent variables then must have been specified. We can formulate a generalized form of eq. (3.4) such that it includes the noise of multiports. We therefore introduce a general noise-source vector and denote it by **Y**. The generalized form of eq. (3.4) then reads

$$\mathbf{L} = \mathbf{MK} + \mathbf{Y}. \tag{3.48}$$

The components of **Y** need not be interpreted as amplitudes of noise *waves* in the sense of sec. 2.1.4, but as amplitudes of general noise quantities. Of course, the components are elements of a Hilbert space. It will be clear that the units of noise-source amplitudes must be identical with those of the corresponding dependent variables.

A general power matrix of the noise sources is defined by

$$\mathbf{G}_M = \mathbf{YY}^\times. \tag{3.49}$$

The definition corresponds to that of eq. (2.191). From eq. (3.49) it is clear that in all coordinate systems noise-source power matrices are Hermitian. The diagonal elements determine the power densities of the constituent sources, the off-diagonal elements the mutual correlations.

Analogously to eqs (3.4) and (3.7), eqs (3.5) and (3.8) can be brought into a general form so as to include noise. It is readily seen that these general forms are given by respectively

$$(\mathbf{I} \mid -\mathbf{M}) \begin{pmatrix} \mathbf{L} \\ \hline \mathbf{K} \end{pmatrix} = \mathbf{Y} \tag{3.50}$$

and

$$(\mathbf{I} \mid -\mathbf{S}) \begin{pmatrix} \mathbf{B} \\ \hline \mathbf{A} \end{pmatrix} = \mathbf{X}. \tag{3.51}$$

Since in the formalism adopted here noise quantities are added to the dependent coordinates of the signal-state space, this type of noise representation will be called the *dependent-coordinate noise representation* (cf. sec. 3.2.3).

3.2.2. *Coordinate transformations*

When all ports of a multiport are terminated by non-reflecting noiseless loads, the noise waves radiated from that multiport and into the terminations can be evaluated in terms of $\mathbf{Y}$ and $\mathbf{M}$. The corresponding power matrix, expressed in $\mathbf{G}_M$ and $\mathbf{M}$, must be the same as the one obtained in the scattering-matrix signal representation which is given by eq. (2.191). Equating the two results, we can derive $\mathbf{G}_M$. However, there is an alternative and more simple procedure for arriving at this result. A coordinate transformation in the signal-state space yields the same result.

Let us perform transformation $\mathbf{\Phi}$ on the scattered-wave coordinate system. Inserting $\mathbf{\Psi\Phi}$ in between the two factors of the left-hand side of eq. (3.51) and premultiplying both sides by $\mathbf{\Gamma}$, we readily see, by virtue of eqs (3.9), (3.31), and (3.50), that

$$\mathbf{Y} = \mathbf{\Gamma X}. \tag{3.52}$$

The noise-source vector is transformed linearly by $\mathbf{\Gamma}$. For this reason, in sec. 3.1.5, $\mathbf{\Gamma}$ has been called the noise-transformation matrix.

By virtue of eq. (2.191), substitution of eq. (3.52) in eq. (3.49) yields

$$\mathbf{G}_M = \mathbf{\Gamma G}_S \mathbf{\Gamma}^{\times}. \tag{3.53}$$

The noise-source power matrix is transformed by the Hermitian congruence transformation defined by $\mathbf{\Gamma}$. Hence, rank and signature of that matrix are preserved under coordinate transformations. Since, furthermore, in sec. 2.5.4 it has been shown that $\mathbf{G}_S$ is positive semi-definite, we may conclude that the noise-source power matrix is positive semi-definite in every coordinate system.

This conclusion implies that the diagonal elements of $\mathbf{G}_M$ are non-negative, as they should be to be in accordance with the definition of eq. (3.49).

3.2.3. *Other noise representations*

The procedure of describing noise performance by adding noise quantities to the dependent coordinates is not the only one possible. For the description of noise properties each member of any arbitrary set of n out of the $2n$ coordinates can be associated with a noise quantity. The n noise quantities represent n noise sources (not necessarily noise-*wave* sources). Two of all the possibilities have a particular character, viz. the one for which noise quantities are added to the dependent coordinates: the dependent-coordinate noise representation, and the one for which the independent coordinates are associated with noise quantities: the independent-coordinate noise representation. All others can be called mixed-coordinate noise representations.

Evidently, most of these possible noise representations have no practical significance, in most cases the dependent type being the only convenient one. Therefore, we shall henceforth restrict ourselves mainly to that particular one as is done in most literature on linear noisy systems. Thus, Ryder and Kirchner [32]), investigating transistors in the impedance-matrix signal representation, describe the noise properties by means of two voltage sources, one at the input terminals and one at the output terminals. In the same way, Becking, Groendijk and Knol [33]), making use of the chain matrix (the *ABCD*-matrix), describe the noise of a 2-port by means of a voltage source in series with and a current source shunted over the input terminals.

3.2.4. *Noise-distribution matrix*

In sec. 2.6.5 it has been derived that, for the scattered-wave coordinate system, the noise-source power matrix $\mathbf{G}_S$ of a passive multiport $\mathbf{S}$ at the uniform absolute temperature T is given by (cf. eq. (2.246))

$$\mathbf{G}_S = kT\,\mathbf{N}_S. \tag{3.54}$$

The matrix $\mathbf{N}_S$, given by (cf. eq. (2.211))

$$\mathbf{N}_S = \mathbf{I} - \mathbf{S}\mathbf{S}^{\times}, \tag{3.55}$$

depends on the signal performance only and has been called noise-distribution matrix. Performing a coordinate transformation $\mathbf{\Phi}$, we obtain by substitution of eq. (3.54) in eq. (3.53)

$$\mathbf{G}_M = \mathbf{\Gamma}\,(kT\,\mathbf{N}_S)\,\mathbf{\Gamma}^{\times} = kT\,\mathbf{\Gamma}\,\mathbf{N}_S\mathbf{\Gamma}^{\times}. \tag{3.56}$$

Hence, for a passive multiport at the uniform temperature T the noise-source power matrix $\mathbf{G}_M$ can be written in the form

$$\mathbf{G}_M = kT\,\mathbf{N}_M, \tag{3.57}$$

where $\mathbf{N}_M$ is obtained from $\mathbf{N}_S$ by the Hermitian congruence transformation

$$\mathbf{N}_M = \mathbf{\Gamma}\,\mathbf{N}_S\mathbf{\Gamma}^\times. \tag{3.58}$$

The matrix $\mathbf{N}_M$ will be called the noise-distribution matrix for the **M**-matrix signal representation. It is a function of **M** only.

By virtue of eq. (3.38), it is readily seen from eq. (3.55) that $\mathbf{N}_S$ can be written in the form *)

$$\mathbf{N}_S = -(\mathbf{I}\,|\,-\mathbf{S})\,\mathbf{Q}_{(S)}{}^{-1}\,(\mathbf{I}\,|\,-\mathbf{S})^\times. \tag{3.59}$$

Inserting the relations

$$\mathbf{\Psi}\mathbf{\Psi}^{-1} = (\mathbf{\Psi}^\times)^{-1}\mathbf{\Psi}^\times = \mathbf{I} \tag{3.60}$$

in between the factors of the right-hand side and substituting the resulting equation for $\mathbf{N}_S$ in eq. (3.58), we obtain, by virtue of eqs (3.31) and the inverted form of eq. (3.40),

$$\mathbf{N}_M = -(\mathbf{I}\,|\,-\mathbf{M})\,\mathbf{Q}_{(M)}{}^{-1}\,(\mathbf{I}\,|\,-\mathbf{M})^\times. \tag{3.61}$$

We will consider this equation to be the definition of the noise-distribution matrix of a multiport in a general coordinate system. Due to the Hermitian character of $\mathbf{Q}_M$ (cf. sec. 3.1.6), the noise-distribution matrix is Hermitian. As it has been stated already above, the noise-distribution matrix depends only on the signal properties of the multiport and it governs the way thermally radiated power is distributed over the ports.

3.2.5. *Noise-temperature matrix*

In the introduction it has been stated that many of the difficulties encountered in theories about noise of linear systems are due to the strong interrelation between signal and noise properties of multiports. Moreover, it has been argued that one of our aims should be the separation, at least formally, of quantities which, respectively, describe signal properties and noise properties. In order to attain such a separation for the scattering-matrix signal representation the concept of noise-temperature matrix has been defined in sec. 2.7.1.

By eq. (2.272) the noise-wave power matrix $\mathbf{G}_S$ is divided, apart from Boltzmann's constant, into two matrix factors, i.e.

$$\mathbf{G}_S = k\,\mathbf{T}_S\mathbf{N}_S. \tag{3.62}$$

One factor, the latter, is the noise-distribution matrix $\mathbf{N}_S$ which has been

*) In the particular case of the scattering-matrix signal representation $\mathbf{Q}_{(S)}{}^{-1} = \mathbf{Q}_{(S)}$. Instead of $\mathbf{Q}_{(S)}{}^{-1}$ in eq. (3.59) $\mathbf{Q}_{(S)}$ could have been written. This, however, is not true in other coordinate systems. Moreover, $\mathbf{Q}_{(S)}$ has not been used because in expressions about the noise, such as the noise-distribution matrix, the Hermitian conjugate matrices appear on the right-hand sides. Bringing the transform of $\mathbf{N}_S$ back to the same general form, we must therefore make use of the inverted form of eq. (3.40) so that $\mathbf{Q}_{(S)}{}^{-1}$ must appear in eq. (3.59).

discussed in the previous section. In analogy to the scalar factor T in the right-hand side of eq. (3.54), the other factor, the former, has been called the noise-temperature matrix $\mathbf{T}_S$.

It is hoped that the introduction of $\mathbf{T}_S$ will lead to quantities which are invariant under coordinate transformations, which do not depend on signal properties and which describe the intensity of the noise uniquely. It should be emphasized that $\mathbf{T}_S$ itself still depends on $\mathbf{S}$ and, as will be shown right away, is not invariant under coordinate transformations.

Substitution of eq. (3.62) in eq. (3.53) yields

$$\mathbf{G}_M = k\,\mathbf{\Gamma}\,(\mathbf{T}_S\mathbf{N}_S)\,\mathbf{\Gamma}^{\times}. \tag{3.63}$$

Since $\mathbf{\Gamma}$ is non-singular, by virtue of eq. (3.58), this can be written in the form

$$\begin{aligned}\mathbf{G}_M &= k\,\mathbf{\Gamma}\mathbf{T}_S\,(\mathbf{\Gamma}^{-1}\mathbf{\Gamma})\,\mathbf{N}_S\mathbf{\Gamma}^{\times} = \\ &= k(\mathbf{\Gamma}\mathbf{T}_S\,\mathbf{\Gamma}^{-1})\,\mathbf{N}_M.\end{aligned} \tag{3.64}$$

Comparing this result with eq. (3.62) we now define a noise-temperature matrix $\mathbf{T}_M$ for the $\mathbf{M}$-matrix coordinate system by

$$\mathbf{G}_M = k\,\mathbf{T}_M\mathbf{N}_M. \tag{3.65}$$

This general noise-temperature matrix can then be obtained from $\mathbf{T}_S$ by the similarity transformation

$$\mathbf{T}_M = \mathbf{\Gamma}\,\mathbf{T}_S\mathbf{\Gamma}^{-1}. \tag{3.66}$$

In matrix algebra it is shown that the eigenvalues of a matrix are invariant under similarity transformations. Hence, the eigenvalues of the noise-temperature matrix of a given multiport depend neither on the coordinate system used nor on the signal-performance matrix in that reference frame. Moreover, it will be shown below that these eigenvalues are real numbers (chapter 4) and that they characterize the intensity of the noise of the multiport in the same way the absolute temperature characterizes the intensity of purely thermal noise of a passive multiport at uniform temperature. In fact, the eigenvalues of the noise-temperature matrix satisfy all conditions which should apply to the noise-describing quantities we are looking for. For these reasons the eigenvalues will be called characteristic noise temperatures of the noisy multiport.

3.2.6. *Justification*

One might still wonder whether the positing of eqs (3.61) and (3.65) for respectively the noise-distribution matrix $\mathbf{N}_M$ and the noise-temperature matrix $\mathbf{T}_M$ needs more elucidation and motivation. With the deduction performed in the preceding sections the consistency of the above equations has been sufficiently verified. However, that does not make the introduction of $\mathbf{N}_M$ and $\mathbf{T}_M$ reasonable, since $\mathbf{G}_M$ itself already describes the noise completely and

in a much shorter fashion. The reason should be found in the fact that $\mathbf{N}_M$ and $\mathbf{T}_M$ offer the possibility of having (scalar) noise-characterizing quantities which do not depend inherently upon the signal performance and which are invariant to coordinate transformations.

Starting from these ideas, the argument can be put the other way round. If one suppose that $\mathbf{G}_M$ can be split up into two matrix factors in the sense of eq. (3.65), one of which depends only upon the signal performance and the other also on the intensity of the noise, it is clear that both factors transform under coordinate transformations, since $\mathbf{G}_M$ does so. Hence, none of those two matrices is invariant. Moreover, it is readily seen that one factor is transformed by a Hermitian congruence transformation and the other by a similarity transformation. Now, if invariant scalar quantities are wished for, they must be the eigenvalues of the latter matrix so that eq. (3.66) can be valid for one of the factor matrices without any restriction being put on the formalism. Then, if it is argued further that for a uniformly warm passive multiport the eigenvalues should be equal to the absolute temperature, eq. (3.61) is readily derived.

3.3. Impedance-matrix signal representation

3.3.1. *Intermezzo*

Before proceeding with the investigation of the classification of linear noisy multiports it may be worth while to illustrate the formalism defined thus far with a transformation to a commonly used coordinate system. Since network theories are often formulated in the impedance-matrix signal representation, in the following intercalary sections we will briefly discuss the transformation from the scattered-waves coordinate system to the voltages and currents reference frame. The implications of a transformation to the admittance-matrix signal representation or to any other signal representation can be found as easily.

3.3.2. *Transformation matrices*

Let $\mathbf{V}$ and $\mathbf{J}$ be the n-dimensional vectors of respectively the voltages over and the currents into the ports of an n-port. Let all components of these vectors be normalized with respect to the appropriate characteristic impedances of the lines. The relation between these vectors is given by

$$\mathbf{V} = \mathbf{Z}\mathbf{J}. \tag{3.67}$$

The matrix $\mathbf{Z}$ is the normalized impedance matrix of the n-port. According to eq. (3.5), in this signal representation the signal-state vector is written in the form

$$\mathbf{W}_{(Z)} = \begin{pmatrix} \mathbf{V} \\ \hline \mathbf{J} \end{pmatrix}. \tag{3.68}$$

Furthermore, if the vectors **A** and **B** represent respectively the incident and reflected waves, the equations

$$\mathbf{V} = \mathbf{A} + \mathbf{B} \tag{3.69}$$

and

$$\mathbf{J} = \mathbf{A} - \mathbf{B} \tag{3.70}$$

apply. From these the coordinate-transformation matrix $\boldsymbol{\Phi}$ is readily derived, i.e.

$$\boldsymbol{\Phi} = \left(\begin{array}{c|c} \mathbf{I} & \mathbf{I} \\ \hline -\mathbf{I} & \mathbf{I} \end{array}\right). \tag{3.71}$$

The inverse matrix $\boldsymbol{\Psi}$ is given by

$$\boldsymbol{\Psi} = \frac{1}{2}\left(\begin{array}{c|c} \mathbf{I} & -\mathbf{I} \\ \hline \mathbf{I} & \mathbf{I} \end{array}\right). \tag{3.72}$$

By virtue of eqs (3.14) and (3.19), the excitation-transformation matrix is found to be

$$\boldsymbol{Y} = \mathbf{I} - \mathbf{S} = 2\,(\mathbf{I} + \mathbf{Z})^{-1} = \boldsymbol{\Lambda}^{-1}. \tag{3.73}$$

Analogously, one obtains from eqs (3.28) and (3.30) for the noise-transformation matrix

$$\boldsymbol{\Gamma} = \mathbf{I} + \mathbf{Z} = 2\,(\mathbf{I} - \mathbf{S})^{-1} = \boldsymbol{\Delta}^{-1}. \tag{3.74}$$

3.3.3. *The characterizing matrices*

The last two expressions of the previous section contain equations from which **Z** can be expressed in **S** and conversely. The same result can be obtained by insertion of eqs (3.71) and (3.72) into eqs (3.17) and (3.20) or into the alternative eqs (3.27) and (3.29). The result is given by

$$\mathbf{Z} = (\mathbf{I} + \mathbf{S})\,(\mathbf{I} - \mathbf{S})^{-1} = (\mathbf{I} - \mathbf{S})^{-1}\,(\mathbf{I} + \mathbf{S}) \tag{3.75}$$

and

$$\mathbf{S} = (\mathbf{Z} - \mathbf{I})\,(\mathbf{Z} + \mathbf{I})^{-1} = (\mathbf{Z} + \mathbf{I})^{-1}\,(\mathbf{Z} - \mathbf{I}). \tag{3.76}$$

That the factors of the products which form the right-hand sides commute, is readily verified.

Substitution of eqs (3.38) and (3.72) in eq. (3.40) yields for the power matrix $\mathbf{Q}_{(Z)}$ in the impedance-matrix signal representation

$$\mathbf{Q}_{(Z)} = \frac{1}{2}\left(\begin{array}{c|c} \mathbf{O} & \mathbf{I} \\ \hline \mathbf{I} & \mathbf{O} \end{array}\right). \tag{3.77}$$

Hence, by virtue of eq. (3.35), the absorbed power is given by the well-known expression

$$P_{\text{abs}} = \frac{1}{2}\left(\begin{array}{c}\mathbf{V}\\ \hline \mathbf{J}\end{array}\right)^{\times}\left(\begin{array}{c|c}\mathbf{O} & \mathbf{I}\\ \hline \mathbf{I} & \mathbf{O}\end{array}\right)\left(\begin{array}{c}\mathbf{V}\\ \hline \mathbf{J}\end{array}\right) = \frac{1}{2}(\mathbf{J}^{\times}\mathbf{V} + \mathbf{V}^{\times}\mathbf{J})\,. \tag{3.78}$$

Inserting eq. (3.77) into eqs (3.42) and (3.61), one obtains for the power-absorption matrix $\mathbf{P}_Z$ and the noise-distribution matrix $\mathbf{N}_Z$ the respective expressions

$$\mathbf{P}_Z = \tfrac{1}{2}(\mathbf{Z} + \mathbf{Z}^{\times}) \tag{3.79}$$

and

$$\mathbf{N}_Z = 2\,(\mathbf{Z} + \mathbf{Z}^{\times}). \tag{3.80}$$

For a passive n-port at the uniform temperature T the noise-source power matrix $\mathbf{G}_Z$ is found to be given by

$$\mathbf{G}_Z = 4\,kT\,\tfrac{1}{2}(\mathbf{Z} + \mathbf{Z}^{\times}). \tag{3.81}$$

This result has been obtained by Twiss [46]). The noise sources have the characters of "voltage" sources in series with the terminals [32]). This is clear from the fact that the noise quantities are added to the voltages over the ports, the latter being the dependent variables.

For the admittance-matrix signal representation one can derive analogously that the noise-source power matrix $\mathbf{G}_Y$ is given by

$$\mathbf{G}_Y = 4\,kT\,\tfrac{1}{2}(\mathbf{Y} + \mathbf{Y}^{\times}), \tag{3.82}$$

where $\mathbf{Y}$ is the admittance matrix. This result, too, has been obtained by Twiss [46]). In this case the noise sources can be represented by "current" sources shunted over the ports.

3.4. Classification of multiports

3.4.1. *Invariants of signal performance*

Before discussing invariant quantities which can classify a multiport, we will enumerate briefly the numbers which characterize square matrices. The number n of rows and columns is called the order of a matrix. It is equal to the number of eigenvalues. The number r of linearly independent rows and columns is the rank. It is equal to the number of non-zero eigenvalues. The difference between n and r, the nullity, defect, or degeneracy, is denoted by d. It gives the number of eigenvalues which are zero. The number p of positive eigenvalues is called the positivity or index and the number q of negative eigenvalues the negativity. The difference between p and q is called the signature and is denoted by s. It will be clear that the relations

$$n = p + q + d, \tag{3.83}$$

$$r = p + q \tag{3.84}$$

and

$$s = p - q \tag{3.85}$$

apply. Only three of the six numbers are independent.

In matrix algebra, Sylvester's law of inertia states that all characteristic numbers are invariant under non-singular Hermitian congruence transformations. However, the eigenvalues themselves are in general not invariant under such transformations. On the other hand, similarity transformations preserve the eigenvalues as well as the characteristic numbers.

Deriving the formulas which describe the transformation of the signal performance of a multiport, we have found in secs 3.1.4 and 3.1.5 that the signal-performance matrix does not transform in an elementary way, that is neither by a similarity nor by a Hermitian congruence transformation. The characteristic numbers of that matrix are not preserved under coordinate transformations. Hence, none of them can be used properly to classify a multiport.

In secs 3.1.7 and 3.2.4 it has been shown that both the power-absorption matrix and the noise-distribution matrix are Hermitian-congruent in different coordinate systems. Hence, the characteristic numbers of these matrices are invariants and can be used to classify a multiport. In sec. 2.6.1 it has been found that, for the scattering-matrix signal representation, the noise-distribution matrix $\mathbf{N}_S$ has the same eigenvalues as the power-absorption matrix $\mathbf{P}_S$ (cf. appendix C, theorem II). This property need not apply in other signal representations as may be clear from the results of the previous section. There it is easy to deduce from eqs (3.79) and (3.80) that the eigenvalues of $\mathbf{N}_Z$ and $\mathbf{P}_Z$ differ by a factor of 4. However, what can be concluded from the eigenvalues of $\mathbf{N}_S$ and $\mathbf{P}_S$ being identical is that the characteristic numbers of the noise-distribution matrix and the power-absorption matrix are identical and, hence, the same for any signal representation. In this respect these matrices need therefore not be investigated separately. Moreover, the numbers n, r, d, p, q, and s of those matrices could also be called, respectively, order, rank, defect, positivity, negativity, and signature of the multiport proper.

When looking for invariant quantities of active 2-ports, Mason [47]) studied the effects of reciprocal transformations *). He has found that the unilateral power gain U is invariant under such transformations. In terms of the elements Z_{ij} of the impedance matrix the unilateral power gain is defined by

$$U = \frac{|Z_{21} - Z_{12}|^2}{4\,(R_{11}R_{22} - R_{12}R_{21})}, \tag{3.86}$$

*) Reciprocal transformations are defined by their equivalent imbedding networks being reciprocal. The equivalence of imbeddings and coordinate transformations will be discussed at length in subsequent sections.

where R_{ij} is the real part of Z_{ij}. Mason distinguished three classes of amplifiers in accordance with respectively $U < 0$, $0 < U < 1$ and $1 < U$. However, if non-reciprocal transformations, too, are considered, the numerical value of U is not an invariant any more while, moreover, the first and the last of the above classes merge and become identical. Only two classes of active 2-ports can then still be distinguished, viz. those amplifiers for which U is in the region $0 < U < 1$ constitute one class and those for which U is out of that region another. Haus and Adler [15]) have shown that these two classes are identical with the classes of active 2-ports for which q is respectively equal to 2 and 1.

If we define a unilateral power "gain" U also for passive 2-ports by applying eq. (3.86) to such 2-ports too, we readily find that U is then restricted to the region $0 \leqslant U \leqslant 1$. In particular, the condition $0 < U < 1$ defines in this case also the class of non-reciprocal passive 2-ports which are purely dissipative *). All reciprocal 2-ports, whether they are dissipative, active, or reactive are classified by $U = 0$. Moreover, $U = 1$ determines the 2-ports which are, partly, or completely, reactive.

The condition $0 < U < 1$ does not only define a certain class of active 2-ports but also a class of passive 2-ports. It will be shown below that the latter class is characterized by a p-value of 2. The p- and q-values therefore distinguish passive multiports from active ones better than the unilateral power gain does. We conclude that p and q form a better classification for 2-ports than U. Since, furthermore, the unilateral power gain is not well suited to networks with less or more than two ports and since we do not wish to confine ourselves to reciprocal transformations only, we will not pay further attention to unilateral power gain.

3.4.2. *Dissipativity, reactivity, and activity*

In the previous section it has been argued that the characteristic numbers of $\mathbf{N}_M$ and $\mathbf{P}_M$ classify a multiport. This will be verified in chapter 4. The physical meaning of those quantities will also become clear only there. Nevertheless, in this and the next section we will anticipate the results and discuss them briefly because that will explain the purpose of further investigations.

When $p = r = n$, $(q = d = 0)$, the matrices $\mathbf{N}_M$ and $\mathbf{P}_M$ are positive definite and the multiport is passive and purely dissipative. For purely active multiports $q = r = n$, $(p = d = 0)$, and $\mathbf{N}_M$ and $\mathbf{P}_M$ are negative definite. When $p = r < n$, $(q = 0, d \neq 0)$, $\mathbf{N}_M$ and $\mathbf{P}_M$ are positive semi-definite and the multiport is passive and mixed dissipative and reactive. When $q = r < n$, $(p = 0, d \neq 0)$, $\mathbf{N}_M$ and $\mathbf{P}_M$ are negative semi-definite and the multiport is lossless and mixed active and reactive. In the general case of p, q, and d all unequal zero, $\mathbf{N}_M$ and

*) The meaning of such concepts as purely dissipative, partly reactive, etc. will become clear during the further course of this treatise (chapter 4).

$\mathbf{P}_M$ are indefinite and the multiport is a mixture of dissipative, reactive, and active elements.

The classification can be elaborated a little further. The values of p, q, and d give the numbers of respectively the positive, negative, and zero eigenvalues of both $\mathbf{P}_M$ and $\mathbf{N}_M$. Since these numbers are invariant, by virtue of eqs (3.41) and (3.83), it is easy to show that, irrespective of the reference frame, for the excitation space a base can be found which consists of n linearly independent excitation vectors in such a way that for p of them the absorbed power is inherently positive (dissipation), for q of them inherently negative (emission), and for d of them inherently zero (complete reflection). Therefore, we will call p the dissipativity, q the activity, and d the reactivity of the multiport. The sum of p and d could be called the passivity of the multiport.

It should be noted that in speaking of activity, we do not refer to spontaneously active multiports but to those multiports which can be active under stimulation only, that is we do not consider generators but only amplifiers (cf. sec. 5.1.2).

3.4.3. *Invariants of noise performance*

Inspecting the matrices which describe the noise of a multiport, we find that the noise-source power matrix $\mathbf{G}_M$ is transformed by a Hermitian congruence transformation and the noise-temperature matrix $\mathbf{T}_M$ by a similarity transformation. Hence, of the former only the characteristic numbers enumerated in sec. 3.4.1 are invariant while of the latter the eigenvalues, too, remain unaltered under coordinate transformations.

In sec. 2.5.4 it has been shown that $\mathbf{G}_M$ is positive semi-definite so that its negativity is inherently zero. The positivity p_G of $\mathbf{G}_M$ will be called the noise index. In chapter 4 it will be shown that this number refers to the number of inherently uncorrelated internal noise sources which are externally manifest. The defect d_G of $\mathbf{G}_M$ will be called the noise defect.

In secs 2.7.2 and 3.2.6, motivating and justifying the introduction of the noise-temperature matrix, we have stated that the eigenvalues of that matrix are real, have the dimension of a temperature, and can serve as unique, unambiguous, and quantitative measures of the noise intensity. In fact, for the larger part the remainder of this treatise will be devoted to the verification of this particular statement.

3.5. The transfer matrix

3.5.1. *The 2-port*

Many practical circuits consist of 2-ports. When 2-ports are cascaded, it is convenient to describe them in terms neither of the impedance matrix nor of the scattering matrix, but instead of it in terms of the chain matrix (general-

circuit-constant matrix or *ABCD*-matrix) or, alternatively, of the transfer matrix (or **T**-matrix). Owing to its practical interest and because, for the transformation of n-ports, we shall later be interested in the transfer matrix of $2n$-ports, we think it useful to pay now some attention to the transfer matrix of 2-ports.

When eq. (3.51) is applied to 2-ports, the signal-state vector, as given by

$$\left(\begin{array}{c}\mathbf{B}\\ \hline \mathbf{A}\end{array}\right)=\left(\begin{array}{c}B_1\\ B_2\\ \hline A_1\\ A_2\end{array}\right), \tag{3.87}$$

has four components. The former pair, the reflected waves B_1 and B_2, is dependent on the latter pair, the incident waves A_1 and A_2. Any fourth-order permutation matrix $\mathbf{\Pi}_{(4)}$ applied to this signal-state vector yields a different choice of dependent coordinates. It is easy to show that the inverse of a permutation matrix is equal to its transpose, i.e.

$$\mathbf{\Pi}_{(p)}{}^{-1}=\mathbf{\Pi}_{(p)}{}^{\mathrm{T}}. \tag{3.88}$$

The particular permutation matrix we are interested in is the one which leads to the transfer-matrix representation. The transfer matrix $\mathbf{\Theta}$ of a 2-port is the matrix which relates the reflected and incident waves B_1 and A_1, respectively, at the input to the incident and reflected waves B_2 and A_2, respectively, at the output *) (note the difference in sequence). Then, the former are the dependent, the latter the independent coordinates **). The resulting signal-state vector is

$$\left(\begin{array}{c}\mathbf{L}\\ \hline \mathbf{K}\end{array}\right)=\left(\begin{array}{c}B_1\\ A_1\\ \hline B_2\\ A_2\end{array}\right). \tag{3.89}$$

Equation (3.50) now reads

$$(\mathbf{I}\,|\,-\mathbf{\Theta})\left(\begin{array}{c}\mathbf{L}\\ \hline \mathbf{K}\end{array}\right)=\mathbf{Y}. \tag{3.90}$$

The two components Y_1 and Y_2 of the noise-source vector $\mathbf{Y}$ are associated with B_1 and A_1, respectively. Hence, the noise is represented by two sources at the input. An illustration of these ideas is given in fig. 18.

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*) Contrary to the practice adopted hitherto, the wave incident upon the output of a 2-port is denoted by B_2, instead of by A_2. By doing so, we shall henceforth be able to equate the excitation vector of a 2-port with the response vector of a following 2-port while, furthermore, if a 1-port is connected to the output of a 2-port, the signal-state vector of the former is equal to the excitation vector of the latter.

**) The choice of the input waves as dependent, made by Belevitch [21]) and in this treatise, is different from the choice made by Dicke [23]) who, defining the T-matrix, considered the output waves to be dependent. The difference of approach is not very serious. Which choice can be better made depends strongly upon the order in which the cascaded 2-ports enter calculations.

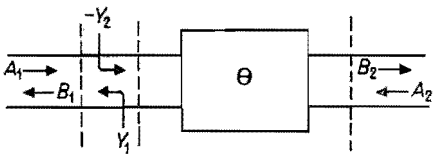


Fig. 18. A noisy 2-port represented by a transfer matrix and two corresponding noise-wave sources.

The transfer matrix is a convenient expedient for handling cascades of 2-ports. The independent output waves $\mathbf{K}_i$ of the ith 2-port of a cascade are equal to the dependent waves $\mathbf{L}_{i+1}$ at the input of the following 2-port.

The permutation matrix $\mathbf{\Pi}_{(4)}$ which leads to the transfer-matrix signal representation of 2-ports is readily shown to be

$$\mathbf{\Phi} = \mathbf{\Pi}_{(4)} = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{array}\right). \tag{3.91}$$

By virtue of eq. (3.88), the inverse transformation matrix $\mathbf{\Psi}$ is given by

$$\mathbf{\Psi} = \mathbf{\Pi}_{(4)}{}^{-1} = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ \hline 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right). \tag{3.92}$$

From eqs (3.38) and (3.40) the power matrix $\mathbf{Q}_{(\Theta)}$ of this coordinate system is found to be

$$\mathbf{Q}_{(\Theta)} = \left(\begin{array}{c|c} -\mathbf{\Omega} & \mathbf{O} \\ \hline \mathbf{O} & \mathbf{\Omega} \end{array}\right), \tag{3.93}$$

where $\mathbf{\Omega}$ is the 2nd-order polarity matrix introduced in sec. 3.1.6 and given by

$$\mathbf{\Omega} = \mathbf{\Omega}^{-1} = \mathbf{\Omega}^{\mathrm{T}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{3.94}$$

Substitution of eq. (3.91) in eq. (3.14) yields the excitation-transformation matrix Y, i.e.

$$Y = \begin{pmatrix} 0 & 1 \\ S_{21} & S_{22} \end{pmatrix}. \tag{3.95}$$

Analogously, substitution of eq. (3.92) in eq. (3.28) yields the noise-transformation matrix $\mathbf{\Gamma}$, i.e.

$$\mathbf{\Gamma} = S_{21}{}^{-1} \begin{pmatrix} S_{21} & -S_{11} \\ 0 & -1 \end{pmatrix}. \tag{3.96}$$

The conditions of non-singularity of the matrices $\boldsymbol{Y}$ and $\boldsymbol{\Gamma}$ are identical (cf. sec. 3.1.5) and require

$$S_{21} \neq 0. \tag{3.97}$$

If the introduction of the transfer matrix and the representation of the noise by means of two noise sources at the input are to be meaningful for a given 2-port, it is necessary that there should exist a coupling from the input to the output [33]).

For the calculation of the elements of the transfer matrix $\boldsymbol{\Theta}$ use can be made of eq. (3.17) or, alternatively, eq. (3.27). The result is

$$\left.\begin{aligned} \Theta_{11} &= S_{12} - S_{11}S_{21}^{-1}S_{22}, & \Theta_{12} &= S_{11}S_{21}^{-1}, \\ \Theta_{21} &= -S_{21}^{-1}S_{22}, & \Theta_{22} &= S_{21}^{-1}. \end{aligned}\right\} \tag{3.98}$$

By virtue of eq. (3.52), the components of the noise vector $\mathbf{Y}$ are found to be

$$Y_1 = X_1 - S_{11}S_{21}^{-1}X_2 \tag{3.99}$$

and

$$Y_2 = -S_{21}^{-1}X_2. \tag{3.100}$$

At this stage we wish to warn against a possible pitfall and call attention to what might seem to be a discrepancy between the mathematical formalism adopted thus far and the physical picture one might have formed.

From the introduction of noise-wave sources applying in the scattering-matrix signal representation, and from the discussion about noise quantities in secs 3.2.1 and 3.2.3, one might have concluded that the latter can always be interpreted as amplitudes of noise sources. This is not quite correct. The noise quantities have been defined such that they must be added to appropriate coordinates of the "noiseless" multiport to get the corresponding coordinates of the noisy multiport. In the above case this means that Y_2 must be added to A_1' in order to obtain A_1. On the other hand, if two noise-wave sources are thought to be present at the input of the "noiseless" 2-port, as illustrated in fig. 18, the amplitude of the inwardly radiating source must be added to A_1 in order to obtain A_1'. The amplitude of that source is therefore not equal to Y_2, but to $-Y_2$.

The casual appearance of minus signs at some noise-source amplitudes could have been avoided. Polarity matrices should then have been inserted into the right-hand sides of such equations as eqs (3.48), (3.49), and (3.50). That procedure, however, would have been more complicated mathematically than the one adopted in secs 3.2.1 and 3.2.3. We will therefore maintain the definitions given thus far and accept minus signs at amplitudes of some noise sources.

The ostensible discrepancy can also be illustrated by the automatic appearance of a minus sign in the right-hand member of eq. (3.100). The equations (3.99) and (3.100) could also have been obtained as the result of a direct inspection

of the configuration (see figs 17 and 18). A noise wave X_2 which emerges from the output port of a 2-port can be accounted for by a noise wave $S_{21}^{-1}X_2$ which is incident upon the input. In this way $-Y_2$ is obtained, leading to eq. (3.100). However, this noise wave is partly reflected at the input, yielding a reflected wave $S_{11}S_{21}^{-1}X_2$ emerging from the input port. Actually, a noise wave X_1 emerges from that port. Hence, an additional noise-wave source $(X_1 - S_{21}^{-1}S_{11}X_2)$, which radiates away from the 2-port, must be present at the input. This is expressed by eq. (3.99).

Insertion of eq. (3.93) into eqs (3.42) and (3.61) yields respectively the power-absorption and noise-distribution matrices. The result is

$$\mathbf{P}_\Theta = \boldsymbol{\Omega} - \boldsymbol{\Theta}^\times\boldsymbol{\Omega}\boldsymbol{\Theta} \tag{3.101}$$

and

$$\mathbf{N}_\Theta = \boldsymbol{\Omega} - \boldsymbol{\Theta}\boldsymbol{\Omega}\boldsymbol{\Theta}^\times. \tag{3.102}$$

These expressions could also have been obtained by applying the Hermitian congruence transformation of eqs (3.45) and (3.58) to the corresponding matrices of the scattering-matrix signal representation. For passive 2-ports both the above matrices are positive semi-definite. For lossless passive 2-ports both are equal to the zero matrix, which results in

$$\boldsymbol{\Theta}^\times\boldsymbol{\Omega}\boldsymbol{\Theta} = \boldsymbol{\Theta}\boldsymbol{\Omega}\boldsymbol{\Theta}^\times = \boldsymbol{\Omega}. \tag{3.103}$$

These two equations follow from each other as can readily be shown by virtue of eq. (3.94).

The noise-source power matrix can be written down in the form

$$\mathbf{G}_\Theta = k\,\mathbf{T}_\Theta\mathbf{N}_\Theta, \tag{3.104}$$

where $\mathbf{T}_\Theta$ can be deduced from $\mathbf{T}_S$ by means of the similarity transformation of eq. (3.66). If the 2-port is passive and homogeneously warm at the temperature T, we have

$$\mathbf{G}_\Theta = kT(\boldsymbol{\Omega} - \boldsymbol{\Theta}\boldsymbol{\Omega}\boldsymbol{\Theta}^\times). \tag{3.105}$$

Starting from the impedance-matrix signal representation, discussed in secs 3.3.2 and 3.3.3, the same permutation matrix $\boldsymbol{\Phi} = \boldsymbol{\Pi}_{(4)}$ leads to the chain-matrix signal representation. Some of the results can be found in refs 15 and 48. In this case the applicability criterion is given by

$$Z_{21} \neq 0. \tag{3.106}$$

Becking et al. [33]) have deduced this as the condition which must be satisfied if the noise of a 2-port is to be represented by a voltage and a current source at the input.

3.5.2. *The 2n-port*

The transfer matrix, in the previous section discussed for the 2-port, can be generalized for $2n$-ports. This subject is broached in the present section because, in what follows, we shall need the results for the discussion of imbeddings. An imbedding is a transformation of an n-port by means of a $2n$-port "transformer".

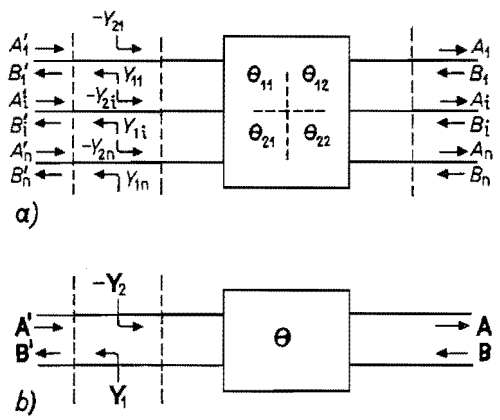


Fig. 19. A noisy $2n$-port represented by a transfer matrix and $2n$ corresponding noise-wave sources.

Of the $2n$-port n ports are considered to be the input ports, the remaining n ports being the output ports. This is illustrated in fig. 19*a*. The waves incident upon and reflected at the input ports, the A_i''s and B_i''s, respectively, together constitute the dependent coordinates. The $2n$-dimensional response vector is defined by

$$\mathbf{L} = \begin{pmatrix} \mathbf{B}' \\ \hline \mathbf{A}' \end{pmatrix}. \tag{3.107}$$

The same waves at the output ports (but in reversed order*)), the B_i''s and the A_i''s, respectively, constitute the independent coordinates. The $2n$-dimensional excitation vector is written in the form **)

*) Here, too, the waves incident upon the output are denoted by B_i, etc. (cf. sec. 3.5.1).

**) There is an alternative method to define the response and excitation vectors, leading to an alternative different definition of the transfer matrix. In fact, it is also possible to define the response vector such that the first pair of components are the reflected and incident waves at the first input port, the second pair the same waves at the second input port, etc. The excitation vector is then formulated analogously. It will be clear that for the definition of the chain matrix of a $2n$-port the same alternative exists. Haus and Adler [15]) have made this alternative choice, adopting an alternating sequence of voltage and current variables.

However, the alternative method is not well suited to the insertion of the incident and reflected waves, that is of the excitation of an imbedding $2n$-port. Therefore, we shall adopt the definitions as they are given in the text such that the first n components of the response vector of the $2n$-port constitute the n reflected waves at the n ports of the input, etc.

$$\mathbf{K} = \left(\begin{array}{c} \mathbf{B} \\ \hline \mathbf{A} \end{array} \right). \tag{3.108}$$

The signal-performance matrix, the transfer matrix $\boldsymbol{\Theta}$, is a $2n$th-order square matrix. The general multiport equation, eq. (3.48), can now conveniently be written in the partitioned form (see fig. 19*b*)

$$\left(\begin{array}{c} \mathbf{B}' \\ \hline \mathbf{A}' \end{array} \right) = \left(\begin{array}{c|c} \boldsymbol{\Theta}_{11} & \boldsymbol{\Theta}_{12} \\ \hline \boldsymbol{\Theta}_{21} & \boldsymbol{\Theta}_{22} \end{array} \right) \left(\begin{array}{c} \mathbf{B} \\ \hline \mathbf{A} \end{array} \right) + \left(\begin{array}{c} \mathbf{Y}_1 \\ \hline \mathbf{Y}_2 \end{array} \right), \tag{3.109}$$

where the $\boldsymbol{\Theta}_{ij}$ are nth-order square submatrices of $\boldsymbol{\Theta}$. The n-dimensional vectors $\mathbf{Y}_1$ and $-\mathbf{Y}_2$ refer to the noise-wave sources which radiate respectively out of and into the input ports.

The scattering-matrix equation, eq. (3.51), can be written in the form

$$(\mathbf{I} \mid -\mathbf{S}) \left(\begin{array}{c} \boldsymbol{\beta} \\ \hline \boldsymbol{\alpha} \end{array} \right) = \boldsymbol{\xi}, \tag{3.110}$$

where

$$\boldsymbol{\alpha} = \left(\begin{array}{c} \mathbf{A}' \\ \hline \mathbf{B} \end{array} \right), \tag{3.111}$$

$$\boldsymbol{\beta} = \left(\begin{array}{c} \mathbf{B}' \\ \hline \mathbf{A} \end{array} \right) \tag{3.112}$$

and

$$\boldsymbol{\xi} = \left(\begin{array}{c} \mathbf{X}' \\ \hline \mathbf{X} \end{array} \right), \tag{3.113}$$

or in the partitioned form, eq. (3.47),

$$\left(\begin{array}{c} \mathbf{B}' \\ \hline \mathbf{A} \end{array} \right) = \left(\begin{array}{c|c} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \hline \mathbf{S}_{21} & \mathbf{S}_{22} \end{array} \right) \left(\begin{array}{c} \mathbf{A}' \\ \hline \mathbf{B} \end{array} \right) + \left(\begin{array}{c} \mathbf{X}' \\ \hline \mathbf{X} \end{array} \right). \tag{3.114}$$

The coordinate transformation is defined by

$$\left(\begin{array}{c} \mathbf{L} \\ \hline \mathbf{K} \end{array} \right) = \boldsymbol{\Phi} \left(\begin{array}{c} \boldsymbol{\beta} \\ \hline \boldsymbol{\alpha} \end{array} \right). \tag{3.115}$$

The transformation matrix $\boldsymbol{\Phi}$ is the $4n$th-order permutation matrix $\boldsymbol{\Pi}_{(4n)}$ (cf. eqs (3.91) and (3.92)):

$$\boldsymbol{\Phi} = \boldsymbol{\Psi}^{\mathrm{T}} = \boldsymbol{\Pi}_{(4n)} = \left(\begin{array}{cc|cc} \mathbf{I} & \mathbf{O} & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} & \mathbf{I} & \mathbf{O} \\ \hline \mathbf{O} & \mathbf{O} & \mathbf{O} & \mathbf{I} \\ \mathbf{O} & \mathbf{I} & \mathbf{O} & \mathbf{O} \end{array} \right). \tag{3.116}$$

We will not repeat all considerations made in the previous section, but write down at once the generalized results. The $2n$th-order polarity matrix $\boldsymbol{\Omega}$ can now be written (cf. eq. (3.94))

$$\mathbf{\Omega} = \mathbf{\Omega}^{-1} = \mathbf{\Omega}^{T} = \left(\begin{array}{c|c} \mathbf{I} & \mathbf{O} \\ \hline \mathbf{O} & -\mathbf{I} \end{array}\right). \tag{3.117}$$

We have

$$\boldsymbol{Y} = \left(\begin{array}{c|c} \mathbf{O} & \mathbf{I} \\ \hline \mathbf{S}_{21} & \mathbf{S}_{22} \end{array}\right), \tag{3.118}$$

$$\boldsymbol{\Gamma} = \left(\begin{array}{c|c} \mathbf{I} & -\mathbf{S}_{11} \\ \hline \mathbf{O} & -\mathbf{S}_{21} \end{array}\right)^{-1} = \left(\begin{array}{c|c} \mathbf{I} & -\mathbf{S}_{11}\mathbf{S}_{21}{}^{-1} \\ \hline \mathbf{O} & -\mathbf{S}_{21}{}^{-1} \end{array}\right), \tag{3.119}$$

$$\mathbf{\Theta} = \left(\begin{array}{c|c} \mathbf{S}_{12} - \mathbf{S}_{11}\mathbf{S}_{21}{}^{-1}\mathbf{S}_{22} & \mathbf{S}_{11}\mathbf{S}_{21}{}^{-1} \\ \hline -\mathbf{S}_{21}{}^{-1}\mathbf{S}_{22} & \mathbf{S}_{21}{}^{-1} \end{array}\right), \tag{3.120}$$

$$\left(\begin{array}{c} \mathbf{Y}_1 \\ \hline \mathbf{Y}_2 \end{array}\right) = \left(\begin{array}{c} \mathbf{X}_1 - \mathbf{S}_{11}\mathbf{S}_{21}{}^{-1}\mathbf{X}_2 \\ \hline -\mathbf{S}_{21}{}^{-1}\mathbf{X}_2 \end{array}\right), \tag{3.121}$$

$$\mathbf{P}_{\Theta} = \mathbf{\Omega} - \mathbf{\Theta}^{\times}\mathbf{\Omega}\mathbf{\Theta}, \tag{3.122}$$

$$\mathbf{N}_{\Theta} = \mathbf{\Omega} - \mathbf{\Theta}\mathbf{\Omega}\mathbf{\Theta}^{\times}. \tag{3.123}$$

The conditions that a $2n$-port can be described by a transfer matrix are furnished by the requirements of non-singular $\boldsymbol{Y}$ and $\boldsymbol{\Gamma}$. These two conditions are identical (cf. sec. 3.1.5) and are satisfied if $\mathbf{S}_{21}$ is non-singular. This means that the coupling from the n input ports to the n output ports must be unambiguous.

In the following the lossless $2n$-port will play an important role as imbedding network or general "transformer". In appendix D it is shown that for such a $2n$-port the condition of non-singular $\mathbf{S}_{21}$ leads to non-singular $\mathbf{S}_{12}$. This means that, when the coupling from input to output of a lossless $2n$-port is unambiguous, the coupling from output to input is so, too. It can further be shown from the condition of freedom of loss,

$$\mathbf{I} - \mathbf{S}\mathbf{S}^{\times} = \mathbf{I} - \mathbf{S}^{\times}\mathbf{S} = \mathbf{O}, \tag{3.124}$$

that in this case one can write for the submatrices of $\mathbf{\Theta}$:

$$\mathbf{\Theta}_{11} = \mathbf{S}_{12} - \mathbf{S}_{11}\mathbf{S}_{21}{}^{-1}\mathbf{S}_{22} = (\mathbf{S}_{12}{}^{\times})^{-1}, \tag{3.125}$$

$$\mathbf{\Theta}_{12} = \mathbf{S}_{11}\mathbf{S}_{21}{}^{-1} = -(\mathbf{S}_{12}{}^{\times})^{-1}\mathbf{S}_{22}{}^{\times}, \tag{3.126}$$

$$\mathbf{\Theta}_{21} = -\mathbf{S}_{21}{}^{-1}\mathbf{S}_{22} = \mathbf{S}_{11}{}^{\times}(\mathbf{S}_{12}{}^{\times})^{-1}, \tag{3.127}$$

$$\mathbf{\Theta}_{22} = \mathbf{S}_{21}{}^{-1} = \mathbf{S}_{21}{}^{\times} - \mathbf{S}_{11}{}^{\times}(\mathbf{S}_{12}{}^{\times})^{-1}\mathbf{S}_{22}{}^{\times}. \tag{3.128}$$

The first and the last of these equations show that the diagonal submatrices $\mathbf{\Theta}_{11}$ and $\mathbf{\Theta}_{22}$ of the transfer matrix are non-singular. This can also be shown from the condition of freedom of loss in the transfer-matrix signal representation. From eqs (3.122) and (3.123) we obtain for that condition the identical equations

$$\Theta\Omega\Theta^{\times} = \Theta^{\times}\Omega\Theta = \Omega. \tag{3.129}$$

In partitioned form they read

$$\Theta_{11}\Theta_{11}^{\times} - \Theta_{12}\Theta_{12}^{\times} = \mathbf{I}, \tag{3.130}$$

$$\Theta_{22}\Theta_{22}^{\times} - \Theta_{21}\Theta_{21}^{\times} = \mathbf{I}, \tag{3.131}$$

$$\Theta_{11}\Theta_{21}^{\times} - \Theta_{12}\Theta_{22}^{\times} = \mathbf{O} \tag{3.132}$$

and

$$\Theta_{11}^{\times}\Theta_{11} - \Theta_{21}^{\times}\Theta_{21} = \mathbf{I}, \tag{3.133}$$

$$\Theta_{22}^{\times}\Theta_{22} - \Theta_{12}^{\times}\Theta_{12} = \mathbf{I}, \tag{3.134}$$

$$\Theta_{11}^{\times}\Theta_{12} - \Theta_{21}^{\times}\Theta_{22} = \mathbf{O}. \tag{3.135}$$

Let us suppose that Θ_{11} is singular. Then there must exist a non-zero vector $\mathbf{V}$ such that $\Theta_{11}\mathbf{V} = \mathbf{O}$ and $\mathbf{V}^{\times}\mathbf{V} > 0$. Premultiplying eq. (3.133) by $\mathbf{V}^{\times}$ and postmultiplying by $\mathbf{V}$ yield

$$-\mathbf{V}^{\times}\Theta_{21}^{\times}\Theta_{21}\mathbf{V} = \mathbf{V}^{\times}\mathbf{V}. \tag{3.136}$$

Since the Gaussian transform $\Theta_{21}^{\times}\Theta_{21}$ is positive (semi)-definite [40]) the left-hand side is non-positive, which contradicts the condition $\mathbf{V}^{\times}\mathbf{V} > 0$. Hence, Θ_{11} must be non-singular, which verifies the above finding. In the same way it can be shown that Θ_{22} is non-singular. When this result is applied to eq. (3.135) or eq. (3.132), is easy to show that Θ_{12} and Θ_{21} are simultaneously singular or simultaneously non-singular.

It is worth while to point at a further interesting consequence. The fact that, when a lossless $2n$-port can be described in the transfer-matrix signal representation, Θ_{11} and Θ_{22} must both be non-singular leads, by virtue of eqs (3.125) and (3.128), to the inference that that $2n$-port can always be described also in the scattering-matrix representation. The corresponding scattering matrix then always has non-singular $\mathbf{S}_{12}$ and $\mathbf{S}_{21}$.

The matrices Y and Γ which govern the transformation of excitation and noise from the transfer matrix to the scattering-matrix signal representation are given respectively by

$$Y_{\Theta\to S} = \left(\begin{array}{c|c} \Theta_{21} & \Theta_{22} \\ \hline \mathbf{I} & \mathbf{O} \end{array}\right), \tag{3.137}$$

$$\Gamma_{\Theta\to S} = \left(\begin{array}{c|c} \mathbf{I} & -\Theta_{12}\Theta_{22}^{-1} \\ \hline \mathbf{O} & -\Theta_{22}^{-1} \end{array}\right). \tag{3.138}$$

Indeed, these matrices are non-singular if Θ_{22} is non-singular and that is always the case when the $2n$-port is lossless. Then, from eq. (3.127), the diagonal submatrices $\mathbf{S}_{11}$ and $\mathbf{S}_{22}$ of the scattering matrix are simultaneously either singular or non-singular. Moreover, $\mathbf{S}_{11}$ and $\mathbf{S}_{22}$ have these properties simultaneously with Θ_{21} and Θ_{12}.

3.6. Imbeddings

3.6.1. *Imbedding of a multiport*

Haus and Adler [15]) have analyzed the noise performance of linear n-ports by imbedding them in lossless $2n$-ports (see sec. 5.1). Since we wish to consider to what extent the formalism developed by these authors agrees with that adopted in this treatise, we will now investigate the general imbedding of an n-port.

When the n output ports of a $2n$-port are terminated by an n-port, the n input ports constitute another n-port. We say: the former, the old, n-port is transformed by the $2n$-port to the latter, the new, n-port. The old n-port is imbedded in the $2n$-port. It is called itself the imbedded or original n-port. The $2n$-port acts as a general "transformer". It is called the imbedding or transforming $2n$-port. The resulting n-port is called the transformed or resultant n-port.

At this stage we are interested in the resultant n-port and the description of its properties in terms of those of the imbedded and imbedding multiports. In fig. 20 an illustration of a general imbedding is given. In fig. 20*a* the actual imbedding is shown while fig. 20*b* shows the equivalent resultant multiport.

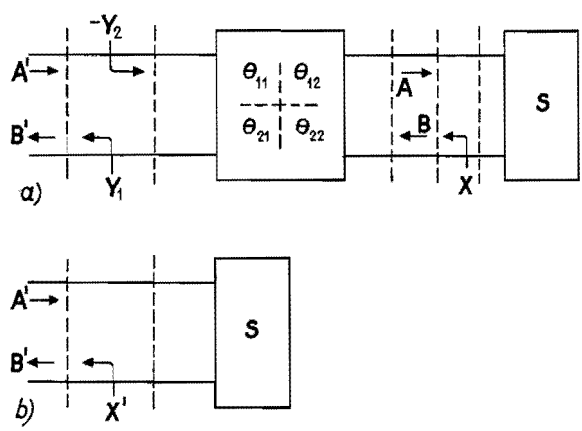


Fig. 20. Imbedding of an n-port in a $2n$-port results in another n-port. The imbedding $2n$-port is represented by a transfer matrix.

Let the scattering matrix and the noise-source vector of the original n-port be denoted respectively by $\mathbf{S}$ and $\mathbf{X}$ and let the corresponding primed symbols denote the same quantities of the resultant n-port. The equations

$$\mathbf{B} = \mathbf{SA} + \mathbf{X} \tag{3.139}$$

and

$$\mathbf{B}' = \mathbf{S}'\mathbf{A}' + \mathbf{X}' \tag{3.140}$$

are valid. The primed and unprimed vectors $\mathbf{A}'$, $\mathbf{B}'$ and $\mathbf{A}$, $\mathbf{B}$ are as defined in

the previous section (cf. figs 19 and 20). Substitution of eq. (3.139) in eq. (3.109) yields *)

$$\mathbf{B}' = (\mathbf{\Theta}_{11}\mathbf{S} + \mathbf{\Theta}_{12})\,\mathbf{A} + \mathbf{\Theta}_{11}\mathbf{X} + \mathbf{Y}_1 \tag{3.141}$$

and

$$\mathbf{A}' = (\mathbf{\Theta}_{21}\mathbf{S} + \mathbf{\Theta}_{22})\,\mathbf{A} + \mathbf{\Theta}_{21}\mathbf{X} + \mathbf{Y}_2. \tag{3.142}$$

Elimination of $\mathbf{A}$ and comparison of the result with eq. (3.140) lead to

$$\mathbf{S}' = (\mathbf{\Theta}_{11}\mathbf{S} + \mathbf{\Theta}_{12})\,(\mathbf{\Theta}_{21}\mathbf{S} + \mathbf{\Theta}_{22})^{-1} \tag{3.143}$$

and

$$\mathbf{X}' = \mathbf{\Gamma}_{\Theta}\mathbf{X} + \mathbf{Y}_1 - \mathbf{S}'\mathbf{Y}_2, \tag{3.144}$$

where (cf. eq. (3.30))

$$\mathbf{\Gamma}_{\Theta} = \mathbf{\Theta}_{11} - \mathbf{S}'\mathbf{\Theta}_{21}. \tag{3.145}$$

The elimination can be carried out only if the matrix Y_{Θ},

$$Y_{\Theta} = \mathbf{\Theta}_{21}\mathbf{S} + \mathbf{\Theta}_{22}, \tag{3.146}$$

is non-singular (cf. eq. (3.14)). To some extent this condition provides a criterion for the imbedding to be meaningful (cf. sec. 3.6.7).

There is a close resemblance between the results obtained here and those found in secs 3.1.4 and 3.1.5. The transfer matrix $\mathbf{\Theta}$ of an imbedding $2n$-port is the analogue of the matrix $\mathbf{\Phi}$ of a coordinate transformation. An imbedding is not necessarily isomorphic, that is there need not exist a one-to-one correspondence between the original and resultant n-ports. This signifies that $\mathbf{\Theta}$ is not necessarily non-singular. Hence, the analogue of the inverse $\mathbf{\Psi}$ of the coordinate-transformation matrix need not exist. Furthermore, the matrices $\mathbf{\Gamma}_{\Theta}$ and Y_{Θ} are analogous to respectively the noise- and signal-transformation matrices $\mathbf{\Gamma}$ and Y. Again the analogues of $\mathbf{\Delta}$ and $\mathbf{\Lambda}$, that is the matrices $\mathbf{\Delta}_{\Theta}$ and $\mathbf{\Lambda}_{\Theta}$, cannot always be defined. Equations analogous to eqs (3.19) and (3.28) need not exist (cf. sec. 3.7.1).

3.6.2. *Power-absorption matrix*

Let the power absorbed in the resultant n-port be denoted by $P_{\mathrm{abs},S'}$. This power is dissipated partly in the imbedding $2n$-port and partly in the original n-port. Let these parts be denoted by respectively $P_{\mathrm{abs},\Theta}$ and $P_{\mathrm{abs},S}$. We can write down

$$P_{\mathrm{abs},S'} = P_{\mathrm{abs},\Theta} + P_{\mathrm{abs},S}. \tag{3.147}$$

Furthermore, by virtue of eqs (3.41), (3.43), and (3.101), we have

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*) Instead of in eq. (3.109) the substitution could also have been made in eq. (3.114). When that is done the transformation formulas are obtained in terms of the elements (or submatrices) of the scattering matrix of the imbedding $2n$-port. This possible procedure has been worked out in appendix D.

$$P_{\text{abs},S'} = \mathbf{A}'^{\times}\mathbf{P}_{S'}\mathbf{A}' = \mathbf{A}'^{\times}\,(\mathbf{I} - \mathbf{S}'^{\times}\mathbf{S}')\,\mathbf{A}', \tag{3.148}$$

$$P_{\text{abs},S} = \mathbf{A}^{\times}\mathbf{P}_{S}\mathbf{A} = \mathbf{A}^{\times}\,(\mathbf{I} - \mathbf{S}^{\times}\mathbf{S})\,\mathbf{A} \tag{3.149}$$

and

$$P_{\text{abs},\Theta} = \left(\begin{array}{c}\mathbf{B}\\ \hline \mathbf{A}\end{array}\right)^{\times}\mathbf{P}_{\Theta}\left(\begin{array}{c}\mathbf{B}\\ \hline \mathbf{A}\end{array}\right) = \mathbf{A}^{\times}\left(\begin{array}{c}\mathbf{S}\\ \hline \mathbf{I}\end{array}\right)^{\times}(\boldsymbol{\Omega} - \boldsymbol{\Theta}^{\times}\boldsymbol{\Omega}\boldsymbol{\Theta})\left(\begin{array}{c}\mathbf{S}\\ \hline \mathbf{I}\end{array}\right)\mathbf{A}. \tag{3.150}$$

If for the time being we neglect noise, by virtue of eqs (3.142) and (3.146), we obtain

$$\mathbf{P}_{S'} = \mathbf{P}_{S}' + \mathbf{P}_{\Theta}', \tag{3.151}$$

where

$$\mathbf{P}_{S}' = (Y_{\Theta}^{-1})^{\times}\,\mathbf{P}_{S}Y_{\Theta}^{-1}, \tag{3.152}$$

$$\mathbf{P}_{\Theta}' = (Y_{\Theta}^{-1})^{\times}\left(\begin{array}{c}\mathbf{S}\\ \hline \mathbf{I}\end{array}\right)^{\times}\mathbf{P}_{\Theta}\left(\begin{array}{c}\mathbf{S}\\ \hline \mathbf{I}\end{array}\right)Y_{\Theta}^{-1}. \tag{3.153}$$

The power-absorption matrix $\mathbf{P}_{S'}$ of the resultant n-port consists of two parts. One part, $\mathbf{P}_{S}'$, governs the power absorbed in the original n-port while the other part, $\mathbf{P}_{\Theta}'$, refers to the power dissipated in the imbedding $2n$-port.

3.6.3. *Noise transformation*

The noise-source power matrix $\mathbf{G}_{S'}$ of the resultant n-port can be written in the form

$$\mathbf{G}_{S'} = k\,\mathbf{T}_{S'}\mathbf{N}_{S'} = k\,\mathbf{T}_{S'}\,(\mathbf{I} - \mathbf{S}'\mathbf{S}'^{\times}). \tag{3.154}$$

We assume that $\mathbf{X}$ and $\mathbf{Y}$ are uncorrelated. Insertion of eq. (3.144) into eq. (3.49) shows that $\mathbf{G}_{S'}$, too, consists of two parts, i.e.

$$\mathbf{G}_{S'} = \mathbf{G}_{S}' + \mathbf{G}_{\Theta}', \tag{3.155}$$

where

$$\mathbf{G}_{S}' = \boldsymbol{\Gamma}_{\Theta}\mathbf{G}_{S}\boldsymbol{\Gamma}_{\Theta}^{\times} \tag{3.156}$$

and

$$\mathbf{G}_{\Theta}' = (\mathbf{I}\,|\,-\mathbf{S}')\,\mathbf{G}_{\Theta}\,(\mathbf{I}\,|\,-\mathbf{S}')^{\times}. \tag{3.157}$$

The first part originates from the original n-port while the second part refers to the noise which descends from the imbedding $2n$-port.

By analogy with the deduction in sec. 3.2.5, the imbedded noise-source power matrix $\mathbf{G}_{S}'$ can be written in the form

$$\mathbf{G}_{S}' = k\,\mathbf{T}_{S}'\mathbf{N}_{S}', \tag{3.158}$$

where

$$\mathbf{T}_{S}' = \boldsymbol{\Gamma}_{\Theta}\mathbf{T}_{S}\boldsymbol{\Gamma}_{\Theta}^{-1} \tag{3.159}$$

and

$$\mathbf{N}_{S}' = \boldsymbol{\Gamma}_{\Theta}\mathbf{N}_{S}\boldsymbol{\Gamma}_{\Theta}^{\times}. \tag{3.160}$$

A necessary condition for this to be permissible is that $\boldsymbol{\Gamma}_{\Theta}$ be non-singular

(cf. sec. 3.6.4). If $\mathbf{\Gamma}_\Theta$ is singular, for certain noise-source vectors, the null vectors of $\mathbf{\Gamma}_\Theta$, the noise of the original n-port cannot be perceived at all at the input of the imbedding $2n$-port, and for other noise-source vectors part of that noise is not detectable there. An analogous division of $\mathbf{G}_\Theta{}'$ into two matrix factors would be artificial and ambiguous. It would not have much physical meaning either.

In general the inequalities

$$\mathbf{N}_S{}' \neq \mathbf{N}_{S'} \tag{3.161}$$

and

$$\mathbf{T}_S{}' \neq \mathbf{T}_{S'} \tag{3.162}$$

apply. We prove the first one only. Let us insert eqs (3.55) and (3.145) into eq. (3.160). When, further, the solution for $\mathbf{S}$ of eq. (3.143), given by

$$\mathbf{S} = -(\mathbf{\Theta}_{11} - \mathbf{S}'\mathbf{\Theta}_{21})^{-1}(\mathbf{\Theta}_{12} - \mathbf{S}'\mathbf{\Theta}_{22}), \tag{3.163}$$

is substituted, we obtain

$$\begin{aligned}\mathbf{N}_S{}' = {} & (\mathbf{\Theta}_{11}\mathbf{\Theta}_{11}{}^\times - \mathbf{\Theta}_{12}\mathbf{\Theta}_{12}{}^\times) - \mathbf{S}'(\mathbf{\Theta}_{21}\mathbf{\Theta}_{11}{}^\times - \mathbf{\Theta}_{22}\mathbf{\Theta}_{12}{}^\times) + \\ & - (\mathbf{\Theta}_{11}\mathbf{\Theta}_{21}{}^\times - \mathbf{\Theta}_{12}\mathbf{\Theta}_{22}{}^\times)\mathbf{S}'{}^\times + \mathbf{S}'(\mathbf{\Theta}_{21}\mathbf{\Theta}_{21}{}^\times - \mathbf{\Theta}_{22}\mathbf{\Theta}_{22}{}^\times)\mathbf{S}'{}^\times.\end{aligned} \tag{3.164}$$

Comparison of this result with eq. (3.154) verifies the inequality. Only when eqs (3.130), (3.131), and (3.132) are valid, that is only when the imbedding $2n$-port is lossless, the two matrices $\mathbf{N}_{S'}$ and $\mathbf{N}_S{}'$ are identical.

It will be clear that all matrix relations deduced in the previous and the present section can also be derived by straightforward matrix-algebraic methods, when expressions like eq. (3.164) are used. However, many calculations are involved.

3.6.4. *Images and reductions*

Two different ways of visualizing the process of an imbedding are possible. First, the resultant n-port can be considered to be an image of the original one as seen through the imbedding $2n$-port. Secondly, the resultant n-port can also be thought to be a reduction of the imbedding $2n$-port, the reduction being effectuated by the original n-port. It should be emphasized that these two ways of visualization are excluding each other.

The matrices $\mathbf{P}_S{}'$, $\mathbf{N}_S{}'$, $\mathbf{G}_S{}'$ and $\mathbf{T}_S{}'$ introduced in the preceding sections are the images of the corresponding matrices of the original n-port. The more or less the imbedding $2n$-port is transparent, the more or less complete is the image of the original n-port. The transparency of the $2n$-port depends on its losses. The smaller the latter are, the better the former is. Only if the $2n$-port is lossless, are the images complete. In that case the matrices mentioned above are equal to those of the resultant n-port, which itself is then the unblurred image of the original n-port (cf. sec. 3.6.6).

The matrices $\mathbf{P}_\Theta{}'$ and $\mathbf{G}_\Theta{}'$ cannot be considered to be images of $\mathbf{P}_\Theta$ and $\mathbf{G}_\Theta$.

Even the orders are not the same. Instead of images we can call the former matrices reductions of the latter. If the reducing original n-port is lossless, it is readily shown that the reduced matrices $\mathbf{P}_\Theta'$ and $\mathbf{G}_\Theta'$ are equal to the corresponding matrices of the resultant n-port. In this particular case similar remarks apply to the noise-temperature and the noise-distribution matrices.

3.6.5. *Homogeneously warm multiports*

In the present and the next section we will investigate two simple cases. First, the case that both, imbedded and imbedding, multiports are at uniform temperatures and, secondly, the case of a lossless imbedding multiport. It will turn out that the latter types of imbeddings are equivalent with certain coordinate transformations (cf. sec. 3.7.1).

Let the temperatures of the imbedded n-port and of the imbedding $2n$-port both be uniform and be denoted by respectively T_S and T_Θ. By virtue of eq. (3.159), we have

$$\mathbf{T}_S' = T_S\mathbf{I} \tag{3.165}$$

and hence, due to eq. (3.158),

$$\mathbf{G}_S' = kT_S\mathbf{N}_S'. \tag{3.166}$$

Furthermore, substituting the $2n$th-order version of eq. (3.105) in eq. (3.157), by virtue of eq. (3.164), we obtain

$$\mathbf{G}_\Theta' = kT_\Theta\,(\mathbf{N}_{S'} - \mathbf{N}_S'). \tag{3.167}$$

Insertion of these results into eq. (3.155) gives

$$\mathbf{G}_{S'} = kT_\Theta\mathbf{N}_{S'} + k\,(T_S - T_\Theta)\,\mathbf{N}_S'. \tag{3.168}$$

If the uniform temperatures are equal, i.e.

$$T_S = T_\Theta = T, \tag{3.169}$$

the obvious equation

$$\mathbf{G}_{S'} = kT\,\mathbf{N}_{S'} \tag{3.170}$$

is obtained.

Equating eqs (3.168) and (3.154) and solving for $\mathbf{T}_{S'}$, we find

$$\mathbf{T}_{S'} = T_\Theta\mathbf{I} + (T_S - T_\Theta)\,\mathbf{N}_S'\mathbf{N}_{S'}^{-1}. \tag{3.171}$$

It is readily seen that, in general, two homogeneously warm multiports at different temperatures do not constitute a resultant multiport at an apparently uniform temperature. That would be so only if

$$\mathbf{N}_{S'} = \mathbf{N}_S' \tag{3.172}$$

is satisfied. As will be shown in the next section, this is the case if the imbedding multiport is lossless. Then the resultant n-port is at the apparently uniform temperature T_S. Of course, the — abstract — temperature of a lossless multiport does not matter at all.

3.6.6. *Lossless imbeddings*

It has already been remarked that many of the results of the preceding sections would be simple if the imbedding $2n$-port were lossless. Introducing the conditions of freedom of loss, as contained in eq. (3.129), into the expressions for $\mathbf{P}_\Theta$ and $\mathbf{G}_\Theta$, by virtue of eqs (3.153) and (3.157), we obtain

$$\mathbf{P}_\Theta' = \mathbf{G}_\Theta' = \mathbf{O}. \tag{3.173}$$

Then, from eqs (3.151) and (3.155), it is readily derived that

$$\mathbf{P}_{S'} = \mathbf{P}_S' \tag{3.174}$$

and

$$\mathbf{G}_{S'} = \mathbf{G}_S'. \tag{3.175}$$

Substitution of eqs (3.130), (3.131), and (3.132) in eq. (3.164) yields

$$\mathbf{N}_{S'} = \mathbf{N}_S', \tag{3.176}$$

so that insertion of eq. (3.175) into eq. (3.158) results in

$$\mathbf{T}_{S'} = \mathbf{T}_S'. \tag{3.177}$$

The relations (3.174) and (3.177) can be deduced in the same matrix-algebraic way in which eq. (3.176) has been derived. This then results in eq. (3.175). In this way we have verified the consistency of the imbedding relations.

3.6.7. *Conditions*

In deriving the imbedding relations, several conditions have to be satisfied for an imbedding to be unambiguous. In sec. 3.6.1 it has been stated that the matrix $\boldsymbol{Y}_\Theta$ must be non-singular and in sec. 3.6.3 the same has been observed for the matrix $\boldsymbol{\Gamma}_\Theta$. A physical interpretation of these conditions is most easily obtained when the imbedding $2n$-port is represented by its scattering matrix. This has been done in appendix D. Recapitulating the investigations made there, the following three conditions can be formulated. Let $\mathbf{S}_{ij}$ be the square submatrices of the scattering matrix of the $2n$-port. We then have:

(a) the scattering matrix $\mathbf{S}'$ of the resultant n-port is determined uniquely by the scattering matrix $\mathbf{S}$ of the original n-port only if the system formed by original n-port and imbedding $2n$-port has no internal undamped resonances, i.e.

$$(\mathbf{I} - \mathbf{S}_{22}\mathbf{S}): \text{ non-singular};$$

(b) the transformation of the noise of the original n-port is unambiguous only if, apart from condition (a), a unique coupling from the output towards the input of the imbedding $2n$-port exists, i.e.

$$\mathbf{S}_{12}\text{: non-singular;}$$

(c) the image of the power-absorption matrix of the original n-port is complete only if, apart from condition (a), the coupling from the input towards the output of the imbedding $2n$-port is unique, i.e.

$$\mathbf{S}_{21}\text{: non-singular.}$$

The last condition is identical with the criterion for the transfer-matrix representation of the imbedding $2n$-port to be meaningful, as it has been derived in sec. 3.5.2. That its transfer matrix exists is therefore a necessary condition for a $2n$-port to be an unambiguous transformer but not yet a sufficient one. Also its transfer matrix in the opposite direction must exist and, furthermore, it is not permitted to constitute an undamped resonant system with the n-port it is transforming.

When the imbedding $2n$-port is lossless these conditions become quite simplified. In fact, as has been shown in sec. 3.5.2, the conditions about the couplings between input and output or the existence of the two transfer matrices are then automatically satisfied. The condition about the non-existence of undamped resonances remains only to be satisfied separately.

In sec. 3.1.5 it has been stated that for coordinate transformations the conditions which warrant unambiguous excitation transformation and noise transformation, respectively, are identical. There, the identity is due to the non-singularity of the transformation. The two unambiguity conditions of a coordinate transformation are equivalent with the above conditions (b) and (c), provided condition (a) has been satisfied. In appendix D it is also shown that, if the transfer matrix is non-singular, the two conditions are identical, so that in essence there is only one.

3.7. Group theory of transformations

3.7.1. *Equivalence of imbeddings to coordinate transformations*

It is easy to prove that the lossless imbedding is equivalent to a non-singular coordinate transformation. Conversely, as will be shown in the next section, the inverse of this statement is not necessarily true.

The transfer matrix $\boldsymbol{\Theta}$ of a lossless $2n$-port is non-singular. Due to eqs (3.129) and (3.117), the inverse of $\boldsymbol{\Theta}$ is then given by

$$\boldsymbol{\Theta}^{-1} = \boldsymbol{\Omega}\boldsymbol{\Theta}^{\times}\boldsymbol{\Omega}. \tag{3.178}$$

In general, a lossless $2n$-port is noiseless. Hence, comparing eq. (3.109) with

the definition of a coordinate transformation, that is with eq. (3.9), we see that $\boldsymbol{\Theta}$ is analogous to the matrix $\boldsymbol{\Phi}$ of a coordinate transformation. The following analogies apply:

$$\boldsymbol{\Theta} \rightarrow \boldsymbol{\Phi}, \tag{3.179}$$

$$\boldsymbol{\Theta}^{-1} \rightarrow \boldsymbol{\Psi}, \tag{3.180}$$

$$\boldsymbol{\Gamma}_{\Theta} \rightarrow \boldsymbol{\Gamma}, \tag{3.181}$$

$$Y_{\Theta} \rightarrow Y. \tag{3.182}$$

In the case of lossless imbeddings the non-singularity of $\boldsymbol{\Gamma}_{\Theta}$ follows from that of Y_{Θ} and vice versa (cf. sec. 3.1.5). Moreover, in that case, matrices $\boldsymbol{\Delta}_{\Theta}$ and $\boldsymbol{\Lambda}_{\Theta}$ can be defined uniquely (cf. sec. 3.6.1) and the analogies

$$\boldsymbol{\Delta}_{\Theta} \rightarrow \boldsymbol{\Delta} \tag{3.183}$$

and

$$\boldsymbol{\Lambda}_{\Theta} \rightarrow \boldsymbol{\Lambda} \tag{3.184}$$

are valid.

The above equivalence of lossless imbeddings to non-singular coordinate transformations can be generalized. Let, in an arbitrary coordinate system, the matrices $\mathbf{M}$ and $\mathbf{M}'$ be the signal-performance matrices of respectively the original and the resultant n-ports. These matrices have the same character, either being both scattering matrices, or being both impedance matrices, etc. We say, the original and the resultant n-ports are described in identical coordinate systems. Hence, to both of them the same power matrix $\mathbf{Q}_{(M)}$ applies (cf. sec. 3.1.6).

Let further $\begin{pmatrix} \mathbf{L} \\ \hline \mathbf{K} \end{pmatrix}$ and $\begin{pmatrix} \mathbf{L}' \\ \hline \mathbf{K}' \end{pmatrix}$ be the vectors of the signal states at respectively the original and the resultant n-ports. Then, the coordinate system in which the imbedding $2n$-port is to be represented should be chosen so that the corresponding signal-performance matrix $\boldsymbol{\Theta}_{(M)}$ relates the above signal-state vectors explicitly by

$$\begin{pmatrix} \mathbf{L}' \\ \hline \mathbf{K}' \end{pmatrix} = \boldsymbol{\Theta}_{(M)} \begin{pmatrix} \mathbf{L} \\ \hline \mathbf{K} \end{pmatrix}. \tag{3.185}$$

Thus, if the $\mathbf{M}$-matrices are the scattering matrices $\mathbf{S}$ of the n-ports, the matrix $\boldsymbol{\Theta}_{(M)}$ is the transfer matrix $\boldsymbol{\Theta}_{(S)}$ of the $2n$-port or, if the former are the impedance matrices $\mathbf{Z}$, the latter is the chain matrix $\boldsymbol{\Theta}_{(Z)}$, etc.

A further remark on the signal representation of the $2n$-port should be made. The choice of a coordinate system for the $2n$-port as made above is not at all the only one possible. Whichever signal representation for the $2n$-port may have been chosen, the properties of the resultant n-port can be calculated from those of the original one. Haus and Adler [15]), for instance, investigated lossless

imbeddings of n-ports, making use of the impedance-matrix signal representation for both the n-ports and the $2n$-port. Analogously, in appendix D, where imbedding effects and conditions are calculated, again the n-ports as well as the $2n$-port are described by their respective scattering matrices. However, only if the choice has been made as described above, the signal-performance matrix $\mathbf{\Theta}_{(M)}$ of the imbedding $2n$-port is equivalent to the matrix $\mathbf{\Phi}$ of a non-singular coordinate transformation. For, in order that this equivalence is to be valid, it is necessary that eq. (3.185) applies.

For a lossless imbedding the power absorbed by the resultant n-port is equal to that dissipated in the original one. Hence, by virtue of eq. (3.35), equation

$$\begin{pmatrix} \mathbf{L}' \\ \hline \mathbf{K}' \end{pmatrix}^{\times} \mathbf{Q}_{(M)} \begin{pmatrix} \mathbf{L}' \\ \hline \mathbf{K}' \end{pmatrix} = \begin{pmatrix} \mathbf{L} \\ \hline \mathbf{K} \end{pmatrix}^{\times} \mathbf{Q}_{(M)} \begin{pmatrix} \mathbf{L} \\ \hline \mathbf{K} \end{pmatrix} \tag{3.186}$$

must hold. Substituting eq. (3.185) in this equation and observing that the result must be valid for any possible original signal state $\begin{pmatrix} \mathbf{L} \\ \hline \mathbf{K} \end{pmatrix}$, we find that for any lossless imbedding $\mathbf{\Theta}_{(M)}$ the equation

$$\mathbf{\Theta}_{(M)}{}^{\times} \mathbf{Q}_{(M)} \mathbf{\Theta}_{(M)} = \mathbf{Q}_{(M)} \tag{3.187}$$

must apply. As will be shown in sec. 3.7.3, this equation defines a group of matrices $\mathbf{\Theta}_{(M)}$.

3.7.2. *Canonical coordinate transformations*

The inverse of the analogy discussed in the previous section need not be true. Not every non-singular coordinate transformation is necessarily equivalent to a lossless imbedding with similar description of original and resultant n-ports. Comparing eq. (3.185) to eq. (3.9), we have concluded that any lossless imbedding is equivalent to a non-singular coordinate transformation. Conversely, when we compare eq. (3.9) to eq. (3.185), by virtue of eq. (3.187), we can observe that only those non-singular coordinate transformations whose matrices $\mathbf{\Phi}$ and $\mathbf{\Psi}$ satisfy such conditions as

$$\mathbf{\Phi}^{\times} \mathbf{Q} \mathbf{\Phi} = \mathbf{Q} \tag{3.188}$$

and

$$\mathbf{\Psi}^{\times} \mathbf{Q} \mathbf{\Psi} = \mathbf{Q}, \tag{3.189}$$

are equivalent to lossless imbeddings. In fact, the matrix $\mathbf{\Phi}$ is equal to that signal-performance matrix $\mathbf{\Theta}$ of the equivalent imbedding $2n$-port, which should be chosen to have both the n-ports being represented in the coordinate system of which $\mathbf{Q}$ is the power matrix.

It should be remembered that not all matrices $\mathbf{Q}$ are permissible. Let a coordinate transformation which transforms from one type of signal representation to another be called a transition transformation. Let, further, the corre-

sponding transformation matrices $\mathbf{\Phi}_t$ and $\mathbf{\Psi}_t$ be denoted by the subscript t and be called transition matrices. From eqs (3.38) and (3.40) it will be clear that for any permissible $\mathbf{Q}$ a pair of non-singular transition matrices $\mathbf{\Phi}_t$ and $\mathbf{\Psi}_t$ must exist so that the equation

$$\mathbf{Q} = -\mathbf{\Psi}_t{}^{\times}\mathbf{\Omega}\mathbf{\Psi}_t \tag{3.190}$$

is valid. This condition signifies that those, and only those, matrices $\mathbf{Q}$ which are Hermitely congruent with $\mathbf{\Omega}$ are permissible. All possible $\mathbf{Q}$'s are mutually Hermitely congruent.

In the next section it will be shown that all coordinate transformations $\mathbf{\Phi}$ which, for a given $\mathbf{Q}$, satisfy eq. (3.188) form a group in the mathematical sense. Thus, any $\mathbf{Q}$ defines a group of coordinate transformations. Such a group is equivalent to the group of lossless imbeddings. In other words, there is a one-to-one correspondence between the members of a $\mathbf{Q}$-defined group of transformations and the members of the group of lossless imbedding $2n$-ports. Such a one-to-one correspondence is called an isomorphism while the transformations and imbedding $2n$-ports are called isomorphous. Furthermore, any group of $\mathbf{\Phi}$'s defined by a given $\mathbf{Q}$ is a subgroup of the group of all non-singular coordinate transformations. In fact, that subgroup is the group whose members leave the particular power matrix $\mathbf{Q}$ unaltered. In mechanics coordinate transformations for which power expressions, e.g. Hamiltonians, are invariant, that means those which leave $\mathbf{Q}$ unaltered, are called canonical coordinate transformations, i.e. canonical with respect to the particular power matrix or the particular signal representation. We shall adopt the same name here *) and call all coordinate transformations, which leave the character of the signal representation unaltered, canonical with respect to that type of representation. The group of these transformations is the appropriate canonical group.

From eq. (3.188), or eq. (3.189), and eq. (3.10) it can be derived that the matrices $\mathbf{\Phi}$ and $\mathbf{\Psi}$ of a canonical coordinate transformation are related by

$$\mathbf{\Psi} = \mathbf{\Phi}^{-1} = \mathbf{Q}^{-1}\mathbf{\Phi}^{\times}\mathbf{Q}. \tag{3.191}$$

3.7.3. *Group theory of transformations and imbeddings*

A collection of elements is a group in the mathematical sense if the following conditions are satisfied:

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*) The use of the word canonical is at variance with that made by Haus and Adler [15]) and, moreover, with common usage in matrix algebra. Haus and Adler use the concept of canonical imbedding for that lossless imbedding which diagonalizes simultaneously signal and noise performance of an n-port. Below, the canonical transformation which diagonalizes both signal and noise performance of an n-port, will be called the diagonalizing transformation of that n-port.

(a) the operation of multiplication of two members of the collection is defined and is unique; the result is again a member;
(b) the operation of multiplication is associative;
(c) a unitary element exists;
(d) the inverse of a member exists and is a member itself.

We next show that, in accordance with eq. (3.188), i.e.

$$\mathbf{\Phi}^{\times}\mathbf{Q}\mathbf{\Phi} = \mathbf{Q}, \tag{3.192}$$

any possible matrix $\mathbf{Q}$ defines a group of elements $\mathbf{\Phi}$.
(a) Let $\mathbf{\Phi}_1$ and $\mathbf{\Phi}_2$ be two members of the collection of all matrices $\mathbf{\Phi}$ which satisfy eq. (3.192). In matrix algebra multiplication of these two matrices is defined. Let $\mathbf{\Phi}_3$ be the product of $\mathbf{\Phi}_1$ and $\mathbf{\Phi}_2$, i.e.

$$\mathbf{\Phi}_3 = \mathbf{\Phi}_1\mathbf{\Phi}_2. \tag{3.193}$$

We then have

$$\mathbf{\Phi}_3{}^{\times}\mathbf{Q}\mathbf{\Phi}_3 = \mathbf{\Phi}_2{}^{\times}\mathbf{\Phi}_1{}^{\times}\mathbf{Q}\mathbf{\Phi}_1\mathbf{\Phi}_2 = \mathbf{\Phi}_2{}^{\times}\mathbf{Q}\mathbf{\Phi}_2 = \mathbf{Q}. \tag{3.194}$$

Hence, $\mathbf{\Phi}_3$ is also a member of the collection of $\mathbf{\Phi}$'s.
(b) In matrix algebra it is shown that multiplication of matrices is an associative operation.
(c) The identity matrix $\mathbf{I}$ is the required unitary element. Indeed, this matrix satisfies eq. (3.192).
(d) We consider only non-singular coordinate transformations. Hence the inverse $\mathbf{\Phi}^{-1}$ of a matrix $\mathbf{\Phi}$ exists. If eq. (3.192) is pre- and post-multiplied by respectively $(\mathbf{\Phi}^{-1})^{\times}$ and $\mathbf{\Phi}^{-1}$, we obtain

$$\mathbf{Q} = (\mathbf{\Phi}^{-1})^{\times}\mathbf{Q}\mathbf{\Phi}^{-1}, \tag{3.195}$$

so that, indeed, $\mathbf{\Phi}^{-1}$ itself is a member of the collection of $\mathbf{\Phi}$'s.

All these observations having been made, the above statement about the $\mathbf{Q}$-defined collection of $\mathbf{\Phi}$'s being a group has been proved.

For completeness we shall now show that all $\mathbf{Q}$-defined subgroups of the group of all non-singular coordinate transformations are isomorphous. This means that there exists a unique transition transformation from one $\mathbf{Q}$-defined group to another. This transformation should define a one-to-one correspondence between the members of the two $\mathbf{Q}$-defined groups such that the transformation, or image, of the product of two members of one group is equal to the product of the images of those two members. Such a transition transformation maps one group onto the other, and vice versa. The transition transformation defines an isomorphism between the two $\mathbf{Q}$-defined groups.

Let $\mathbf{\Phi}_{(1)}$ and $\mathbf{\Phi}_{(2)}$ denote members of the $\mathbf{\Phi}$-groups defined by respectively $\mathbf{Q}_{(1)}$ and $\mathbf{Q}_{(2)}$. Further, let $\mathbf{\Phi}_t$ and $\mathbf{\Psi}_t$ be the matrices of the representation tran-

sition which relates $\mathbf{Q}_{(1)}$ and $\mathbf{Q}_{(2)}$ in accordance with eq. (3.40), i.e.

$$\mathbf{Q}_{(2)} = \mathbf{\Psi}_t{}^{\times}\mathbf{Q}_{(1)}\mathbf{\Psi}_t. \tag{3.196}$$

Substitution of this equation in the defining equation of the $\mathbf{Q}_{(2)}$-group, i.e. in

$$\mathbf{\Phi}_{(2)}{}^{\times}\mathbf{\Phi}_{(2)}\mathbf{\Phi}_{(2)} = \mathbf{Q}_{(2)}, \tag{3.197}$$

followed by pre- and post-multiplication of the result by respectively $(\mathbf{\Psi}_t{}^{-1})^{\times}$ and $\mathbf{\Psi}_t{}^{-1}$, yields

$$(\mathbf{\Psi}_t\mathbf{\Phi}_{(2)}\mathbf{\Psi}_t{}^{-1})^{\times}\mathbf{Q}_{(1)}(\mathbf{\Psi}_t\mathbf{\Phi}_{(2)}\mathbf{\Psi}_t{}^{-1}) = \mathbf{Q}_{(1)}. \tag{3.198}$$

It states that the matrix obtained from $\mathbf{\Phi}_{(2)}$ by the similarity transformation

$$\mathbf{\Phi}_{(1)} = \mathbf{\Psi}_t\mathbf{\Phi}_{(2)}\mathbf{\Psi}_t{}^{-1} \tag{3.199}$$

is a member of the $\mathbf{\Phi}$-group defined by $\mathbf{Q}_{(1)}$. Conversely, the similarity transformation

$$\mathbf{\Phi}_{(2)} = \mathbf{\Phi}_t\mathbf{\Phi}_{(1)}\mathbf{\Phi}_t{}^{-1} \tag{3.200}$$

maps the $\mathbf{Q}_{(1)}$-group onto the $\mathbf{Q}_{(2)}$-group. Since the discussions are restricted to non-singular matrices $\mathbf{\Phi}_t$ and $\mathbf{\Psi}_t$, the mapping is unique. That this mapping is an isomorphism is clear from the following observation. Let $\mathbf{A}_{(1)}$, $\mathbf{B}_{(1)}$, and $\mathbf{C}_{(1)}$ be members of the $\mathbf{Q}_{(1)}$-group and $\mathbf{A}_{(2)}$, $\mathbf{B}_{(2)}$, and $\mathbf{C}_{(2)}$ the respective images in the $\mathbf{Q}_{(2)}$-group. Let further the equation

$$\mathbf{C}_{(1)} = \mathbf{A}_{(1)}\mathbf{B}_{(1)} \tag{3.201}$$

hold. Pre- and post-multiplying this equation by respectively $\mathbf{\Phi}_t$ and $\mathbf{\Phi}_t{}^{-1}$ and inserting $\mathbf{\Phi}_t{}^{-1}\mathbf{\Phi}_t = \mathbf{I}$ into the right-hand side, we obtain

$$\mathbf{C}_{(2)} = \mathbf{A}_{(2)}\mathbf{B}_{(2)}. \tag{3.202}$$

This proves the isomorphism.

All non-singular coordinate transformations can serve as transition transformations. This is also true for members of **Q**-defined subgroups. Let $\mathbf{\Phi}_{(M)}$ be a member of the $\mathbf{\Phi}$-group defined by $\mathbf{Q}_{(M)}$. The matrix $\mathbf{\Psi}_{(M)} = \mathbf{\Phi}_{(M)}{}^{-1}$ can serve as a transition matrix $\mathbf{\Psi}_t$ for other representations, e.g. characterized by the power matrix **Q**. However, if **Q** is equal to $\mathbf{Q}_{(M)}$, that is if $\mathbf{\Psi}_{(M)}$ is applied to its own signal representation or $\mathbf{Q}_{(M)}$-defined group, by virtue of

$$\mathbf{\Psi}_t{}^{\times}\mathbf{Q}_{(M)}\mathbf{\Psi}_t = (\mathbf{\Phi}_{(M)}{}^{-1})^{\times}\mathbf{Q}_{(M)}\mathbf{\Phi}_{(M)}{}^{-1} = \mathbf{Q}_{(M)}, \tag{3.203}$$

the signal representation is transformed onto itself. Indeed, it is readily verified that the result, say $\mathbf{\Phi}_{(2)}$, of the similarity transformation $\mathbf{\Psi}_t$ of a member of the $\mathbf{Q}_{(M)}$-defined group, say $\mathbf{\Phi}_{(1)}$, i.e.

$$\mathbf{\Phi}_{(2)} = \mathbf{\Psi}_t\mathbf{\Phi}_{(1)}\mathbf{\Psi}_t{}^{-1} = \mathbf{\Phi}_{(M)}{}^{-1}\mathbf{\Phi}_{(1)}\mathbf{\Phi}_{(M)}, \tag{3.204}$$

is again a member of the $\mathbf{Q}_{(M)}$-defined group. Such a transformation of a group onto itself is called an automorphism. Hence, members of a $\mathbf{\Phi}$-group constitute automorphisms to their own group. To other $\mathbf{\Phi}$-groups they define isomorphisms.

The findings of the present section can be illustrated with those of secs 3.3.2 and 3.3.3. In the latter the transition from the scattering-matrix to the impedance-matrix signal representation has been investigated. The power matrices of the two signal representations are given by eqs (3.38) and (3.77), i.e.

$$\mathbf{Q}_{(S)} = \left(\begin{array}{c|c} -\mathbf{I} & \mathbf{O} \\ \hline \mathbf{O} & \mathbf{I} \end{array}\right) \tag{3.205}$$

and

$$\mathbf{Q}_{(Z)} = \frac{1}{2}\left(\begin{array}{c|c} \mathbf{O} & \mathbf{I} \\ \hline \mathbf{I} & \mathbf{O} \end{array}\right). \tag{3.206}$$

The matrix $\mathbf{\Psi}_t$ of the transition is given by eq. (3.72), i.e.

$$\mathbf{\Psi}_t = \frac{1}{2}\left(\begin{array}{c|c} \mathbf{I} & -\mathbf{I} \\ \hline \mathbf{I} & \mathbf{I} \end{array}\right).$$

The transition equation, eq. (3.196), in this case

$$\mathbf{Q}_{(Z)} = \mathbf{\Psi}_t{}^{\times}\mathbf{Q}_{(S)}\mathbf{\Psi}_t, \tag{3.207}$$

is readily verified.

All non-singular matrices $\mathbf{\Phi}_{(S)}$, which satisfy

$$\mathbf{\Phi}_{(S)}{}^{\times}\mathbf{Q}_{(S)}\mathbf{\Phi}_{(S)} = \mathbf{Q}_{(S)}, \tag{3.208}$$

are members of the $\mathbf{Q}_{(S)}$-defined subgroup of coordinate-transformation matrices. Due to eq. (3.129), they can be interpreted as transfer matrices of lossless imbedding $2n$-ports. In the same way chain matrices of lossless imbedding $2n$-ports satisfy

$$\mathbf{\Phi}_{(Z)}{}^{\times}\mathbf{Q}_{(Z)}\mathbf{\Phi}_{(Z)} = \mathbf{Q}_{(Z)}. \tag{3.209}$$

Furthermore, the matrices $\mathbf{\Phi}_{(S)}$ and $\mathbf{\Phi}_{(Z)}$ are similar due to the isomorphism of eq. (3.200), i.e.

$$\mathbf{\Phi}_{(Z)} = \mathbf{\Psi}_t{}^{-1}\mathbf{\Phi}_{(S)}\mathbf{\Psi}_t. \tag{3.210}$$

This relation between the chain matrix and the transfer matrix of a $2n$-port is easy to prove.

Summarizing, we can formulate the result of the present section as follows. All lossless imbedding $2n$-ports form a group in the mathematical sense. This group is isomorphous with subgroups of non-singular coordinate transformations. The matrices of these coordinate transformations can be interpreted as signal-performance matrices of $2n$-ports in specific signal representations. These special representations are adapted to the signal representations in which the

imbedded n-ports are described. The latter representations define power matrices $\mathbf{Q}_{(M)}$. In their turn these power matrices define subgroups of coordinate transformations or, what is the same thing, groups of special signal-performance matrices appropriate to the $2n$-ports in the adapted representation. Transition from one signal representation for the n-ports to another determines a regular matrix $\mathbf{\Phi}_t$ which, by means of a similarity transformation, maps the one corresponding subgroup of coordinate transformations onto the other.

Concluding, we can state that, when investigating transformations or imbeddings of n-ports, it is irrelevant whether we actually consider lossless transforming networks, lossless imbedding relations, or canonical coordinate transformations, all three being identical in abstracto. Therefore, henceforth we will simply speak of transformations.

4. DIAGONALIZATION

4.1. The diagonalizing transformation

4.1.1. *Diagonalization and paired representations*

When we classify an n-port and describe its noise in numerical values, it would be both convenient and instructive if we could describe the n-port in an elementary coordinate system. The adjective "elementary" should be understood with reference to the particular n-port under consideration. In other words, we put the question whether, for a given type of signal representation and for any arbitrary n-port, there exists a canonical transformation which diagonalizes simultaneously all characterizing matrices. If so, the resultant matrices $\mathbf{M}$, $\mathbf{P}_M$, $\mathbf{N}_M$, $\mathbf{G}_M$, and $\mathbf{T}_M$, all being diagonal, have the most elementary form. In this chapter we will investigate this problem. Doing so, we will confine ourselves to particular types of signal representations.

Signal representations where corresponding components of the excitation and response vectors refer to one and the same port will be called paired representations. The $2n$ coordinates of the signal-state space are divided into n pairs of corresponding dependent and independent coordinates. Each pair refers to one port and, conversely, each port is provided with one pair. Impedance-matrix, admittance-matrix, and scattering-matrix signal representations are examples. Of all possible signal representations the paired species are of particular interest *).

When a transition transformation is performed from one paired signal representation to another one, the two members of a new pair are independent linear combinations of the two members of a corresponding old pair. If in a given paired representation the diagonalizing canonical transformation of a given n-port has been found, it is known in all other paired representations. This is so because in a paired representation a diagonal signal-performance matrix implies that the n-port consists of n uncoupled 1-ports. In sec. 2.6.4 such an n-port has been called diagonal. It can be shown that the power-absorption and the noise-distribution matrices are then also diagonal. This involves some considerations about signal-state power-absorption matrices of paired representations. We will not make those considerations here.

If, moreover, the noise-source power matrix, too, is diagonal, the noise

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*) All other signal representations could be called unpaired. We admit that these names are a little arbitrary. Such coordinate systems as for example the transfer-matrix signal representation, where corresponding independent and dependent coordinates refer to the incident wave at one port and the reflected wave at a corresponding other port, respectively, could also have been called paired.
In certain cases, e.g. with amplifiers and circulators, such other signal representations are also of importance [33]).

sources at the n separate 1-ports are uncorrelated. This means that the n uncoupled 1-ports can be considered as n different independent 1-ports and that is then true in all paired representations. The lossless imbedding $2n$-port which is the isomorphous image (cf. sec. 3.7) of the set of diagonalizing canonical transformations — there is, apart from those obtained by simple permutations of the ports, one for each canonical group (cf. sec. 4.1.2) — is the *diagonalization transformer* of the n-port under consideration. Such a "transformer" transforms the n-port into a set of n uncoupled independent 1-ports with uncorrelated noise-power sources.

In the present chapter this "principal-axis" problem will be investigated. It will be done only for one particular paired signal representation, for the scattering-matrix signal representation. This is not a real restriction. When the diagonalization is obtained in the scattering-matrix representation, only a transition transformation gives the diagonal forms of the characterizing matrices in another paired representation *).

The diagonalization problem will be dealt with by constructive methods. This means that, when the existence of a diagonalizing canonical transformation is proved, the matrix of that transformation is obtained in explicit form at the same time. Haus and Adler [15]) have investigated the same problem in the impedance-matrix version (see chapter 5). They, however, have not used a constructive procedure and, consequently, they have not obtained expressions for the imbedding $2n$-port. In fact, they have proved only the existence of a diagonalizing imbedding.

It will turn out that the question as to the possibility of canonical diagonalizations can be answered in the affirmative with only one exception. In the cases that an n-port can be brought in a diagonal form in such a manner only that it contains at least one pair of one active 1-port and one dissipative 1-port of which the noise sources are completely correlated, the noise-source power matrix and the signal-performance matrix of that n-port cannot be diagonalized simultaneously.

4.1.2. *Diagonalization of signal performance* **)

The Gaussian transforms $\mathbf{S}\mathbf{S}^{\times}$ and $\mathbf{S}^{\times}\mathbf{S}$ of an arbitrary square matrix $\mathbf{S}$ are positive definite or positive semi-definite Hermitian matrices. They have the same real non-negative eigenvalues.

*) The diagonalization in unpaired representations can be obtained by essentially the same methods, no additional operations being required. Of course the diagonal result does then not represent n separate 1-ports. Nevertheless, in many of these cases the result can also be interpreted in terms of simple physical structures.

**) Many of the considerations made in this section are dealt with extensively in good textbooks on matrix algebra; e.g. ref. 40, pp. 194-203.

Let ξ_i be the ith eigenvalue of $\mathbf{SS}^\times$ and the column vector $\mathbf{U}_i^{(1)}$ the corresponding normalized eigenvector *). If we take $\mathbf{U}_i^{(1)}$ to be the ith column of a square matrix $\mathbf{U}^{(1)}$ we have

$$\mathbf{U}^{(1)\times}\mathbf{SS}^\times\mathbf{U}^{(1)} = \mathbf{\Xi}, \tag{4.1}$$

where $\mathbf{\Xi}$ is the diagonal eigenvalue matrix:

$$\mathbf{\Xi} = \text{diag}\,(\xi_i). \tag{4.2}$$

When the multiplicity, or degeneracy, p_i of an eigenvalue ξ_i is larger than unity, due to the fact that $\mathbf{SS}^\times$ is Hermitian, we can always find a set of p_i orthogonal eigenvectors $\mathbf{U}_i^{(1)}$ which correspond to ξ_i. The matrix $\mathbf{U}^{(1)}$ is the unitary modal matrix of $\mathbf{SS}^\times$, i.e.

$$\mathbf{U}^{(1)}\mathbf{U}^{(1)\times} = \mathbf{U}^{(1)\times}\mathbf{U}^{(1)} = \mathbf{I}. \tag{4.3}$$

Because the eigenvalues ξ_i are non-negative, we can write them in the form

$$\xi_i = r_i r_i^*. \tag{4.4}$$

For the time being, the argument φ_i of the complex number r_i can still be chosen arbitrarily. Let us now introduce a diagonal matrix $\mathbf{R}$ by

$$\mathbf{R} = \text{diag}\,(r_i). \tag{4.5}$$

We then have

$$\mathbf{\Xi} = \mathbf{RR}^\times = \mathbf{R}^\times\mathbf{R}. \tag{4.6}$$

Next, we associate with each column vector $\mathbf{U}_i^{(1)}$ a further column vector $\mathbf{V}_i^{(1)}$. When ξ_i is unequal zero, $\mathbf{V}_i^{(1)}$ is defined by

$$r_i^*\mathbf{V}_i^{(1)} = \mathbf{S}^\times\mathbf{U}_i^{(1)}. \tag{4.7}$$

However, when ξ_i and, hence, r_i are zero, this equation cannot be used. We then proceed as follows. The $\mathbf{U}_i^{(1)}$'s which correspond to zero ξ_i are not only null vectors of $\mathbf{SS}^\times$ but also of $\mathbf{S}^\times$. They form a base of the null space of $\mathbf{S}^\times$. Corresponding to this base of orthonormal $\mathbf{U}_i^{(1)}$'s we can define an orthonormal base of the null space of $\mathbf{S}$. These two bases consist of an equal number of orthonormal vectors. The base vectors of the null space of $\mathbf{S}$ are taken as the $\mathbf{V}_i^{(1)}$'s which are to be associated with the null vectors $\mathbf{U}_i^{(1)}$.

In matrix algebra it is shown that the square matrix $\mathbf{V}^{(1)}$ formed by the n column vectors $V_i^{(1)}$, in the appropriate order, is unitary, i.e.

$$\mathbf{V}^{(1)}\mathbf{V}^{(1)\times} = \mathbf{V}^{(1)\times}\mathbf{V}^{(1)} = \mathbf{I}, \tag{4.8}$$

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*) In this and following sections a number of canonical transformations will be applied consecutively. The superscript, between brackets, refers to the serial number of these transformations.

and that the equations

$$\mathbf{S}^{\times}\mathbf{U}^{(1)} = \mathbf{V}^{(1)}\mathbf{R}^{\times}, \tag{4.9}$$

$$\mathbf{S}\,\mathbf{V}^{(1)} = \mathbf{U}^{(1)}\mathbf{R} \tag{4.10}$$

and

$$\mathbf{V}^{(1)\times}\mathbf{S}^{\times}\mathbf{S}\,\mathbf{V}^{(1)} = \mathbf{R}^{\times}\mathbf{R} = \boldsymbol{\Xi} \tag{4.11}$$

are valid [40]). By virtue of eq. (4.3), premultiplication of eq. (4.10) by $\mathbf{U}^{(1)\times}$ gives

$$\mathbf{U}^{(1)\times}\mathbf{S}\,\mathbf{V}^{(1)} = \mathbf{R}. \tag{4.12}$$

We now define a first transformation $\boldsymbol{\Phi}^{(1)}$ by

$$\boldsymbol{\Phi}^{(1)} = \left(\begin{array}{c|c} \mathbf{U}^{(1)\times} & \mathbf{O} \\ \hline \mathbf{O} & \mathbf{V}^{(1)\times} \end{array}\right). \tag{4.13}$$

By virtue of eqs (4.3) and (4.8), a test with the condition of eq. (3.178) (cf. eqs (3.188) and (3.189)) verifies that $\boldsymbol{\Phi}^{(1)}$ is canonical in that the character of the signal representation is retained. The inverse matrix $\boldsymbol{\Psi}^{(1)}$ is given by

$$\boldsymbol{\Psi}^{(1)} = \left(\begin{array}{c|c} \mathbf{U}^{(1)} & \mathbf{O} \\ \hline \mathbf{O} & \mathbf{V}^{(1)} \end{array}\right). \tag{4.14}$$

Application of this transformation to the n-port $\mathbf{S}$ diagonalizes the signal performance as follows. By virtue of eqs (4.3) and (4.12), substitution of eq. (4.14) in eq. (3.27) gives for the resultant scattering matrix $\mathbf{S}^{(1)}$ the diagonal result

$$\mathbf{S}^{(1)} = \mathbf{U}^{(1)\times}\mathbf{S}\,\mathbf{V}^{(1)} = \mathbf{R}. \tag{4.15}$$

The reflection coefficient ϱ_i of the ith uncoupled 1-port is given by

$$\varrho_i = r_i. \tag{4.16}$$

Substituting eq. (4.13) in eqs (3.14) and (3.30), we obtain the excitation and noise-transformation matrices. They are respectively given by

$$\boldsymbol{Y}^{(1)} = \boldsymbol{\Lambda}^{(1)\times} = \mathbf{V}^{(1)\times} \tag{4.17}$$

and

$$\boldsymbol{\Gamma}^{(1)} = \boldsymbol{\Delta}^{(1)\times} = \mathbf{U}^{(1)\times}. \tag{4.18}$$

Inserting this result into eqs (3.45) and (3.58), we obtain, by virtue of eqs (4.1), (4.3), (4.8), and (4.11), for the resultant absorption and noise-distribution matrices the respective expressions

$$\mathbf{P}_S^{(1)} = \mathbf{V}^{(1)\times}\,(\mathbf{I} - \mathbf{S}^{\times}\mathbf{S})\,\mathbf{V}^{(1)} = \mathbf{I} - \mathbf{R}^{\times}\mathbf{R} = \mathbf{I} - \boldsymbol{\Xi} \tag{4.19}$$

and

$$\mathbf{N}_S^{(1)} = \mathbf{U}^{(1)\times}\,(\mathbf{I} - \mathbf{S}\mathbf{S}^{\times})\,\mathbf{U}^{(1)} = \mathbf{I} - \mathbf{R}\mathbf{R}^{\times} = \mathbf{I} - \boldsymbol{\Xi}. \tag{4.20}$$

Both these matrices are diagonal. Moreover, they are identical, the ith diagonal element being given by

$$\mathbf{P}_{S,ii}^{(1)} = \mathbf{N}_{S,ii}^{(1)} = 1 - \varrho_i \varrho_i^* = 1 - \xi_i. \tag{4.21}$$

This result is consistent with eq. (4.15). Hence, recapitulating, we can conclude that the canonical transformation $\mathbf{\Phi}^{(1)}$ diagonalizes the signal performance. The matrices $\mathbf{S}$, $\mathbf{P}_S$, and $\mathbf{N}_S$ are diagonalized simultaneously.

The way in which the columns $\mathbf{U}_i^{(1)}$ are ordered to constitute the matrix $\mathbf{U}^{(1)}$ does not affect the above result. The sequence of the columns of $\mathbf{U}^{(1)}$ can be chosen arbitrarily if only the order of the columns of the matrix $\mathbf{V}^{(1)}$ is taken correspondingly. Furthermore, because the transformation $\mathbf{\Phi}^{(1)}$ does not alter the rank and the signature of the matrices $\mathbf{P}_S$ and $\mathbf{N}_S$ (cf. sec. 3.4.1), the numbers of positive, negative, and zero values of $(1 - \xi_i)$ are invariants. For future derivations it is convenient to take the sequence of the columns of $\mathbf{U}^{(1)}$ and, hence, of $\mathbf{V}^{(1)}$ so that the first p diagonal elements of $\mathbf{P}_S^{(1)}$ and $\mathbf{N}_S^{(1)}$ are positive, the next q negative, and the remaining d zero. The absolute values of the corresponding diagonal elements of $\mathbf{R}$ are smaller than, larger than, and equal to unity, respectively. Choosing the sequence of the columns as indicated means that the ports of the diagonal resultant n-port $\mathbf{S}^{(1)}$ are arranged with respect to their dissipative, active, and reactive characters.

If the column vectors $\mathbf{U}_i^{(1)}$ and $\mathbf{V}_i^{(1)}$ are multiplied by complex scalars with absolute values equal to unity and with arbitrary arguments α_i and β_i and if, moreover, the argument φ_i of r_i is augmented with $(\beta_i - \alpha_i)$, all above equations maintain their validity. This signifies that, for all values of i, the arguments α_i and β_i can be chosen so that r_i is real and non-negative. Physically, this means that the reference planes in the input and output lines of a diagonalizing imbedding network can always be positioned such that the reflection coefficients of the n resultant 1-ports are real and non-negative. Hereafter, the choice of reference planes will be presumed to have been made accordingly.

In general the transformation $\mathbf{\Phi}^{(1)}$ does not diagonalize the noise. In most cases the resultant noise-source power matrix and noise-temperature matrix, respectively given by

$$\mathbf{G}_S^{(1)} = \mathbf{\Gamma}^{(1)} \mathbf{G}_S \mathbf{\Gamma}^{(1)\times} = \mathbf{U}^{(1)\times} \mathbf{G}_S \mathbf{U}^{(1)} \tag{4.22}$$

and

$$\mathbf{T}_S^{(1)} = \mathbf{\Gamma}^{(1)} \mathbf{T}_S (\mathbf{\Gamma}^{(1)})^{-1} = \mathbf{U}^{(1)\times} \mathbf{T}_S \mathbf{U}^{(1)}, \tag{4.23}$$

are not diagonal. The question whether further canonical transformations can be found in such a way that the diagonality of the signal performance is maintained and the noise is brought into an elementary, perhaps a diagonal, form will be broached below.

4.1.3. *Normalization of the diagonal signal performance*

Before proceeding with the problem of noise diagonalization, it is convenient to have the diagonal signal performance in a normalized form. This can also be accomplished by means of a canonical transformation: the second one. The appropriate matrix $\mathbf{\Phi}^{(2)}$ can be written in the partitioned form

$$\mathbf{\Phi}^{(2)} = \left(\begin{array}{ccc|ccc} \Phi_{11,p}^{(2)} & & & \Phi_{12,p}^{(2)} & & \\ & \Phi_{11,q}^{(2)} & & & \Phi_{12,q}^{(2)} & \\ & & \Phi_{11,d}^{(2)} & & & \Phi_{12,d}^{(2)} \\ \hline \Phi_{21,p}^{(2)} & & & \Phi_{22,p}^{(2)} & & \\ & \Phi_{21,q}^{(2)} & & & \Phi_{22,q}^{(2)} & \\ & & \Phi_{21,d}^{(2)} & & & \Phi_{22,d}^{(2)} \end{array}\right). \quad (4.24)$$

The vacant places are supposed to be identically zero blocks. The other blocks are defined as follows:

(a) The blocks with subscript p are of order p. They act upon the p dissipative 1-ports. For $i = 1, \ldots, p$, we have $r_i < 1$ and

$$\Phi_{11,p}^{(2)} = \Phi_{22,p}^{(2)} = \text{diag}\,[1/(1 - r_i^2)^{1/2}], \quad (4.25)$$

$$\Phi_{12,p}^{(2)} = \Phi_{21,p}^{(2)} = \text{diag}\,[-r_i/(1 - r_i^2)^{1/2}]. \quad (4.26)$$

This portion of the transformation can be interpreted either as a set of p matching transformers which match the 1-ports to the input lines or, alternatively, as a set of p ideal jumps in the wave impedances of the input lines at the reference planes.

(b) The blocks with subscript q are of order q and transform the q active 1-ports. For $i = p + 1, \ldots, p + q$, we have $r_i > 1$ and

$$\Phi_{11,q}^{(2)} = \Phi_{22,q}^{(2)} = \text{diag}\,[(r_i\sqrt{2} - 1)/(r_i^2 - 1)^{1/2}], \quad (4.27)$$

$$\Phi_{12,q}^{(2)} = \Phi_{21,q}^{(2)} = \text{diag}\,[(r_i - \sqrt{2})/(r_i^2 - 1)^{1/2}]. \quad (4.28)$$

The part of the imbedding $2n$-port that corresponds to these blocks is also a set of q transformers or impedance jumps at the reference planes. In this case they are chosen such that the active 1-ports deliver as much power as is incident upon them. The resulting reflection coefficients $\varrho_i^{(2)}$ are equal to $\sqrt{2}$. The resultant 1-ports are reflective amplifiers with a power gain of 2 (3 dB). This normalization is quite arbitrary, but it has been chosen in this way because the corresponding elements of the power-absorption and noise-distribution matrices are equal to —1.

(c) The blocks with subscript d are of order d. They correspond to the d reactive 1-ports. Normalization of these 1-ports has not much sense, if any. For

$i = p + q + 1, \ldots, n$, we have $r_i = 1$ and

$$\mathbf{\Phi}_{11,d}^{(2)} = \mathbf{\Phi}_{22,d}^{(2)} = \mathbf{I}_d, \tag{4.29}$$

$$\mathbf{\Phi}_{12,d}^{(2)} = \mathbf{\Phi}_{21,d}^{(2)} = \mathbf{O}_d. \tag{4.30}$$

The corresponding part of the imbedding $2n$-port consists of d direct connections.

Due to the canonicity of the transformation, the matrix of the inverse transformation can easily be calculated from eq. (3.178). However, since we do not need $\mathbf{\Psi}^{(2)}$ for our considerations, we will not write it down explicitly. Moreover, as the new signal-performance matrix is known in advance, for the determination of the noise-transformation matrix we can use eqs (3.30) and (3.33) instead of eq. (3.28). Similar remarks can be made for the canonical transformations which will be applied in the following sections.

It is easy to verify that $\mathbf{\Phi}^{(2)}$ as defined above satisfies the canonicity condition of eq. (3.178). Performing the transformation we obtain for the resulting characterizing matrices the following normalized expressions:

$$\mathbf{S}^{(2)} = \begin{pmatrix} \mathbf{O}_p & & \\ & 1/2\,\mathbf{I}_q & \\ & & \mathbf{I}_d \end{pmatrix} \tag{4.31}$$

and

$$\mathbf{P}_S^{(2)} = \mathbf{N}_S^{(2)} = \begin{pmatrix} \mathbf{I}_p & & \\ & -\mathbf{I}_q & \\ & & \mathbf{O}_d \end{pmatrix}. \tag{4.32}$$

The noise-transformation matrix $\mathbf{\Gamma}^{(2)}$ is given by

$$\mathbf{\Gamma}^{(2)} = \begin{pmatrix} \mathbf{\Gamma}_p^{(2)} & & \\ & \mathbf{\Gamma}_q^{(2)} & \\ & & \mathbf{\Gamma}_d^{(2)} \end{pmatrix}, \tag{4.33}$$

where

$$\mathbf{\Gamma}_p^{(2)} = \operatorname{diag}\,[1/(1 - r_i^2)^{1/2}], \tag{4.34}$$

$$\mathbf{\Gamma}_q^{(2)} = \operatorname{diag}\,[1/(r_i^2 - 1)^{1/2}], \tag{4.35}$$

$$\mathbf{\Gamma}_d^{(2)} = \mathbf{I}_d. \tag{4.36}$$

The resulting noise-source power matrix is given by

$$\mathbf{G}_S^{(2)} = \mathbf{\Gamma}^{(2)}\mathbf{G}_S^{(1)}\mathbf{\Gamma}^{(2)\times} = (\mathbf{\Gamma}^{(2)}\mathbf{\Gamma}^{(1)})\,\mathbf{G}_S\,(\mathbf{\Gamma}^{(2)}\mathbf{\Gamma}^{(1)})^{\times}. \tag{4.37}$$

Comparing $\mathbf{G}_S^{(2)}$ with $\mathbf{G}_S^{(1)}$ it is clear that, due to the fact that $\mathbf{\Gamma}^{(2)}$ is diagonal, the transformation $\mathbf{\Phi}^{(2)}$ does not change the correlations of the various noise sources, the powers of these sources being altered only.

For future investigations it is convenient to partition $\mathbf{G}_S^{(2)}$ with respect to the three types of 1-ports. We then have

$$\mathbf{G}_S^{(2)} = \begin{pmatrix} \mathbf{G}_{11}^{(2)} & \mathbf{G}_{12}^{(2)} & \mathbf{G}_{13}^{(2)} \\ \mathbf{G}_{21}^{(2)} & \mathbf{G}_{22}^{(2)} & \mathbf{G}_{23}^{(2)} \\ \mathbf{G}_{31}^{(2)} & \mathbf{G}_{32}^{(2)} & \mathbf{G}_{33}^{(2)} \end{pmatrix}, \tag{4.38}$$

where the subscripts 1, 2, and 3 now refer to the dissipative, active, and reactive 1-ports, respectively.

4.1.4. *Removal of noise-source correlation from the reactive 1-ports*

In this section we will remove the correlations of the noise sources of the reactive 1-ports. This can be done completely by a third canonical transformation.

The mutual correlations of the noise sources of the reactive 1-ports are expressed by the off-diagonal elements of the square submatrix $\mathbf{G}_{33}^{(2)}$. The correlations of these noise sources with those of the dissipative and the active 1-ports are respectively described by the matrices $\mathbf{G}_{13}^{(2)} = \mathbf{G}_{31}^{(2)\times}$ and $\mathbf{G}_{23}^{(2)} = \mathbf{G}_{32}^{(2)\times}$.

The matrix $\mathbf{\Phi}^{(3)}$ of the third canonical transformation is of the form

$$\mathbf{\Phi}^{(3)} = \left(\begin{array}{ccc|ccc} \mathbf{I}_p & \mathbf{O} & \mathbf{A} & \mathbf{O} & \mathbf{O} & -\mathbf{A} \\ \mathbf{O} & \mathbf{I}_q & -\gamma\mathbf{B} & \mathbf{O} & \mathbf{O} & \gamma\mathbf{B} \\ -\mathbf{C}\mathbf{A}^{\times} & \gamma\mathbf{C}\mathbf{B}^{\times} & \mathbf{C} & \mathbf{C}\mathbf{A}^{\times} & -\gamma\mathbf{C}\mathbf{B}^{\times} & \mathbf{O} \\ \hline \mathbf{O} & \mathbf{O} & \mathbf{A} & \mathbf{I}_p & \mathbf{O} & -\mathbf{A} \\ \mathbf{O} & \mathbf{O} & -\gamma\mathbf{B} & \mathbf{O} & \mathbf{I}_q & \gamma\mathbf{B} \\ -\mathbf{C}\mathbf{A}^{\times} & \gamma\mathbf{C}\mathbf{B}^{\times} & \mathbf{O} & \mathbf{C}\mathbf{A}^{\times} & -\gamma\mathbf{C}\mathbf{B}^{\times} & \mathbf{C} \end{array}\right), \tag{4.39}$$

where the scalar factor γ is equal to $(1 + \sqrt{2})$. The $p \times d$-submatrix $\mathbf{A}$, the $q \times d$-submatrix $\mathbf{B}$, and the square submatrix $\mathbf{C}$ of order d must still be specified. We will now show that this specification can always be made such that the correlation of the reactive 1-ports is removed and the diagonal signal performance is not changed.

If the matrix $\mathbf{C}$ is unitary,

$$\mathbf{C}\mathbf{C}^{\times} = \mathbf{C}^{\times}\mathbf{C} = \mathbf{I}_d, \tag{4.40}$$

it is easy to verify that $\mathbf{\Phi}^{(3)}$ satisfies the canonicity condition of eq. (3.178) and that the equation

$$\begin{pmatrix} \mathbf{I}_p & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{I}_q & \mathbf{O} \\ \mathbf{C}\mathbf{A}^{\times} & \mathbf{C}\mathbf{B}^{\times} & \mathbf{C} \end{pmatrix} \begin{pmatrix} \mathbf{I}_p & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{I}_q & \mathbf{O} \\ -\mathbf{A}^{\times} & -\mathbf{B}^{\times} & \mathbf{C}^{\times} \end{pmatrix} = \mathbf{I}_n \tag{4.41}$$

holds for all $\mathbf{A}$ and $\mathbf{B}$. Substitution of eqs (4.31) and (4.39) in eq. (3.17) shows

that the signal-performance matrix $\mathbf{S}^{(2)}$ is then invariant to $\mathbf{\Phi}^{(3)}$. Hence, we have

$$\mathbf{S}^{(3)} = \mathbf{S}^{(2)}. \tag{4.42}$$

By virtue of eqs (3.14), (3.28), and (4.41), we find for the excitation and noise-transformation matrices, respectively,

$$\mathbf{\Lambda}^{(3)} = \begin{pmatrix} \mathbf{I}_p & \mathbf{O} & \mathbf{O} \\ \mathbf{O} & \mathbf{I}_q & \mathbf{O} \\ -\mathbf{A}^{\times} & -\mathbf{B}^{\times} & \mathbf{C}^{\times} \end{pmatrix} \tag{4.43}$$

and

$$\mathbf{\Gamma}^{(3)} = \begin{pmatrix} \mathbf{I}_p & \mathbf{O} & \mathbf{A} \\ \mathbf{O} & \mathbf{I}_q & \mathbf{B} \\ \mathbf{O} & \mathbf{O} & \mathbf{C} \end{pmatrix}. \tag{4.44}$$

Application of $\mathbf{\Phi}^{(3)}$ to the power-absorption matrix and the noise-distribution matrix by means of eqs (3.45) and (3.58) shows that these matrices, too, are invariant to $\mathbf{\Phi}^{(3)}$.

We now have found that $\mathbf{\Phi}^{(3)}$ is canonical and that it does not disturb the diagonality of the signal performance. We must still determine **A**, **B**, and **C** in such a way that the correlations from the noise sources of the reactive 1-ports will be removed.

Inserting eqs (4.44) and (4.37) into eq. (3.53), we obtain for the transformed noise-source power matrix $\mathbf{G}_S^{(3)}$ the partitioned expression (4.45),

see the next page.

Since $\mathbf{G}_S^{(2)}$ is Hermitian and positive (semi-)definite (cf. sec. 3.5.4), the principal submatrix $\mathbf{G}_{33}^{(2)}$ is Hermitian and positive (semi-)definite. Hence, if **C** is chosen to be the unitary modal matrix of $\mathbf{G}_{33}^{(2)}$, eq. (4.40) is satisfied and $\mathbf{G}_{33}^{(2)}$ is transformed to the diagonal form

$$\mathbf{G}_{33}^{(3)} = \mathbf{C}\mathbf{G}_{33}^{(2)}\mathbf{C}^{\times} = \operatorname{diag}(G_i), \quad i = p + q + 1, \ldots, n, \tag{4.46}$$

where the G_i's are the non-negative real eigenvalues of $\mathbf{G}_{33}^{(2)}$. In fact, G_i expresses the power of the noise source of the ith resultant 1-port which is still reactive after the correlation has been removed by the third transformation $\mathbf{\Phi}^{(3)}$.

Before we determine the matrices **A** and **B** explicitly, it is useful to make the following observations. Let the ith diagonal element G_{ii} of a positive semi-definite matrix **G** be equal to zero:

$$G_{ii} = 0. \tag{4.47}$$

The positive semi-definite character of **G** implies among other things that the determinant of the 2nd-order principal submatrix, which is formed by elements

$$\mathbf{G_S}^{(3)} =$$

$$\begin{pmatrix} \mathbf{G_{11}}^{(2)} + \mathbf{AG_{31}}^{(2)} + \mathbf{G_{13}}^{(2)}\mathbf{A}^{\times} + \mathbf{AG_{33}}^{(2)}\mathbf{A}^{\times} & \mathbf{G_{12}}^{(2)} + \mathbf{AG_{32}}^{(2)} + \mathbf{G_{13}}^{(2)}\mathbf{B}^{\times} + \mathbf{AG_{33}}^{(2)}\mathbf{B}^{\times} & \mathbf{G_{13}}^{(2)}\mathbf{C}^{\times} + \mathbf{AG_{33}}^{(2)}\mathbf{C}^{\times} \\ \mathbf{G_{21}}^{(2)} + \mathbf{BG_{31}}^{(2)} + \mathbf{G_{23}}^{(2)}\mathbf{A}^{\times} + \mathbf{BG_{33}}^{(2)}\mathbf{A}^{\times} & \mathbf{G_{22}}^{(2)} + \mathbf{BG_{32}}^{(2)} + \mathbf{G_{23}}^{(2)}\mathbf{B}^{\times} + \mathbf{BG_{33}}^{(2)}\mathbf{B}^{\times} & \mathbf{G_{23}}^{(2)}\mathbf{C}^{\times} + \mathbf{BG_{33}}^{(2)}\mathbf{C}^{\times} \\ \mathbf{CG_{31}}^{(2)} + \mathbf{CG_{33}}^{(2)}\mathbf{A}^{\times} & \mathbf{CG_{32}}^{(2)} + \mathbf{CG_{33}}^{(2)}\mathbf{B}^{\times} & \mathbf{CG_{33}}^{(2)}\mathbf{C}^{\times} \end{pmatrix} \tag{4.45}$$

of the ith and jth columns and rows, is non-negative, i.e.

$$\begin{vmatrix} G_{ii} & G_{ij} \\ G_{ji} & G_{jj} \end{vmatrix} = G_{ii}G_{jj} - G_{ij}G_{ji} \geqslant 0. \tag{4.48}$$

Due to the Hermiticity of $\mathbf{G}$, substitution of eq. (4.47) yields

$$-G_{ij}G_{ij}^{*} = -|G_{ij}|^2 \geqslant 0. \tag{4.49}$$

This inequality can be satisfied only if

$$G_{ij} = G_{ji} = 0 \qquad \text{for all } j. \tag{4.50}$$

All cross-products which involve noise sources with zero power are identically zero. This means physically that the correlation of a noise source with a zero noise source is inherently zero. This is confirmed by the definitions of secs 2.1.4 and 2.1.5.

When $\mathbf{G}_{33}^{(3)}$ has diagonal elements G_i which are zero, the corresponding columns of $\mathbf{C}^{\times}$ are null vectors of $\mathbf{G}_{33}^{(2)}$. The corresponding columns of $\mathbf{G}_{33}^{(2)}\mathbf{C}^{\times}$ are zero vectors and so, therefore, are those of $\mathbf{A}\mathbf{G}_{33}^{(2)}\mathbf{C}^{\times}$. Because, by virtue of what has been observed above, the corresponding columns of $(\mathbf{G}_{13}^{(2)}\mathbf{C}^{\times} + \mathbf{A}\mathbf{G}_{33}^{(2)}\mathbf{C}^{\times})$ are zero vectors, the corresponding columns of $\mathbf{G}_{13}^{(2)}\mathbf{C}^{\times}$ must be zero vectors.

After these observations having been made, we proceed as follows. We wish to determine $\mathbf{A}$ in such a way that the equation

$$\mathbf{G}_{13}^{(3)} = \mathbf{G}_{13}^{(2)}\mathbf{C}^{\times} + \mathbf{A}\mathbf{G}_{33}^{(2)}\mathbf{C}^{\times} = \mathbf{O} \tag{4.51}$$

holds. By virtue of eqs (4.40) and (4.46), this can be written in the form

$$\mathbf{G}_{13}^{(3)} = \mathbf{G}_{13}^{(2)}\mathbf{C}^{\times} + \mathbf{A}\mathbf{C}^{\times}\mathbf{G}_{33}^{(3)} = \mathbf{O}. \tag{4.52}$$

For those columns which correspond to zero diagonal elements of $\mathbf{G}_{33}^{(3)}$ this equation is automatically satisfied, as has been observed above. Corresponding columns of $\mathbf{A}\mathbf{C}^{\times}$ can be chosen arbitrarily. Those columns of $\mathbf{A}\mathbf{C}^{\times}$ which do not correspond to zero diagonal elements of $\mathbf{G}_{33}^{(3)}$ must be taken proportional to corresponding columns of $\mathbf{G}_{13}^{(2)}\mathbf{C}^{\times}$. The reciprocals of the non-zero diagonal elements of $\mathbf{G}_{33}^{(3)}$ are the proportionality constants. Hence, if we define $\mathbf{A}$ by

$$\mathbf{A} = -\mathbf{G}_{13}^{(2)}\mathbf{C}^{\times}\mathbf{D}\mathbf{C} \tag{4.53}$$

with the diagonal matrix $\mathbf{D}$ given by

$$\mathbf{D} = \operatorname{diag}(D_i), \tag{4.54}$$

where

$$D_i = \begin{cases} G_i^{-1}, & \text{for } G_i \neq 0, \\ \text{arbitrary scalar}, & \text{for } G_i = 0, \end{cases} \tag{4.55}$$

eq. (4.51) is satisfied.

The matrix **B** can be defined in the same way by

$$\mathbf{B} = -\mathbf{G}_{23}{}^{(2)}\mathbf{C}^{\times}\mathbf{D}\mathbf{C}. \tag{4.56}$$

The arbitrary elements of **D** need not be the same as for the definition of **A**. We obtain

$$\mathbf{G}_{23}{}^{(3)} = \mathbf{G}_{23}{}^{(2)}\mathbf{C}^{\times} + \mathbf{B}\mathbf{G}_{33}{}^{(2)}\mathbf{C}^{\times} = \mathbf{O}. \tag{4.57}$$

The number of zero elements of $\mathbf{G}_{33}{}^{(3)}$ is denoted by d_r. It is called the defect of the noise of the reactive part, i.e. of the noise of the reactive 1-ports.

Substituting eqs (4.46), (4.51), and (4.57) in eq. (4.45), we obtain for the resultant noise-source power matrix $\mathbf{G}_S{}^{(3)}$ the form in demand, i.e.

$$\mathbf{G}_S{}^{(3)} = \begin{pmatrix} \mathbf{G}_{11}{}^{(2)} - \mathbf{A}\mathbf{G}_{33}{}^{(2)}\mathbf{A}^{\times} & \mathbf{G}_{12}{}^{(2)} - \mathbf{A}\mathbf{G}_{33}{}^{(2)}\mathbf{B}^{\times} & \mathbf{O} \\ \mathbf{G}_{21}{}^{(2)} - \mathbf{B}\mathbf{G}_{33}{}^{(2)}\mathbf{A}^{\times} & \mathbf{G}_{22}{}^{(2)} - \mathbf{B}\mathbf{G}_{33}{}^{(2)}\mathbf{B}^{\times} & \mathbf{O} \\ \mathbf{O} & \mathbf{O} & \mathbf{G}_{33}{}^{(3)} \end{pmatrix}. \tag{4.58}$$

This result shows that the choice of **A**, **B**, and **C** made above indeed removes all correlations from the noise sources of the reactive 1-ports. Physically, this is accomplished by an imbedding $2n$-port which, while diagonalizing the noise of the reactive part of the multiport, subtracts appropriate portions of that noise from the noise of the active and dissipative parts. It has also been shown above that the proper choice of **A**, **B**, and **C** can always be made. This is so because the addition of a reactive element to a dissipative 1-port or to an active 1-port does not alter the dissipative or active character.

4.1.5. *Noise of active-dissipative, diagonal n-ports*

The dissolution of the correlation of the noise of the reactive part is unilateral and rather simple. Since a combination of a dissipative and an active element may be either dissipative or active, the suppression of mutual correlation of noise of active parts and dissipative parts is however not equally simple. As will be shown below, in some borderline cases it even is not possible.

We can restrict further investigations to n-ports with non-singular signal-performance matrices. Because the reactive part of an n-port has been dealt with exhaustively above we further need consider only those n-ports which include only activity and dissipativity.

If the reactive 1-ports are omitted, the noise-source power matrix of the diagonalized n-port can be written in the form

$$\mathbf{G}_S{}^{(3)} = \begin{pmatrix} \mathbf{G}_{11}{}^{(3)} & \mathbf{G}_{12}{}^{(3)} \\ \mathbf{G}_{21}{}^{(3)} & \mathbf{G}_{22}{}^{(3)} \end{pmatrix}. \tag{4.59}$$

Due to the positive semi-definite character of $\mathbf{G}_S{}^{(3)}$, the principal submatrices $\mathbf{G}_{11}{}^{(3)}$ and $\mathbf{G}_{22}{}^{(3)}$ are positive semi-definite. Hence, unitary matrices $\mathbf{\Gamma}_{11}{}^{(4)}$ and $\mathbf{\Gamma}_{22}{}^{(4)}$ can be found such that the matrices

$$\mathbf{G}_{11}^{(4)} = \mathbf{\Gamma}_{11}^{(4)}\mathbf{G}_{11}^{(3)}\mathbf{\Gamma}_{11}^{(4)\times} \tag{4.60}$$

and

$$\mathbf{G}_{22}^{(4)} = \mathbf{\Gamma}_{22}^{(4)}\mathbf{G}_{22}^{(3)}\mathbf{\Gamma}_{22}^{(4)\times} \tag{4.61}$$

are diagonal with the non-negative eigenvalues of $\mathbf{G}_{11}^{(3)}$ and $\mathbf{G}_{22}^{(3)}$ as the respective diagonal elements.

Let us now define a fourth transformation $\mathbf{\Phi}^{(4)}$ by

$$\mathbf{\Phi}^{(4)} = \left(\begin{array}{cc|cc} \mathbf{\Gamma}_{11}^{(4)} & & & \\ & \mathbf{\Gamma}_{22}^{(4)} & & \\ \hline & & \mathbf{\Gamma}_{11}^{(4)} & \\ & & & \mathbf{\Gamma}_{22}^{(4)} \end{array}\right). \tag{4.62}$$

This transformation satisfies the canonicity condition and does not alter the diagonal form of the signal performance. Moreover, it removes the internal correlations within the dissipative part and within the active part of the multiport.

Let d_d and d_a denote the defects of the submatrices $\mathbf{G}_{11}^{(3)}$ and $\mathbf{G}_{22}^{(3)}$. These numbers will be called the defects of the noise of the dissipative part and of the active part, respectively. They refer to the numbers of zero diagonal elements of $\mathbf{G}_{11}^{(4)}$ and $\mathbf{G}_{22}^{(4)}$, respectively, and, hence, to the numbers of noiseless dissipative and noiseless active 1-ports of the resultant diagonal n-port $\mathbf{S}^{(4)}$. By virtue of what has been observed in the previous section, the corresponding rows and columns of $\mathbf{G}_{12}^{(4)}$ and $\mathbf{G}_{21}^{(4)}$ are identically zero. Again, the corresponding 1-ports can be omitted from further considerations. We are thus left with a reduced, diagonal, and non-reactive multiport. It is of order $(n - d + d_d - d_a)$. All its 1-ports are noisy, while correlations are present only between noise sources of active and dissipative 1-ports. Neither noise sources of two active 1-ports, nor noise sources of two dissipative 1-ports are correlated any more. The reduced matrices are given by

$$\mathbf{S}^{(4)} = \begin{pmatrix} \mathbf{O}_{p'} & \\ & \sqrt{2}\,\mathbf{I}_{q'} \end{pmatrix}, \tag{4.63}$$

$$\mathbf{N}_S^{(4)} = \mathbf{P}_S^{(4)} = \begin{pmatrix} \mathbf{I}_{p'} & \\ & -\mathbf{I}_{q'} \end{pmatrix} \tag{4.64}$$

and

$$\mathbf{G}_S^{(4)} = \begin{pmatrix} \mathbf{G}_{11}^{(4)} & \mathbf{G}_{12}^{(4)} \\ \mathbf{G}_{21}^{(4)} & \mathbf{G}_{22}^{(4)} \end{pmatrix}. \tag{4.65}$$

The square submatrices $\mathbf{G}_{11}^{(4)}$ and $\mathbf{G}_{22}^{(4)}$ are non-singular, positive, and diagonal. Their orders p' and q' are respectively given by

$$p' = p - d_d \tag{4.66}$$

and

$$q' = q - d_a. \tag{4.67}$$

In sec. 3.4.3 the defect d_G of the original noise-source power matrix $\mathbf{G}_S$ has been called the noise defect. This number is invariant to transformations. The sum of the noise defects of the different parts of the multiport will be called the normal noise defect. It will be denoted by $d_{G,N}$ and is given by

$$d_{G,N} = d_r + d_d + d_a. \tag{4.68}$$

We can now distinguish between two different cases. They are respectively discriminated by

$$d_{G,N} = d_G \tag{4.69}$$

and

$$d_{G,N} < d_G. \tag{4.70}$$

These relations refer to n-ports which have respectively purely normal noise defects and normal plus anomalous noise defects. The anomaly of the noise defect can be defined by

$$d_{G,A} = d_G - d_{G,N}. \tag{4.71}$$

The number $d_{G,A}$ will be called the anomalous noise defect.

4.1.6. *Normal noise defect*

The first of the above two cases is simple. If eq. (4.69) applies, it is easy to show that $\mathbf{G}_S^{(4)}$ in its reduced form of eq. (4.65) is non-singular and, hence, positive definite. It is not then difficult to perform the further diagonalization of the noise.

In matrix algebra it is shown [40]) that two Hermitian matrices, one of which is positive definite, can be transformed to diagonal forms simultaneously by the same Hermitian congruence transformation. Solving the general eigenvalue problem of the two matrices, we find the appropriate transformation matrix as follows. The matrices $\mathbf{N}_S^{(4)}$ and $\mathbf{G}_S^{(4)}$ are the two matrices that are to be diagonalized simultaneously. In the present case $\mathbf{N}_S^{(4)}$ is diagonal already, but that fact does not alter the considerations. The vectors $\mathbf{v}_i$ normalized with respect to $\mathbf{N}_S^{(4)}$ and satisfying the general eigenvalue problem *) [40])

$$(\mathbf{G}_S^{(4)} - \lambda_i \mathbf{N}_S^{(4)})\, \mathbf{v}_i = \mathbf{o}, \tag{4.72}$$

*) Equation (4.72) is stated in a slightly unconventional form. Usually, the positive definite matrix is taken to be the second term in the left-hand side. We do not follow this practice here because we want to have the ultimately resulting noise-distribution matrix in a normalized form (cf. sec. 4.1.3). Furthermore, it will be seen that the eigenvalues of the general problem as stated in eq. (4.72) are equal to the characteristic noise temperatures (multiplied by a factor k). This is thought to be a convenient result. The small alteration in the formulation of the eigenvalue problem affects neither the validity nor the generality of the results.

constitute the columns of the matrix $\mathbf{\Gamma}^{(5)}$ in demand. The numbers λ_i are the eigenvalues pertaining to the $\mathbf{v}_i$'s.

Interpreting $\mathbf{\Gamma}^{(5)}$ as the noise-transformation matrix of a fifth canonical transformation, we find

$$\mathbf{N}_S^{(5)} = \mathbf{\Gamma}^{(5)}\mathbf{N}_S^{(4)}\mathbf{\Gamma}^{(5)\times} = \begin{pmatrix} \mathbf{I}_{p'} & \\ & -\mathbf{I}_{q'} \end{pmatrix} \tag{4.73}$$

and

$$\mathbf{G}_S^{(5)} = \mathbf{\Gamma}^{(5)}\mathbf{G}_S^{(4)}\mathbf{\Gamma}^{(5)\times} = \operatorname{diag}(|\lambda_i|). \tag{4.74}$$

If $\mathbf{\Gamma}^{(5)}$ is partitioned into blocks of order p' and q', i.e.

$$\mathbf{\Gamma}^{(5)} = \begin{pmatrix} \mathbf{\Gamma}_{11}^{(5)} & \mathbf{\Gamma}_{12}^{(5)} \\ \mathbf{\Gamma}_{21}^{(5)} & \mathbf{\Gamma}_{22}^{(5)} \end{pmatrix}, \tag{4.75}$$

substitution into eq. (4.73) shows that the submatrices satisfy

$$\mathbf{\Gamma}_{11}^{(5)}\mathbf{\Gamma}_{11}^{(5)\times} - \mathbf{\Gamma}_{12}^{(5)}\mathbf{\Gamma}_{12}^{(5)\times} = \mathbf{I}_{p'}, \tag{4.76}$$

$$\mathbf{\Gamma}_{22}^{(5)}\mathbf{\Gamma}_{22}^{(5)\times} - \mathbf{\Gamma}_{21}^{(5)}\mathbf{\Gamma}_{21}^{(5)\times} = \mathbf{I}_{q'} \tag{4.77}$$

and

$$\mathbf{\Gamma}_{11}^{(5)}\mathbf{\Gamma}_{21}^{(5)\times} = \mathbf{\Gamma}_{12}^{(5)}\mathbf{\Gamma}_{22}^{(5)\times}. \tag{4.78}$$

The matrix $\mathbf{\Phi}^{(5)}$ of the transformation can now be defined by

$$\mathbf{\Phi}^{(5)} = \left(\begin{array}{cc|cc} \mathbf{\Gamma}_{11}^{(5)} & \mathbf{\Gamma}_{12}^{(5)} & \mathbf{O}_{p'} & -\sqrt{2}\,\mathbf{\Gamma}_{12}^{(5)} \\ -\mathbf{\Gamma}_{21}^{(5)} & \mathbf{\Gamma}_{22}^{(5)} & \sqrt{2}\,\mathbf{\Gamma}_{21}^{(5)} & \mathbf{O}_{q'} \\ \hline \mathbf{O}_{p'} & \sqrt{2}\,\mathbf{\Gamma}_{12}^{(5)} & \mathbf{\Gamma}_{11}^{(5)} & -\mathbf{\Gamma}_{12}^{(5)} \\ -\sqrt{2}\,\mathbf{\Gamma}_{21}^{(5)} & \mathbf{O}_{q'} & \mathbf{\Gamma}_{21}^{(5)} & \mathbf{\Gamma}_{22}^{(5)} \end{array}\right). \tag{4.79}$$

By virtue of the above equations, insertion of $\mathbf{\Phi}^{(5)}$ into eq. (3.178) proves the canonicity of the transformation.

From eq. (3.17) we obtain for the resulting signal-performance matrix $\mathbf{S}^{(5)}$ the unaltered form

$$\mathbf{S}^{(5)} = \begin{pmatrix} \mathbf{O}_{p'} & \\ & \sqrt{2}\,\mathbf{I}_{q'} \end{pmatrix}. \tag{4.80}$$

By means of eq. (3.14) the excitation-transformation matrix $\mathbf{Y}^{(5)}$ is found to be given by

$$\mathbf{Y}^{(5)} = \begin{pmatrix} \mathbf{\Gamma}_{11}^{(5)} & \mathbf{\Gamma}_{12}^{(5)} \\ \mathbf{\Gamma}_{21}^{(5)} & \mathbf{\Gamma}_{22}^{(5)} \end{pmatrix} = \mathbf{\Gamma}^{(5)}. \tag{4.81}$$

This matrix is identical to the noise-transformation matrix. It is easy to show that the inverses of these matrices are given by

$$\mathbf{\Lambda}^{(5)} = \mathbf{\Delta}^{(5)} = \begin{pmatrix} \mathbf{\Gamma}_{11}^{(5)\times} & -\mathbf{\Gamma}_{21}^{(5)\times} \\ -\mathbf{\Gamma}_{12}^{(5)\times} & \mathbf{\Gamma}_{22}^{(5)\times} \end{pmatrix}. \tag{4.82}$$

The resulting power-absorption matrix $\mathbf{P}_S^{(5)}$ is now readily evaluated, i.e.

$$\mathbf{P}_S^{(5)} = \mathbf{\Lambda}^{(5)\times}\mathbf{P}_S^{(4)}\mathbf{\Lambda}^{(5)} = \begin{pmatrix} \mathbf{I}_{p'} & \\ & -\mathbf{I}_{q'} \end{pmatrix}. \tag{4.83}$$

All these results show that the canonical transformation defined by eq. (4.79) satisfies all required conditions.

We have now arrived at the point where the question of the simultaneous diagonalization of both the signal performance and the noise of an n-port in paired representations, as has been posed in sec. 4.1.1, can be answered in the affirmative. The canonical transformation $\mathbf{\Phi}_D$, which is the product of the five canonical transformations introduced in this and the preceding sections and the matrix of which is given by

$$\mathbf{\Phi}_D = \mathbf{\Phi}^{(5)}\mathbf{\Phi}^{(4)}\mathbf{\Phi}^{(3)}\mathbf{\Phi}^{(2)}\mathbf{\Phi}^{(1)}, \tag{4.84}$$

where the subscript D stands for diagonalizing, has the necessary and sufficient properties to do the job. Thus, the Haus and Adler theorem [15]) is valid. We restate it here in a form which includes reactive 1-ports:

"At any particular frequency, every linear n-port can be reduced by lossless imbedding into a diagonal form consisting of n separate 1-ports which are dissipative, reactive, or active, and which possess mutually uncorrelated noise-power sources."

Thus far the considerations have, however, been restricted to n-ports with normal noise defects. In the next section it will be shown that, if the noise defect contains an anomalous part, the theorem is not valid in its strict form of all noise-power sources being mutually uncorrelated.

4.1.7. *Anomalous noise defect*

If the anomalous noise defect $d_{G,A}$ of an n-port is not equal to zero, i.e.

$$d_{G,A} \neq 0, \tag{4.85}$$

the reduced noise-source power matrix $\mathbf{G}_S^{(4)}$, as given by eq. (4.65), is singular. That matrix is then not positive definite, but positive semi-definite. In that case a fifth canonical transformation $\mathbf{\Phi}^{(5)}$ cannot be derived by solving the general eigenvalue problem of the matrices $\mathbf{G}_S^{(4)}$ and $\mathbf{N}_S^{(4)}$. That eigenvalue problem cannot then be solved because the matrix pair $\mathbf{G}_S^{(4)}$ and $\mathbf{N}_S^{(4)}$ does not have n linearly independent eigenvectors.

In the case of anomalous noise defect we must look for an alternative

derivation of the noise-transformation matrix $\mathbf{\Gamma}^{(5)}$ *). That derivation is given in appendix E. The matrix $\mathbf{\Gamma}^{(5)}$ derived there carries $\mathbf{G}_S^{(4)}$ into an elementary form, which is not now necessarily diagonal, while it leaves the noise-distribution matrix $\mathbf{N}_S^{(4)}$ unaltered. The result of this noise transformation is expressed by

$$\mathbf{N}_S^{(5)} = \mathbf{\Gamma}^{(5)}\mathbf{N}_S^{(4)}\mathbf{\Gamma}^{(5)\times} = \mathbf{N}_S^{(4)} = \begin{pmatrix} \mathbf{I}_{r_d} & & \\ & \boxed{\begin{matrix} \mathbf{I}_{d_{G,A}} & \\ & -\mathbf{I}_{d_{G,A}} \end{matrix}} & \\ & & -\mathbf{I}_{r_a} \end{pmatrix} \tag{4.86}$$

and

$$\mathbf{G}_S^{(5)} = \mathbf{\Gamma}^{(5)}\mathbf{G}_S^{(4)}\mathbf{\Gamma}^{(5)\times} = \begin{pmatrix} \mathbf{G}_{r_d} & & \\ & \boxed{\begin{matrix} \mathbf{I}_{d_{G,A}} & \mathbf{I}_{d_{G,A}} \\ \mathbf{I}_{d_{G,A}} & \mathbf{I}_{d_{G,A}} \end{matrix}} & \\ & & \mathbf{G}_{r_a} \end{pmatrix}. \tag{4.87}$$

The principal submatrices $\mathbf{G}_{r_d}$ and $\mathbf{G}_{r_a}$ are positive and diagonal. Their orders are respectively given by

$$r_d = p' - d_{G,A} = p - d_d - d_{G,A} \tag{4.88}$$

and

$$r_a = q' - d_{G,A} = q - d_a - d_{G,A}. \tag{4.89}$$

The noise-transformation matrix $\mathbf{\Gamma}^{(5)}$ can again be written in the partitioned form (cf. eq. (4.75))

$$\mathbf{\Gamma}^{(5)} = \begin{pmatrix} \mathbf{\Gamma}_{11}^{(5)} & \mathbf{\Gamma}_{12}^{(5)} \\ \mathbf{\Gamma}_{21}^{(5)} & \mathbf{\Gamma}_{22}^{(5)} \end{pmatrix}, \tag{4.90}$$

where, again, the partitioning relates to the dissipative and the active parts. By virtue of eq. (4.64), substitution in eq. (4.86) shows that the submatrices satisfy eqs (4.76), (4.77), and (4.78). Therefore, also in this case the canonical transformation looked for is defined by eq. (4.79). In the previous section it has been shown that this transformation does not alter the diagonalized form of the signal performance. Moreover, it transforms the noise into an elementary form which, however, in the case of anomalous noise defect is not diagonal.

The framed part of eq. (4.87) cannot be diagonalized under invariance of the framed part of eq. (4.86). This can explicitly be shown by the example of one active and one dissipative 1-port which have mutually completely correlated noise sources. In fact, the framed parts of eqs (4.86) and (4.87) refer to a

*) The matrix $\mathbf{\Gamma}^{(5)}$ derived in appendix E and discussed in the present section replaces that matrix derived and discussed in the previous section.
In fact, when the noise defect is normal, both derivations yield identical results. The derivation of the previous section is then much the simpler.

collection of $d_{G,A}$ pairs of such completely correlated active and dissipative 1-ports.

The restriction under which the Haus and Adler theorem is valid can now be formulated as follows:

> "The theorem of the simultaneous diagonalization of the signal performance and the noise is valid only if the noise of an n-port does not contain complete correlations between the active and the dissipative parts, i.e. only if the anomalous noise defect is zero."

If this condition is not fulfilled, the most elementary form obtainable for $\mathbf{G}_S$, while maintaining a diagonal signal performance, is expressed by the right-hand side of eq. (4.87).

Below it will be shown that the "amplitude" of the framed part in the elementary form of $\mathbf{G}_S$ is arbitrary although it must be finite and unequal zero. This complication can be understood from the fact that a dissipative and an active 1-port can be combined to form a reactive 1-port *). Because the noise sources of the two 1-ports are mutually completely correlated, the noise source of the combination is completely correlated with both of them. Hence, by combining the two 1-ports with such a combination the two original noise sources can be given any non-zero finite power. Under this operation the characters of the two 1-ports and the complete correlation of their noise sources are maintained.

4.1.8. *The general diagonalization transformation*

Now we have arrived at the point where it is possible to define the general diagonalizing canonical transformation and the subsidiary transformation matrices. This transformation is defined as the product of the non-reduced versions of the particular canonical transformations formulated in the preceding sections. Thus, when the scattering-matrix coordinate system in which a given n-port $\mathbf{S}$ is represented is transformed to another scattering-matrix coordinate system by means of the canonical transformation $\mathbf{\Phi}_D$, defined by eq. (4.84), the resulting scattering-matrix representation yields the most elementary forms that pertain to that particular n-port. This means that the canonical transformation $\mathbf{\Phi}_D$ transforms the signal performance to a normalized diagonal form while the noise-source power matrix attains its most elementary form. The matrix $\mathbf{\Phi}_D$ itself can be interpreted as the transfer matrix $\mathbf{\Theta}_D$ of the diagonalizing lossless imbedding $2n$-port.

It is not difficult to show that the signal- and noise-transformation matrices $\mathbf{\Lambda}_D$ and $\mathbf{\Gamma}_D$ which pertain to $\mathbf{\Phi}_D$ are given by the products of the corresponding matrices pertaining to the particular transformations introduced in the preceding sections, i.e.

*) The combination of the two 1-ports is highly unstable, at least over a small band of frequencies. The synthesized reactive 1-port may oscillate internally, so that its signal performance and even its noise performance is degraded.

$$\mathbf{\Lambda}_D = \mathbf{\Lambda}^{(5)}\,\mathbf{\Lambda}^{(4)}\,\mathbf{\Lambda}^{(3)}\,\mathbf{\Lambda}^{(2)}\,\mathbf{\Lambda}^{(1)} \tag{4.91}$$

and

$$\mathbf{\Gamma}_D = \mathbf{\Gamma}^{(5)}\,\mathbf{\Gamma}^{(4)}\,\mathbf{\Gamma}^{(3)}\,\mathbf{\Gamma}^{(2)}\,\mathbf{\Gamma}^{(1)}\,. \tag{4.92}$$

The matrices $\mathbf{S}_D$, $\mathbf{P}_{\mathrm{S},D}$ and $\mathbf{G}_{\mathrm{S},D}$ characterizing the resulting diagonalized n-port are given by the expressions

$$\mathbf{S}_D = \mathbf{R} = \begin{pmatrix} \mathbf{O}_{r_d} & & & & & & & \\ & \mathbf{O}_{d_d} & & & & & & \\ & & \mathbf{O}_{d_{G,A}} & \sqrt{2}\mathbf{I}_{d_{G,A}} & & & & \\ & & & & \sqrt{2}\mathbf{I}_{r_a} & & & \\ & & & & & \sqrt{2}\mathbf{I}_{d_a} & & \\ & & & & & & \mathbf{I}_{r_r} & \\ & & & & & & & \mathbf{I}_{d_r} \end{pmatrix}, \tag{4.93}$$

$$\mathbf{P}_{\mathrm{S},D} = \mathbf{N}_{\mathrm{S},D} = \begin{pmatrix} \mathbf{I}_{r_d} & & & & & & & \\ & \mathbf{I}_{d_d} & & & & & & \\ & & \mathbf{I}_{d_{G,A}} & -\mathbf{I}_{d_{G,A}} & & & & \\ & & & & -\mathbf{I}_{r_a} & & & \\ & & & & & -\mathbf{I}_{d_a} & & \\ & & & & & & \mathbf{O}_{r_r} & \\ & & & & & & & \mathbf{O}_{d_r} \end{pmatrix}, \tag{4.94}$$

$$\mathbf{G}_{\mathrm{S},D} = \begin{pmatrix} \mathbf{G}_{r_d} & & & & & & & \\ & \mathbf{O}_{d_d} & & & & & & \\ & & \mathbf{I}_{d_{G,A}} & \mathbf{I}_{d_{G,A}} & & & & \\ & & \mathbf{I}_{d_{G,A}} & \mathbf{I}_{d_{G,A}} & & & & \\ & & & & \mathbf{G}_{r_a} & & & \\ & & & & & \mathbf{O}_{d_a} & & \\ & & & & & & \mathbf{G}_{r_r} & \\ & & & & & & & \mathbf{O}_{d_r} \end{pmatrix}. \tag{4.95}$$

In the right-hand side of the last equation the matrices $\mathbf{G}_{r_d}$, $\mathbf{G}_{r_a}$, and $\mathbf{G}_{r_r}$ are positive and diagonal.

Equations (4.93) and (4.94) contain information about the different characters of the signal properties of the elementary 1-ports of an n-port, about the number of times these different types of 1-ports occur, and about the way the noise is distributed over them. Equation (4.95), moreover, contains additional information about the intensity and, as far as anomalous noise defect is concerned, about the unremovable correlation of the noise. Table I presents a summary of this information.

TABLE I

reflection coeffi-cient ϱ_i	power-absorp-tion coeffi-cient $P_{ii} = N_{ii}$	nature of signal perform-ance	number of 1-ports		power of elementary noise source G_{ii}	character-istic noise tempera-ture T_i	noisiness of 1-ports
0	1	dissipa-tive	p	r_d	>0	>0	noisy
				d_d	0	0	noiseless
				$d_{G,A}$	>0	$\to 0$	completely correlated noisy pair
$\sqrt{2}$	-1	active	q	r_a	>0	<0	noisy
				d_a	0	0	noiseless
1	0	reactive	d	r_r	>0	∞	noisy
				d_r	0	?	noiseless

TABLE II

number of 1-ports	mathematical definition	character of the 1-ports	classification of the n-port
p	index of $\mathbf{P_S}$ and $\mathbf{N_S}$ (number of pos. eigenvalues)	dissipative	dissipativity
q	number of negative eigenvalues of $\mathbf{P_S}$ and $\mathbf{N_S}$	active	activity
d	defect of $\mathbf{P_S}$ and $\mathbf{N_S}$ (number of zero eigenvalues)	reactive	reactivity

Table I comprises also information about the characteristic noise temperatures. This will be discussed in the next section.

The left-hand side of the column which lists the numbers of 1-ports contains the numbers p, q, and d which classify the signal performance of the n-port. Table II presents a summary of the mathematical definitions of these numbers together with the characters of the corresponding elementary 1-ports and the classification of the n-port (cf. sec. 3.4).

The right-hand side of the above-mentioned column of table I contains the numbers of various types of 1-ports which are noisy or noiseless. The mathematical definitions of these numbers and the characters of the corresponding elementary 1-ports are given in table III.

TABLE III

number of 1-ports	mathematical definition		character of 1-ports
r_d	rank of dissipative part		noisy dissipative
d_d	defect of dissipative part		noiseless dissipative
r_a	rank of active part		noisy active
d_a	defect of active part	of $\mathbf{G}_S$	noiseless active
r_r	rank of reactive part		noisy reactive
d_r	defect of reactive part		noiseless reactive
$d_{G,A}$	anomalous defect		completely correlated pair of one noisy active and one noisy dissipative 1-port

4.2. The temperature matrix

4.2.1. *The diagonal temperature matrix*

In the preceding section we have derived the matrix of the canonical transformation which puts the n-port in its most elementary form. Moreover, we have determined the corresponding signal- and noise-transformation matrices $\mathbf{\Lambda}_D$ and $\mathbf{\Gamma}_D$, respectively. Since $\mathbf{\Gamma}_D$ is a non-singular matrix, the noise-temperature matrix of the diagonalized n-port can in principle now be determined by means of eq. (3.66), i.e.

$$\mathbf{T}_{S,D} = \mathbf{\Gamma}_D \mathbf{T}_S \mathbf{\Gamma}_D^{-1}. \tag{4.96}$$

This matrix $\mathbf{T}_{S,D}$ will be called the diagonal temperature matrix of the n-port *).

*) It should be noted that in the particular case of anomalous noise defect the matrix $\mathbf{T}_{S,D}$ itself is not diagonal.

For the determination of $\mathbf{T}_{S,D}$ one can also advantageously make a direct use of the definition of temperature matrix as contained in eqs (3.62) and (3.65). This definition remains valid under transformations. Hence, by virtue of eqs (4.94) and (4.95), for $\mathbf{T}_{S,D}$ one can write down the expression

$$\mathbf{T}_{S,D} = k^{-1} \begin{pmatrix} \mathbf{G}_{r_d} & & & \\ & \mathbf{O}_{d_d} & & \\ & & \boxed{\begin{matrix} \mathbf{I}_{d_{G,A}} & -\mathbf{I}_{d_{G,A}} \\ \mathbf{I}_{d_{G,A}} & -\mathbf{I}_{d_{G,A}} \end{matrix}} & \\ & & & -\mathbf{G}_{r_a} \\ & & \mathbf{O}_{d_a} & \\ & \boxed{\infty} & & \\ \boxed{?} & & & \end{pmatrix}, \tag{4.97}$$

where the framed parts still need explanation. Apart from these framed parts, which will be discussed in following sections, $\mathbf{T}_{S,D}$ is the diagonal matrix of the eigenvalues of $\mathbf{T}_S$, i.e. of the characteristic noise temperatures of the n-port. Thus, if for the time being we leave the reactive part of the n-port and the part which has anomalous noise defect out of discussion, we have

$$\mathbf{T}_{S,D} = \text{diag}\,(T_i). \tag{4.98}$$

More precisely, the diagonal elements of $\mathbf{T}_{S,D}$ are the ratios of the corresponding diagonal elements of $\mathbf{G}_{S,D}$ and $\mathbf{N}_{S,D}$, divided by Boltzmann's constant k, i.e.

$$T_i = G_{ii}/kN_{ii} = \text{sign}(N_{ii})\, k^{-1} G_{ii}. \tag{4.99}$$

Since G_{ii} is non-negative, the sign of T_i is determined by that of N_{ii}. Hence, dissipativity is related to positive, activity to negative characteristic noise temperatures (cf. sec. 2.3.1). Moreover, it will be clear that characteristic noise temperatures can be interpreted as the equivalent noise temperatures of the corresponding 1-ports of the diagonalized n-port. As mentioned earlier (sec. 2.7.2), the characteristic noise temperatures determine the intensities of the elementary noise sources. Hence, as a set, they form a measure of the intensity of the noise of a multiport.

Concluding, we can state that the concepts of temperature matrix and characteristic noise temperatures, together with the noise-distribution matrix and its diagonal version, are quite adequate to describe the noise of a multiport. This is true anyway in so far as noisy active and noisy dissipative parts, which are mutually completely correlated, and reactive parts are not considered. In fact, most actual multiports can be split into uncorrelated dissipative and active parts only.

4.2.2. *The reactive part of a multiport*

When a multiport contains a reactive part, its noise-distribution matrix $\mathbf{N}_S$ is singular. Then, therefore, the defining relation of eqs (2.272) and (3.65) cannot be used to determine the temperature matrix. Since, furthermore, the relevant elements of $\mathbf{N}_{S,D}$ are then zero, it is equally impossible to resort to eq. (4.99) to calculate the characteristic noise temperatures pertaining to the reactive 1-ports. If the latter method could be exploited, from the characteristic noise temperatures then obtainable by it, the original temperature matrix could be determined by means of the inverse of eq. (4.96). We can conclude that, strictly spoken, the concept of temperature matrix does not apply to multiports which have reactive parts. However, to circumvent this failure we assume that, then, eq. (4.99) is still formally applicable.

Two cases can be distinguished. The reactive 1-ports of a diagonalized n-port are either noisy or noiseless.

The elements of $\mathbf{G}_{S,D}$ relevant to noiseless reactive 1-ports are zero, so that, due to eq. (4.99) the corresponding T_i are indeterminate. This means that the characteristic noise temperatures of noiseless reactive 1-ports may be given arbitrary values. In the corresponding framed part of eq. (4.97) this character of the T_i's has been indicated by an interrogation mark. Indeed, the physical temperature of a normal reactive network component, whatever it may be, is not relevant to its noise performance. In fact, a capacitor, an inductor, a short circuit, etc. does not radiate noise power.

If the concept of temperature matrix is also to be maintained for reactive 1-ports which do radiate noise power, i.e. for which $G_{ii} \neq 0$, the corresponding characteristic noise temperatures T_i must be thought to be infinitely high. In eq. (4.97) this has been indicated by the infinity symbol. At low frequencies a saturated diode, such as used as a noise source for measuring purposes, is an example of a noisy reactive network component. Thus, in so far as we are concerned with noise performance, we can ascribe to such a component an infinitely high temperature.

4.2.3. *Dependent active and dissipative parts of a multiport*

The part of the diagonal temperature matrix $\mathbf{T}_{S,D}$ that relates to those active and dissipative 1-ports of the diagonalized n-port which are mutually completely correlated, can be rearranged into non-zero diagonal blocks of order two and of the form

$$\mathbf{T}_{jj} = k^{-1} \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}. \tag{4.100}$$

It is readily shown that these principal submatrices cannot be diagonalized. This corresponds to the fact that, in this case, the corresponding parts of $\mathbf{N}_S$ and $\mathbf{G}_S$ cannot be diagonalized simultaneously (cf. sec. 4.1.7).

The rearrangement into second-order blocks means physically that the 1-ports involved are regrouped into pairs of one active and one dissipative 1-port which have completely correlated noise. Between the noise of two 1-ports of different pairs there is then no correlation. Hence, we can restrict further considerations to one such a pair.

The noise-distribution matrix, the noise-source power matrix, and the noise-temperature matrix of a pair are respectively given by

$$\mathbf{N}_{P,D} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{4.101}$$

$$\mathbf{G}_{P,D} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \tag{4.102}$$

and

$$\mathbf{T}_{P,D} = k^{-1} \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}. \tag{4.103}$$

The two roots of the characteristic equation of $\mathbf{T}_{P,D}$ are both equal to zero. This might suggest that the characteristic noise temperature of a pair could be defined as equal to zero.

As has been mentioned already in sec. 4.1.7, it has been found in appendix E that the imbedding multiport (a 4-port in the present case), which removes the mutual correlation of the active and dissipative parts as far as possible, can be chosen in several different ways.

It can be done such that the elements of that part of the diagonal noise-source power matrix, and hence of the diagonal temperature matrix, which pertains to the anomalous noise defect, are multiplied by an arbitrary positive, but finite, and non-zero scalar number, while all other quantities remain unaltered.

Starting from the result presented in the above equations the same arbitrary scalar multiplication can also be accomplished by a further, a sixth, canonical transformation. Let α be an arbitrary, finite, and non-zero complex number. Let further $\mathbf{\Phi}^{(\alpha)}$ be the matrix of this sixth canonical transformation. The matrix $\mathbf{\Phi}^{(\alpha)}$ can be interpreted as the transfer matrix of a lossless 4-port, which imbeds the pair of mutually dependent 1-ports. We can now define the sixth transformation by

$$\mathbf{\Phi}^{(\alpha)} = \left(\begin{array}{cc|cc} \sigma & \tau & 0 & -\sqrt{2\tau} \\ -\tau & \sigma & \sqrt{2\tau} & 0 \\ \hline 0 & \sqrt{2\tau} & \sigma & -\tau \\ -\sqrt{2\tau} & 0 & \tau & \sigma \end{array}\right), \tag{4.104}$$

where

$$\sigma = \tfrac{1}{2}(\alpha + 1/\alpha^*) \tag{4.105}$$

and

$$\tau = \tfrac{1}{2}(\alpha - 1/\alpha^*). \tag{4.106}$$

It is easy to show that this transformation satisfies the canonicity requirement. The corresponding noise-transformation matrix $\mathbf{\Gamma}^{(\alpha)}$ is given by

$$\mathbf{\Gamma}^{(\alpha)} = \begin{pmatrix} \sigma & \tau \\ \tau & \sigma \end{pmatrix}. \tag{4.107}$$

Furthermore, it is readily shown that the signal performance is invariant to $\mathbf{\Phi}^{(\alpha)}$ and that $\mathbf{\Phi}^{(\alpha)}$ scalarly multiplies the matrices $\mathbf{G}_{P,D}$ and $\mathbf{T}_{P,D}$ by a factor $|\alpha|^2$. Hence, the result of the canonical transformation $\mathbf{\Phi}^{(\alpha)}$ is given by

$$\mathbf{S}_{P,D}{}^{(\alpha)} = \mathbf{S}_{P,D} = \begin{pmatrix} 0 & 0 \\ 0 & 1/2 \end{pmatrix}, \tag{4.108}$$

$$\mathbf{P}_{P,D}{}^{(\alpha)} = \mathbf{P}_{P,D} = \mathbf{N}_{P,D}{}^{(\alpha)} = \mathbf{N}_{P,D} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{4.109}$$

$$\mathbf{G}_{P,D}{}^{(\alpha)} = |\alpha|^2\, \mathbf{G}_{P,D} = \begin{pmatrix} \alpha\alpha^* & \alpha\alpha^* \\ \alpha\alpha^* & \alpha\alpha^* \end{pmatrix}, \tag{4.110}$$

$$\mathbf{T}_{P,D}{}^{(\alpha)} = |\alpha|^2\, \mathbf{T}_{P,D} = \alpha\alpha^*\, k^{-1} \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}. \tag{4.111}$$

The above transformation can physically be realized as follows. By means of circulators and transformers the imbedding 4-port combines the active 1-port with the dissipative one in such a way that the net result is still a dissipative 1-port, and that the noise waves add or subtract depending on whether $|\alpha|$ is larger or smaller than unity. The same thing is done to obtain an active 1-port with either a larger or a smaller noise source. The results are again transformed to normalized impedances (cf. sec. 4.1.3). It will be clear that the coupling of the two 1-ports for reduction or augmentation of the noise of the dissipative 1-port must be different from the coupling for the reduction or augmentation of the noise of the active 1-port. Indeed, it is easy to show that $\mathbf{\Phi}^{(\alpha)}$ refers to a non-reciprocal 4-port.

The condition that α be non-zero and finite is equivalent to the requirement that the combination of original 2-port and imbedding 4-port must be stable. This can be understood from the fact that all elements S_{jj} of the scattering matrix of the 4-port are given by

$$S_{jj} = \frac{(|\alpha| - |\alpha|^{-1})^2}{1/2\,(|\alpha|^2 + |\alpha|^{-2})}, \qquad j = 1, 2, 3, 4. \tag{4.112}$$

If $|\alpha|$ tends either to zero or to infinity, the combination of the pair of 1-ports and the imbedding 4-port becomes internally unstable (cf. appendix D), i.e. internal oscillations start and degrade the signal performance. This means physically among others that, if the noise of the dissipative 1-port is to be completely eliminated by that of the active 1-port, such an intense combination

of the two 1-ports is required that, instead of being a dissipative 1-port, the result is a reactive, and hence internally unstable, 1-port.

When one does not bother about stability requirements, one can choose $|\alpha|$ arbitrarily small, but finite. Hence, the intensity of the noise can be reduced to very small values, while the signal performance is kept unaltered. Consequently, the eigenvalues of the temperature matrix, which both are equal to zero, can be interpreted as the lower limit of the characterizing noise temperatures, so that the above assumption about the eigenvalues is valid.

Examples of active-dissipative 2-ports, which have completely correlated noise, are presented by ideal grounded-grid triodes and by transistors in which no recombination occurs, both devices being used in the lower-frequency regions. In both devices the noise is pure shot noise, which is completely correlated at the input and output sides. Indeed, pure shot noise is not characterized by a "temperature" and it can be reduced below any finite value by means of suitable "feed-back". In the next section we will give a more detailed discussion of the second example.

4.2.4. *Transistor shot noise at low frequencies* *)

Polder and Baelde [49]) gave an equivalent circuit [50]) for a transistor (see fig. 21) and derived an expression for its noise, i.e.

$$\mathbf{G}_Y = 2\,kT\,(\mathbf{Y} + \mathbf{Y}^{\times}) - 2\,q\mathbf{J}, \tag{4.113}$$

where $\mathbf{G}_Y$ is the power-density matrix of the noise-current sources at input and output, T is the ambient absolute temperature, $\mathbf{Y}$ is the admittance matrix, q is the charge of the carrier, and $\mathbf{J}$ is the matrix

$$\mathbf{J} = \begin{pmatrix} J_{in} & 0 \\ 0 & J_{out} \end{pmatrix}, \tag{4.114}$$

J_{in} and J_{out} being the input and output bias currents, respectively.

For low frequencies the storage capacitance can be neglected. If, furthermore, it is assumed that the carrier lifetime is infinitely large and that built-in drift fields are negligible, it can readily be shown, from the formulae given by the above authors [49]), that the admittance matrix $\mathbf{Y}$ is given by

$$\mathbf{Y} = \begin{pmatrix} g & 0 \\ -g & 0 \end{pmatrix}, \tag{4.115}$$

where

$$g = q\,J_s/kT \tag{4.116}$$

and

$$J_{in} = -J_{out} = J_s. \tag{4.117}$$

*) The discussion of the present section is given in the admittance-matrix signal representation to be in conformity with the notation of ref. 49.

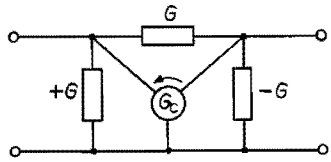


Fig. 21. Equivalent circuit of a transistor (ref. 49).

For the noise-distribution matrix $\mathbf{N}_Y$ one finds

$$\mathbf{N}_Y = 2\,(\mathbf{Y} + \mathbf{Y}^{\times}) = 2\begin{pmatrix} 2g & -g \\ -g & 0 \end{pmatrix}. \tag{4.118}$$

Substitution of these results in eq. (4.100) yields for the noise-source power matrix

$$\mathbf{G}_Y = 2\,q\,J_s\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{4.119}$$

The noise-temperature matrix is then found to be given by

$$\mathbf{T}_Y = k^{-1}\,\mathbf{G}_Y\mathbf{N}_Y{}^{-1} = T\begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}. \tag{4.120}$$

This matrix has two zero eigenvalues and cannot be diagonalized. Moreover, the transistor is a dissipative-active 2-port. It is easy to verify that $\mathbf{N}_Y$ has one positive and one negative eigenvalue for all values of g, except zero.

When the frequency increases, it is not permitted to neglect the storage capacitance anymore. The degeneracy of the eigenvalues of the temperature matrix is then removed. One of the two characteristic temperatures becomes slightly positive and the other slightly negative. Their absolute values are equally large and proportional with frequency.

The degeneracy of the characteristic temperatures is also removed if the generation and recombination of charge carriers cannot be neglected, i.e. if the carrier lifetime is finite. This mechanism is of a purely dissipative nature, and therefore an amount of thermal noise is introduced. A first-order approximation shows that of the two eigenvalues of the temperature matrix one shifts again to small positive and the other to small negative values. The absolute values are again equally large and, in this case, inversely proportional to the square root of the carrier lifetime.

These last remarks make it clear that in any actual device the peculiar singularity of the temperature matrix as discussed in this and the previous section does not occur. In all actual cases distributed reactive elements and/or pure

dissipation (ohmic losses, etc.) are present. Hence, in practice, the concept of characteristic noise temperature is substantial in all cases.

However, if most of the noise of an active-dissipative device consists of shot noise, originating from one real physical shot-noise source only, some or all of the characteristic noise temperatures can be very small. This implies that by proper imbedding (by means of a non-reciprocal feedback system) the output noise power can be made very small, while the signal performance remains unaffected.

5. OPTIMIZATION OF NOISE PERFORMANCE

5.1. The Haus and Adler theory

5.1.1. *Motivation*

While developing the mathematical formalism of noisy linear systems, described in the preceding chapters, we found that many of our considerations parallel several parts of the circuit theory of linear noisy networks, which has been presented by Haus and Adler in several papers [19,48,51–54]. They have summarized that theory into a comprehensive monograph [15]. To some extent several of the concepts introduced and discussed in the preceding chapters appeared to be equivalent to ideas put forward by those authors. We therefore think it desirable to devote a brief discussion to the Haus and Adler theory. Although these authors have presented most of their theory in terms of impedance matrices, we will adhere to the scattering-matrix signal representation.

5.1.2. *Exchangeable power of signal sources*

A 1-port signal source can be considered to be a spontaneously active 1-port (cf. sec. 3.4.2). It can be represented by a cascade of a normal 1-port and a signal-wave source [30]. The 1-port can be dissipative, reactive, or active (by stimulation). Let ϱ_1 be the reflection coefficient of the 1-port and x the amplitude of the signal-wave source (cf. sec. 2.1.2). When a load, which is not spontaneously active, is connected to the source, power is extracted from or delivered to the source. Let the reflection coefficient of the termination be denoted by ϱ_2. A diagram is given in fig. 22. The signal waves towards and from the termination are represented by b and a, respectively. The power P_{21} extracted from or, when negative, delivered to the source can easily be calculated:

$$P_{21} = bb^* - aa^* = \frac{(1 - \varrho_2\varrho_2^*)\, xx^*}{(1 - \varrho_1\varrho_2)\,(1 - \varrho_1^*\varrho_2^*)}. \tag{5.1}$$

For a given signal source, this quantity is a real function of ϱ_2. When ϱ_2 is equal to the inverse of ϱ_1, i.e.

$$\varrho_2 = \varrho_1^{-1}, \tag{5.2}$$

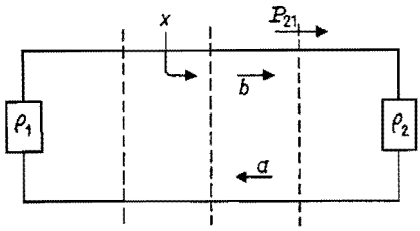


Fig. 22. Power extracted from (or delivered to) a 1-port signal source.

the function $P_{21}(\varrho_2)$ becomes infinite. Everywhere in the neighbourhood of this point at infinity, P_{21} has the same sign, being positive when $|\varrho_1| > 1$ and negative when $|\varrho_1| < 1$.

When ϱ_2 is equal to the complex conjugate of ϱ_1, i.e.

$$\varrho_2 = \varrho_1^*, \tag{5.3}$$

the function $P_{21}(\varrho_2)$ has a relative extremum P_e:

$$P_e = xx^*/(1 - \varrho_1\varrho_1^*). \tag{5.4}$$

Haus and Adler call this quantity the exchangeable power of the signal source [51]). The sign of the exchangeable power is opposite to that at the point at infinity. When the source 1-port is dissipative, i.e. $|\varrho_1| < 1$, P_e is positive. The exchangeable power is then identical to what conventionally is called the available power.

When the source 1-port is active, i.e. $|\varrho_1| > 1$, P_e is negative. Haus and Adler suggest [15]) that the magnitude of P_e then represents the maximum of power that can be pushed into the source. The termination, active for this case, then performs as a stimulated source. In this case source and load may form an unstable system in which the signal-wave source x does not govern the exchange of power any more. Hence, for sources with active 1-ports the concept of exchangeable power may have no practical meaning.

When the source 1-port is reactive, i.e. $|\varrho_1| = 1$ — this is a very unpractical case —, the two conditions of eqs (5.2) and (5.3) coincide, yielding

$$\varrho_2 = \varrho_1^{-1} = \varrho_1^*. \tag{5.5}$$

The function $P_{21}(\varrho_2)$ becomes infinite for this termination, but it has not now the same sign everywhere in the immediate neighbourhood. The meaning of exchangeable power collapses for this case.

5.1.3. *Exchangeable noise power*

Analogously to the observations of the previous section we can define an exchangeable noise power with respect to a noisy 1-port. By virtue of the definitions in secs 2.1.4 and 2.1.6, the definition is given by

$$P_e = \mathsf{X}\mathsf{X}^+\delta f/(1 - \varrho_1\varrho_1^*). \tag{5.6}$$

Dividing this expression by δf, we obtain the density of the exchangeable noise power:

$$P_e/\delta f = \mathsf{X}\mathsf{X}^+/(1 - \varrho_1\varrho_1^*). \tag{5.7}$$

In sec. 2.3.1 it has been shown that this quantity does not depend on the signal properties of the 1-port. Due to eq. (2.71) we obtain

$$P_e/\delta f = F(T, f), \tag{5.8}$$

or, with eq. (2.93), approximately

$$P_e/\delta f = kT. \tag{5.9}$$

The density of the exchangeable noise power of a noisy 1-port is equal to the product of Boltzmann's constant and the characteristic noise temperature of the 1-port.

5.1.4. *Exchangeable noise power of a multiport*

Generalizing the concept of exchangeable noise power of a 1-port, Haus and Adler [15,19]) define the exchangeable noise power of an n-port. According to their definition, it is the stationary value of the noise power drawn from or pushed into the n-port when the load, which is an n-port itself, is varied arbitrarily *).

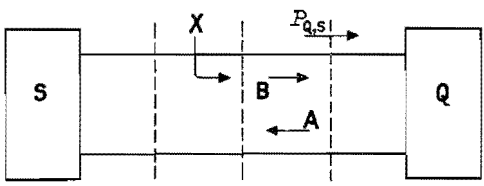


Fig. 23. Noise power radiated from a multiport into its load.

Referring to fig. 23, let a noisy n-port **S** with noise-source vector **X** be terminated by a noiseless n-port **Q**. The noise waves **B** and **A** which travel respectively to and from the load **Q** are given by (cf. sec. 2.5.3)

$$\mathbf{B} = (\mathbf{I} - \mathbf{SQ})^{-1}\,\mathbf{X} \tag{5.10}$$

and

$$\mathbf{A} = \mathbf{Q}\,(\mathbf{I} - \mathbf{SQ})^{-1}\,\mathbf{X}. \tag{5.11}$$

The noise power $P_{Q,S}$ which is extracted from **S** simultaneously via all the ports is given by

$$P_{Q,S} = (\mathbf{B}^{\times}\mathbf{B} - \mathbf{A}^{\times}\mathbf{A})\,\delta f. \tag{5.12}$$

With some algebraic manipulation we obtain

$$P_{Q,S} = \mathbf{X}^{\times}\,(\mathbf{I} - \mathbf{SS}^{\times})^{-1}\,\mathbf{X}\,\delta f + \\ - \mathbf{X}^{\times}\,(\mathbf{I} - \mathbf{Q}^{\times}\mathbf{S}^{\times})^{-1}\,(\mathbf{S} - \mathbf{Q}^{\times})\,(\mathbf{I} - \mathbf{S}^{\times}\mathbf{S})^{-1}\,(\mathbf{S}^{\times} - \mathbf{Q})\,(\mathbf{I} - \mathbf{SQ})^{-1}\,\mathbf{X}\,\delta f. \tag{5.13}$$

The right-hand side of the last equation is a homogeneous quadratic form

*) In fact, Haus and Adler do not define and determine the exchangeable power by varying the load, but by means of variations of the currents at the terminals. These procedures however are wholly equivalent.

of the components of **X**. The first term does not depend on **Q** and can be considered constant. The second term has a zero stationary value when the vector **V**, given by

$$\mathbf{V} = (\mathbf{S}^{\times} - \mathbf{Q})\,(\mathbf{I} - \mathbf{SQ})^{-1}\,\mathbf{X}, \tag{5.14}$$

is equal to the zero vector. This is the case when

$$\mathbf{Q} = \mathbf{S}^{\times}. \tag{5.15}$$

This equation is a generalized form of eq. (5.3) and it defines a general matching condition. The corresponding stationary value of $P_{Q,S}$ is given by

$$P_{e,S} = \mathbf{X}^{\times}\,(\mathbf{I} - \mathbf{SS}^{\times})^{-1}\,\mathbf{X}\,\delta f. \tag{5.16}$$

This equation is the generalized form of eq. (5.6). Haus and Adler define $P_{e,S}$ to be the exchangeable noise power of **S**.

When **S** is a purely dissipative n-port, i.e. when the matrices $(\mathbf{I} - \mathbf{SS}^{\times})$ and $(\mathbf{I} - \mathbf{S}^{\times}\mathbf{S})$ are positive definite, $P_{e,S}$ is positive. From eq. (5.13) it can be seen that $P_{e,S}$ then constitutes a maximum. It could be called the total available power. Analogously, when **S** is purely active, $P_{e,S}$ is a negative minimum. When **S** is of a mixed nature, i.e. when the matrices $(\mathbf{I} - \mathbf{SS}^{\times})$ and $(\mathbf{I} - \mathbf{S}^{\times}\mathbf{S})$ are indefinite, but non-singular, $P_{e,S}$ constitutes a saddle point. Whether the value of this saddle point is positive or negative depends on both **S** and **X**.

When the n-port **S** contains a reactive part, the matrices $(\mathbf{I} - \mathbf{SS}^{\times})$ and $(\mathbf{I} - \mathbf{S}^{\times}\mathbf{S})$ are singular. The partition of the exchanged noise power $P_{Q,S}$ into two terms in accordance with eq. (5.13) cannot then be made. The quantity $P_{Q,S}$ does not then have a stationary value and, hence, an exchangeable power $P_{e,S}$ cannot then be defined.

From eqs (5.10) and (5.11) or, alternatively, from eqs (5.13) and (5.14) it can be seen that $P_{Q,S}$ becomes infinite for

$$\mathbf{Q} = \mathbf{S}^{-1}. \tag{5.17}$$

This equation is the general form of eq. (5.2).

We shall show in following sections that the above concept of overall exchangeable noise power of a multiport as defined by Haus and Adler is of restricted use and can be substituted by definitions which are of a more universal applicability. Usually, one is not interested at all in the largest amount of noise power that can be drawn from or injected into a multiport via all its ports simultaneously. What is often of much more interest is the maximum or minimum noise power that can be exchanged with a multiport via one port only, while that multiport is imbedded in a general lossless feedback system. In the next section we will therefore discuss this type of noise-power exchange and evaluate the extreme values. Doing so, we will arrive at the definition of a whole set of exchangeable noise powers pertaining to a given noisy multiport. There

is another reason that the set of exchangeable noise powers is of more importance than the overall exchangeable noise power. They are better suited for the estimation of optimum noise performance than the above overall concept. As a matter of fact, Haus and Adler too make use of such a set of extreme values for characterizing noise performance. When these authors investigate the optimum noise performance of an n-port amplifier, they show that the eigenvalues of the characteristic-noise matrix (see next section) are equal to the extreme values the noise measure (see sec. 20.5) attains, if the n-port is imbedded in a variable loss less $(n + 2)$-port.

5.1.5. *Alternative definition; exchangeable noise powers of a multiport* [15,43])

Imbedding a noisy n-port in a noiseless and loss-free $(n + 1)$-port results in a noisy 1-port. The exchangeable noise power of the resultant 1-port can be determined. Due to the freedom of loss of the $(n + 1)$-port, this extreme value denotes the maximum of noise power that can be exchanged with the n-port when the latter is combined with the particular imbedding network. This exchangeable noise power clearly depends on the properties of the $(n + 1)$-port, which, under the condition of freedom of loss, can however be varied at will.

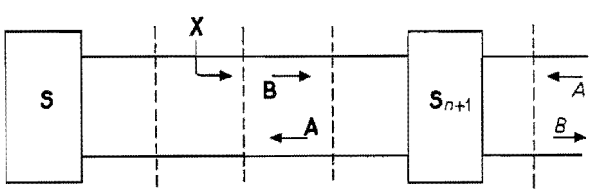


Fig. 24. Reduction of an n-port to a 1-port by imbedding in an $(n + 1)$-port.

Figure 24 shows a sketch of the imbedding which reduces the number of ports to one. Let the scattering matrix $\mathbf{S}_{n+1}$ of the $(n + 1)$-port be given by

$$\mathbf{S}_{n+1} = \left(\begin{array}{c|c} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \hline \mathbf{S}_{21} & \mathbf{S}_{22} \end{array} \right), \tag{5.18}$$

where $\mathbf{S}_{11}, \mathbf{S}_{12}, \mathbf{S}_{21}$, and $\mathbf{S}_{22}$ are (1×1), $(1 \times n)$, $(n \times 1)$, and $(n \times n)$-submatrices. If A and B are the incident and reflected (noise) waves at the resultant 1-port, the equations

$$B = \mathbf{S}_{11} A + \mathbf{S}_{12} \mathbf{B}, \tag{5.19}$$

$$\mathbf{A} = \mathbf{S}_{21} A + \mathbf{S}_{22} \mathbf{B} \tag{5.20}$$

and

$$\mathbf{B} = \mathbf{S}\mathbf{A} + \mathbf{X} \tag{5.21}$$

hold. An algebraic calculation of the type presented in secs 3.6.1 and 3.6.3 and in appendix D yields, by virtue of the freedom of loss of $\mathbf{S}_{n+1}$, the exchangeable power P_e of the resultant 1-port. It is given by the general Rayleigh quotient [40]) of the quadratic forms $\mathbf{X}\mathbf{X}^\times$ and $(\mathbf{I}-\mathbf{S}\mathbf{S}^\times)$, i.e.

$$P_e = \{\boldsymbol{\gamma}\mathbf{X}\mathbf{X}^\times\boldsymbol{\gamma}^\times/\boldsymbol{\gamma}(\mathbf{I}-\mathbf{S}\mathbf{S}^\times)\boldsymbol{\gamma}^\times\}\,\delta f, \tag{5.22}$$

where $\boldsymbol{\gamma}$ is the row vector ($1\times n$-matrix)

$$\boldsymbol{\gamma} = \mathbf{S}_{12}(\mathbf{I}-\mathbf{S}\mathbf{S}_{22})^{-1}. \tag{5.23}$$

The derivation can be performed only if the matrix $(\mathbf{I}-\mathbf{S}\mathbf{S}_{22})$ is non-singular. This means that the system formed by n-port and $(n+1)$-port is not allowed to have internal oscillations.

Variation of the $(n+1)$-port means variation of $\boldsymbol{\gamma}$. Hence, looking for extreme values of P_e, considered as a function of $\mathbf{S}_{n+1}$, is the same as finding extreme values of the above Rayleigh quotient under variation of $\boldsymbol{\gamma}$. In matrix algebra [40]) it is shown that the latter extreme values are equal to the solutions of the general eigenvalue equation

$$[\mathbf{X}\mathbf{X}^\times - \lambda(\mathbf{I}-\mathbf{S}\mathbf{S}^\times)]\boldsymbol{\gamma}^\times = \mathbf{o}. \tag{5.24}$$

They are attained when the vectors $\boldsymbol{\gamma}^\times$ in eq. (5.22) are equal to the corresponding eigenvectors of the matrix pair $\mathbf{X}\mathbf{X}^\times$ and $(\mathbf{I}-\mathbf{S}\mathbf{S}^\times)$. The eigenvalue problem of eq. (5.24), i.e. the problem of the simultaneous diagonalization of the matrices $\mathbf{G}_S$ and $\mathbf{N}_S$ by means of the same Hermitian congruence transformation, has been stated in sec. 4.1.1. It has been discussed at length and solved in subsequent sections. We need not repeat that discussion here.

It has been shown that, although the eigenvalues may be degenerate, one can always find n independent eigenvectors, except in the case of an anomalously singular matrix $\mathbf{G}_S$. Thus, in normal cases, P_e has n extreme values $P_{e,i}$ which are given by

$$P_{e,i} = kT_i\,\delta f. \tag{5.25}$$

This equation is a direct generalization of eq. (5.9). For each anomalous singularity of $\mathbf{G}_S$ the exchangeable noise power P_e has a pair of extreme values which approach zero from opposite sides. The pair refers to an active and a dissipative 1-port, respectively (cf. sec. 4.2.3). In all cases $P_{e,i}$ is equal to the exchangeable noise power of the ith elementary 1-port of the diagonalized n-port. We should therefore like to call the set of n extreme values $P_{e,i}$ the set of exchangeable noise powers of the noisy n-port. Thus, with regard to a noisy n-port, we prefer to define a set of n exchangeable noise powers $P_{e,i}$ instead of only one overall exchangeable noise power $P_{e,S}$.

It will be clear that the set of exchangeable noise powers characterizes the noise of an n-port in exactly the same way as the set of characteristic temper-

atures. The $P_{e,i}$'s are the eigenvalues of the matrix $\mathbf{G_S N_S}^{-1}\,\delta f$, at least if this matrix exists, which need not always be the case. For this reason Haus and Adler [19]) call the negative of that matrix the characteristic-noise matrix and denote it by $\mathbf{N}$, i.e. *)

$$\mathbf{N} = -\mathbf{G_S N_S}^{-1}\,\delta f. \tag{5.26}$$

From eq. (2.273) it is clear that, if it exists, the characteristic-noise matrix $\mathbf{N}$ is equivalent to the noise-temperature matrix $\mathbf{T_S}$. This equivalence is expressed by

$$\mathbf{N} = -k\,\mathbf{T_S}\,\delta f. \tag{5.27}$$

When the multiport contains a reactive part, the matrix $\mathbf{N_S}$ is singular [62]). The definition of eq. (5.26) cannot then be applied. However, by means of eq. (5.27), this difficulty can be circumvented, since the matrix $\mathbf{T_S}$ can still be determined in that case as has been discussed in sec. 4.2.2. Haus and Adler do not consider the case of a singular noise-distribution matrix.

Haus and Adler [19]) have shown that the sum of the exchangeable noise powers $P_{e,i}$ equals the overall exchangeable noise power $P_{e,S}$. Hence, due to eq. (5.16), we have

$$\sum_{i=1}^{i=n} P_{e,i} = P_{e,S} = \mathbf{X}^{\times}\,(\mathbf{I} - \mathbf{S}\mathbf{S}^{\times})^{-1}\,\mathbf{X}\,\delta f = -\,\text{trace}\,(\mathbf{N}) \tag{5.28}$$

or, alternatively,

$$k\sum_{i=1}^{i=n} T_i = \mathbf{X}^{\times}\,(\mathbf{I} - \mathbf{S}\mathbf{S}^{\times})^{-1}\,\mathbf{X} = k\,\text{trace}\,(\mathbf{T_S}). \tag{5.29}$$

In general, a noisy n-port has both positive and negative exchangeable noise powers $P_{e,i}$. The characteristic temperatures T_i, too, are generally also both positive and negative. Let $P_{e,1}$ and T_1 be the smallest positive exchangeable noise power and the smallest positive characteristic noise temperature, respectively. Let further $P_{e,n}$ and T_n be the corresponding negative quantities with the smallest absolute values. Investigating the Rayleigh quotients of eq. (5.22) as a function of $\boldsymbol{\gamma}$, Haus and Adler found that the Rayleigh quotient, i.e. the exchangeable noise power P_e of the resulting 1-port, cannot attain values which

---

*) In the notation of Haus and Adler the factor δf does not appear, because there it is contained in the matrix $\mathbf{G_S}$. Furthermore it should be emphasized that the characteristic-noise matrix $\mathbf{N}$ is not at all equivalent to the noise-distribution matrix $\mathbf{N_S}$ which was introduced in sec. 2.6.1 and discussed subsequently at several places. The reason of the minus sign in eq. (5.26) will become clear in sec. 5.2.6.

lie between the smallest positive and the largest negative extremum. Hence, one of the inequalities

$$P_e \geqslant P_{e,1} \geqslant 0 \quad \text{or} \quad T \geqslant T_1 \geqslant 0, \tag{5.30}$$

$$P_e \leqslant P_{e,n} \leqslant 0 \quad \text{or} \quad T \leqslant T_n \leqslant 0 \tag{5.31}$$

applies. This result will be shown to be most important for the determination of the optimum noise performance of an n-port (see sec. 5.2.)

5.1.6. *Diagonalization*

Haus and Adler [15]) also show that by imbedding an n-port in a lossless $2n$-port, the n-port can be brought into a diagonal form which consists of n independent 1-ports. The exchangeable noise powers of these elementary 1-ports are equal to the extrema $P_{e,i}$ of P_e. The n noise sources pertaining to the n 1-ports are uncorrelated.

Haus and Adler prove only the existence of the diagonalizing transformation. However, their proof is not constructive. They do not obtain the signal-performance matrix of the diagonalizing $2n$-port. Moreover, they argue [15]) that the cases of singular matrices $\mathbf{G}_S$ and/or $\mathbf{N}_S$ can be handled as simple limiting cases of n-ports with non-singular matrices and with small losses which tend to zero. They therefore conclude that the simultaneous diagonalization of both the signal performance and the noise can always be obtained. This is not true. Penfield [62]) has described a relatively simple case in which this procedure does not lead to unambiguous results. It is even more serious that Haus and Adler have overlooked the cases of what we have called anomalously singular $\mathbf{G}_S$. As has been shown in sec. 4.1.7, simultaneous diagonalization cannot be obtained in those cases. The theory of Haus and Adler does not include systems where noise is pure shot noise. From a theoretical point of view such systems are important [49]).

5.1.7. *Some remarks*

Haus and Adler call the simultaneously diagonal transform of an n-port the canonical form of that n-port. Above, we have denoted this form by the adjective diagonal because the adjective canonical had been used to indicate those coordinate transformations which preserve the character of a given signal representation (cf. sec. 3.7.2).

Determining the extreme values of the exchangeable power P_e of a reduced n-port, Haus and Adler construct the reducing $(n+1)$-port by leaving $(n-1)$ ports of a general lossless $2n$-port open-circuited. They could afford this simple procedure because they made their considerations in the impedance-matrix signal representation. This is due to the fact that noise-voltage sources of open-circuited terminal pairs do not manifest themselves at the other terminal pairs.

However, in most other signal representations this is not the case. For example, in the scattering-matrix signal representation waves emitted by noise-wave sources at open ports are reflected by these ports and transmitted to other ports. In order that a noise-wave source does not produce effects at other ports, its associated port must be terminated by a non-reflecting load. Such terminations, however, introduce losses into the $(n + 1)$-port and that is not permitted. Hence, for many signal representations one cannot make use of the simple Haus and Adler procedure.

On the other hand, if an extreme value of the exchangeable noise power P_e has been obtained, the relevant $(n + 1)$-port can be the diagonalizing $2n$-port with $(n - 1)$ ports left open-circuited. The 1-ports of the diagonal form being mutually uncoupled, the noise-wave sources at the $(n - 1)$ open ports do not then produce effects at the port under consideration.

Haus and Adler [19,47]) also consider the possibility of representation of a multiport in coordinate systems other than the impedance- and admittance-matrix signal representations. They also show that the eigenvalues of the characteristic-noise matrix **N** are invariant under transformations from one signal representation to another. Like the temperature matrix, the characteristic-noise matrix is transformed by a similarity transformation. We need not discuss these properties here again because, above, that has been done at length. However, Haus and Adler did not find explicitly that the eigenvalues can be interpreted in terms of temperatures in all coordinate systems.

At last, it should be notified that the vectors γ which produce the extreme values $P_{e,i}$ constitute the rows of the diagonalizing noise-transformation matrix $\mathbf{\Gamma}_D$ as defined by eq. (4.92).

5.2. Optimization of noise performance of amplifiers

5.2.1. *Possible interpretation of exchangeable noise powers*

Haus and Adler [15]) have the opinion that in any commonly used signal representation a special interpretation can be given of the resultant exchangeable noise power P_e or, better, of its extreme values $P_{e,i}$, i.e. of the negatives of the eigenvalues of the characteristic-noise matrix **N**. Moreover, they believe that such interpretations can give a deeper insight into the noise performance of multiports.

In order to verify this assumption in a special case they investigate the general-circuit-constant-matrix — the *ABCD*-matrix — signal representation of 2-port amplifiers. They find that, in that case, the exchangeable noise powers $P_{e,i}$ of a 2-port are related to the noise measure of that 2-port. The noise measure is a scalar quantity which has been defined by them elsewhere [19]) (see sec. 5.2.5) to characterize noise performance of 2-port amplifiers. The noise measure, as the noise figure, is still a function of the manner in which the 2-port is utilized

or imbedded. In fact, Haus and Adler find that the extreme values of the noise measure are proportional to the exchangeable noise powers $P_{e,i}$. The proportionality factor depends on how a standard noise source is being defined. Usually the latter is defined as a dissipative 1-port which radiates noise power only thermally and which is at a uniform temperature of 290 °K.

In following sections we shall discuss the observations made on this subject by Haus and Adler and which have been extended by Rummler [26]).

5.2.2. *Multiport signal source and multiport amplifier*

For the discussion of the noise properties of signal sources and amplifiers we will make use of a method which Rummler [26,56,57]) has introduced to investigate the general interconnection of a noisy n-port signal source and a noisy m-port amplifier. The interconnection is obtained by imbedding the n-port and m-port together in a lossless $(n + m + 1)$-port. A noisy 1-port signal source then results. It is this 1-port which is to be optimized with respect to both signal and noise properties.

Rummler makes his investigations in the impedance-matrix signal representation. Again, we will present the derivations in the scattering-matrix signal representation.

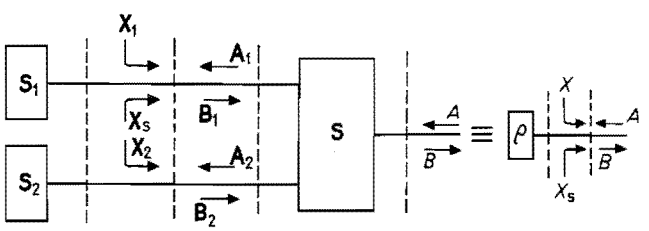


Fig. 25. A combination of an n-port signal source, an m-port amplifier, and a general connecting $(n + m + 1)$-port is equivalent to a 1-port signal source.

Let $\mathbf{S}_1$ in fig. 25 be the scattering matrix of the signal source, $\mathbf{S}_2$ of the amplifier, and $\mathbf{S}$ of the imbedding network, while ϱ is the reflection coefficient of the resultant 1-port. Let $\mathbf{X}_1$ and $\mathbf{X}_2$ be the noise-wave-source vectors of signal source and amplifier, respectively, and X be the noise-wave source which is associated with the resultant 1-port. Moreover, $\mathbf{X}_s$ and X_s refer to the harmonic-wave sources of the original signal source and the resultant signal source, respectively (cf. sec. 3.1.1). Let, furthermore, the **A**'s and **B**'s denote the noise waves and/or harmonic waves at the connections as is indicated in fig. 25. For all these waves the relevant equations are

$$\left(\begin{array}{c} \mathbf{A}_1 \\ \hline \mathbf{A}_2 \\ \hline B \end{array}\right) = \left(\begin{array}{ccc} \mathbf{S}_{11} & \mathbf{S}_{12} & \mathbf{S}_{13} \\ \mathbf{S}_{21} & \mathbf{S}_{22} & \mathbf{S}_{23} \\ \mathbf{S}_{31} & \mathbf{S}_{32} & \mathbf{S}_{33} \end{array}\right) \left(\begin{array}{c} \mathbf{B}_1 \\ \hline \mathbf{B}_2 \\ \hline A \end{array}\right), \tag{5.32}$$

$$\begin{pmatrix} \mathbf{B}_1 \\ \hline \mathbf{B}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{S}_1 & \mathbf{O} \\ \mathbf{O} & \mathbf{S}_2 \end{pmatrix} \begin{pmatrix} \mathbf{A}_1 \\ \hline \mathbf{A}_2 \end{pmatrix} + \begin{pmatrix} \mathbf{X}_1 \\ \hline \mathbf{X}_2 \end{pmatrix} + \begin{pmatrix} \mathbf{X}_s \\ \hline \mathbf{O} \end{pmatrix}, \tag{5.33}$$

$$B = \varrho A + X + X_s. \tag{5.34}$$

The matrix **S** has been written in an appropriately partitioned form.

The quantities which determine the signal and noise properties of the resultant 1-port can now be calculated from these equations. We obtain

$$\varrho = \mathbf{S}_{33} + (\boldsymbol{\sigma}_1 \mid \boldsymbol{\sigma}_2) \left(\begin{array}{c|c} \mathbf{S}_1 & \mathbf{O} \\ \hline \mathbf{O} & \mathbf{S}_2 \end{array} \right) \begin{pmatrix} \mathbf{S}_{13} \\ \hline \mathbf{S}_{23} \end{pmatrix}, \tag{5.35}$$

$$X = (\boldsymbol{\sigma}_1 \mid \boldsymbol{\sigma}_2) \begin{pmatrix} \mathbf{X}_1 \\ \hline \mathbf{X}_2 \end{pmatrix} \tag{5.36}$$

and

$$X_s = \boldsymbol{\sigma}_1 \mathbf{X}_s, \tag{5.37}$$

where

$$(\boldsymbol{\sigma}_1 \mid \boldsymbol{\sigma}_2) = (\mathbf{S}_{31} \mid \mathbf{S}_{32}) \left[\begin{pmatrix} \mathbf{I}_n & \mathbf{O} \\ \mathbf{O} & \mathbf{I}_m \end{pmatrix} - \begin{pmatrix} \mathbf{S}_1 & \mathbf{O} \\ \mathbf{O} & \mathbf{S}_2 \end{pmatrix} \begin{pmatrix} \mathbf{S}_{11} & \mathbf{S}_{12} \\ \mathbf{S}_{21} & \mathbf{S}_{22} \end{pmatrix} \right]^{-1}. \tag{5.38}$$

The $(n + m)$-dimensional row vector $(\boldsymbol{\sigma}_1 \mid \boldsymbol{\sigma}_2)$ has been introduced by Rummler and can be called the reduction vector [56]). It describes how the $(n + m + 1)$-port **S** reduces the combination of the n-port $\mathbf{S}_1$ and the m-port $\mathbf{S}_2$ to the 1-port ϱ. This reduction is unique only if the matrix between square brackets in the right-hand side of eq. (5.38) is non-singular. Physically this means that the system formed by **S**, $\mathbf{S}_1$, and $\mathbf{S}_2$ is not allowed to have internal resonances (cf. appendix B).

The powers *) of the resultant noise-wave source and the resultant harmonic-wave source are respectively given by

$$. XX^+ = (\boldsymbol{\sigma}_1 \mid \boldsymbol{\sigma}_2) \left(\begin{array}{c|c} \mathbf{G}_{S_1} & \mathbf{O} \\ \hline \mathbf{O} & \mathbf{G}_{S_2} \end{array} \right) \begin{pmatrix} \boldsymbol{\sigma}_1{}^{\times} \\ \hline \boldsymbol{\sigma}_2{}^{\times} \end{pmatrix} = \boldsymbol{\sigma}_1 \mathbf{G}_{S_1} \boldsymbol{\sigma}_1{}^{\times} + \boldsymbol{\sigma}_2 \mathbf{G}_{S_2} \boldsymbol{\sigma}_2{}^{\times} \tag{5.39}$$

and

$$X_s X_s{}^+ = \boldsymbol{\sigma}_1 \mathbf{X}_s \mathbf{X}_s{}^{\times} \boldsymbol{\sigma}_1{}^{\times}, \tag{5.40}$$

where $\mathbf{G}_{S_1}$ and $\mathbf{G}_{S_2}$ are the noise-source-power matrices of signal source and amplifier, respectively. By virtue of eq. (5.34) and due to the condition that **S** is free of loss, i.e. eq. (2.210), the noise-distribution matrix $\mathbf{N}_\varrho$ of the resultant 1-port is given by

$$\mathbf{N}\varrho = 1 - \varrho\varrho^* = \boldsymbol{\sigma}_1 \mathbf{N}_{S_1} \boldsymbol{\sigma}_1{}^{\times} + \boldsymbol{\sigma}_2 \mathbf{N}_{S_2} \boldsymbol{\sigma}_2{}^{\times}, \tag{5.41}$$

where $\mathbf{N}_{S_1}$ and $\mathbf{N}_{S_2}$ are the noise-distribution matrices of signal source and amplifier, respectively.

*) It should be recalled that physically these quantities give power per unit bandwidth (power density).

The object is now to determine how the imbedding must be arranged in order to have a possibly low resultant exchangeable noise power under the condition that the resultant exchangeable signal power remains high.

The restriction that the resultant exchangeable signal power must be high is indeed a real one. The optimum noise performance is not obtained when the imbedding is arranged in such a manner that the resultant exchangeable noise power is minimum without any restriction, i.e. so that the resultant 1-port has the smallest positive characteristic noise temperature possible. The latter quantity is equal to the smallest positive eigenvalue of the noise-temperature matrix of the $(n + m)$-port which is formed by the combination of original signal source and original amplifier. It is therefore identical either to that quantity of the signal source or to that quantity of the amplifier. If this case were realized, either the amplifier or the signal source would be cut off from the output. This means that either the amplifier would not be used at all or the signal would not any longer be present at the output.

The paramount question is what is the highest exchangeable signal- to noise-power ratio obtainable at high signal powers. Rummler argues [57]) that this ratio is a good measure for the noise performance of the above system. A good solution to the optimization problem is obtained if, for a given amplification of the exchangeable signal power, one looks for the smallest decrease of the signal- to noise-power ratio. Rummler has investigated this problem with good success for the case that both signal source and amplifier are multiports.

For the time being we will restrict our considerations to the case of a 1-port signal source and a 2-port amplifier because that case contains all items we wish to emphasize. A further restriction is made by the practical assumption that the signal source is dissipative. This implies that the magnitude of its reflection coefficient ϱ_1 satisfies

$$|\varrho_1| < 1. \tag{5.42}$$

Consequently the noise-distribution matrix $\mathbf{N}_{s_1}$, or $\mathbf{N}_{\varrho_1}$ in this case, of the signal source given by

$$\mathbf{N}_{s_1} = \mathbf{N}_{\varrho_1} = 1 - \varrho_1 \varrho_1^* \tag{5.43}$$

is a positive scalar.

5.2.3. *Noise performance of a 2-port amplifier*

Usually observations on noisy systems are centered around two quantities. The first is the amplification of the signal power. This is defined as the quotient of the signal powers at output and input of the amplifier, respectively. However, these powers depend strongly on the coupling of the source and the amplifier and on the load which terminates the amplifier output. To circumvent these difficulties the considerations are mostly made on the amplification of the ex-

changeable signal power. This is defined as the quotient of the exchangeable signal power at the output, when the amplifier is connected to the source, and the exchangeable signal power of the source. Let g be the exchangeable signal-power amplification and let $X_{s,1}$ be the signal-wave source of the 1-port signal source. By virtue of eqs (5.40), (5.41), and (5.43), we then have

$$g = \frac{X_s X_s^+/(1-\varrho\varrho^*)}{X_{s,1}X_{s,1}^+/(1-\varrho_1\varrho_1^*)} = \frac{1}{1+\boldsymbol{\sigma}_2\mathbf{N}_{S_2}\boldsymbol{\sigma}_2^\times/\sigma_1\sigma_1^*(1-\varrho_1\varrho_1^*)}. \tag{5.44}$$

It should be noted that for the case of a 1-port source and a 2-port amplifier $\boldsymbol{\sigma}_1$ reduces to a scalar σ_1 and $\boldsymbol{\sigma}_2$ to a 2-dimensional row vector.

The second quantity of interest is the quotient of the signal- to noise-power ratios at input and output, respectively. Conventionally, this quantity is called the noise figure [58,59]). The ratio of the signal and noise powers at the input, both powers being emitted by the source into the amplifier, is in the present case equal to the ratio of the corresponding exchangeable powers of the source. The latter ratio R_i is given by

$$R_i = \frac{X_{s,1}X_{s,1}^+}{X_1X_1^+} = \frac{X_{s,1}X_{s,1}^+}{kT_1\,(1-\varrho_1\varrho_1^*)}, \tag{5.45}$$

where T_1 is the characteristic noise temperature (there is only one) of the source. The subscript i refers to the input of the amplifier. Analogously, the same ratio at the output R_o is found, by virtue of eqs (5.39) and (5.40), to be given by

$$R_o = \frac{\sigma_1\, X_{s,1}X_{s,1}^+\sigma_1^*}{\sigma_1\, kT_1\,(1-\varrho_1\varrho_1^*)\;\sigma_1^* + \boldsymbol{\sigma}_2\mathbf{G}_{S_2}\boldsymbol{\sigma}_2^\times}. \tag{5.46}$$

The subscript o refers to the amplifier output. The noise figure F is now readily found to be given by

$$F = \frac{R_i}{R_o} = 1 + \frac{\boldsymbol{\sigma}_2\mathbf{G}_{S_2}\boldsymbol{\sigma}_2^\times}{kT_1\;\sigma_1\sigma_1^*\,(1-\varrho_1\varrho_1^*)}. \tag{5.47}$$

Let us now introduce for normalization purposes a new transformation vector $\boldsymbol{\gamma}_2$ defined by

$$\boldsymbol{\gamma}_2 = \boldsymbol{\sigma}_2/\sigma_1\,(1-\varrho_1\varrho_1^*)^{1/2}. \tag{5.48}$$

The exchangeable power amplification g and the noise figure F can then be written

$$g = (1+\boldsymbol{\gamma}_2\,\mathbf{N}_{S_2}\boldsymbol{\gamma}_2^\times)^{-1} \tag{5.49}$$

and

$$F = 1 + \boldsymbol{\gamma}_2\,\mathbf{G}_{S_2}\boldsymbol{\gamma}_2^\times/kT_1. \tag{5.50}$$

Both these quantities depend not only on the properties of the amplifier, that is on $\mathbf{N}_{S_2}$ and $\mathbf{G}_{S_2}$, respectively, but also on those of the imbedding 4-port, that is on $\boldsymbol{\gamma}_2$.

If the use of the amplifier is to be profitable, g must be larger than unity. Hence, the imbedding network must be chosen such that the vector γ_2 fulfils the condition

$$-1 < \gamma_2 \, \mathbf{N}_{s_2} \gamma_2^{\times} < 0. \tag{5.51}$$

This inequality implies that the amplifier must be operated in the negative region of the quadratic form $\gamma_2 \, \mathbf{N}_{s_2} \gamma_2^{\times}$. This can be done only if the matrix $\mathbf{N}_{s_2}$ has at least one negative eigenvalue. This requirement is equivalent to the quite trivial condition that the amplifier has at least an active part, i.e. it must be a "true" amplifier.

A small degradation of the signal- to noise-power ratio R implies a small value of the noise figure F. Since, owing to the positive (semi)-definite character of $\mathbf{G}_{s_2}$, the second term of the right-hand side of eq. (5.50) is non-negative, this term should be made as small as possible in order to obtain an optimum noise performance. In general, an arbitrary decrease of this term, e.g. by diminishing the norm of γ_2, decreases however also the magnitude of $\gamma_2 \, \mathbf{N}_{s_2} \gamma_2^{\times}$ so that the amplification g is also diminished. This is therefore not a good approach. The minimum value of $\gamma_2 \, \mathbf{G}_{s_2} \gamma_2^{\times}$ must be determined by variation of γ_2 while the amplification is kept constant, i.e. in accordance with eq. (5.49) subject to the constraint

$$-\gamma_{s_2} \, \mathbf{N}_{s_2} \gamma_2^{\times} = 1 - g^{-1} = \text{constant}. \tag{5.52}$$

This problem can be solved by the method of Lagrange multipliers. Since eq. (5.51) must hold, this is equivalent to the determination of the smallest positive eigenvalue of the general eigenvalue problem *)

$$[\mathbf{G}_{s_2} + \lambda \, \mathbf{N}_{s_2}] \, \gamma_2^{\times} = \mathbf{0}. \tag{5.53}$$

This is again the same problem as has been investigated thoroughly in chapter 4.

Let $\gamma_{2,0}$ be the eigenvector that corresponds to the smallest positive eigenvalue λ_2. The relevant equation is then

$$\lambda_2 = \frac{-\gamma_{2,0} \mathbf{G}_{s_2} \gamma_{2,0}^{\times}}{\gamma_{2,0} \mathbf{N}_{s_2} \gamma_{2,0}^{\times}} = -kT_2, \tag{5.54}$$

where T_2 is the negative eigenvalue of the noise-temperature matrix of the amplifier, which has the smallest magnitude. This solution is in agreement with the positive semi-definite character of $\mathbf{G}_{s_2}$ and the requirement that $\gamma_{2,0} \mathbf{N}_{s_2} \gamma_{2,0}^{\times}$ must be negative for amplifiers.

If the imbedding is effected in accordance with this solution, the noise figure is optimum and the best noise performance has been obtained. By virtue of

*) Contrary to the practice in sec. 4.1.6, here the plus sign has been used in order to be in agreement with the formalism adopted by Haus and Adler.

eqs (5.50), (5.52), and (5.54), the optimum noise figure F_{opt} is given by

$$F_{opt} = 1 + (\lambda_2/kT_1)(1 - g^{-1}) = 1 - (T_2/T_1)(1 - g^{-1}). \qquad (5.55)$$

Conventionally, the noise temperature T_1 of the signal source is standardized [60]) at room temperature (290 °K). In point of fact the deterioration of the signal-to noise-power ratio depends as well on the actual characteristic noise temperature T_1 of the signal source as on that of the amplifier T_2.

At this stage the following remarks may be made. The derivation in the present section is largely due to Haus and Adler [15]). However, these authors have not presented the results in terms of characteristic noise temperatures. In fact, the results contained in eqs (5.54) and (5.55) are given by them in terms of λ_2, and not of T_2. When the chain-matrix signal representation is used, as they have done, the quantity (λ_2/kT_1) can be interpreted as a noise measure (see sec. 5.2.5). The noise measure however contains implicitly the temperature of the source. Moreover, from a physical point of view, the concept of noise measure is lucid only for the chain-matrix signal representation. On the other hand, when the results are given in terms of characteristic noise temperatures, as we have done, not only the relative importance of signal source and amplifier temperatures is emphasized but also is the interpretation valid and lucid in all signal representations (cf.s sec. 5.2.6).

The result of eq. (5.55) is also valid for dissipative and reactive 2-ports. An amplification g smaller than unity can be interpreted as an attenuation. When this occurs, by virtue of eqs (5.49) and (5.54), both the quantities $\boldsymbol{\gamma}_2 \mathbf{N}_{s_2} \boldsymbol{\gamma}_2^{\times}$ and T_2 are positive. Hence, for attenuating 2-ports, the noise figure F is also larger than unity. To make optimum use of an attenuating 2-port from the point of view of noise performance, the vector $\boldsymbol{\gamma}_2$ should be chosen as the eigenvector that corresponds to the negative eigenvalue λ with the smallest magnitude. This agrees with the smallest positive characteristic noise temperature T_2 of the "amplifier". If a 2-port is reactive, or if only its reactive part is used, the quantity $\boldsymbol{\gamma}_2 \mathbf{N}_{s_2} \boldsymbol{\gamma}_2^{\times}$ is equal to zero. The gain g and, hence, the noise figure F are then both equal to unity, so that then there is no amplification or attenuation and no degradation of the noise performance.

5.2.4. *Equivalent noise temperature*

By virtue of eqs (5.48) and (5.52), substitution of eq. (5.54) in eq. (5.39) yields for the power of the noise-wave source X of the resultant 1-port the expression

$$XX^{+} = \sigma_1 \sigma_1^{*} (1 - \varrho_1 \varrho_1^{*}) [kT_1 - kT_2 (1 - g^{-1})]. \qquad (5.56)$$

The first term between the square brackets refers to the intensity of the noise which originates from the signal source. The second term indicates the amplifier noise.

In literature the concept of equivalent noise temperature T_{eq} of a noisy

amplifier or attenuator is often used (cf. sec. 2.3.1). This is the temperature the input termination should have in order to deliver the same amount of noise power at the output of an equivalent noiseless amplifier as the actual noisy amplifier does, if its input is connected to the same, but noiseless, termination. From eq. (5.56) it is readily seen that the equivalent noise temperature T_{eq} is given by

$$T_{eq} = -T_2 (1 - g^{-1}). \tag{5.57}$$

For large amplifications the approximation

$$T_{eq} \simeq |T_2| \tag{5.58}$$

applies.

5.2.5. *Noise measure*

In sec. 5.2.3 it has been shown that optimization of noise performance is not obtained, if the noise figure F is minimized straightforwardly, but that the latter figure must be reduced to a minimum subject to a constraint. The quantity which can be minimized straightforwardly is the Rayleigh quotient:

$$\lambda = \frac{-\mathbf{Y}_2 \mathbf{G}_{S_2} \mathbf{Y}_2^{\times}}{\mathbf{Y}_2 \mathbf{N}_{S_2} \mathbf{Y}_2^{\times}}. \tag{5.59}$$

Dividing both members by kT_1 and inserting eqs (5.50) and (5.52), we obtain

$$\lambda / kT_1 = (F - 1)/(1 - g^{-1}) = M. \tag{5.60}$$

Haus and Adler [19]) call this quantity the noise measure of the imbedded amplifier. Optimization of the noise performance is achieved if the noise measure M is made minimum. This optimization does not depend on the amplification. Hence, the noise measure offers a criterion for the noise performance of an amplifier, which is independent of the manner in which the signal properties are utilized. This is the most important feature of that quantity.

Haus and Adler *) show that, if two amplifiers are cascaded, the amplifier having the smaller noise measure should be placed first in order to achieve the best noise performance of the cascade. Furthermore, they show [15]) that the noise measure of an arbitrary interconnection of 2-port amplifiers, which yields an overall 2-port amplifier, cannot be smaller than the optimum noise measure of the individual amplifier with the smallest value of that quantity.

Rummler introduces the concept of noise-performance plane [57]). He plots the noise- to signal-power ratio at the output of the amplifier against the reciprocal of the exchangeable power at that output. He shows that, when a source and an amplifier are imbedded, the negative slope of the straight line in that plane which determines the linear relation between the two output quantities

*) Haus and Adler discuss only "true" amplifiers. When losses are present, this may cause difficulties [62]). It should be noted that the noise measure is negative for dissipating 2-ports (see sec. 5.2.7).

mentioned is proportional to the noise measure. It is then readily shown that the noise measure determines the quotient of the increase of noise- to signal-power ratio and the corresponding decrease of the reciprocal of the exchangeable output power.

5.2.6. *The optimum noise measure*

The minimum value of the noise measure M is obtained, if the amplifier is imbedded properly as has been discussed in sec. 5.2.3. This minimum value is called the optimum noise measure M_{opt} and is given by

$$M_{\text{opt}} = \lambda_2/kT_1 = -T_2/T_1. \tag{5.61}$$

The quantity $\lambda_2 \delta f$ is the smallest positive eigenvalue of the characteristic-noise matrix **N** (cf. sec. 5.1.5), so that $\lambda_2 \delta f$ is equal to the negative of that negative exchangeable power $P_{e,i}$ of the amplifier, which has the smallest absolute value. Since they consider it convenient that both M_{opt} and λ_2 are positive numbers for true amplifiers, Haus and Adler have introduced the minus sign into the definition of the characteristic-noise matrix **N** (cf. secs 5.1.5 and 5.2.3).

Although the optimum noise measure is a good criterion for the noise performance of an amplifier, it has some drawbacks. To understand the first, one should remember that the ability of an amplifier to perform well with respect to noise depends not only on the intensity of the noise it emits itself, i.e. on λ_2, but also on its signal properties, i.e. on g, and on the noise which accompanies the signal at the input, i.e. on kT_1. In our opinion a good noise criterion should refer to the first aspect only. The other two should be known separately. The optimum noise measure is indeed independent of the signal performance.This is not true for the noise figure. The optimum noise measure, however, still depends on the noise of the signal source, even though the latter is standardized usually at 290 °K. In some situations such as in radio astronomy and space communication the standardization, however, is impractical. To avoid the dependence on the source temperature we might suggest to characterize noise performance not by M_{opt}, as Haus and Adler have done, but by λ_2. We then have an absolute measure (in terms of exchangeable power) instead of a relative measure.

The second drawback of the optimum noise measure concerns the physical interpretation. In the definition of Haus and Adler [15]), eq. (5.60), the noise measure is only an algebraic combination of other quantities, viz. of F and g. It is not an elementary physical quantity. Only in the chain-matrix or transfer-matrix signal representation it can be given a straightforward physical interpretation as the negative of the exchangeable noise power at the output of the optimized amplifier relative to the exchangeable power of the standard signal source. Again, to avoid the dependence on the temperature of the signal source, we can take directly the negative of the exchangeable noise power instead of

the relative quantity, i.e. the noise measure. However, the physical meaning is then still straight forward only when the amplifier is used as a transfer 2-port, described in the chain-matrix or transfer-matrix signal representation. In other signal representations neither the noise measure nor the exchangeable noise power can be interpreted in elementary physical quantities relevant for the system.

Haus and Adler have introduced the minus sign in the definition of the characteristic-noise matrix, eq. (5.26), in order that the noise measure be positive for amplifiers. The consequence is that for dissipative components the noise measure is negative. For systems which consist of both active and dissipative components this does not seem logical.

For all these reasons we prefer to characterize noise performance by characteristic noise temperatures. The optimum noise performance is then determined by the characteristic noise temperature with the smallest magnitude. It is positive for dissipative components and negative for active components. The fact that the optimum noise temperature is negative for amplifiers can be considered an advantage, because for other reasons, too, it has become practice to characterize active components by means of negative temperatures (cf. secs 2.3.1 and 2.3.3). The characteristic noise temperatures are tied up intimately with the temperature matrix which has been defined as the general analogue of the scalar temperature of a homogeneously warm passive multiport. In this context the characteristic temperatures refer directly to the internal "temperature" distribution of a multiport.

5.2.7. *The optimum noise temperature*

From what has been said in the previous section, it will be clear that we choose the negative characteristic noise temperature with the smallest magnitude to be the measure of the best noise performance obtainable with a given amplifier. For an attenuating component it is the smallest positive of these eigenvalues.

Let T_{opt} be the best noise temperature. When expressed in terms of this quantity, we find from eqs (5.61) and (5.54) for the optimum noise measure and the negative of the optimum exchangeable noise-power density

$$M_{opt} = -T_{opt}/T_1, \tag{5.62}$$

and

$$\lambda_{opt} = -k\,T_{opt}. \tag{5.63}$$

The optimum noise temperature can also serve as a criterion for the selection of the best amplifier out of a set. It can furthermore be used to determine the most advantageous sequence for a cascade of, both amplifiying and attenuating, 2-ports. By virtue of eq. (5.56), it can be shown that, if possible, the "hottest" — in the sense of sec. 2.3.3 — 2-port should go first and the "coldest" last. This means that amplifiers should come before attenuators

and, moreover, that the best amplifier should lead and the best attenuator bring up the rear.

When compared to the noise temperature of the signal source, the magnitude of the optimum noise temperature can also be used as a criterion for the suitability of an amplifier to accomplish a certain task. The value of $|T_2|$ should be smaller than but of the same order of magnitude as T_1. Inspection of eq. (5.56) shows that, if $|T_2| \ll T_1$, the amplifier is too good and, if $|T_2| \gg T_1$, it is too bad for amplifying the signal.

As a last argument for our choice, we would put forward the fact that the concept of temperature is a physically well established measure of noise intensity.

5.3. A simplified amplifier model

5.3.1. *Matching of input and output*

Without violating the generality of the discussion, some simplifying assumptions can be made in the subject matter of the preceding sections. These simplifications lead to an obvious interpretation of some significant quantities.

First, we again consider a 2-port amplifier connected to a 1-port signal source, both components being noisy. The 4-port which reduces the combination of signal source and amplifier to one resultant 1-port, can be thought to contain at the port to which the signal source is connected a lossless 2-port. This 2-port imbeds the signal source. If it is chosen to be of the normalizing type discussed in sec. 4.1.3, due to the assumed dissipative character of the signal source, the resulting signal source can be made non-reflecting. Mathematically, this means that σ_1 (cf. eq. (5.48)) is split into two factors. One factor given by $(1 - \varrho_1 \varrho_1^*)^{-1/2}$ pertains to the lossless 2-port, while the other factor can be considered to be the new transformed version of σ_1, which appertains to the remaining 4-port. In the expression of secs 5.2.2 and 5.2.3 $X_{s,1}$ is multiplied by the first factor, while ϱ_1 is made equal to zero. From eqs (5.44), (5.45), and (5.46) it is clear that the amplification g and the signal- to noise-power ratios R_i and R_o are not altered numerically.

A second simplification is obtained if the output of the reducing 4-port is provided with a further lossless 2-port which now normalizes the resultant 1-port. From eqs (5.41), (5.48), and (5.51), together with the assumption of a dissipative signal source, we can conclude that the resultant 1-port too is dissipative. Hence, normalization of the output makes it also reflectionless. Mathematically this second normalization implies that σ_1 and $\boldsymbol{\sigma}_2$ (cf. sec. 5.2.3) must both be multiplied by the scalar $(1 - \varrho\varrho^*)^{-1/2}$. The resultant reflection coefficient, the new value of ϱ, is made equal to zero. Again, the amplification g and the ratios R_i and R_o are not altered.

The above invariance of amplification and signal- to noise-power ratios is due to the fact that these quantities are quotients of available powers of 1-ports,

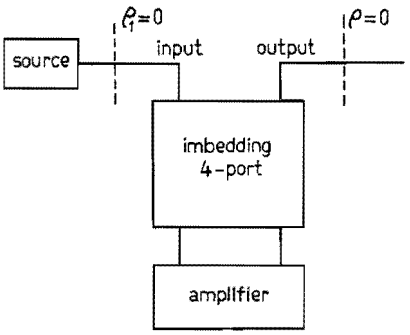


Fig. 26. Diagram of an amplifier model.

which are not changed by loss-free imbedding of 1-ports. In the normalized situation the available power is actually emitted into the line to which the 1-port has been matched, and it is really available at the termination at the other end of that line. On the other hand, waves reflected, emitted, or transmitted into that line by the termination are absorbed completely in the matched 1-port.

In what follows it is assumed that both above-mentioned normalizations have been carried out and that $X_{s,1}$, σ_1, and $\boldsymbol{\sigma}_2$ have the correspondingly altered values. The new 4-port can be considered to be a normal imbedding for the 2-port amplifier. Hence, we may restrict the investigations to the noise performance of a losslessly imbedded 2-port amplifier to the input of which a non-reflecting noisy 1-port signal source has been connected. An illustration of this system has been given in fig. 26.

The equations (5.39), (5.40) and, (5.41) can now be written in the forms

$$XX^+ = \sigma_1\sigma_1^* (kT_1 + \boldsymbol{\gamma}_2 \mathbf{G}_{s_2}\boldsymbol{\gamma}_2^\times), \tag{5.64}$$

$$X_sX_s^+ = \sigma_1\sigma_1^* X_{s,1}X_{s,1}^+ \tag{5.65}$$

and

$$\sigma_1\sigma_1^* (1 + \boldsymbol{\gamma}_2 \mathbf{N}_{s_2}\boldsymbol{\gamma}_2^\times) = 1, \tag{5.66}$$

where kT_1 is the power of the noise-wave source of the signal source and $\boldsymbol{\gamma}_2$ is given by (cf. eq. (5.48))

$$\boldsymbol{\gamma}_2 = \sigma_1^{-1}\boldsymbol{\sigma}_2. \tag{5.67}$$

From eq. (5.65) the power amplification g is equal to $\sigma_1\sigma_1^*$. By virtue of eq. (5.66), it can be written in the form

$$g = \sigma_1\sigma_1^* = (1 + \boldsymbol{\gamma}_2 \mathbf{N}_{s_2}\boldsymbol{\gamma}_2^\times)^{-1}. \tag{5.68}$$

This equation agrees with eq. (5.49).

As has already been discussed above, the best noise performance is obtained if the resultant noise-wave-source power XX^+ is minimized while keeping the amplification $\sigma_1\sigma_1^*$ unaltered. Hence, from eqs (5.64) it is seen that the optimum noise performance is achieved if the quadratic from $\boldsymbol{\gamma}_2 \mathbf{G}_{s_2}\boldsymbol{\gamma}_2^\times$ is

minimized under the constraint of $\gamma_2 \mathbf{N}_{s_2} \gamma_2^{\times}$ being constant. Above it was found that the imbedding 4-port must and can be chosen such that $\sigma_1 \sigma_1^*$ has the value desired and γ_2 is the eigenvector of the matrix pair $\mathbf{G}_{s_2}$ and $\mathbf{N}_{s_2}$, which corresponds to the eigenvalue with the smallest magnitude.

Two cases may occur. First, the 2-port is used as an amplifier. In this case γ_2 belongs to the negative eigenvalue with the smallest magnitude. Then, $\gamma_2 \mathbf{N}_{s_2} \gamma_2^{\times}$ is inherently negative so that, due to eq. (5.68), the amplification g is larger than unity. Secondly, the 2-port is used as an attenuating component. Then, γ_2 belongs to the smallest positive eigenvalue and $\gamma_2 \mathbf{N}_{s_2} \gamma_2^{\times}$ is inherently positive. In this case the "amplification" g is smaller than unity, denoting an attenuation.

The norm of an eigenvector is not prescribed by the eigenvalue problem. Hence, the vector γ_2 can be multiplied by any finite scalar so that the condition of eq. (5.68) can be satisfied for any arbitrary value of g which is larger or smaller than unity in the two respective cases. In both cases, however, the equation

$$\gamma_2 \mathbf{G}_{s_2} \gamma_2^{\times} = kT_2 (\gamma_2 \mathbf{N}_{s_2} \gamma_2^{\times}) = -kT_2 (1 - g^{-1}) \tag{5.69}$$

applies. Substitution in eq. (5.64) yields

$$XX^+ = gkT_1 + (1 - g) kT_2. \tag{5.70}$$

The last result is a well-known formula for the total noise-power output of a passive system consisting of a non-reflecting 1-port at the uniform temperature T_1 in cascade with a matched 2-port at the uniform temperature T_2 and with an output- to input-power ratio g. In our formalism, eq. (5.70) has a much more general validity if only the T's are appropriate characteristic noise temperatures. Since for g-values larger than unity T_2 is non-positive, the second term in the right-hand member of eq. (5.70) is always non-negative. This means that in all cases, the 2-port contributes essentially to the total noise-power output.

It should be noted that T_2 is the characteristic noise temperature of the imbedded 2-port which refers to the particular direction of transmission. The properties of the other direction, from output towards signal source, are not interesting since the input termination is non-reflecting.

5.3.2. *The optimum imbedding of a 2-port amplifier*

In order that the complete system will have optimum noise performance, the 4-port which imbeds and reduces the combination of signal source and amplifier must fulfil certain requirements. The imbedding network that satisfies these conditions can be thought to consist of some particular parts which we describe next.

Figure 27 gives a diagram of the system. The signal source is imbedded in a 2-port t_1 which normalizes it to a non-reflecting 1-port. The 2-port amplifier is transformed to its elementary, perhaps diagonal, form by means of the

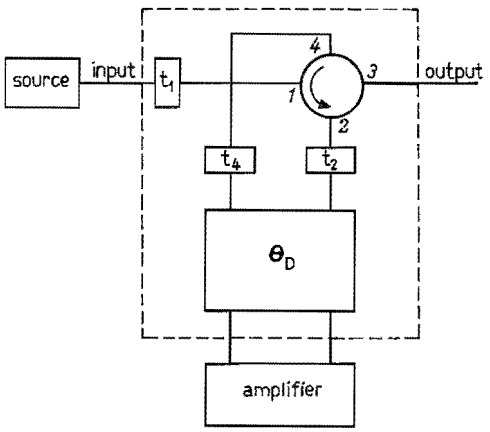


Fig. 27. Diagram of the essential parts of an amplifier circuit.

4-port Θ_D (cf. sec. 4.1.8). If the 2-port is an amplifier, at least one of the two resulting elementary 1-ports is active. The one with the negative characteristic noise temperature with the smallest absolute value is connected to one of the ports (2) of an ideal 4-port circulator. This connection is made via a transformer t_2. The other elementary 1-port, which may be active, reactive, or dissipative, is connected to the opposite port (4) of the circulator. This connection is or is not made via a transformer t_4. The normalized signal source is connected to the circular port (1) that precedes the port (2) to which the least noisy active elementary 1-port has been connected. The remaining circulator port (3) is the output of the system.

Figure 28 gives a simplified diagram. The reflection coefficient ϱ_1 of the imbedded signal source is made equal to zero by the transformer t_1. Hence, any wave incident upon the output and reflected by the elementary 1-port ϱ_4 as well as the noise waves emitted by that 1-port, are absorbed completely by the signal source. Thus, neither ϱ_3 nor T_3 are of much importance, at least with respect to the noise performance. The same can be said of ϱ_4 and T_4 so that the transformer t_4, too, is not interesting. For stability criteria all these quantities, however, may be significant, and in general they are.

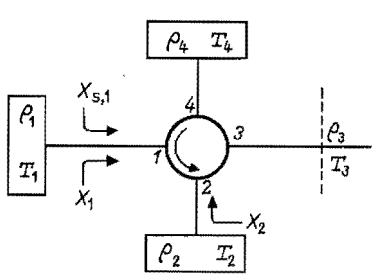


Fig. 28. Simplified diagram of an amplifier circuit.

The signal wave $X_{s,1}$, emitted by the source ϱ_1, is incident upon the elementary 1-port ϱ_2. The power of the signal wave reflected by the latter 1-port, that is of the amplified signal wave, is given by (cf. eq. (5.65))

$$X_s X_s^+ = \varrho_2 \varrho_2^* X_{s,1} X_{s,1}^+ \tag{5.71}$$

so that the amplification g is given by

$$g = \sigma_1 \sigma_1^* = \varrho_2 \varrho_2^*. \tag{5.72}$$

This quantity can be varied at will in the region $g > 1$ by means of the transformer t_2. However, increasing the amplification increases also the power of the noise wave emitted by the 1-port ϱ_2. This contribution to the power of the output noise wave is given by

$$X_2 X_2^+ = kT_2 (1 - \varrho_2 \varrho_2^*) = kT_2 (1 - g). \tag{5.73}$$

The contribution of the signal source to the power of the output noise wave is found to be

$$\varrho_2 \varrho_2^* X_1 X_1^+ = kT_1 (1 - \varrho_1 \varrho_1^*) \varrho_2 \varrho_2^* = kT_1 g. \tag{5.74}$$

Summation of the two contributions yields again eq. (5.70), i.e.

$$XX^+ = kT_1 g + kT_2 (1 - g). \tag{5.75}$$

5.3.3. *Discussion*

The best noise performance of a given amplifier is obtained if its contribution to the total output noise power is made relatively as small as possible. It has been shown above that, independently of the actual amplification, T_2 indeed refers to this minimum. Therefore, T_2 is a good measure of the best noise performance possible. On the other hand, the optimum noise figure F_{opt}, which is given by

$$F_{opt} = 1 + X_2 X_2^+ / \varrho_2 \varrho_2^* X_1 X_1^+ = 1 + T_2 (1 - g)/T_1 g, \tag{5.76}$$

depends still on the amplification g. Being prescribed or wanted, g may be different for the same amplifier in different situations, so that F_{opt} is not a unique number for a given amplifier.

Some remarks must still be made. First, in conventional noise theories [33,34]) and in practice the noise performance of an amplifier is often optimized by variation of the input termination. In terms of our formalism, this can be accomplished by connecting a lossless 2-port between the signal source and the amplifier and varying it till a relative minimum of the amplifier contribution to the output noise power is obtained. However, since no feedback, etc. is present, a lossless 2-port at the input is not the most general imbedding of an amplifier. In this way one cannot then hope to achieve the optimum solution.

A second remark concerns the waves emitted by the noise-wave source of an

amplifier at its input. Whether having been amplified or not, these waves can be reflected by the components which precede the amplifier. The amplifier then further amplifies them and transmits them towards the output, where they contribute substantially to the total output noise power. In our formalism these input noise-wave sources have already been accounted for. First, while transforming an amplifier to its elementary form, reflected portions of the input noise waves are used at the output ports to minimize the output noise waves by interference effects. This means that completely correlated portions of the output noise waves are made zero by subtracting corresponding portions of the input noise waves. Secondly, making the input termination non-reflecting causes the remaining part of the input noise waves to be absorbed completely by that termination.

In the preceding sections we have discussed noise performance of 2-port amplifiers only. However, in this discussion all considerations which apply to 1-port amplifiers have inherently been made. To obtain the best noise performance of a 2-port amplifier, it is reduced to a set of two uncoupled elementary 1-ports. Then, only one of these 1-ports, the least noisy, is used for the amplification process. Hence, for 1-port amplifiers the transformation to the diagonal form need not be performed because such an amplifier is diagonal already. All further considerations apply without any restriction. It is then readily seen that, in terms of characteristic noise temperature, the noise performance of a 1-port amplifier cannot be improved or deteriorated by lossless imbedding. When a 1-port amplifier is imbedded in a lossy 2-port, the noise performance can worsen only.

5.4. Multiport signal sources and amplifiers

In this section we will give a short survey of the results of Rummler [26,56,57]. As mentioned in sec. 5.2.2, Rummler investigated the general lossless imbedding of an n-port signal source together with an m-port amplifier in a lossless $(n + m + 1)$-port so as to obtain a 1-port signal source with an enlarged exchangeable power. The signal- to noise-power ratio of this ultimate 1-port should be made as large as possible for a given value of the resultant signal power.

Rummler mentions two examples of significant practical multiport systems. With one, diversity reception, a set of receiving antennae is used. This collection of antennae can be considered as a multiport source. The problem is how the signals and noise of all these antennae must be combined to yield the best overall performance at one single output port. The other example refers to a multi-frequency parametric system. Although the derivations of Rummler, as well as our formalism, are not yet adapted for multi-frequency systems, he shows that such systems which satisfy the Manley and Rowe power relations can easily be

brought into a form which can be treated by that formalism. In particular, he investigates the noise performance of a double-sideband degenerate parametric amplifier of which the sidebands can be distinguished.

Rummler shows that any lossless $(n + m + 1)$-port imbedding can be realized as follows. First, the n-port signal source is reduced to a 1-port signal source. Secondly, the m-port amplifier is reduced to a 2-port amplifier. At last, the resulting 1- and 2-port are cascaded.

From what has been derived above, the best reduction of the amplifier is readily determined. At first, the m-port is transformed to its diagonal form. If it is a "true" amplifier, the resulting diagonal m-port consists of at least one active 1-port. Then, that active 1-port of which the negative characteristic noise temperature has the smallest magnitude is connected to one of the ports of a circulator. If the preceding and the next ports of that circulator constitute respectively the input and output ports of the reduced amplifier, the 2-port amplifier with the best possible noise performance has been obtained. This result agrees with the findings of Rummler.

The determination of the best reduction of the signal source is a little more intricate. If a 1-port signal source is losslessly imbedded in a 2-port, the signal- to noise-power ratio does not depend on that transformation, since the available powers are invariant. This, however, is not true for the signal- to noise-power ratio of the 1-port which results when an n-port signal source is reduced. By varying the reducing $(n + 1)$-port, of that 1-port both the resulting exchangeable signal power $P_{s,i}$, given by

$$P_{s,i} = X_{s,i} X_{s,i}^{+} \delta f (1 - \varrho_1 \varrho_1^{*})^{-1}, \tag{5.77}$$

and the exchangeable noise power $P_{n,i}$, given by

$$P_{n,i} = X_1 X_1^{+} \delta f (1 - \varrho_1 \varrho_1^{*})^{-1}, \tag{5.78}$$

can be varied over certain restricted regions.

Rummler introduces the noise-performance plane of which the coordinates are respectively the reciprocal of the exchangeable signal power, $1/P_{s,i}$, and the noise- to signal-power ratio, $1/R_i$. The latter is given by eq. (5.45), i.e.

$$1/R_i = P_{n,i}/P_{s,i}. \tag{5.79}$$

In this plane the points, the source points, which denote the possible values of $1/P_{s,i}$ and $1/R_i$ of the imbedded source form a closed region, the source region. This is illustrated in fig. 29. Rummler shows that the boundary of this region is a convex curve or, at least, a semi-convex curve. If the power matrix $\mathbf{X_s X_s}^{\times}$ of the signal-wave-source vector $\mathbf{X_s}$ is positive semi-definite, the source region can extend to infinity. Moreover, if the noise-distribution matrix $\mathbf{N_{s_1}}$ of the signal source is indefinite, the source region includes also negative values of the coordinate $1/P_{s,i}$.

It will be clear that, for a given value of $P_{s,i}$ the noise- to signal-power ratio

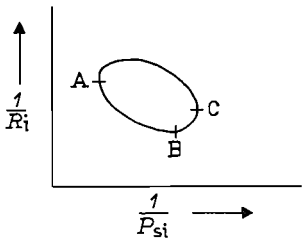


Fig. 29. The noise-performance plane (ref. 26).

$1/R_i$ can be varied only over a restricted range of values. The lower limit of this range determines the maximum value $R_{i,max}$ of R_i attainable at that particular value of $P_{s,i}$. If $P_{s,i}$ should be kept at a constant value, the best reducing $(n + 1)$-port, the best with respect to noise performance, would be the one that yields this constraint maximum $R_{i,max}$. However, this maximum itself is a function of $P_{s,i}$. We can write down the functional relation

$$1/R_{i,max} = f(1/P_{s,i}). \tag{5.80}$$

In fig. 29 this function is represented by the curve ABC. Due to the convexity of the boundary of the source region, this function has a minimum (point B in fig. 29), which determines the absolute maximum of the signal- to noise-power ratio R_i. This absolute maximum need neither coincide with a maximum of $P_{s,i}$ nor with a minimum of $P_{n,i}$.

Adjustment of the $(n + 1)$-port so as to yield the absolute maximum of R_i does not guarantee the best noise performance of the complete signal-source and amplifier system. That is obtained only if the signal- to noise-power ratio R_o at the output is as large as possible. The reciprocal of this quantity, given by

$$1/R_o = 1/R_i + [(1 - g)/g]\, kT_2\delta f/P_{s,i}, \tag{5.81}$$

must be a minimum.

It will be clear that, whatever the value of $P_{s,i}$ may be, the quantity $1/R_i$ should be as small as possible, that is equal to $1/R_{i,max}$. This requirement being satisfied, by virtue of eq. (5.80), we can rewrite eq. (5.81) in the form

$$1/R_o = f(1/P_{s,i}) + [(1 - g)/g]\, kT_2\delta f/P_{s,i}. \tag{5.82}$$

This equation determines R_o as a function of $P_{s,i}$. The maximum of this function, that is the minimum of eq. (5.82), determines the best, not the largest, exchangeable signal power $P_{s,i}$ at the input and, hence, the best reduction of the n-port signal source.

Rummler describes the relationship of eq. (5.82) in the noise-performance plane where, now, the coordinates are formed by respectively the reciprocal of exchangeable signal power at the output, $1/P_{s,o}$, and the noise- to

signal-power ratio, $1/R_o$. He then shows that in this plane the relation of eq. (5.82) is represented by straight lines of which the slope is given by $kT_2\delta f$ or $-M_{opt}kT_1\delta f$ (cf. sec. 5.2.5). Hence the increase of the noise- to signal-power ratio due to the use of the amplifier is proportional to the decrease of the reciprocal of the exchangeable signal power, the proportionality constant being equal to $-kT_2\delta f$.

For a noiseless amplifier, $T_2 = 0$, the maximum of $R_{i,max}$ represents the best total noise performance. If the amplifier is noisy, it is readily seen from eq. (5.82) that it is as a rule more advantageous to adjust at a slightly larger value of $P_{s,i}$ and to allow for a somewhat smaller value of $R_{i,max}$. The second term of the right-hand member of eq. (5.82) can then be made even smaller. As a consequence of these observations, we find that the best reduction of an n-port signal source depends not only on the noise and signal properties of the source itself but also on the intensity of the amplifier noise, that is on T_2.

5.5. Recapitulation

We have seen above that the noise intensity of any multiport without signal sources can be determined uniquely by a set of invariant numbers. This characterization does not depend on the loading which happens to terminate the multiport. Moreover, in all coordinate systems it is the same set of numbers.

In the formalism we have adopted the invariant numbers have temperature dimension and are called characteristic noise temperatures. They are found to be the eigenvalues of the temperature matrix, which is defined as a generalized matrix analogue of the absolute temperature of a homogeneously warm passive multiport. The eigenvalues of any other matrix which is scalarly proportional to this temperature matrix can serve equally well as characterizing quantities. Such a scalarly proportional matrix, the characteristic-noise matrix, has been defined by Haus and Adler. The negatives of the eigenvalues of the latter matrix are the extrema of the exchangeable noise power of the reduced multiport. Those eigenvalues themselves are equal to extreme values of the noise measure of the amplifier which results if the multiport is reduced to a transfer 2-port.

The best possible noise performance of a multiport without signal sources is determined by the characteristic noise temperature with the smallest absolute value. That characteristic noise temperature must be negative when the multiport is used as an amplifier, and positive when used as an attenuator or an attenuating component.

For signal sources matters are a little more complicated. Again, the intensity of the noise of a multiport signal source is determined uniquely by a set of characteristic noise temperatures. However, with signal sources we are not primarily interested in the noise intensity, but more in the signal- to noise-intensity ratio. Therefore, in most cases the signal and noise properties of a spontane-

ously active multiport are described by both the exchangeable signal power and the ratio of the exchangeable signal and noise powers. If the latter cannot be varied by external means, the best noise performance is obtained by a simple matching procedure, as is mostly the case with 1-port signal sources. If the signal- to noise-power ratio can be varied, as may be the case with multiport signal sources, the adjustment for the best noise performance depends not only on the properties of the signal source itself but also on the remainder of the system. This is quite contrary to the amplifier case. There, the best noise performance is determined only by the amplifier itself.

In order to judge, with regard to noise, the ability of a given amplifier to handle the signal of a particular source a comparison of the significant characteristic noise temperatures of both source and amplifier can be made. The characteristic noise temperature of the latter should preferably be smaller than, but of the same order of magnitude as, the characteristic noise temperature of the former. Apart from this criterion, it has been shown that the characteristic noise temperatures determine the intensity of the noise irrespective of the signal properties. Moreover, they can readily be interpreted in terms of a, well-established and absolute, physical quantity, viz. the concept of absolute temperature.

As a result of all these observations, we wish to conclude this study with the inference that the characteristic noise temperatures are absolute, independent, and unique measure numbers of the spot-noise intensity of all kinds of linear systems.

APPENDIX A

An integral for broadband operation

In sec. 2.4.3, where the noise-power exchange of two 1-ports connected respectively to the two ends of a line is investigated, the integral

$$J = \int_{f_1}^{f_2} [1 + R_1^2 R_2^2 - 2\, R_1 R_2 \cos(\varphi_1 + \varphi_2 + 2\,\beta L)]^{-1} \mathrm{d}f \qquad \text{(A.1)}$$

occurs. Several quantities are defined as follows:

$R_1 \exp(i\varphi_1) = \varrho_1$ = reflection coefficient of one 1-port,
$R_2 \exp(i\varphi_2) = \varrho_2$ = reflection coefficient of the other 1-port,
L = length of the line,
$\beta = \beta(f)$ = propagation constant of the line at the frequency f,
$f_2 - f_1 = \Delta f$ = total bandwidth considered.

The integral J can be evaluated easily when the following conditions apply:

(1) ϱ_1 and ϱ_2 do not vary with frequency, or at least not much,
(2) βL varies strongly with frequency,
(3) L is large.

Let us introduce a new integration variable Ψ given by

$$\Psi = \tfrac{1}{2}(\varphi_1 + \varphi_2) + \beta L. \qquad \text{(A.2)}$$

The integral J then reads

$$J = \int_{\Psi_1}^{\Psi_2} \frac{L^{-1}\,(\mathrm{d}f/\mathrm{d}\beta)\,\mathrm{d}\Psi}{1 + R_1^2 R_2^2 - 2\, R_1 R_2 \cos(2\,\Psi)}, \qquad \text{(A.3)}$$

where

$$\Psi_1 = \tfrac{1}{2}(\varphi_1 + \varphi_2) + L\beta(f_1) = \tfrac{1}{2}(\varphi_1 + \varphi_2) + L\beta_1 \qquad \text{(A.4)}$$

and

$$\Psi_2 = \tfrac{1}{2}(\varphi_1 + \varphi_2) + L\beta(f_2) = \tfrac{1}{2}(\varphi_1 + \varphi_2) + L\beta_2. \qquad \text{(A.5)}$$

We now divide the integration interval into a large number of sub-intervals, each of length π, in such a way that the integral J can be written in the form

$$J = \sum_{j=N}^{N+n-1} \int_{\Psi_j}^{\Psi_{j+1}} \frac{L^{-1}\,(\mathrm{d}f/\mathrm{d}\beta)\,\mathrm{d}\Psi}{1 + R_1^2 R_2^2 - 2\, R_1 R_2 \cos 2\Psi} = \sum_{j=N}^{N+n-1} J_j, \qquad \text{(A.6)}$$

with

$$\Psi_j = (2j + 1)\,\pi/2. \tag{A.7}$$

To meet the condition of eq. (A.7), in eq. (A.6) the limits of integration are shifted slightly but this has a negligible influence only on the ultimate result, as can be verified easily. The length of the complete integration interval is taken to be

$$\Psi_{N+n-1} - \Psi_N = \{2\,(N + n - 1) + 1\}\,\pi/2 - (2\,N + 1)\,\pi/2 = n\pi, \tag{A.8}$$

in which, on account of condition (2), n is a large number. Since condition (3) requires that L is large, the variation of β over a sub-interval, which is given by

$$\beta_{j+1} - \beta_j = \pi/L, \tag{A.9}$$

is small. Hence in a sub-interval the derivative $\mathrm{d}f/\mathrm{d}\beta$ can be taken constant, i.e.

$$\left(\frac{\mathrm{d}f}{\mathrm{d}\beta}\right)_j = \frac{(\Delta f)_j}{(\Delta\beta)_j} = \frac{f_{j+1} - f_j}{\beta_{j+1} - \beta_j}. \tag{A.10}$$

Substitution of the approximation and changing to a new integration variable t, given by

$$t = (1 + R_1R_2)\,(1 - R_1R_2)^{-1} \tan \Psi, \tag{A.11}$$

yield for the jth term of the series of eq. (A.6)

$$J_j = \frac{1}{L}\,\frac{f_{j+1} - f_j}{\beta_{j+1} - \beta_j}\,\frac{1}{1 - R_1^2R_2^2}\int_{-\infty}^{\infty}\frac{\mathrm{d}t}{1 + t^2}. \tag{A.12}$$

Substituting eq. (A.9) and evaluating the remaining integral, we obtain

$$J_j = (f_{j+1} - f_j)\,(1 - R_1^2R_2^2)^{-1}. \tag{A.13}$$

Insertion of this result in eq. (A.6) gives for the integral J the expression

$$J = (f_2 - f_1)\,(1 - R_1^2R_2^2)^{-1} = \Delta f\,(1 - R_1^2R_2^2)^{-1}. \tag{A.14}$$

APPENDIX B

Resonances and oscillations

When two n-ports **S** and **Q** are connected to each other, the appropriate noise- and signal-wave equations can be solved only, if the matrices $(\mathbf{I} - \mathbf{SQ})$ and $(\mathbf{I} - \mathbf{QS})$ are non-singular. In matrix algebra [40]) it is shown that, if one of these matrices is singular, the other is also singular. In this appendix we will investigate the physical meaning of the singularity of, say, $(\mathbf{I} - \mathbf{SQ})$.

Not bothering about their origin, let us assume that a set of, perhaps coherent, waves **X** are incident upon **Q**. Then the reflected waves **Y** are given by

$$\mathbf{Y} = \mathbf{QX}. \tag{B.1}$$

These are incident upon **S**, resulting in the reflected waves **Z**, which are given by

$$\mathbf{Z} = \mathbf{SY} = \mathbf{SQX} \tag{B.2}$$

and which are again incident upon **Q**. Thus, after one turn through the two n-ports a set of waves **X** becomes the set of waves **Z**.

First we shall restrict ourselves to passive n-ports **Q** and **S**. The condition of passivity requires that the total power of **Z** be smaller than, or at most equal to, the total power of **X**, i.e.

$$\mathbf{X}^{\times}\mathbf{X} - \mathbf{Z}^{\times}\mathbf{Z} = \mathbf{X}^{\times}\,(\mathbf{I} - \mathbf{Q}^{\times}\mathbf{S}^{\times}\mathbf{SQ})\,\mathbf{X} \geqslant 0. \tag{B.3}$$

This implies that $(\mathbf{I} - \mathbf{Q}^{\times}\mathbf{S}^{\times}\mathbf{SQ})$ is a positive semi-definite matrix. In sec. 2.6.1 it is shown that, if **Q** and **S** are passive n-ports, the matrices $(\mathbf{I} - \mathbf{Q}^{\times}\mathbf{Q})$ and $(\mathbf{I} - \mathbf{S}^{\times}\mathbf{S})$, the power-absorption matrices $\mathbf{P}_Q$ and $\mathbf{P}_S$, respectively, are positive semi-definite. In appendix C it will be shown that, then, the matrix $(\mathbf{I} - \mathbf{Q}^{\times}\mathbf{S}^{\times}\mathbf{SQ})$ is positive semi-definite. When **Q** and **S** are lossless, that is when both matrices **Q** and **S** are unitary, the equality sign only is valid. From eq. (B.3) we can conclude that any wave present in the closed system **SQ** is decaying or at least not growing.

For some particular wave vectors **X**, the eigenvectors of the matrix **SQ**, the equation

$$\mathbf{Z} = \lambda\mathbf{X} \tag{B.4}$$

holds, where λ is an eigenvalue of **SQ**. Insertion of eq. (B.4) in eq. (B.3) shows that for passive n-ports

$$\lambda\lambda^{*} \leqslant 1. \tag{B.5}$$

The investigation made thus far does not correspond to a physical reality. In a closed passive system no signal source is present to sustain the considered

waves. Only in the situation

$$\mathbf{Z} = \mathbf{X}, \tag{B.6}$$

or when $\lambda = 1$ is an eigenvalue of **SQ**, can the set of (coherent) waves **X** be self-sustaining. Physically this means that **SQ** is resonant. Substitution of eq. (B.6) in eq. (B.2) yields

$$(\mathbf{I} - \mathbf{SQ})\,\mathbf{X} = \mathbf{O}, \tag{B.7}$$

which equation shows that, when $(\mathbf{I} - \mathbf{SQ})$ and, hence, $(\mathbf{I} - \mathbf{QS})$ are singular at a particular frequency, the closed circuit **SQ** is resonant without attenuation at that frequency.

As losses, however small they may be, are always inherent in physical systems, in eq. (B.5) only the inequality symbol has physical meaning. This implies that for a physical system **SQ** cannot have eigenvalues with magnitudes equal to unity and, consequently, that the matrix $(\mathbf{I} - \mathbf{SQ})$ cannot be singular. Hence, the assumptions made in secs 2.2.1 and 2.5.3 are quite sensible.

However, the mathematical difficulty is still present. To meet this one can suppose, among others, that small losses are incorporated in the lines connecting the two n-ports or in the entrance lines or ports of one of the n-ports, say **Q**. This can be effectuated by means of a slightly lossy transmission network, a $2n$-port, which is inserted between **S** and **Q**. The transmission of this network can be described by an nth-order diagonal matrix $\mathbf{I}_\varepsilon$ of which the ith diagonal element $I_{\varepsilon,i}$ is given by

$$I_{\varepsilon,i} = 1 - \varepsilon_i, \tag{B.8}$$

where ε_i is real and satisfies

$$0 < \varepsilon_i < 1. \tag{B.9}$$

Though it is not necessary, to facilitate the calculations ε_i may have the same value ε for all i, i.e.

$$\varepsilon_i = \varepsilon. \tag{B.10}$$

The introduction of $\mathbf{I}_\varepsilon$ in this way can be realized by replacing **Q** by **Q**′, the latter being given by

$$\mathbf{Q}' = \mathbf{I}_\varepsilon \mathbf{Q} \mathbf{I}_\varepsilon = (1 - \varepsilon)^2\,\mathbf{Q}. \tag{B.11}$$

Then, as an approximation, for small ε we have

$$\mathbf{X}^\times(\mathbf{I} - \mathbf{Q}'^\times \mathbf{S}^\times \mathbf{S}\mathbf{Q}')\,\mathbf{X} \approx \mathbf{X}^\times\,(\mathbf{I} - \mathbf{Q}^\times \mathbf{S}^\times \mathbf{S}\mathbf{Q})\,\mathbf{X} + 4\,\varepsilon \mathbf{X}^\times \mathbf{Q}^\times \mathbf{S}^\times \mathbf{S}\mathbf{Q}\mathbf{X}. \tag{B.12}$$

Since $\mathbf{Q}^\times \mathbf{S}^\times \mathbf{SQ}$ is a positive semi-definite Hermitian matrix, the second term in the right-hand side is non-negative for any **X**. For those non-zero **X**, if any, which make the second term zero, the first term of the right-hand side is positive. Hence, for all non-zero **X**, the relation

$$\mathbf{X}^{\times} (\mathbf{I} - \mathbf{Q}'^{\times}\mathbf{S}^{\times}\mathbf{S}\mathbf{Q}') \mathbf{X} > 0 \qquad \text{(B.13)}$$

applies, so that we can conclude that the matrix $(\mathbf{I} - \mathbf{S}\mathbf{Q}')$ cannot be singular.

When at a particular frequency noise- or signal-wave equations cannot be solved on account of singularity of $(\mathbf{I} - \mathbf{S}\mathbf{Q})$, it is always possible to solve the equations obtained by introducing a modified matrix $\mathbf{Q}'$. The ultimate result for $\mathbf{Q}$ can then be obtained by taking the limit of the results for ε tending to zero.

When the n-ports $\mathbf{S}$ and $\mathbf{Q}$ are not both passive, matters are more complicated. The inequality of eq. (B.3) need then not apply and the eigenvalues λ of $\mathbf{S}\mathbf{Q}$ can then have magnitudes larger than unity. The power of the $\mathbf{Z}$ waves can then be larger than that of the $\mathbf{X}$ waves. This means that amplification occurs and that the waves are growing. If some phase conditions, which will not be discussed here, are also fulfilled, the system then oscillates. When this is not the case at the frequency at which the actual considerations are being made, it probably happens at some nearby frequency. Such oscillations which, due to saturation effects, do not grow beyond any limit deteriorate the signal performance of the system. Anyhow, the system is highly unstable and not a very practical one. Thus, closed systems $\mathbf{S}\mathbf{Q}$ of which the matrices $(\mathbf{I} - \mathbf{Q}^{\times}\mathbf{S}^{\times}\mathbf{S}\mathbf{Q})$ are indefinite, or perhaps negative definite, are of suspicious practical value due to instabilities.

APPENDIX C

Some matrix-algebraic theorems

In this appendix some more or less familiar theorems of matrix algebra used in the present treatise are given, together with their proofs.

Theorem I. The trace of a product **AB** of two arbitrary quadratic matrices **A** and **B** is equal to the trace of the commuted product **BA** and, hence, the sums of the eigenvalues of these products are the same.

This theorem can be proved by changing the order of summation in the evaluation of the trace of one of these products, say of **AB**. We have

$$\text{trace}\,(\mathbf{AB}) = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} B_{ji} =$$

$$= \sum_{j=1}^{n} \sum_{i=1}^{n} B_{ji} A_{ij} = \text{trace}\,(\mathbf{BA}), \tag{C.1}$$

q.e.d.

Theorem II. For any pair of quadratic matrices **A** and **B** the matrix $(\mathbf{I} - \mathbf{AB})$ has the same eigenvalues as the matrix $(\mathbf{I} - \mathbf{BA})$ and with the same multiplicity.

A proof of this theorem is given in ref. 61. We shall give here an alternative proof, using different matrix-algebraic methods which are more in accordance with the content of this treatise. When both matrices **A** and **B** are symmetrical or when at least one of them is non-singular, the theorem can be proved easily. In the more general case, however, the proof is a bit more intricate.

Let λ_i be an eigenvalue of $(\mathbf{I}-\mathbf{AB})$ and the column vector $\mathbf{X}_i$ the corresponding eigenvector. The equation

$$(\mathbf{I} - \mathbf{AB})\,\mathbf{X}_i = \lambda_i \mathbf{X}_i \tag{C.2}$$

is then valid. It is assumed that $\mathbf{X}_i$ is not identical with the zero vector, for that would lead to a trivial result. Premultiplying both sides of eq. (C.2) with **B** yields

$$(\mathbf{I} - \mathbf{BA})\,\mathbf{BX}_i = \lambda_i \mathbf{BX}_i, \tag{C.3}$$

which with

$$\mathbf{Y}_i = \mathbf{BX}_i \tag{C.4}$$

results in

$$(\mathbf{I} - \mathbf{BA})\,\mathbf{Y}_i = \lambda_i \mathbf{Y}_i. \tag{C.5}$$

This equation implies that λ_i is also an eigenvalue of $(\mathbf{I} - \mathbf{BA})$, now with the eigenvector $\mathbf{Y}_i$, except when $\mathbf{Y}_i$ is the zero vector, for then eq. (C.5) is trivial. If $\mathbf{Y}_i$ is identical with the zero vector, due to eq. (C.4), $\mathbf{X}_i$ satisfies

$$\mathbf{BX}_i = \mathbf{O}. \tag{C.6}$$

Substitution in eq. (C.2) reduces that equation to

$$\mathbf{X}_i = \lambda_i \mathbf{X}_i \tag{C.7}$$

or

$$\lambda_i = 1. \tag{C.8}$$

Now, we can conclude that any eigenvalue λ_i of $(\mathbf{I} - \mathbf{AB})$, which is not equal to unity, is also an eigenvalue of $(\mathbf{I} - \mathbf{BA})$.

That the multiplicities $p_{i(\mathbf{AB})}$ and $p_{i(\mathbf{BA})}$ of such an eigenvalue corresponding to the two matrices, respectively, are the same can be shown as follows. When the multiplicity $p_{i(\mathbf{AB})}$ of the eigenvalue λ_i of $(\mathbf{I} - \mathbf{AB})$ is larger than unity, more than one eigenvector $\mathbf{X}_i$ correspond to λ_i. Since singularity of $\mathbf{B}$ is not excluded, two or more of these $\mathbf{X}_i$'s may be transformed by $\mathbf{B}$ to the same $\mathbf{Y}_i$, resulting in

$$p_{i(\mathbf{AB})} \geqslant p_{i(\mathbf{BA})}. \tag{C.9}$$

To compensate for the loss in multiplicity of λ_i the matrix $(\mathbf{I} - \mathbf{BA})$ cannot have another eigenvalue which is unequal to unity and which is not an eigenvalue of the matrix $(\mathbf{I} - \mathbf{AB})$ at the same time, for, as the reverse of the above observation may show, such another eigenvalue must be an eigenvalue of $(\mathbf{I} - \mathbf{AB})$, too. Reversing the above considerations (starting with $(\mathbf{I} - \mathbf{BA})$) is even more profitable, as it leads to

$$p_{i(\mathbf{AB})} \leqslant p_{i(\mathbf{BA})}. \tag{C.10}$$

The equations (C.9) and (C.10) can be satisfied simultaneously only when the equality sign is valid. Hence, we have

$$p_{i(\mathbf{AB})} = p_{i(\mathbf{BA})}. \tag{C.11}$$

When one or both of the matrices $(\mathbf{I} - \mathbf{AB})$ and $(\mathbf{I} - \mathbf{BA})$ have an eigenvalue equal to unity, which is the case when at least one of the matrices $\mathbf{A}$ and $\mathbf{B}$ is singular, we still have to investigate the relation between the two respective multiplicities $p_{1(\mathbf{AB})}$ and $p_{1(\mathbf{BA})}$. Application of theorem I about the equality of the sums of eigenvalues leads to

$$p_{1(\mathbf{AB})} \cdot 1 + \sum_i p_{i(\mathbf{AB})}\lambda_i = p_{1(\mathbf{BA})} \cdot 1 + \sum_i p_{i(\mathbf{BA})}\lambda_i. \tag{C.12}$$

By virtue of the conclusions drawn above, this reduces to

$$p_{1(\mathbf{AB})} = p_{1(\mathbf{BA})}, \tag{C.13}$$

with which the theorem is proved.

Theorem III. For any pair of quadratic matrices **A** and **B**, on the assumption of non-singularity of $(\mathbf{I} - \mathbf{AB})$, the matrix $(\mathbf{I} - \mathbf{AB})^{-1}\mathbf{A}$ is equal to the matrix $\mathbf{A}(\mathbf{I} - \mathbf{BA})^{-1}$.

Premultiplying both sides of

$$(\mathbf{I} - \mathbf{AB})^{-1}\mathbf{A} = \mathbf{A}\,(\mathbf{I} - \mathbf{BA})^{-1} \tag{C.14}$$

by $(\mathbf{I} - \mathbf{AB})$ and postmultiplying them by $(\mathbf{I} - \mathbf{BA})$ yields

$$\mathbf{A} - \mathbf{ABA} = \mathbf{A} - \mathbf{ABA}, \tag{C.15}$$

which proves the equality.

Theorem IV. For any pair of quadratic matrices **A** and **B**, providing the matrix $(\mathbf{I} - \mathbf{AB})$ be non-singular, the matrix $(\mathbf{I} - \mathbf{AB})^{-1}$ has the same eigenvalues and the same trace as the matrix $(\mathbf{I} - \mathbf{BA})^{-1}$.

When $(\mathbf{I} - \mathbf{AB})$ is non-singular, it has only non-zero eigenvalues. Since according to theorem II, $(\mathbf{I} - \mathbf{BA})$ has the same non-zero eigenvalues, that matrix is non-singular too. Hence, both the inverse matrices $(\mathbf{I} - \mathbf{AB})^{-1}$ and $(\mathbf{I} - \mathbf{BA})^{-1}$ exist. On the other hand, the eigenvalues of an inverse matrix are the inverses of the eigenvalues of the inverted matrix. Hence, $(\mathbf{I} - \mathbf{AB})^{-1}$ and $(\mathbf{I} - \mathbf{BA})^{-1}$ have the same eigenvalues, from which we may conclude that the sums of their eigenvalues, that is their traces, are equal.

Theorem V. For any pair of symmetric matrices **A** and **B**, providing the matrix $(\mathbf{I} - \mathbf{AB})$ be non-singular, the matrix $(\mathbf{I} - \mathbf{AB})^{-1}$ has the same diagonal elements as the matrix $(\mathbf{I} - \mathbf{BA})^{-1}$.

In the same way as in the preceding theorem the existence of $(\mathbf{I} - \mathbf{AB})^{-1}$ and $(\mathbf{I} - \mathbf{BA})^{-1}$ can be proved. Furthermore, transposition of a matrix does not alter the diagonal elements. Thus

$$\{(\mathbf{I} - \mathbf{AB})^{-1}\}_{ii} = \{[(\mathbf{I} - \mathbf{AB})^{-1}]^{\mathrm{T}}\}_{ii} = \{(\mathbf{I} - \mathbf{B}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}})^{-1}\}_{ii}, \tag{C.16}$$

which with the symmetry of **A** and **B** proves the theorem.

Theorem VI. For a diagonal matrix $\mathbf{D} = \mathrm{diag}\,(d_i)$ and an arbitrary quadratic matrix **A**, on the assumption of the non-singularity of $(\mathbf{I} - \mathbf{DA})$, the relation $d_i\{(\mathbf{I} - \mathbf{DA})^{-1}\}_{ij} = d_j\{(\mathbf{I} - \mathbf{AD})^{-1}\}_{ij}$ is valid.

From theorem III we have

$$\mathbf{D}\,(\mathbf{I} - \mathbf{AD})^{-1} = (\mathbf{I} - \mathbf{DA})^{-1}\,\mathbf{D}. \tag{C.17}$$

This implies

$$[\mathbf{D}\,(\mathbf{I} - \mathbf{AD})^{-1}]_{ij} = [(\mathbf{I} - \mathbf{DA})^{-1}\,\mathbf{D}]_{ij}, \tag{C.18}$$

which can be written in the form

$$\sum_{k=1}^{n} D_{ik} \{(\mathbf{I} - \mathbf{AD})^{-1}\}_{kj} = \sum_{l=1}^{n} \{(\mathbf{I} - \mathbf{DA})^{-1}\}_{il} D_{lj}. \tag{C.19}$$

By virtue of the diagonality of **D**, this leads to

$$d_i \{(\mathbf{I} - \mathbf{AD})^{-1}\}_{ij} = \{(\mathbf{I} - \mathbf{DA})^{-1}\}_{ij} d_j, \tag{C.20}$$

q.e.d.

For $i = j$ this yields

$$(\mathbf{I} - \mathbf{DA})_{ii}^{-1} = (\mathbf{I} - \mathbf{AD})_{ii}^{-1}, \tag{C.21}$$

so that we can formulate the corollary (cf. theorem V):

The matrices $(\mathbf{I} - \mathbf{DA})^{-1}$ and $(\mathbf{I} - \mathbf{AD})^{-1}$ have the same diagonal elements.

Theorem VII. Positive semi-definiteness of $(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A})$ implies positive semi-definiteness of $(\mathbf{I} - \mathbf{AA}^{\times})$.

We first observe that both matrices are Hermitian and that, by virtue of theorem II, they have the same eigenvalues. Due to the positive semi-definiteness of $(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A})$, these eigenvalues are non-negative. Conversely, this implies that $(\mathbf{I} - \mathbf{AA}^{\times})$ is positive semi-definite.

Theorem VIII. If the Hermitian matrices **M** and $(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A})$ are positive semi-definite, the trace of the matrix $(\mathbf{M} - \mathbf{AMA}^{\times})$ is non-negative.

In matrix algebra it is well known that a positive semi-definite Hermitian matrix can be written as a Gaussian transform of a quadratic matrix. Hence, **M** can be written in the form

$$\mathbf{M} = \mathbf{BB}^{\times}, \tag{C.22}$$

from which we have

$$\mathbf{M} - \mathbf{AMA}^{\times} = \mathbf{BB}^{\times} - \mathbf{ABB}^{\times}\mathbf{A}^{\times} = \mathbf{BB}^{\times} - (\mathbf{AB})(\mathbf{AB})^{\times}. \tag{C.23}$$

By virtue of theorem I, we then find

$$\begin{aligned} \text{trace}\,(\mathbf{M} - \mathbf{AMA}^{\times}) &= \text{trace}\,\{\mathbf{B}^{\times}\mathbf{B} - (\mathbf{AB})^{\times}(\mathbf{AB})\} = \\ &= \text{trace}\,\{\mathbf{B}^{\times}(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A})\,\mathbf{B}\}. \end{aligned} \tag{C.24}$$

Since $(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A})$ is positive semi-definite, the matrix $\mathbf{B}^{\times}(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A})\,\mathbf{B}$ is positive semi-definite and, consequently, has a non-negative trace. Hence, $(\mathbf{M} - \mathbf{AMA}^{\times})$ has a non-negative trace, q.e.d.

If **M** is positive definite, the matrix **B** is non-singular. If, furthermore, $(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A})$ is positive definite, the matrix $\mathbf{B}^{\times}(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A})\,\mathbf{B}$ is positive so that the trace of $(\mathbf{M} - \mathbf{AMA}^{\times})$ is then positive.

Theorem IX. If the matrices $(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A})$ and $(\mathbf{I} - \mathbf{B}^{\times}\mathbf{B})$ are positive semi-definite, the matrix $(\mathbf{I} - \mathbf{A}^{\times}\mathbf{B}^{\times}\mathbf{BA})$ is positive semi-definite.

For any vector $\mathbf{X}$ one can write down

$$\mathbf{X}^{\times}(\mathbf{I} - \mathbf{A}^{\times}\mathbf{B}^{\times}\mathbf{BA})\,\mathbf{X} = \mathbf{X}^{\times}(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A} + \mathbf{A}^{\times}\mathbf{A} - \mathbf{A}^{\times}\mathbf{B}^{\times}\mathbf{BA})\,\mathbf{X} = \\ = \mathbf{X}^{\times}(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A})\,\mathbf{X} + \mathbf{X}^{\times}\mathbf{A}^{\times}(\mathbf{I} - \mathbf{B}^{\times}\mathbf{B})\,\mathbf{AX}. \quad \text{(C.25)}$$

Positive semi-definiteness of $(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A})$ implies

$$\mathbf{X}^{\times}(\mathbf{I} - \mathbf{A}^{\times}\mathbf{A})\,\mathbf{X} \geqslant 0 \quad \text{(C.26)}$$

and positive semi-definiteness of $(\mathbf{I} - \mathbf{B}^{\times}\mathbf{B})$ yields

$$(\mathbf{AX})^{\times}(\mathbf{I} - \mathbf{B}^{\times}\mathbf{B})\,(\mathbf{AX}) \geqslant 0. \quad \text{(C.27)}$$

Hence, both terms of the last side of eq. (C.25) are non-negative, so that

$$\mathbf{X}^{\times}(\mathbf{I} - \mathbf{A}^{\times}\mathbf{B}^{\times}\mathbf{BA})\,\mathbf{X} \geqslant 0, \quad \text{(C.28)}$$

which equation is a necessary and sufficient condition for the positive semi-definiteness of $(\mathbf{I} - \mathbf{A}^{\times}\mathbf{B}^{\times}\mathbf{BA})$.

Theorem X. For any pair of quadratic matrices $\mathbf{A}$ and $\mathbf{B}$, providing the matrix $(\mathbf{I} - \mathbf{AB})$ be non-singular, the equation
$(\mathbf{I} - \mathbf{A})\,(\mathbf{I} - \mathbf{BA})^{-1}(\mathbf{I} - \mathbf{B}) = (\mathbf{I} - \mathbf{B})\,(\mathbf{I} - \mathbf{AB})^{-1}(\mathbf{I} - \mathbf{A})$ is valid.

For the existence and the non-singularity of the matrices $(\mathbf{I} - \mathbf{BA})^{-1}$ and $(\mathbf{I} - \mathbf{AB})^{-1}$ we may refer to theorem IV. Pre- and postmultiplication of the above equation by respectively $(\mathbf{I} - \mathbf{BA})$ and $(\mathbf{I} - \mathbf{AB})$ yields

$$(\mathbf{I} - \mathbf{B})\,(\mathbf{I} - \mathbf{AB}) - (\mathbf{I} - \mathbf{BA})\,\mathbf{A}\,(\mathbf{I} - \mathbf{BA})^{-1}(\mathbf{I} - \mathbf{AB}) + \\ + (\mathbf{I} - \mathbf{BA})\,\mathbf{A}\,(\mathbf{I} - \mathbf{BA})^{-1}\mathbf{B}\,(\mathbf{I} - \mathbf{AB}) = \\ = (\mathbf{I} - \mathbf{BA})\,(\mathbf{I} - \mathbf{B}) - (\mathbf{I} - \mathbf{BA})\,(\mathbf{I} - \mathbf{AB})^{-1}\mathbf{A}\,(\mathbf{I} - \mathbf{AB}) + \\ + (\mathbf{I} - \mathbf{BA})\,\mathbf{B}\,(\mathbf{I} - \mathbf{AB})^{-1}\mathbf{A}\,(\mathbf{I} - \mathbf{AB}). \quad \text{(C.29)}$$

Application of theorem III shows that the middle terms of both sides are equal while the last terms can be simplified. The result is

$$(\mathbf{I} - \mathbf{B} - \mathbf{AB} + \mathbf{BAB}) + (\mathbf{AB} - \mathbf{BAAB}) = \\ = (\mathbf{I} - \mathbf{BA} - \mathbf{B} + \mathbf{BAB}) + (\mathbf{BA} - \mathbf{BAAB}), \quad \text{(C.30)}$$

which equation is readily seen to be valid.

APPENDIX D

Transformation and imbedding conditions

D.1. *Identity of the two transformation conditions*

In sec. 3.1.5 it has been stated that the excitation- and noise-transformation matrices are simultaneously either singular or non-singular. We will here give a proof of that statement.

Let $\mathbf{\Phi}$ and $\mathbf{\Psi}$ be the two non-singular transformation matrices. They are each others inverses: eq. (3.10). In the detailed form that relationship reads

$$\mathbf{\Phi}_{11}\mathbf{\Psi}_{11} + \mathbf{\Phi}_{12}\mathbf{\Psi}_{21} = \mathbf{I}\,, \tag{D.1}$$

$$\mathbf{\Phi}_{11}\mathbf{\Psi}_{12} + \mathbf{\Phi}_{12}\mathbf{\Psi}_{22} = \mathbf{O}, \tag{D.2}$$

$$\mathbf{\Phi}_{21}\mathbf{\Psi}_{11} + \mathbf{\Phi}_{22}\mathbf{\Psi}_{21} = \mathbf{O}, \tag{D.3}$$

$$\mathbf{\Phi}_{21}\mathbf{\Psi}_{12} + \mathbf{\Phi}_{22}\mathbf{\Psi}_{22} = \mathbf{I}\,. \tag{D.4}$$

Let the sub-matrix $\mathbf{\Psi}_{21}$ be singular and let $\mathbf{v}_i$ be its set of column null vectors. Operation of the matrix of eq. (D.1) upon one of the $\mathbf{v}_i$'s gives

$$(\mathbf{\Phi}_{11}\mathbf{\Psi}_{11} + \mathbf{\Phi}_{12}\mathbf{\Psi}_{21})\,\mathbf{v}_i = \mathbf{\Phi}_{11}\mathbf{\Psi}_{11}\mathbf{v}_i = \mathbf{v}_i. \tag{D.5}$$

The $\mathbf{v}_i$'s cannot be null vectors of $\mathbf{\Psi}_{11}$. Operation of the matrix of eq. (D.3) upon one of the $\mathbf{v}_i$'s gives

$$(\mathbf{\Phi}_{21}\mathbf{\Psi}_{11} + \mathbf{\Phi}_{22}\mathbf{\Psi}_{21})\,\mathbf{v}_i = \mathbf{\Phi}_{21}\,(\mathbf{\Psi}_{11}\mathbf{v}_i) = \mathbf{O}. \tag{D.6}$$

From this result it can be seen that the non-zero vector $\mathbf{\Psi}_{11}\mathbf{v}_i$ must be a null vector of $\mathbf{\Phi}_{21}$ so that $\mathbf{\Phi}_{21}$ must be singular. Conversely, making use of the alternative form of eq. (3.10), i.e. $\mathbf{\Psi\Phi} = \mathbf{I}$, it can be shown that singularity of $\mathbf{\Phi}_{21}$ implies singularity of $\mathbf{\Psi}_{21}$. Furthermore, the fact that the matrices $\mathbf{\Phi}_{21}$ and $\mathbf{\Psi}_{21}$ are simultaneously singular implies that, if any of those matrices is non-singular, the other one must be so.

The excitation- and noise-transformation matrices are given respectively by

$$\boldsymbol{Y} = \mathbf{\Phi}_{21}\mathbf{S} + \mathbf{\Phi}_{22} \tag{D.7}$$

and

$$\mathbf{\Delta} = \mathbf{\Psi}_{11} - \mathbf{S}\mathbf{\Psi}_{21}. \tag{D.8}$$

Multiplying $\mathbf{\Delta}$ from the left by $\mathbf{\Phi}_{21}$ and $\boldsymbol{Y}$ from the right by $\mathbf{\Psi}_{21}$ and adding the results, we obtain, by virtue of eq. (D.3) (cf. eq. (3.34)),

$$\mathbf{\Phi}_{21}\mathbf{\Delta} + \boldsymbol{Y}\mathbf{\Psi}_{21} = \mathbf{\Phi}_{21}\mathbf{\Psi}_{11} - \mathbf{\Phi}_{21}\mathbf{S}\mathbf{\Psi}_{21} + \mathbf{\Phi}_{21}\mathbf{S}\mathbf{\Psi}_{21} + \mathbf{\Phi}_{22}\mathbf{\Psi}_{21} = \mathbf{O}. \tag{D.9}$$

From this equation it is readily seen that, when both $\mathbf{\Phi}_{21}$ and $\mathbf{\Psi}_{21}$ are non-singular, the matrices $\boldsymbol{Y}$ and $\mathbf{\Delta}$ are either simultaneously singular or simultaneously non-singular.

When both $\mathbf{\Phi}_{21}$ and $\mathbf{\Psi}_{21}$ are singular, matters are a little more complicated. As a basis of n-dimensional vector space we take the eigenvectors of $\mathbf{\Psi}_{21}$. Let the $\mathbf{v}_k$'s be the null vectors and the $\mathbf{w}_p$'s the other eigenvectors. Let further $\boldsymbol{Y}$ be non-singular. When the matrix of eq. (D.9) operates on one of the $\mathbf{v}_k$'s, we have

$$(\mathbf{\Phi}_{21}\mathbf{\Delta} + \boldsymbol{Y}\mathbf{\Psi}_{21})\,\mathbf{v}_k = \mathbf{\Phi}_{21}\mathbf{\Delta}\mathbf{v}_k = \mathbf{O}. \tag{D.10}$$

Moreover, we find, by virtue of eq. (D.8),

$$\mathbf{\Delta}\mathbf{v}_k = \mathbf{\Psi}_{11}\mathbf{v}_k - \mathbf{S}\mathbf{\Psi}_{21}\mathbf{v}_k = \mathbf{\Psi}_{11}\mathbf{v}_k. \tag{D.11}$$

Above it has been shown that this is unequal to the zero vector, and that eq. (D.10) is satisfied. We conclude that the $\mathbf{v}_k$'s are not null vectors of $\mathbf{\Delta}$.When eq. (D.9) operates upon one of the $\mathbf{w}_p$'s, we have

$$\mathbf{\Phi}_{21}\mathbf{\Delta}\mathbf{w}_p = -\boldsymbol{Y}\mathbf{\Psi}_{21}\mathbf{w}_p. \tag{D.12}$$

Since $\boldsymbol{Y}$ is non-singular, this vector is unequal the zero vector. Hence, $\mathbf{w}_p$ cannot be a null vector of $\mathbf{\Delta}$.

We can conclude that, if $\boldsymbol{Y}$ is non-singular, $\mathbf{\Delta}$ is so. The converse is also true and can be proved analogously. Moreover, as a corrollary, it is easy to prove that $\boldsymbol{Y}$ and $\mathbf{\Delta}$ are both singular when one of them is so.

D.2. *Imbedding relations*

In sec. 3.6 the relations that govern the imbedding of an n-port in a $2n$-port have been derived. The $2n$-port has there been described by its transfer matrix. Instead of by its transfer matrix, the $2n$-port can be represented as well by its scattering matrix or by any other signal-performance matrix. In the scattering-matrix signal representation the formulation of imbedding conditions is very lucid. We therefore will now formulate the imbedding relations in that signal representation.

The imbedding is again illustrated in fig. 30. The meaning of the symbols is the same as in sec. 3.6. The original n-port $\mathbf{S}$ and the resultant n-port $\mathbf{S}'$ are governed respectively by (cf. eqs (3.139) and (3.140))

$$\mathbf{B} = \mathbf{S}\mathbf{A} + \mathbf{X} \tag{D.13}$$

and

$$\mathbf{B}' = \mathbf{S}'\mathbf{A}' + \mathbf{X}'. \tag{D.14}$$

The scattering-matrix version of the corresponding equation of the $2n$-port can at once be written in the partitioned form

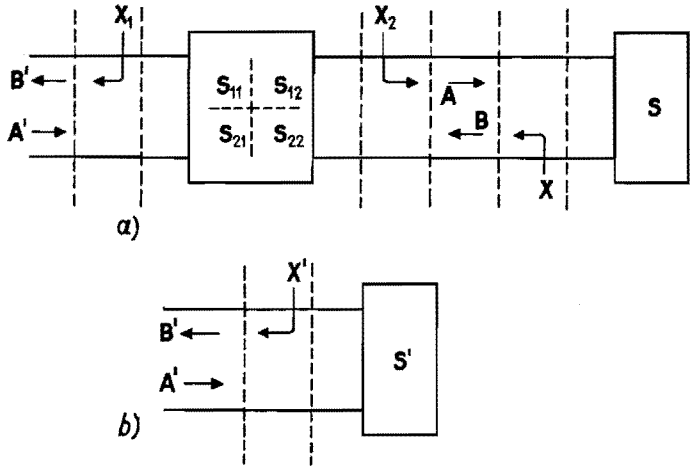


Fig. 30. Imbedding of an n-port in a $2n$-port. The imbedding $2n$-port is represented by a scattering matrix.

$$\mathbf{B}' = \mathbf{S}_{11}\mathbf{A}' + \mathbf{S}_{12}\mathbf{B} + \mathbf{X}_1 \tag{D.15}$$

and

$$\mathbf{A} = \mathbf{S}_{21}\mathbf{A}' + \mathbf{S}_{22}\mathbf{B} + \mathbf{X}_2. \tag{D.16}$$

We now wish to solve for $\mathbf{S}'$ and $\mathbf{X}'$. Eliminating $\mathbf{A}$ and $\mathbf{B}$ from eqs (D.13), (D.15), and (D.16) and equating the result to eq. (D.14), we obtain

$$\mathbf{S}' = \mathbf{S}_{11} + \mathbf{S}_{12}(\mathbf{I} - \mathbf{S}\mathbf{S}_{22})^{-1}\mathbf{S}\mathbf{S}_{21} \tag{D.17}$$

and

$$\mathbf{X}' = \mathbf{\Gamma}_S\mathbf{X} + \mathbf{\Gamma}_S\mathbf{S}\mathbf{X}_2 + \mathbf{X}_1. \tag{D.18}$$

The noise-transformation matrix $\mathbf{\Gamma}_S$ is given by

$$\mathbf{\Gamma}_S = \mathbf{S}_{12}(\mathbf{I} - \mathbf{S}\mathbf{S}_{22})^{-1}. \tag{D.19}$$

If the noise is neglected, substitution of eq. (D.13) in eq. (D.16) gives

$$\mathbf{A}' = \mathbf{Y}_S\mathbf{A}, \tag{D.20}$$

where the excitation-transformation matrix $\mathbf{Y}_S$ is given by

$$\mathbf{Y}_S = \mathbf{S}_{21}{}^{-1}(\mathbf{I} - \mathbf{S}_{22}\mathbf{S}). \tag{D.21}$$

The power absorbed in the original n-port $\mathbf{S}$ is

$$P_{\text{abs},S} = \mathbf{A}^{\times}\mathbf{P}_S\mathbf{A} = \mathbf{A}'^{\times}(\mathbf{Y}_S^{-1})^{\times}\mathbf{P}_S\mathbf{Y}_S^{-1}\mathbf{A}'. \tag{D.22}$$

The power dissipated in the imbedding $2n$-port $\mathbf{S}_{2n}$ is

$$P_{\text{abs},S_{2n}} = \left(\frac{\mathbf{A}'}{\mathbf{B}}\right)^{\times} \mathbf{P}_{S_{2n}} \left(\frac{\mathbf{A}'}{\mathbf{B}}\right) =$$

$$= \mathbf{A}'^{\times} \left(\frac{\mathbf{I}}{\mathbf{S}\boldsymbol{Y}_S^{-1}}\right)^{\times} \mathbf{P}_{S_{2n}} \left(\frac{\mathbf{I}}{\mathbf{S}\boldsymbol{Y}_S^{-1}}\right) \mathbf{A}', \tag{D.23}$$

where $\mathbf{P}_{S_{2n}}$ is the power-absorption matrix of the $2n$-port. The power dissipated in the resultant n-port S′ is equal to the sum of the above powers. It can be written in the form

$$P_{\text{abs},S'} = \mathbf{A}'^{\times} \mathbf{P}_{S'} \mathbf{A}'. \tag{D.24}$$

Hence, the resultant power-absorption matrix $\mathbf{P}_{S'}$ is found to be

$$\mathbf{P}_{S'} = \mathbf{P}_S' + \mathbf{P}_{S_{2n}}', \tag{D.25}$$

where

$$\mathbf{P}_S' = (\boldsymbol{Y}_S^{-1})^{\times} \mathbf{P}_S \boldsymbol{Y}_S^{-1} \tag{D.26}$$

and

$$\mathbf{P}_{S_{2n}}' = \left(\frac{\mathbf{I}}{\mathbf{S}\boldsymbol{Y}_S^{-1}}\right)^{\times} \mathbf{P}_{S_{2n}} \left(\frac{\mathbf{I}}{\mathbf{S}\boldsymbol{Y}_S^{-1}}\right). \tag{D.27}$$

From eq. (D.18) we obtain for the noise-source power matrix $\mathbf{G}_{S'}$ of the resultant n-port the equation

$$\mathbf{G}_{S'} = \mathbf{G}_S' + \mathbf{G}_{S_{2n}}', \tag{D.28}$$

where

$$\mathbf{G}_S' = \boldsymbol{\Gamma}_S \mathbf{G}_S \boldsymbol{\Gamma}_S^{\times} \tag{D.29}$$

and

$$\mathbf{G}_{S_{2n}}' = (\mathbf{I} \,\vdots\, \boldsymbol{\Gamma}_S \mathbf{S})\, \mathbf{G}_{S_{2n}}\, (\mathbf{I} \,\vdots\, \boldsymbol{\Gamma}_S \mathbf{S})^{\times}. \tag{D.30}$$

All further considerations made in sec. 3.6 are here, too, valid. The statements, made in sec. 3.7, about the analogy of the transfer matrix of a lossless $2n$-port and the matrix of a canonical coordinate transformation do not apply. Of course, a lossless imbedding remains equivalent to a canonical coordinate transformation, but the scattering matrix itself cannot be identified with the matrix of a canonical coordinate transformation.

All above equations could also have been deduced by direct matrix-algebraic means. However, the calculations involved would have been very laborious. Furthermore, it is easy to show that the above equations are identical with the corresponding ones in sec. 3.6. For example, by virtue of eq. (3.120), it is readily found that

$$\boldsymbol{\Gamma}_S = \boldsymbol{\Gamma}_\Theta \tag{D.31}$$

and

$$\boldsymbol{Y}_S = \boldsymbol{Y}_\Theta. \tag{D.32}$$

D.3. *Conditions*

In the preceding section we have not bothered about the conditions that must be fulfilled for the observations to be valid. Below we will pay some attention to these conditions; the more so since they can be physically interpreted much more easily when the scattering-matrix representation of the $2n$-port is employed than when use is made of its transfer-matrix representation.

D.3.1. Resonances: $(\mathbf{I}-\mathbf{S}_{22}\mathbf{S})$ non-singular

Elimination of **A** and **B** can be carried out only if the matrix $(\mathbf{I}-\mathbf{S}_{22}\mathbf{S})$ is non-singular. Referring to appendix B, we can observe that the physical meaning of this condition is that the system as formed by the original and imbedding multiports must not have undamped internal resonances. Such resonances cannot be detected at the input of the imbedding multiport, so that, when resonances are present, the signal state of the original multiport does not depend uniquely upon that of the resultant multiport.

This condition does not prohibit either the original or the imbedding multiport from having internal resonances themselves, for, if present, such resonances cannot be detected by any means and, therefore, they do not influence the results of investigations at the ports. Of course, such internal resonances can influence the properties of a multiport, and mostly they do, but that cannot be verified from the outside.

D.3.2. Output-input coupling: $\mathbf{S}_{12}$ non-singular

The transformation of the noise of the original multiport towards the input of the imbedding multiport is unambiguous only if the noise-transformation matrix $\mathbf{\Gamma}_S$ is non-singular (cf. eq. (D.18)). Hence, from eq. (D.19) it is clear that in addition to the above condition the sub-matrix $\mathbf{S}_{12}$ must be non-singular. This signifies that the coupling from the output towards the input of the imbedding multiport must be unique.

When this condition is not fulfilled, $\mathbf{\Gamma}_S$ has eigenvectors with zero eigenvalues. This means that those noise vectors of original multiports, or those parts of the noise vectors, which are equal to these eigenvectors, have no images at the input of the imbedding multiport. Consequently, the corresponding noise, or part of it, cannot be perceived at the input of the imbedding multiport.

For a complete unambiguous reduction of the noise of a $2n$-port a further condition applies. From eq. (D.18) it is easy to see that the scattering matrix **S** of a reducing n-port (the original n-port) must be non-singular. This additional condition can be interpreted in the same way as the previous one.

D.3.3. Input-output coupling: $\mathbf{S}_{21}$ non-singular

The powers absorbed respectively in the imbedded and the imbedding multi-

ports should be determined uniquely by the signal state of the resultant multiport. This is true only if the excitation-transformation matrix $\boldsymbol{Y}_S$ is non-singular. Hence, if again the first condition is satisfied, by virtue of eq. (D.21), the matrix $\mathbf{S}_{21}$ must be non-singular. This means that the coupling from the input towards the output of the imbedding multiport must be unambiguous. The physical interpretation of this condition is analogous to that of the preceding condition.

It may be worth recalling that the present condition is equivalent to the criterion that makes representing the imbedding $2n$-port by a transfer matrix make sense (cf. sec. 3.5.2). For lossless imbeddings this equivalence is quite logical, because the transfer matrix can then be identified with the corresponding canonical coordinate-transformation matrix. A complete and unique image obtainable with one method must be obtainable with the other, and vice versa. Hence, if the present condition applies, both a coordinate-transformation matrix and a transfer matrix make sense and are unambiguous.

When the imbedding $2n$-port is lossless, i.e.

$$\mathbf{S}_{2n}\,\mathbf{S}_{2n}{}^{\times} = \mathbf{I}, \tag{D.33}$$

it can analogously to the derivation in sec. D.1 be shown that non-singularity of $\mathbf{S}_{12}$ implies non-singularity of $\mathbf{S}_{21}$, and conversely (cf. sec. 3.5.2). The conditions D.3.2 and D.3.3 are then identical.

From eq. (D.23), it will be clear that for the reduction of a $2n$-port to be unique the scattering matrix $\mathbf{S}$ of the reducing n-port must again be non-singular.

APPENDIX E

The diagonalizing noise-transformation matrix *)

For a noisy n-port with a diagonalized and normalized signal performance, as in sec. 4.1.3, a noise-transformation matrix can be derived which transforms the noise-source power matrix into an elementary form and to which the diagonal normalized noise-distribution matrix is invariant. In this appendix a derivation of this noise-transformation matrix is presented for the general case that the noise-source-power matrix has also anomalous defect (cf. sec. 4.1.5).

In accordance with eq. (4.32), let the diagonal and normalized noise-distribution matrix $\mathbf{N}_S$ be given by

$$\mathbf{N}_S = \begin{pmatrix} \mathbf{I}_p & & \\ & -\mathbf{I}_q & \\ & & \mathbf{O}_d \end{pmatrix}. \tag{E.1}$$

Further, let the positive semi-definite noise-source power matrix $\mathbf{G}_S$ have a rank r_G and a defect d_G. A non-singular matrix $\mathbf{C}^{(1)}$ can be found such that $\mathbf{G}_S$ is brought into a diagonal and normalized form $\mathbf{G}^{(1)}$ by a Hermitian congruence transformation, i.e.

$$\mathbf{G}^{(1)} = \mathbf{C}^{(1)}\mathbf{G}_S\mathbf{C}^{(1)\times} = \left(\begin{array}{c|c} \mathbf{I}_{r_G} & \\ \hline & \mathbf{O}_{d_G} \end{array}\right). \tag{E.2}$$

The corresponding Hermitely congruent transform of $\mathbf{N}_S$ is an Hermitian matrix which can be written in the form

$$\mathbf{N}^{(1)} = \mathbf{C}^{(1)}\mathbf{N}_S\mathbf{C}^{(1)\times} = \left(\begin{array}{c|c} \mathbf{N}_{11} & \mathbf{N}_{12} \\ \hline \mathbf{N}_{21} & \mathbf{N}_{22} \end{array}\right), \tag{E.3}$$

where $\mathbf{N}_{11}{}^{(1)}$ and $\mathbf{N}_{22}{}^{(1)}$ are Hermitian matrices of order r_G and d_G, respectively.

We shall now apply subsequently a number of Hermitian congruence transformations $\mathbf{C}^{(\nu)}$ in such a way that the matrix $\mathbf{G}^{(1)}$ is left unaltered and the matrix $\mathbf{N}^{(1)}$ is carried back again into an elementary form. It is not difficult to show that the appropriate transformation matrices $\mathbf{C}^{(\nu)}$ must all be of the form

$$\mathbf{C}^{(\nu)} = \left(\begin{array}{c|c} \mathbf{U}^{(\nu)} & \mathbf{A}^{(\nu)} \\ \hline \mathbf{O} & \mathbf{B}^{(\nu)} \end{array}\right), \tag{E.4}$$

*) For much of the content of this appendix the author is indebted to Prof. Dr N. G. de Bruijn.

where $\mathbf{U}^{(\nu)}$ is a unitary matrix of order r_G, $\mathbf{B}^{(\nu)}$ a non-singular square matrix of order d_G, and $\mathbf{A}^{(\nu)}$ an arbitary $r_G \times d_G$-matrix.

The first of these transformations is chosen so that the coupling between the ports which correspond to the zero noise sources is removed. The matrix $\mathbf{C}^{(2)}$ is then defined according to

$$\mathbf{C}^{(2)} = \left(\begin{array}{c|c} \mathbf{I} & \\ \hline & \mathbf{B}^{(2)} \end{array} \right) \tag{E.5}$$

and

$$\mathbf{B}^{(2)}\mathbf{N}_{22}{}^{(1)}\mathbf{B}^{(2)\times} = \left(\begin{array}{cc} \mathbf{\Pi}_t & \\ & \mathbf{O}_s \end{array} \right), \quad s = d_G - t, \tag{E.6}$$

where the polarity matrix $\mathbf{\Pi}_t$ is defined by

$$\mathbf{\Pi}_t = \left(\begin{array}{c|c} \mathbf{I}_{d_d} & \\ \hline & -\mathbf{I}_{d_a} \end{array} \right), \quad t = d_d + d_a. \tag{E.7}$$

The resulting matrix $\mathbf{N}^{(2)}$ is given in partitioned form by

$$\mathbf{N}^{(2)} = \left(\begin{array}{c|cc} \mathbf{N}_{11}{}^{(2)} & \mathbf{N}_{12}{}^{(2)} & \mathbf{N}_{13}{}^{(2)} \\ \hline \mathbf{N}_{12}{}^{(2)\times} & \mathbf{\Pi}_t & \\ \mathbf{N}_{13}{}^{(2)\times} & & \mathbf{O}_s \end{array} \right), \tag{E.8}$$

where

$$\mathbf{N}_{11}{}^{(2)} = \mathbf{N}_{11}{}^{(1)} \tag{E.9}$$

and

$$(\mathbf{N}_{12}{}^{(2)} \mid \mathbf{N}_{13}{}^{(2)}) = \mathbf{N}_{12}{}^{(1)}\mathbf{B}^{(2)\times}. \tag{E.10}$$

The following transformation is chosen so as to remove the coupling between the noisy ports which correspond to $\mathbf{N}_{11}{}^{(2)}$ and the noise-free, active and dissipative, ports which correspond to $\mathbf{\Pi}_t$. The corresponding matrix $\mathbf{C}^{(3)}$ is given in partitioned form by

$$\mathbf{C}^{(3)} = \left(\begin{array}{c|cc} \mathbf{I} & -\mathbf{N}_{12}{}^{(2)}\mathbf{\Pi}_t & \\ \hline & \mathbf{I}_t & \\ & & \mathbf{I}_s \end{array} \right). \tag{E.11}$$

The sub-matrices $\mathbf{N}_{12}{}^{(2)}$ and $\mathbf{N}_{12}{}^{(2)\times}$ are removed and the resulting matrix $\mathbf{N}^{(3)}$ is given by

$$\mathbf{N}^{(3)} = \left(\begin{array}{c|cc} \mathbf{N}_{11}{}^{(3)} & & \mathbf{N}_{13}{}^{(3)} \\ \hline & \mathbf{\Pi}_t & \\ \mathbf{N}_{13}{}^{(3)\times} & & \mathbf{O}_s \end{array} \right), \tag{E.12}$$

where

$$\mathbf{N}_{11}{}^{(3)} = \mathbf{N}_{11}{}^{(2)} - \mathbf{N}_{12}{}^{(2)}\mathbf{\Pi}_t\mathbf{N}_{12}{}^{(2)\times} \tag{E.13}$$

and

$$\mathbf{N}_{13}{}^{(3)} = \mathbf{N}_{13}{}^{(2)}. \tag{E.14}$$

The next transformation is intended to bring the coupling between the noiseless reactive ports, which correspond to $\mathbf{O}_s$, and the noisy ports, which correspond to $\mathbf{N}_{11}{}^{(3)}$, in as simple a form as possible. The matrix $\mathbf{N}_{13}{}^{(3)}$ should be transformed into the most elementary form possible. It is not always possible to make $\mathbf{N}_{13}{}^{(3)}$ equal to a zero matrix. The object set before can be achieved, if the transformation matrix $\mathbf{C}^{(4)}$ is of the form

$$\mathbf{C}^{(4)} = \left(\begin{array}{c|c} \mathbf{U}^{(4)} & \\ \hline & \begin{array}{cc} \mathbf{I}_t & \\ & \mathbf{B}_s{}^{(4)} \end{array} \end{array}\right). \tag{E.15}$$

The matrix $\mathbf{N}^{(3)}$ is then transformed into $\mathbf{N}^{(4)}$, i.e.

$$\mathbf{N}^{(4)} = \left(\begin{array}{c|c} \mathbf{U}^{(4)}\mathbf{N}_{11}{}^{(3)}\mathbf{U}^{(4)\times} & \mathbf{U}^{(4)}\mathbf{N}_{13}{}^{(3)}\mathbf{B}_s{}^{(4)\times} \\ \hline \mathbf{B}_s{}^{(4)}\mathbf{N}_{13}{}^{(3)\times}\mathbf{U}^{(4)\times} & \begin{array}{cc} \mathbf{\Pi}_t & \\ & \mathbf{O}_s \end{array} \end{array}\right). \tag{E.16}$$

The matrices $\mathbf{U}^{(4)}$ and $\mathbf{B}_s{}^{(4)}$ must now be determined so that the matrix $\mathbf{U}^{(4)}\mathbf{N}_{13}{}^{(3)}\mathbf{B}_s{}^{(4)\times}$ is as elementary as possible. We therefore proceed as follows.

The Gaussian transform $\mathbf{N}_{13}{}^{(3)\times}\mathbf{N}_{13}{}^{(3)}$ is a positive (semi-)definite Hermitian matrix of order s. Hence, a non-singular square matrix $\mathbf{F}$ of order s can be found such that the equation *)

$$\mathbf{N}_{13}{}^{(3)\times}\mathbf{N}_{13}{}^{(3)} = \mathbf{F}\begin{pmatrix} \mathbf{I}_{d_{G,A}} & \\ & \mathbf{O}_{d_r} \end{pmatrix}\mathbf{F}^{\times} \tag{E.17}$$

holds, where $d_{G,A}$ is the rank of $\mathbf{N}_{13}{}^{(3)\times}\mathbf{N}_{13}{}^{(3)}$ and d_r its defect. Let us now define an $r_G \times s$-matrix $\mathbf{K}$ by

$$\mathbf{K} = \mathbf{N}_{13}{}^{(3)}\,(\mathbf{F}^{\times})^{-1}. \tag{E.18}$$

By virtue of eq. (E.17), the relation

$$\mathbf{K}^{\times}\mathbf{K} = \begin{pmatrix} \mathbf{I}_{d_{G,A}} & \\ & \mathbf{O}_{d_r} \end{pmatrix} \tag{E.19}$$

then is valid. The first $d_{G,A}$ columns of $\mathbf{K}$ constitute an orthonormal set of r_G-dimensional vectors. This set can be augmented to form a complete set, i.e.

*) Even in the case of $r_G < s$ it can be shown that $d_{G,A} < r_G$ (see below). For the time being the definitions of $d_{G,A}$ and d_r are as made here.

an orthonormal base of r_G-dimensional vector space. These base vectors form the columns of a unitary matrix $\mathbf{V}$ of order r_G. The matrix $\mathbf{K}$ can then be written in the form

$$\mathbf{K} = \mathbf{V}\,\mathbf{E}, \tag{E.20}$$

where $\mathbf{E}$ is the $r_G \times s$-matrix which has zero elements except for the upper left-hand square sub-matrix of order $d_{G,A}$ which is equal to $\mathbf{I}_{d_{G,A}}$, i.e.

$$\mathbf{E} = \begin{pmatrix} \mathbf{I}_{d_{G,A}} & \\ & \mathbf{O}_l \end{pmatrix}, \tag{E.21}$$

where $\mathbf{O}_l$ is an $r_{G,N} \times d_r$-matrix with only zero elements. The numbers $r_{G,N}$ and d_r are given by

$$r_{G,N} = r_G - d_{G,A} \tag{E.22}$$

and

$$d_r = s - d_{G,A}. \tag{E.23}$$

We now define the sub-matrices $\mathbf{U}^{(4)}$ and $\mathbf{B}_s{}^{(4)}$ of the transformation matrix $\mathbf{C}^{(4)}$ by (cf. eq. (E.15))

$$\mathbf{U}^{(4)} = \mathbf{V}^{\times} \tag{E.24}$$

and

$$\mathbf{B}_s{}^{(4)} = \tfrac{1}{2}\,\mathbf{F}^{-1}. \tag{E.25}$$

By virtue of eqs (E.18) and (E.20) and of the matrix $\mathbf{V}$ being unitary, we then obtain

$$\begin{aligned} \mathbf{N}_{13}{}^{(4)} &= \mathbf{U}^{(4)}\mathbf{N}_{13}{}^{(3)}\mathbf{B}_s{}^{(4)\times} = \mathbf{V}^{\times}\mathbf{N}_{13}{}^{(3)}\,(\tfrac{1}{2}\,\mathbf{F}^{-1})^{\times} = \\ &= \tfrac{1}{2}\,\mathbf{V}^{\times}\mathbf{V}\,\mathbf{E} = \tfrac{1}{2}\,\mathbf{E}. \end{aligned} \tag{E.26}$$

The matrix $\mathbf{N}^{(4)}$ can now be written in the partitioned form

$$\mathbf{N}^{(4)} = \left(\begin{array}{cc|c} \mathbf{N}_{11}{}^{(4)} & \mathbf{N}_{12}{}^{(4)} & \tfrac{1}{2}\,\mathbf{I}_{d_{G,A}} \\ \mathbf{N}_{12}{}^{(4)\times} & \mathbf{N}_{22}{}^{(4)} & \\ \hline & & \\ \tfrac{1}{2}\,\mathbf{I}_{d_{G,A}} & & \begin{array}{cc} \mathbf{\Pi}_t & \\ & \mathbf{O}_{d_{G,A}} \\ & \mathbf{O}_{d_r} \end{array} \end{array}\right). \tag{E.27}$$

The matrices $\mathbf{N}_{11}{}^{(4)}$ and $\mathbf{N}_{22}{}^{(4)}$ are Hermitian sub-matrices of orders $d_{G,A}$ and $r_{G,N}$, respectively.

A further transformation is performed to remove the coupling between the noisy ports which correspond to $\mathbf{N}_{11}{}^{(4)}$. The appropriate matrix $\mathbf{C}^{(5)}$ is defined by

$$\mathbf{C}^{(5)} = \left(\begin{array}{cc|c} \mathbf{I}_{d_{G,A}} & & -\mathbf{N}_{11}{}^{(4)} \\ & \mathbf{I}_{r_{G,N}} & -2\mathbf{N}_{12}{}^{(4)\times} \\ \hline & & \mathbf{I}_t \\ & & \mathbf{I}_{d_{G,A}} \\ & & \mathbf{I}_{d_r} \end{array}\right). \tag{E.28}$$

The resulting matrix $\mathbf{N}^{(5)}$ is given by

$$\mathbf{N}^{(5)} = \left(\begin{array}{cc|c} \mathbf{O}_{d_{G,A}} & & \tfrac{1}{2}\mathbf{I}_{d_{G,A}} \\ & \mathbf{N}_{22}{}^{(4)} & \\ \hline & & \mathbf{\Pi}_t \\ \tfrac{1}{2}\mathbf{I}_{d_{G,A}} & & \mathbf{O}_{d_{G,A}} \\ & & \mathbf{O}_{d_r} \end{array}\right). \tag{E.29}$$

A last transformation of the type of eq. (E.4) is applied to remove the coupling of the noisy ports which correspond to $\mathbf{N}_{22}{}^{(4)}$. This can be accomplished when the appropriate matrix $\mathbf{C}^{(6)}$ is defined by

$$\mathbf{C}^{(6)} = \left(\begin{array}{cc|c} \mathbf{I}_{d_{G,A}} & & \\ & \mathbf{U}_{22}{}^{(6)} & \\ \hline & & \mathbf{I}_{d_G} \end{array}\right), \tag{E.30}$$

where $\mathbf{U}_{22}{}^{(6)}$ is the unitary modal matrix of $\mathbf{N}_{22}{}^{(4)}$. The equation

$$\mathbf{N}_{22}{}^{(6)} = \mathbf{U}_{22}{}^{(6)}\mathbf{N}_{22}{}^{(4)}\mathbf{U}_{22}{}^{(6)\times} = \operatorname{diag}(\lambda_i{}^{(6)}) \tag{E.31}$$

applies, where the $\lambda_i{}^{(6)}$'s are the real eigenvalues of $\mathbf{N}_{22}{}^{(4)}$.

Furthermore, we apply a normalization transformation $\mathbf{C}^{(7)}$ (cf. sec. 4.1.3) given by

$$\mathbf{C}^{(7)} = \left(\begin{array}{cc|c} \mathbf{I}_{d_{G,A}} & & \\ & \mathbf{G}_{r_{G,N}}{}^{1/2} & \\ \hline & & \mathbf{I}_{d_G} \end{array}\right), \tag{E.32}$$

with

$$\mathbf{G}_{r_{G,N}}{}^{1/2} = \operatorname{diag}(g_i{}^{1/2}), \tag{E.33}$$

$$g_i = \begin{cases} |\lambda_i{}^{(6)}|^{-1}, & \text{for} \quad \lambda_i{}^{(6)} \neq 0, \\ 1 \quad , & \text{for} \quad \lambda_i{}^{(6)} = 0, \end{cases} \tag{E.34}$$

and a permutation transformation $\mathbf{C}^{(8)}$ which rearranges the rows and columns of $\mathbf{N}^{(7)}$ and $\mathbf{G}^{(7)}$ so as to correspond again to the arrangement in $\mathbf{N}_S$ given in eq. (E.1). The result is contained in the equations

$$\mathbf{N}^{(8)} = \begin{pmatrix} \mathbf{I}_{r_d} & & & & & & & \\ & \mathbf{I}_{d_d} & & & & & & \\ & & \tfrac{1}{2}\mathbf{I}_{d_{G,A}} & & & & & \\ & & \tfrac{1}{2}\mathbf{I}_{d_{G,A}} & & & & & \\ & & & -\mathbf{I}_{r_a} & & & & \\ & & & & -\mathbf{I}_{d_a} & & & \\ & & & & & \mathbf{O}_{r_r} & & \\ & & & & & & \mathbf{O}_{d_r} \end{pmatrix} \tag{E.35}$$

and

$$\mathbf{G}^{(8)} = \begin{pmatrix} \mathbf{G}_{r_d} & & & & & & & \\ & \mathbf{O}_{d_d} & & & & & & \\ & & \mathbf{I}_{d_{G,A}} & & & & & \\ & & & \mathbf{O}_{d_{G,A}} & & & & \\ & & & & \mathbf{G}_{r_a} & & & \\ & & & & & \mathbf{O}_{d_a} & & \\ & & & & & & \mathbf{G}_{r_r} & \\ & & & & & & & \mathbf{O}_{d_r} \end{pmatrix}. \tag{E.36}$$

The matrices $\mathbf{G}_{r_d}$, $\mathbf{G}_{r_a}$ and $\mathbf{G}_{r_r}$ are positive and diagonal. The orders of the diagonal principal sub-matrices satisfy the following algebraic equations:

$$\left.\begin{aligned} r_G &= r_{G,N} + d_{G,A}, \\ d_G &= d_{G,N} + d_{G,A}, \\ p &= r_d + d_d + d_{G,A}, \\ q &= r_a + d_a + d_{G,A}, \\ d &= r_r + d_r, \\ r_{G,N} &= r_d + r_a + r_r, \\ d_{G,N} &= d_d + d_a + d_r. \end{aligned}\right\} \tag{E.37}$$

To carry $\mathbf{N}^{(8)}$ back into the form of $\mathbf{N}_S$ of eq. (E.1), a last transformation $\mathbf{C}^{(9)}$ must be applied, viz. the one given by

$$\mathbf{C}^{(9)} = \begin{pmatrix} \mathbf{I} & & & \\ & \mathbf{I}_{d_{G,A}} & \mathbf{I}_{d_{G,A}} & \\ & \mathbf{I}_{d_{G,A}} & \mathbf{I}_{d_{G,A}} & \\ & & & \mathbf{I} \end{pmatrix}. \tag{E.38}$$

The framed sub-matrix corresponds to the framed sub-matrices of $\mathbf{N}^{(8)}$ and $\mathbf{G}^{(8)}$, being the only sub-matrices which are altered by the transformation. The ultimate result is expressed by the modified forms of eqs (E.35) and (E.36), i.e.

$$\mathbf{N}^{(9)} = \begin{pmatrix} & & & \\ & \mathbf{I}_{d_{G,A}} & & \\ & & -\mathbf{I}_{d_{G,A}} & \\ & & & \end{pmatrix} = \mathbf{N}_S \tag{E.39}$$

and

$$\mathbf{G}^{(9)} = \begin{pmatrix} \boxed{\begin{matrix} \mathbf{I}_{d_{G,A}} & \mathbf{I}_{d_{G,A}} \\ \mathbf{I}_{d_{G,A}} & \mathbf{I}_{d_{G,A}} \end{matrix}} \end{pmatrix} = \mathbf{G}_{S,D}. \tag{E.40}$$

Summarizing, we have found that the transformation $\mathbf{C}$ which is the product of all the above transformations, i.e.

$$\mathbf{C} = \mathbf{C}^{(9)}\,\mathbf{C}^{(8)}\,\mathbf{C}^{(7)}\,\mathbf{C}^{(6)}\,\mathbf{C}^{(5)}\,\mathbf{C}^{(4)}\,\mathbf{C}^{(3)}\,\mathbf{C}^{(2)}\,\mathbf{C}^{(1)}, \tag{E.41}$$

transforms $\mathbf{G}_S$ into the elementary form $\mathbf{G}_{S,D}$, while leaving $\mathbf{N}_S$ unaltered.

There is left one ambiguity. In the right-hand side of eq. (E.25) a scalar factor $\frac{1}{2}$ has been introduced rather arbitrarily. Via eq. (E.26), etc., this factor is transported unto eq. (E.35) where it is again removed from $\mathbf{N}$ by a proper choice of the transformation matrix $\mathbf{C}^{(9)}$ in eq. (E.38). Instead of $\frac{1}{2}$ any other non-zero finite scalar factor γ could have been introduced in eq. (E.25), if only the matrix $\mathbf{C}^{(9)}$ had been chosen appropriately so as to make the ultimate form of $\mathbf{N}^{(9)}$ equal to $\mathbf{N}_S$ (cf. eqs (E.1) and (E.39)). The latter is accomplished by multiplying the identity matrices $I_{d_{G,A}}$ in the framed part of $\mathbf{C}^{(9)}$ by $1/2\gamma$. As a result the framed part of $\mathbf{G}^{(9)}$ of eq. (E.40) is multiplied by $1/2\gamma$, too. The scalar γ cannot be chosen infinite or zero because then the matrices $\mathbf{C}^{(9)}$ or $\mathbf{C}^{(4)}$ would, respectively, become singular, which is not permitted. Hence, the framed part of $\mathbf{G}_{S,D}$ is non-zero and finite, but its amplitude is not determined uniquely.

In secs 4.1.6 and 4.1.7 we were interested in transforming the reduced matrices $\mathbf{N}_S$ and $\mathbf{G}_S$. In the sections which preceded sec. 4.1.6, those matrices were reduced to $\mathbf{N}_S^{(4)}$ and $\mathbf{G}_S^{(4)}$ as given by respectively eqs (4.64) and (4.65). The reduction implied that several of the orders of the principal sub-matrices of $\mathbf{N}_S$ and $\mathbf{G}_S$ were made zero, i.e.

$$d = d_d = d_a = d_r = d_{G,N} = 0 \tag{E.42}$$

and

$$r_r = 0. \tag{E.43}$$

Under these conditions the above considerations signify that the transformation matrix $\mathbf{C}$ of eq. (E.41) is a noise-transformation matrix $\mathbf{\Gamma}^{(5)}$ which does not alter $\mathbf{N}_S^{(4)}$ and which brings $\mathbf{G}_S^{(4)}$ into an elementary form. This has been expressed by eqs (4.86) and (4.87).

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Summary

Exchange of energy between multiports which form the components of linear noisy systems can, in the frequency domain, be described by relations between signals present at the ports of the multiports. Two types of signals are distinguished: harmonic signals and noisy signals. The latter are not as precisely defined as the former. As harmonic signals are denoted by complex numbers, so noisy signals are denoted by elements of an infinitely dimensional Hilbert space.

In most treatments of electrical noisy systems signals are described in terms of voltages and currents. However, many systems lend themselves better to a description in other variables, e.g. amplitudes of incident and reflected waves. The larger part of chapter 2 is therefore devoted to the formulation of the scattering-matrix signal representation of noisy multiports. In analogy to a harmonic wave the concept of noise wave is introduced. In essence, it is a power-density wave which is scattered by a multiport in the same manner as a harmonic wave. Interference effects are, however, not determined simply by relative phases. A general product of two elements of the Hilbert space is introduced to define correlation.

To calculate the transmission of power between multiports a set of noise-wave equations is introduced. A noisy n-port is then equivalent to a noiseless n-port with n noise-wave sources at the ports. It is shown that this formalism is consistent. Moreover, it is shown that for systems which consist of passive multiports at uniform temperatures the requirements of the laws of thermodynamics are satisfied. The contribution of a multiport to the noise in a system is completely determined by a noise-power-density matrix. This matrix is factorized into two matrices. One of these, the noise-distribution matrix, depends on the signal properties only, while the other, the noise-temperature matrix, determines the intensity of the noise.

The formalism is inherently restricted to spot-noise theory. Requirements for applications over finite bandwidths are considered.

In chapter 3 the representation of signals and of noise contributions of multiports in arbitrary variables is discussed. When a set of variables is complete, it constitutes a set of coordinates for the signal-state space. The transformation from one coordinate system to another is investigated. The numbers of positive, zero, and negative eigenvalues of the noise-distribution matrix, and of the power-absorption matrix, are invariants. They can serve to classify a multiport with respect to dissipativity, reactivity, and activity. Moreover, the eigenvalues of the noise-temperature matrix, i.e. the characteristic noise temperatures, are invariants. They form unique measures of the noise intensity.

When an n-port is imbedded in a $2n$-port, it is transformed to another n-port of the same nature. Under certain conditions there is a one-to-one correspondence between the signal-performance matrices of the transforming $2n$-ports and certain subgroups of coordinate-transformation matrices; these define the groups of canonical coordinate transformations.

In chapter 4 it is proved that in any so-called paired signal representation and for any multiport there exists a canonical transformation which diagonalizes simultaneously the signal performance and the noise (Haus and Adler [15])). The proof is constructive so that the diagonalizing transformation is found explicitly. There is one exception only. In the case of two completely correlated noisy 1-ports which are active and dissipative, respectively, the simultaneous diagonalization cannot be performed. An example of such a pair is a grounded-grid triode with only shot noise and at low frequencies. Another example, a transistor at low frequencies, is discussed.

A diagonalized n-port consists of n independent noisy 1-ports. The equivalent noise temperatures of these 1-ports are equal to the characteristic noise temperatures. They are positive for dissipative and negative for active 1-ports.

In the last chapter the theory of Haus and Adler is reviewed and discussed. Following the work of Rummler [26]) the optimum performance of a multiport amplifier is determined. This is uniquely defined by the negative characteristic noise temperature which has the smallest magnitude. A simple model of a combination of a signal source and an amplifier, containing the essential features of source, amplifier, and associated circuit, is given.

Finally, it is concluded that the noise-temperature matrix and the corresponding characteristic noise temperatures give in any signal representation a unique and absolute measure of the noise performance of a multiport.

Samenvatting

De energie-uitwisseling tussen de veelpoorten, waaruit lineaire systemen met ruis zijn opgebouwd, kan als functie van de frekwentie worden beschreven door betrekkingen tussen de signalen, die aanwezig zijn aan de poorten van die veelpoorten. Er wordt onderscheid gemaakt tussen twee soorten signalen, te weten, harmonische signalen en ruis-signalen. De laatste zijn niet zo nauwkeurig gedefiniëerd als de eerste. Zoals harmonische signalen worden voorgesteld door complexe getallen, zo worden ruis-signalen voorgesteld door elementen van een oneindig dimensionale Hilbert-ruimte.

In het merendeel der verhandelingen over elektrische systemen met ruis worden signalen beschreven in termen van elektrische spanningen en stromen. Vele systemen lenen zich echter beter tot een beschrijving in andere variabelen, zoals bij voorbeeld amplitudes van invallende en gereflecteerde golven. Hoofdstuk 2 is daarom grotendeels gewijd aan het formuleren van de verstrooiïngsmatrix-signaal-representatie van ruisende veelpoorten. Het begrip ruisgolf wordt ingevoerd naar analogie met het begrip harmonische golf. Een ruisgolf is in wezen een vermogensdichtheid-golf, die door een veelpoort op dezelfde manier wordt verstrooid als een harmonische golf. De gevolgen van interferentie worden echter niet op eenvoudige wijze door relatieve fasen bepaald. Er wordt daarom een algemeen product van twee elementen van de Hilbertruimte ingevoerd om correlatie te kunnen beschrijven.

Om de overdracht van energie tussen veelpoorten te berekenen wordt een stelsel ruisgolf-vergelijkingen ingevoerd. Een n-poort met ruis is dan gelijkwaardig aan een ruisvrije n-poort met n ruisgolf-bronnen aan zijn poorten. Er wordt aangetoond, dat het formalisme consistent is. Bovendien wordt bewezen, dat er wordt voldaan aan de eisen, die de wetten van de thermodynamica stellen voor systemen, die bestaan uit veelpoorten met uniforme temperaturen. De bijdrage van een veelpoort aan de ruis in een systeem wordt geheel bepaald door een ruis-vermogensdichtheid-matrix. Deze matrix kan worden geschreven als een produkt van twee andere matrices. Eén van deze, de ruis-verdelingmatrix, is slechts afhankelijk van de signaal-eigenschappen van de veelpoort, terwijl de andere, de ruis-temperatuur-matrix, de intensiteit van de ruis vastlegt.

Het formalisme is in wezen beperkt tot „spot noise", d.w.z. zij is slechts geldig voor één frekwentie. De uitbreiding, die noodzakelijk is voor toepassing over eindige frekwentie-banden, wordt beschouwd.

In hoofdstuk 3 wordt de beschrijving van signalen en van de bijdragen aan ruis van veelpoorten in willekeurige grootheden besproken. Als een stelsel variabelen volledig is, kan het worden beschouwd als de coördinaten voor de signaaltoestand-ruimte. De transformatie van één coördinatensysteem naar een ander wordt onderzocht. De aantallen positieve, nul zijnde en negatieve eigenwaarden van de ruis-verdeling-matrix, en van de vermogensabsorptie-

matrix, zijn invariant. Zij kunnen dienen om een veelpoort te klassificeren met betrekking tot dissipativiteit, reactiviteit en activiteit. De eigenwaarden van de ruis-temperatuur-matrix, de karakteristieke ruistemperaturen, zijn bovendien invariant. Zij vormen een eenduidige maat voor de intensiteit van de ruis.

Als een n-poort wordt ingebed in een $2n$-poort, wordt hij getransformeerd tot een andere n-poort van dezelfde soort. Indien er aan zekere voorwaarden wordt voldaan, bestaat er een een-eenduidige betrekking tussen de signaal-overdracht-matrices van de transformerende $2n$-poorten en bepaalde deelgroepen van coördinaten-transformatie-matrices; deze definiëren de groepen van kanonieke coördinaten-transformaties.

In hoofdstuk 4 wordt bewezen, dat er voor iedere zogenaamd gepaarde signaal-representatie en voor iedere veelpoort een kanonieke transformatie bestaat, die de signaal-eigenschappen en de ruis-eigenschappen van die veelpoort gelijktijdig in diagonale vorm brengt (Haus en Adler [15]). Het bewijs is constructief van aard, zodat de diagonaliserende transformatie expliciet gevonden wordt. Er is slechts één uitzondering. Voor het geval van twee 1-poorten met volledig gecorreleerde ruis, die respectievelijk dissipatief en actief zijn, kan de gelijktijdige diagonalisering niet worden uitgevoerd. Een voorbeeld van zo'n paar is een triode, die slechts hagelruis heeft, in een geaarde roosterschakeling en bij lage frekwenties. Een ander voorbeeld, een transistor bij lage frekwenties, wordt besproken.

Een gediagonaliseerde n-poort bestaat uit n onafhankelijke 1-poorten met ruis. De equivalente ruistemperaturen van die 1-poorten zijn gelijk aan de karakteristieke ruistemperaturen. Zij zijn positief voor dissipatieve en negatief voor actieve 1-poorten.

In het laatste hoofdstuk wordt een kort overzicht gegeven van de theorie van Haus en Adler en deze wordt besproken. De optimale prestatie van een veelpoort-versterker wordt bepaald analoog aan het werk van Rummler [26]). Die optimale prestatie wordt eenduidig vastgelegd door de karakteristieke ruistemperatuur, die de kleinste absolute waarde heeft. Een model van een kombinatie van een signaalbron en een versterker, die alle wezenlijke kenmerken van bron, de versterker en het verbindingscircuit bevat, wordt besproken.

Tenslotte stellen we, dat de ruis-temperatuur-matrix en de daarbij behorende karakteristieke ruistemperaturen voor iedere signaal-representatie een eenduidige en absolute maat vormen voor het ruisgedrag van een veelpoort.

Levensbericht

Op 21 april 1931 werd Hendrik Bosma geboren te Grootegast. In 1947 behaalde hij het diploma MULO-B en in 1949 aan de 2de Christelijke HBS te Groningen het diploma HBS-B.

Van 1949 tot 1956 studeerde hij aan de Technische Hogeschool te Delft, waar hij van 1953 tot 1956 student-assistent was in het Laboratorium voor Theoretische Elektriciteitsleer. In 1956 behaalde hij met lof het diploma elektrotechnisch ingenieur.

Na zijn militaire dienstplicht bij de Koninklijke Luchtmacht te hebben vervuld, trad hij in 1958 in dienst bij de N.V. Philips' Gloeilampenfabrieken te Eindhoven als wetenschappelijk medewerker in het Natuurkundig Laboratorium.

In 1966 werd hem door de Microwave Theory and Techniques Group van The Institute of Electrical and Electronics Engineers de "1965 Microwave Prize" toegekend.

STELLINGEN

H. Bosma, 31 Januari 1967

I

In een actieve veelpoort of in een veelpoort met ruis kunnen, bij één frequentie, met behulp van metingen aan de poorten niet meer onafhankelijke signaal- respectievelijk ruis-bronnen worden onderscheiden dan er poorten zijn.

Dit proefschrift, paragraaf 2.5.

II

Het door Haus en Adler ingevoerde begrip uitwisselbaar vermogen, gedefinieerd als het eindige vermogen, dat maximaal aan een één-poort kan worden onttrokken (beschikbaar vermogen) of er aan kan worden toegevoerd, heeft in het laatste geval slechts wiskundige betekenis.

H. A. Haus en R. B. Adler, Proc. Inst. Radio Engrs **45**, 690-691, 1957.

III

Zoals in dit proefschrift beschreven, kunnen reflectievrije twee-poorten, waarvan de ruisbronnen aan de ingang en de uitgang niet gecorreleerd zijn, met betrekking tot het aan de uitgang afgegeven ruisvermogen worden gekarakteriseerd door een karakteristieke ruistemperatuur. Voor versterkende twee-poorten is deze „temperatuur" negatief en voor verzwakkende positief. In dit verband kan een versterker worden opgevat als „heter dan heet".

De karakteristieke ruistemperaturen bepalen de met betrekking tot de ruiseigenschappen gunstigste volgorde van zulke twee-poorten in een keten in die zin, dat de „heetste" voorop en de „koudste" achteraan geschakeld dient te worden.

Dit proefschrift, hoofdstuk 5.

IV

De uitspraak van Haus en Adler: „Results, pertaining to the singular case can be obtained in the limit of vanishing added loss", betrekking hebbend op veelpoorten met singuliere absorptie- en ruisverdeling-matrices, is in zijn algemeenheid niet juist.

H. A. Haus en R. B. Adler, "Circuit theory of linear noisy networks", John Wiley, New York, 1959, p. 16.

V

Penfield berekent op twee manieren de „noise measure" van een twee-poort versterker, die bestaat uit een verliesvrije drie-poort, waarvan één poort is afgesloten met een „negatieve" weerstand. Met een directe methode vindt hij voor alle mogelijke drie-poorten een zelfde, door de „negatieve" weerstand bepaalde, positieve waarde. In tegenspraak hiermee concludeert hij uit de resultaten van een door Haus en Adler aangegeven methode, dat alle waarden groter dan de bovengenoemde positieve of kleiner dan een andere negatieve mogelijk zouden zijn.

Deze conclusie is niet juist en er is geen tegenstrijdigheid tussen de uitkomsten van de twee methoden.

P. Penfield, I.R.E. Trans. Circuit Theory 7, 166-170, 1960.

VI

Het gebruik van het begrip equivalente ruisweerstand ter karakterisering van de ruis van een actieve elektrische netwerk-komponent, zoals bijv. een pentode, dient op didactische gronden te worden ontraden.

VII

Het verdient aanbeveling bij de opleiding tot elektrotechnisch ingenieur aandacht te besteden aan de thermodynamische eigenschappen van elektrische, magnetische en elektromagnetische systemen.

VIII

Van de formule van Kittel voor de frequentie van ferromagnetische resonantie in kleine ellipsoïden wordt veelvuldig een verkeerd gebruik gemaakt.

IX

De ingangssignalen van een drievoudig rotatie-symmetrisch netwerk met drie ingangen, zoals die worden vastgelegd door de drie eigenvektoren van bijv. de admittantie-matrix, vormen respectievelijk een stilstaand, een rechtsom draaiend en een linksom draaiend stelsel. De door Konishi afgeleide uitdrukking voor het verschil van die eigenwaarden van de admittantie-matrix, die betrekking hebben op de linksom en de rechtsom draaiende eigenvektoren, is niet juist.

Y. Konishi, NHK Techn. Monograph, Nr. 6, Techn. Res. Lab., Japan Broadcasting Corporation, May 1965.
Y. Konishi, Tech. J. Japan Broadc. Corp. **17**, Nr. 2, 87-121, 1965.

X

De suggestie van Bosma, transmissie-filters te maken met twee, of meer, op onderling korte afstand (kort t.o.v. de golflengte) in een rechthoekige golfpijp geplaatste capacitieve schermen met kleine spleten, leidt niet tot praktisch bruikbare resultaten.

H. Bosma, Appl. Scient. Res. (B) 7, 131-144, 1959.

XI

De militaire „exercitie" is verouderd en overbodig.

XII

Indien binnen de Societas Studiosorum Reformatorum, of binnen een andere studentenvereniging met een levensbeschouwelijke grondslag, de eigen levensbeschouwing niet onderwerp van discussie is, dan dient de bestaanszin van die vereniging te worden betwijfeld.

XIII

Het leveren van stellingen bij een proefschrift is niet in overeenstemming met de huidige praktijk van wetenschapsbeoefening.